

Co-Design and Evaluation of a Smartphone-based Mental Health Promotion and Anti-stigma Embodied Conversational Agent for Church-affiliated Black Adults

by

Teresa Kenyon O’Leary

To the Faculty of the Graduate School
of the Khoury College of Computer Sciences
in partial fulfillment of the requirements for the degree of
Doctor of Philosophy
(Personal Health Informatics)
Northeastern University
Boston, Massachusetts
August 2023

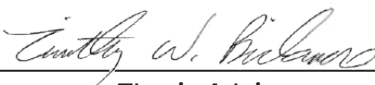
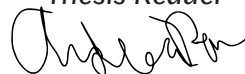
Doctoral Committee:
Professor Timothy Bickmore, Advisor
Professor Alisa Lincoln
Associate Professor Matthew Goodwin
Associate Professor Andrea G. Parker, Georgia Tech

Thesis Title: Co-Design and Evaluation of a Smartphone-based Mental Health Promotion and Anti-stigma Embodied Conversational Agent for Church-affiliated Black Adults

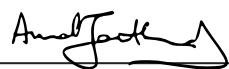
Author: Teresa K. O'Leary

PhD Program: ☐ Computer Science ☐ Cybersecurity ☒ Personal Health Informatics

PhD Thesis Approval to complete all degree requirements for the above PhD program.

 _____ Thesis Advisor	<u>11/14/23</u> _____ Date
 _____ Thesis Reader	<u>11/15/23</u> _____ Date
 _____ Thesis Reader	<u>11/15/23</u> _____ Date
 Goodwin, Matthew Digitally signed by Goodwin, Matthew Date: 2023.11.14 18:04:29 -05'00' _____ Thesis Reader	<u>11/14/23</u> _____ Date
_____ Thesis Reader	_____ Date

KHOURY COLLEGE APPROVAL:

 _____ Associate Dean for Graduate Programs	<u>11/28/2023</u> _____ Date
--	------------------------------------

COPY RECEIVED BY GRADUATE STUDENT SERVICES:

 _____ Recipient's Signature	<u>29 November 2023</u> _____ Date
---	--

Abstract

Depression is the leading cause of disability worldwide, with significant racial and ethnic disparities in diagnosis and treatment. Although Black adults in the United States are disproportionately exposed to psychosocial stressors, only one in four Black adults who need mental healthcare receive it, partly due to mental health stigma. This dissertation investigates using a smartphone-based Embodied Conversational Agent (ECA) counselor to reduce mental health stigma in the community to support mental health care-seeking intentions. Given the importance of religion in the lives of many African Americans and pre-existing close social relationships, this work was centered in predominately Black and African American faith communities. My dissertation research is the culmination of four years of community-engaged work in predominantly Black churches in Boston.

During initial formative work, I created unique co-design methodologies to facilitate participant design contributions as a mechanism for designing technical innovations that reflect community goals, priorities, and cultural values. During this phase, participants identified their communities' assets and their most pressing health concerns, contributed to early design concepts and modalities, and emphasized the importance of creating features that address multiple facets of well-being. This work culminated in a field deployment that evaluated an initial version of an ECA-based mHealth application intervening on physical, social, and spiritual well-being.

Leveraging these findings from earlier phases of this research, I synthesized a set of guiding design principles for developing digital mental health promotion interventions for predominantly Black church communities. In conjunction with principles from stigma theory and counseling psychology, I developed and evaluated an ECA-facilitated depression and stigma-reduction intervention in a 14-day field deployment with 12 church members. The primary aim of this work was to develop a smartphone-based ECA to promote mental health and build a culture of community support for those experiencing mental illness. Participants found the mental health and stigma-reduction intervention acceptable, providing a range of satisfaction ratings significantly above neutral. Qualitative findings revealed how the application facilitated shifts in attitudes toward mental health and illness, primed care-seeking behaviors, and was used to support members experiencing mental illness.

Acknowledgements

Dear reader, if I have learned one thing during this academic journey, it would be that research and dissertations are never done alone. Over the last five years, I have had the invaluable support of mentors, lab mates, family, friends and participants. I'm lucky to have such a large community of people who have rallied around me.

First, this work was conducted alongside and in collaboration with a team of investigators, support staff, and other PhD students. This work could not have come to fruition without these folks, and I'm grateful for their support and insight.

To my advisor, Timothy Bickmore, you gave me the resources and support required to complete this PhD successfully. There were many moments during this PhD where I didn't believe I could make a deadline or execute my ideas. You encouraged and believed in my work, even when I doubted it. I achieved more because you believed I could.

To my committee, your unending support allowed me to investigate an area of research I am passionate about. Alisa, I have learned an incredible amount from you, and I thank you for encouraging me to follow this research path. Blending my love for public health, health tech, and sociology has been more than rewarding. Andrea, I cannot describe how much your support has meant to me. There were critical moments during this program when you encouraged me to keep pushing forward. Every time I write an article, I smile because I hear you tell me to 'unpack' that thought or 'say more.' Your expertise has made me a better scholar. Matthew, not only did you serve on my committee and provide me with great alternative viewpoints to consider, but your love of theory is a present thread I hope you recognize throughout this work.

To my lab mates, old and new—Stefan, Lily, Eve, Shuo, Ameneh, Dhaval, Prasanth, Juan, Hye Sun, Yunus, Farnaz, Mina, Ian, Armin, and Madison. Through your expertise, feedback, laughter, and encouragement—you all have played an essential role in my achieving this degree.

To my Mom, in the words of Cass Elliot, you encouraged me to *"make my own kind of music, sing my own special song. Even if nobody else sings along."* You seemingly knew my path would be unique, with each career step leading to some new twist. Graduate school was a huge risk as I stepped away from a comfortable career, where I didn't feel well understood, to follow a new academic dream at 31. I'm not sure anyone else understood why I did it, including myself at times, but I know you did. Ultimately, Mom, you were right; I finally found others who could appreciate, challenge, and champion my work and finally *sing along with me.*

To my brother, Jason, your weekly phone calls were something I came to rely on and look forward to over the past two years. You were always there with advice. Our paths are different, but you have always supported my passion for helping others.

To my dearest life-long friends, Kate and Jeff, Crys and Benji, Dara and JJ, Kai and Emily, Monique, Marissa, Rachel, and June. You have been lifelong friends who have taken my calls for help day and night. You all believed I could do this and never wavered. At every turn, you volunteered your help. I thank you.

To my stepchildren, Marshall and Carlyn, I cannot describe what it has meant to have you support this dream of mine while you were in college. We were all students, and even though you each graduated before me (yay, kids!), I'll always treasure the time we spent on the phone discussing school life and the new ways I could relate to your experiences. I'm sorry that this degree took time away from you. Please know that all the special trips we still managed to take, the regular moments we shared, and the holidays we celebrated were the moments that kept me going and sustained me. You both have inspired me.

To my husband, Bruce, you were undoubtedly my biggest daily supporter. You cooked our meals and cleaned the house; you reminded me to take breaks. You planned trips so that I could relax. You were my alpha tester when I needed it and my practice audience for every presentation. Multiple times, you were there without a word to pick me up from my lab in Boston at 1 a.m. when I realized I had missed the last train to Worcester. There were numerous times when I told you I couldn't complete this degree; you would smile and say you knew I could. Your consistency and steadfast love have been everything I needed and more. I thank you for taking this journey with me.

Finally, this work was done in partnership with participants. To my participants, I cannot fully express the privilege of working with you. I hope this work does justice to all the lessons you taught and the stories you shared with me during these five years. This work will always be more than a research project to me.

Table of Contents

ABSTRACT	II
ACKNOWLEDGEMENTS.....	III
CHAPTER 1: INTRODUCTION	1
1.1 CONTRIBUTIONS.....	3
1.2 ORGANIZATION	4
CHAPTER 2: RELATED WORK.....	7
2.1 ROLE OF THE BLACK CHURCH	7
2.2 FAITH-BASED HEALTH INITIATIVES FOR CHURCH-AFFILIATED BLACK ADULTS	8
2.3 FAITH-BASED DIGITAL HEALTH TOOLS	9
2.4 CULTURAL TAILORING FOR HEALTH PROMOTION INTERVENTIONS	9
<i>2.4.1 Recommendations for Cultural Tailoring Mental Health Promotion Interventions.....</i>	<i>11</i>
2.5 MENTAL HEALTH STIGMA THEORY.....	11
<i>2.5.1 The Life-limiting Impact of Stigma.....</i>	<i>11</i>
<i>2.5.2 Intersectional Complexities of Race and Religiosity on Mental Health Stigma and Treatment Utilization</i>	<i>12</i>
<i>2.5.3 Self-stigma: a stigma variant.....</i>	<i>15</i>
<i>2.5.4 Public stigma: a stigma variant</i>	<i>15</i>
2.6 STIGMA-REDUCTION INTERVENTIONS	16
2.7 DIGITAL STIGMA-REDUCTION INTERVENTIONS.....	19
2.8 TAKEAWAYS FROM THE STIGMA-REDUCTION INTERVENTION LITERATURE	20
2.9 DIGITAL INTERVENTIONS TO ENHANCE READINESS FOR THERAPY	20
2.10 EMBODIED CONVERSATIONAL AGENTS IN MENTAL HEALTHCARE	21
<i>2.10.1 Screening</i>	<i>21</i>
<i>2.10.2 Psychoeducation and Skill-building.....</i>	<i>21</i>
<i>2.10.3 Detection</i>	<i>22</i>

2.10.4 Stigma Reduction.....	22
2.10.5 Takeaways from the ECA Mental Health Promotion Literature.....	22
CHAPTER 3: PHASE I – UNDERSTANDING COMMUNITY CONTEXT	23
3.1 PARTICIPANT RECRUITMENT PHASE I & PHASE II	23
3.2 PHASE I PARTICIPANT OVERVIEW.....	24
3.3 QUALITATIVE ANALYTIC PROCEDURES COMMON TO PHASE I AND PHASE II.....	24
3.4 FOCUS GROUP 1: UNDERSTANDING ORGANIZATIONAL MISSION & HISTORY – “PEOPLE FOLLOW THE PASTORS LEAD”	25
3.4.1 Purpose.....	25
3.4.2 Method	25
3.4.3 Findings.....	25
3.5 FOCUS GROUP 2: EXPLORING THE EXCHANGE OF SOCIAL SUPPORT “YOU CAN GET WHATEVER YOU NEED IN THE CHURCH.”	27
3.5.1 Purpose	27
3.5.2 Method	28
3.5.3 Findings.....	28
3.6 FOCUS GROUP 3: DESCRIBING PERSONAL USE AND CHURCH TECHNOLOGY UTILIZATION “I WAS SHOCKED. THAT’S ONE MODERN SCREEN APP. I COULDN’T HAVE IMAGINED IT.”	31
3.6.1 Method	32
3.6.2 Findings.....	32
3.6.3 Conclusion.....	34
CHAPTER 4: PHASE II PARTICIPATORY DESIGN FOCUS GROUPS.....	36
4.1 METHODS COMMON TO PHASE II FOCUS GROUPS.....	37
4.2 FOCUS GROUP 1: ESTABLISHING HEALTH PRIORITIES – “IF YOU DON’T TALK ABOUT MENTAL HEALTH, YOU AREN’T DESIGNING FOR OUR COMMUNITY.”	37
4.2.1 Purpose	37
4.2.2 Method	38
4.2.3 Findings.....	38

4.2.4 Implications for Design	41
4.3 FOCUS GROUP 2: CO-DESIGNING A CULTURALLY-TAILORED ECA – “SHE’S SPEAKING MY LANGUAGE!”	41
4.3.1 Method	42
4.3.2 Findings.....	44
4.3.3 Design Implications.....	46
4.4 FOCUS GROUP 2B: TAILORING EMBODIED CONVERSATIONAL AGENT DIALOGUE	47
4.4.1 Purpose	47
4.4.2 Method	48
4.4.3 Findings.....	49
4.4.4 Design Implications.....	52
4.5 FOCUS GROUP 3: EXPLORING FEATURES FOR IN-APP SOCIAL SUPPORT USING MODULAR STORYBOARDS.....	52
4.5.1 Procedure.....	53
4.5.2 Findings.....	54
4.6 FOCUS GROUP 4: EXAMINING THE INTERSECTION OF RACE, RELIGION, AND TECHNOLOGY	56
4.6.1 Purpose	56
4.6.2 Method	57
4.6.3 Findings.....	59
4.6.4 Design Implications.....	68
CHAPTER 5: PHASE III CHURCH CONNECT FIELD DEPLOYMENT.....	69
5.1 FIELD DEPLOYMENT	69
5.1.1 Church Connect Field Deployment System Design.....	70
5.1.2 Procedure.....	72
5.1.3 Findings.....	73
5.1.4 Mental Health Design Implications.....	78
CHAPTER 6: PHASE IV MENTAL HEALTH FEATURE SELECTION	80
6.1 PURPOSE	80

6.2 PROCEDURE	80
6.3 FINDINGS.....	82
6.3.1 Participant Overview	82
6.3.2 Current Church-based Practices for Supporting Mental Well-being	82
6.3.3 Feature Ranking.....	84
6.3.4 Actionable Strategies for Coping	84
6.3.5 Crowdsourcing Structured Informational Social Support.....	84
6.3.6 Potential Future Impact	85
6.4 SYNTHESIS OF PHASES I-IV	86
CHAPTER 7: MENTAL HEALTH AND STIGMA REDUCTION SYSTEM DESIGN	87
7.1 JUSTIFICATION FOR TWO USER TYPES	87
7.2 USER TYPE INTERVENTION FRAMING	88
7.3 USER TYPE ASSIGNMENT	88
7.4 EMBODIED CONVERSATIONAL AGENT DESIGN	90
7.5 SYSTEM ARCHITECTURE	91
7.6 STIGMA-REDUCTION AND DEPRESSION INTERVENTION	91
7.6.1 Stigma-Reduction Intervention.....	92
7.6.2 Depression Intervention.....	95
CHAPTER 8: SUMMATIVE EVALUATION	99
8.1 PARTICIPANT SAMPLING	99
8.2 MEASURES.....	99
8.2.1 Multidimensional Measure of Religious Involvement in African Americans (MMRI-AA).....	100
8.2.2 Patient Health Questionnaire-8 (PHQ-8)	100
8.2.3 Mental Help Seeking Intention Scale (MHSIS)	101
8.2.4 Inventory of Attitudes Toward Seeking Mental Health Services (IASMHS).....	101
8.2.5 Confidence in Discussing Mental Health Topics with Church Members.....	101

8.2.6 Depression Stigma Scale (DSS).....	101
8.2.7 Agent and System Usability	102
8.3 PROCEDURE	102
8.3.1 Initial Study Appointment	102
8.3.2 Fourteen-Day Field Study.....	102
8.3.3 Follow-up Study Appointment and Interview	102
8.4 RESULTS	103
8.5 QUANTITATIVE FINDINGS	103
8.5.1 Participant Overview	103
8.5.2 System Use.....	104
8.5.3 Depression Symptoms.....	105
8.5.4 Formal Help-seeking Intentions and Attitudes towards Formal Treatment Providers.....	106
8.5.5 Confidence	107
8.5.6 Depression Stigma	107
8.5.7 Agent and System Usability	107
8.6 QUALITATIVE FINDINGS	108
8.6.1 Theme 1: Clara as a Partner	109
8.6.2 Theme 2: Starting to Consider Stigma	111
8.6.3 Theme 3: Intervention Approaches for Engaging Church Members with Depression (meeting them where they are)	116
8.6.4 Theme 4: “Feeding the flock,” Community Approaches for Mental Health Promotion in the Church.	119
8.7 DISCUSSION	123
8.7.1 Limitations	123
8.7.2 Feasibility and Acceptability of an ECA-administered Mental Health Promotion Intervention	123
8.7.3 Impact on Care-seeking	125
8.7.4 Responses to Self-disclosure Planning	128
8.7.5 Impact on Confidence in Discussing Mental Wellness in the Church	129

8.7.6 <i>Method Reflections</i>	130
8.7.7 <i>A Case for Community Embeddedness</i>	132
8.8 CONCLUSION	135
CHAPTER 9: CONCLUSIONS	136
9.1 PHASE I: DISCUSSION-BASED FOCUS GROUPS	136
9.1.1 <i>Phase I Contributions</i>	136
9.2 PHASE II: PARTICIPATORY CO-DESIGN.....	136
9.2.1 <i>Phase II Contributions</i>	137
9.3 PHASE III: CHURCH CONNECT FIELD DEPLOYMENT.....	138
9.3.1 <i>Phase III Contributions</i>	138
9.4 PHASE IV: DESIGNING FOR MENTAL WELLNESS.....	138
9.4.1 <i>Phase IV Contributions</i>	139
9.5 PHASE I–IV DESIGN IMPLICATIONS	139
9.6 SUMMATIVE EVALUATION	139
9.6.1 <i>Summative Evaluation Contributions</i>	140
9.7 CONCLUDING REMARKS	140
REFERENCES	142
APPENDIX A – SUMMATIVE EVALUATION MEASURES	157
APPENDIX B - SUMMATIVE EVALUATION JOURNAL QUESTIONS	174
APPENDIX C - SUMMATIVE EVALUATION INTERVIEW GUIDES	174
APPENDIX E – SUMMATIVE EVALUATION PROTOCOL.....	178
APPENDIX E - DEPRESSION FEATURE STORYBOARD	180
APPENDIX F – DEPRESSION FEATURE INTERVIEW GUIDE	189

Chapter 1: Introduction

According to the World Health Organization, depression is the leading cause of disability worldwide (W.H.O. et al., 2017). In the United States, the National Alliance of Mental Illness estimates that about 19% of Black and African Americans have mental health conditions (Ormond et al., 2019). Further, they reported that of those African Americans diagnosed with a mental health condition, only 25% are engaged in formal treatment compared to 53% of white Americans with similar rates of mental illness (Ormond et al., 2019). Research suggests that formal treatment engagement in Black communities is limited due to significant barriers, including stigma (Conner et al., 2010).

Religion is a prominent fixture in the lives of many Black and African Americans, with 47% of African Americans reporting that they attend religious services at least once a week and 36% attending at least once a month (Religious Landscape Study, 2015). Of those Black and African Americans who attend religious services regularly, more than half (60%) attend churches that serve predominantly Black memberships (Mitchell, 2021). Predominantly Black churches have historically been integral in addressing race-based health inequities, serving as sites for community-based health promotion events and interventions (Peterson et al., 2002). Churches provide affordances of community spaces, health ministry initiatives, social support, and local community-based resources, making churches useful sites in establishing and hosting health promotion programs (M. K. Campbell et al., 2007a). The scope of these church-based health initiatives is vast, addressing a broad range of health topics, including chronic disease management (i.e., diabetes self-management (Gutierrez et al., 2014)), behavior change initiatives (i.e., increasing physical activity (Wilcox et al., 2007)), and disease prevention (i.e., vaccine promotion (Lahijani et al., 2021)).

These initiatives are typically limited to bodily well-being. As evidence, a 2016 systematic review of church-based mental health initiatives found only five articles out of 627 where researchers implemented and empirically evaluated faith-based mental health interventions for African Americans (Hays & Aranda, 2016). Recent work implementing a group-based mental health and behavioral activation intervention in historically Black churches found that the initiative was negatively impacted by low intervention adherence, due to a high prevalence of mental health stigma within the community (Bryant-Moore et al., 2019). This finding was echoed in a 2021 survey administered at a church-sponsored mental health awareness event, which reported that the most significant barrier to care-seeking is the fear of “being labeled” (Ormond et al., 2021).

While health promotion programs in predominantly Black church communities have demonstrated success, these programs face numerous constraints. They are often limited by the church's capacity (e.g., funding, volunteer burn-out, limited

access to resources). They are accessible only to those members who can regularly attend events at the church (L. C. Brewer et al., 2018). Effectiveness of and engagement in health interventions set in the physical world, compared to virtual settings, may also be diminished by church member privacy and confidentiality concerns. Such concerns may be even more necessary to consider when implementing health initiatives that focus on sensitive or stigmatized topics like mental health (Bryant-Moore et al., 2019). Together, these barriers significantly limit the ability of in-person-only programs to sustainably and effectively address church member mental wellness.

Digital health tools, such as smartphone-based interventions, offer opportunities for addressing these challenges, presenting cost-effective ways of enabling greater access to and engagement with health promotion initiatives while potentially addressing issues of privacy and stigma. Indeed, churches and their community members have increasingly adopted technologies inside and outside of formal worship settings, as is reflected in a growing body of techno-spiritual research, i.e., technologies that support individual and/or collective spiritual practices or experiences (Bell, 2006). Since the COVID-19 pandemic, some faith communities have adopted remote worship practices using Facebook Live, YouTube, or Zoom video conference software (Imber-Black, 2020). Some of these aforementioned remote practices have allowed church members to continue small group meetings typically only found within the church's walls. These trends, together with increasing smartphone ownership (Mobile Fact Sheet, 2017) and the historically central role of churches in Black communities, present opportunities for addressing health disparities in church contexts using health technologies. While some work has evaluated digital health interventions for religious populations (L. C. Brewer et al., 2018; Jordan et al., 2021; Newton et al., 2018), none of these interventions have addressed mental well-being.

In 2019, my research team and I embarked on a multi-year community-engaged research effort to design, develop, and evaluate a smartphone-based Embodied Conversational Agent (ECA), a virtual human designed to simulate face-to-face conversation, to assist church members in collectively solving health-relevant problems they identified as important. Many in-person church-based health initiatives have been delivered using health liaisons, often called peer health advisors (M. K. Campbell et al., 2007). Such lay advisors are community members trained by health experts to effectively deliver intervention content to their peers. However, these interventions remain challenging to maintain, given that they rely on a sustained volunteer commitment and often suffer from the same previously mentioned issues of privacy and confidentiality. ECAs have been shown to effectively deliver health promotion interventions and leverage health counseling techniques (T. Bickmore et al., 2020b; T. Bickmore & O'Leary, 2023; Jack et al., 2020; King et al., 2020). Thus, a smartphone-based ECA could serve as a valuable modality for automating church-based mental health promotion interventions and may be effective in reducing racial and ethnic health disparities in these contexts.

In year one of our project, I facilitated a *health priorities* participatory design workshop with two church communities. I asked church members to identify their highest priority health concerns for themselves and their community. All focus group

participants ranked mental well-being as their highest community health priority but stated they were uncertain how their church could effectively support mental health due to the high perceived stigma.

To address this gap, I developed and evaluated a digital mental health and anti-stigma intervention to reduce mental health stigma, increase community support around mental health and well-being, and increase care-seeking intentions with 12 church-affiliated Black adults in a 2-week quasi-experimental field deployment. This intervention was facilitated by an ECA modeled as a peer health advisor and included content for both church members who self-identify or screen as depressed and those who do not.

The mental health promotion and stigma-reduction content was designed as a longitudinal intervention—supporting multiple conversations, and responsive to the following social factors: (1) the user type (defined as the user’s mental health status), (2) the user’s stigma status (defined as whether or not the user confirms they have experienced stigma) and (3) culturally-tailored for churches with predominantly Black memberships. This dissertation work is guided by the research questions enumerated in Figure 1.

Findings from both quantitative and qualitative analyses indicated that participants were satisfied with the intervention content and found the ECA-based intervention acceptable. While the intervention was too brief to yield a significant decrease in depression stigma, interviews revealed a positive shift in participant attitudes toward mental illness which led to changes in participant intentions and care-seeking behaviors. Participants who received intervention content tailored for persons with depression provided rich descriptions of the ways in which their care-seeking intentions were primed. Those who received the community ally intervention discussed the numerous ways they facilitated and supported the care-seeking intentions of other church members not currently engaged in the study. While not the focus of the study, study results indicated strong support and preferences for hybrid approaches to digital mental health promotion in place-based religious communities. Participants explored opportunities for synergistic place-based community health initiatives with the purpose of increasing awareness and complementing church member use of the application. In addition, participants used the current system as a design probe—brainstorming opportunities for in-app community features to increase intervention relevance and sustain intervention engagement.

1.1 Contributions

This research contributes to personal health informatics, digital mental health, intelligent and virtual agents, mental health stigma research, sociology, and human-centered computing. Contributions of this work are as follows:

(1) Description of the health priorities, community strengths, and intervention opportunities for predominantly Black churches. Black and African-descended people of faith are a population that remains underrepresented in the design, development, and evaluation of both digital mental health and digital anti-stigma tools (Rivera et al., 2021). This work centers this priority population.

- (2) Description of how ECAs can be used for culturally-tailored, responsive, and acceptable interventions in predominantly Black churches, including design guidelines for tailoring community-based ECAs and methods for tailoring dialogue.
- (3) Demonstration of efficacy of an ECA-administered health intervention that takes a trifold focus on well-being—supporting bodily, spiritual, and social well-being in predominantly Black churches.
- (4) Description of digital mental health promotion intervention opportunities for predominantly African American church communities.
- (5) Design of an ECA-based intervention for mental health promotion and stigma reduction that takes a multi-level approach by supporting church members as active agents in improving their community's mental health and well-being.
- (6) Demonstration of the efficacy of an ECA-based mental health and stigma-reduction intervention. The process of care-seeking is an inherently social process where an individual's social contacts influence an individual's care-seeking intentions (P. W. Corrigan et al., 2014; Pescosolido et al., 2013; Pescosolido & Olafsdottir, 2013). This work advances our knowledge of the feasibility and acceptability of a digital anti-stigma tool designed to reduce stigma in the community to support individual care-seeking intentions.

1.2 Organization

This dissertation contains nine chapters and is organized as follows.

In Chapter 2, I provide an overview of related work starting with the role of the Black church in American society, faith-based health interventions, and an introduction to stigma theory. Next, I cover in-person intervention approaches for decreasing stigma, digital stigma-reduction efforts, digital tools designed to enhance readiness for therapy and ECAs designed to improve user mental wellness.

In Chapter 3, I present findings from Phase I of this work, where I conducted three discussion-based focus groups with church members to both build rapport with community members and learn about their historical context, current practices, and community assets.

In Chapter 4, I conducted four participatory design focus groups to explore preliminary concepts and modalities for a church-based mHealth application, during which participants identified their most pressing health priorities.

In Chapter 5, I present results from a quasi-experimental 28-day field deployment, where 25 church members from Churches A and B used the first version of a smartphone-based ECA system, designed for their church.

Chapter 6 presents findings from the final participatory design activity which captured church leaders' perspectives on and expectations for features that would be most desirable when addressing church member experiences of depression using digital interventions.

In Chapter 7, using key intervention design principles established in Chapters 3 – 6, I describe the design and implementation of the ECA-based mental health and stigma reduction smartphone application including the design of the curriculum.

In Chapter 8, I report the results of my summative evaluation of the mental health and stigma-reduction system. This chapter ends with a discussion of findings and future work opportunities.

In Chapter 9, I present my conclusions and draw connections between all phases of my prior work and the summative evaluation, as well as future directions for research.

Figure 1: Dissertation Research Questions

Phase I: Understanding Community Context

RQ1: How do participants describe their church communities and define their church's current goals and missions?

RQ2: In what ways are church members currently providing social support to one another?

RQ3: What are current church and church member technology practices?

Phase II: Co-Design

RQ1: What health topics would church members identify as most essential for themselves and their community for a church-based mHealth app to address?

RQ1b: What types of church-based care networks, programming, and initiatives currently support these member-identified health topics? If at all?

RQ2: What role and persona should a community-based, culturally-tailored ECA employ to support health and wellness in faith communities with predominantly Black and African American memberships?

RQ2b: How might participants adapt a health counseling conversation to include matters related to their faith and spirituality?

RQ3: What resources would church members expect or seek in a church-based health app?

RQ4: How can we design digital health tools responsive to the intersectional identities of race and religion in multicultural church communities?

Phase III: Field Deployment

RQ: How can digital health technologies be designed to support bodily, spiritual, and social wellbeing within predominantly Black church communities?

Phase IV: Mental Health Feature Selection

RQ1: How do predominantly Black churches and their members currently support members experiencing mental illness?

RQ2: What features would church members affiliated with predominately Black churches find most meaningful for supporting individuals with depression?

Summative Evaluation

RQ1: Is it feasible and acceptable to provide a stigma reduction and care-seeking encouragement intervention using a smartphone-based ECA modeled after a peer health advisor?

RQ2: What effect does a smartphone-based ECA that utilizes stigma reduction strategies to enhance readiness for care-seeking have on African American and Black church-affiliated users?

RQ3: What impact does introducing the topic of disclosure have on Black church member disclosure intentions?

RQ4: How does a smartphone-based ECA that utilizes stigma reduction strategies increase confidence in discussing the topic of mental illness and mental health?

Chapter 2: Related Work

Until recently, faith-based health promotion interventions have not been translated into digital health interventions. My work explores how stigma reduction content delivered by a smartphone-based ECA may positively impact an individual's mental wellness by enhancing their readiness for professional mental healthcare and reducing community stigma. To properly situate this work, I provide an overview of the role of the Black church in American society, faith-based health interventions, and stigma theory. Next, I cover interventions implemented to positively impact mental health: anti-stigma interventions, digital anti-stigma interventions, digital tools to enhance readiness for therapy, and ECAs designed to deliver mental health interventions. Terminology that refers to racial and ethnic identities of study participants used within this related work is consistent with the respective citations (e.g., Black, African American, African-descended people, etc.).

2.1 Role of the Black Church

To better understand the context of this dissertation, it is essential to recognize the historical and contemporary significance of the Black Church to Black and African American communities. The Black Church is a term that refers to seven major Protestant denominations that serve predominantly Black members (PBS, n.d.). In using the term Black Church, I do not suggest that all churches serving predominantly Black and African American people are monolithic or singular in their religious practices (Murphy, 2000). Instead, when I refer to the Black Church, I do so to acknowledge its institutional, political, and societal importance and its role in the liberation of Black and African American people (Murphy, 2000). Built outside of white society, Black churches were built, organized, and run by Black people (Gates, 2021), and while their role is indeed to serve as a house of worship, the cultural and social significance of these institutions is undeniable. Black Churches have acted as social service institutions and places of safety and refuge, protecting enslaved Black people (Gates, 2021).

In contemporary society, the Black Church remains a powerful organization for believers and nonbelievers alike and a “cultural repository” (Gates, 2021). The rich foundation that comprises how these churches and their members connect in service to each other and the larger community. The tenants of fellowship and ministry are cornerstones of these faith communities and thus shape church member values and expectations for technology in service of community well-being. The current reach of the Black church remains high, with 79% of Black and African Americans stating that they are religiously affiliated (Mohamed, n.d.). Given current racial and ethnic health disparities and the churches' presence in Black and African American communities, predominantly Black Churches are well-situated as sites for the provision of care to Black and Indigenous People of Color (Peterson et al., 2002).

2.2 Faith-based Health Initiatives for Church-affiliated Black Adults

In 2001, United States (U.S.) President George W. Bush established the White House Office of Faith-based and Neighborhood Partnerships focused on serving individuals from historically marginalized communities (Peterson et al., 2002). The U.S. Biden-Harris administration has continued this work by establishing programs addressing racial and ethnic health inequities (e.g., vaccination programs, food security, equity, fair housing, behavioral and mental health during COVID-19, etc.) These partnerships have funded numerous faith-based health initiatives in Black church communities throughout the U.S. (House, 2022).

The scope of these church-based health initiatives is vast and includes a broad range of health topics, including but not limited to physical activity and nutrition (Resnicow, Wallace et al., 2000), cardiac health (Lancaster et al., 2014), advanced care planning (Hendricks Sloan et al., 2016), smoking cessation (Schorling et al., 1997), vaccine promotion (Lahijani et al., 2021), cancer screening (Derose et al., 2000), and diabetes self-monitoring (McNabb et al., 1997; Samuel-Hodge et al., 2006). Most faith-based health initiatives are delivered in the church, typically through health liaisons and peer health advocates, often referred to as peer health advisors (M. K. Campbell et al., 2007b). Such lay advisors are trained by professional research staff or program experts to deliver intervention content effectively. In some instances, peer health advisors will work as co-creators assisting project staff with tailoring interventions to the concerns, needs, and assets of their church community (M. K. Campbell et al., 2007b). Such interventions rely on the availability of church members willing and able to be trained as peer health advisors to spearhead initiatives. In addition, like most interventions that rely on brick-and-mortar buildings for service delivery, in-person interventions anchored in a physical setting can only be accessed by those who regularly attend church activities (L. C. Brewer & Williams, 2019).

Previous studies have found that African Americans are more likely to leverage systems of care within the walls of their church in place of professional mental healthcare (Blank et al., 2002; Bopp et al., 2013). Thus, like other health-based initiatives, mental health initiatives embedded within the church targeting professional treatment intentions may be effective; however, a 2016 systematic review of church-based mental health promotion initiatives among African Americans found that only five studies included evaluations, as well as details surrounding methodological rigor and cultural adaptations (Hays & Aranda, 2016). Findings from this review reveal that future interventions should focus on mental health awareness and engagement as prevention strategies (Hays & Aranda, 2016). In addition, prior work indicates a tremendous amount of complexity and nuance surrounding how religious orientations affect formal care-seeking. These works suggest that church-based mental health initiatives should partner directly with church members and leadership to include religious framings that promote mental health care-seeking. Recently, a 2021 community-engaged survey-based research project stressed the importance of community involvement in the co-development of specifications for a mental promotion intervention embedded within African

American faith communities (Berkley-Patton et al., 2021). While this project has not published final outcomes, its findings name the following community-identified intervention components: 1. screening; 2. resource referral; 3. church leader involvement.

Interventions set in the physical world (compared to virtual settings) also create potential privacy and confidentiality issues, especially when the health initiative addresses a sensitive or stigmatized health topic like mental health. Haynes et al. 2018 created a depression intervention for church-affiliated African American adults living in the rural South. Their intervention was a multi-session small group program led by trained peer health advisors. However, due to the prevalence of stigma, some church members voiced discomfort in attending group-based workshops, fearing that their concealable stigmatized identity (a person with depressive symptoms) would be made public simply through attendance. In addition, the authors state that their programming required more cultural tailoring, specifically addressing barriers that stem from cultural misunderstandings of depression, treatment, and medication options (Haynes et al., 2018). In order to improve intervention efficacy, they suggest that researchers include anti-stigma messaging within any mental health intervention (Haynes et al., 2018).

2.3 Faith-based Digital Health Tools

Digital faith-based health interventions have been shown to successfully improve health outcomes in Black church communities in the area of cardiovascular health (L. P. C. Brewer et al., 2019). The technology used in these interventions has primarily been individually-focused and low touch, e.g., short messaging service (SMS), activity logging, or ecological momentary assessments through notifications. Behavior change strategies have included education, self-monitoring activities, feedback, and goal-setting. In their CBT4CBT health program, authors administered a web-based substance use program via tablet (Jordan et al., 2021). Their trial was an efficacy trial and thus required users to come to the church to use the prototype system. Newton et al. 2018, evaluated a hybrid program that used SMS to supplement in-person components for a diabetes self-management program administered in Black church communities (Newton et al., 2018). Finally, FAITH! developed by Brewer et al. incorporated both individually-focused features and computer-mediated social support facilitated through a group forum (L. P. C. Brewer et al., 2019). In this project, the app provided a nutrition and physical activity promotion program to affect cardiovascular health positively. While most of these aforementioned digital health interventions tailored their health content by connecting Scripture and Bible stories to health, none explicitly addressed mental well-being or mental health stigma.

2.4 Cultural Tailoring for Health Promotion Interventions

The concept of tailoring encompasses several approaches and strategies for creating individualized messages for message receivers with the expectation that this type of individuation will increase the effectiveness of the communication on the intended audience (Hawkins et al., 2008a). Tailoring interventions often utilize multiple approaches, including personalization, feedback, and content-matching (Hawkins et al., 2008b). There are numerous attributes and theoretical concepts on which researchers have tailored health communications. These include audience demographic characteristics, perceived self-efficacy, goal-setting,

current health behaviors, and intentions (Noar et al., 2007). Tailored health messages have led to greater intervention engagement, improved behavior intentions, increased motivation, and positive health behavior outcomes (Noar et al., 2007; Van Der Veen et al., 2010).

Cultural-tailoring is a type of tailoring defined as adapting intervention content to recognize, respond to, and reinforce the intended audiences' cultural beliefs and values (Van Der Veen et al., 2010). Furthermore, culturally-tailored health content responds to users' "cultural values, beliefs, historical factors, and social norms" by providing additional context and meaning to the delivered health content (Van Der Veen et al., 2010). Cultural-tailoring has been shown to increase user engagement by making information more personally salient, relevant, and meaningful to the individual (O'Leary et al., 2020).

Resnicow, Solar et al. (2000) developed a framework for developing, evaluating, and implementing culturally-sensitive and tailored interventions for diverse populations. Authors define two levels for tailoring interventions: surface structure and deep structure tailoring. Surface structure involves matching intervention content and messaging to the observable characteristics of a priority population (Resnicow et al., 1999). For technology-enabled materials, surface structure may involve using people, locations, language, music, and food preferred by the priority population. Though some critics may believe that surface structure is superficial, incorporating surface-structure into an intervention has been shown to increase initial intervention engagement and thus plays an essential role. Deep structure cultural-tailoring recognizes the priority population's sociocultural norms, historical factors, values, and beliefs. This approach uses these *deeper* characteristics to catalyze and sustain behavior change (Joseph et al., 2021). In order to establish deep structure factors, researchers recommend utilizing a community-engaged approach, whereby the community's expertise spearheads cultural-tailoring efforts creating a more inclusive and effective health promotion intervention (Costanza-Chock, 2020).

Culturally-tailored interventions have outperformed non-adapted interventions, positioning culturally-tailored interventions as superior to non-adaptive interventions in countering racial and ethnic health disparities. An exhaustive review of culturally-tailored mHealth interventions is outside the scope of this dissertation. However, a recent 2016 systematic review found that African American participation in digital health research is far behind white populations (D. C. James et al., 2017). One reason for this disparity is that a limited number of culturally-tailored interventions designed with surface-structure and deep-structure factors currently exist (D. C. S. James et al., 2017). In fact, out of 56 studies included in the systematic review of African American digital health research participation, only eight were explicitly tailored for African American populations. Furthermore, only two studies focused on mental health promotion explicitly. A 2019 systematic review of digital mental health tools for underserved communities found that no digital mental health interventions as of 2019 had been specifically tailored for African Americans (Schueller et al., 2019).

Electronic approaches for cultural-tailoring hold promise for more flexible and dynamic digital health communication and education technologies that respond to levels of certain characteristics. For instance, of those interventions currently

reviewed, authors almost unequivocally support the use of spiritual messaging in behavior change interventions designed for African Americans. However, recent work indicates a continuum of needs and desires regarding culturally-tailored messaging (Shirley M et al., 2022). Continuing with spirituality as an example characteristic, research has found that individual preferences can range from no spiritual messaging, inspirational secular messaging, Biblical references paraphrased, or direct quotes from Scripture (O’Leary et al., 2020; Shirley M et al., 2022). Thus, technology offers opportunities for creating more flexible and dynamic cultural-tailored interventions across various characteristics that respond to individual preferences within the prioritized cultural group.

2.4.1 Recommendations for Cultural Tailoring Mental Health Promotion Interventions

Using this literature, I present three recommendations for culturally tailoring a digital mental health tool for African American adults in service of reducing disparities in mental health service utilization.

(1) Interventions should consider incorporating anti-stigma messaging explicitly when addressing mental health (Haynes et al., 2018). Doing so may increase engagement and address barriers that stem from cultural misunderstandings and medical racial bias resulting from deeply entrenched racial prejudice and discriminatory healthcare practices. Section 2.5 will discuss this topic further.

(2) Researchers should consider taking a faith-centered approach to mental health promotion by responding to cultural beliefs and religious practices that work to either discourage or encourage mental health promotion (Neely-Fairbanks et al., 2018). Section 2.5.2 will continue to explore this topic.

(3) Experts suggest mental health promotion interventions prioritizing Black and African American community members should include the following components: mental health literacy and education, screening, resource referral, leadership involvement, and activating motivation to seek professional mental healthcare (Ormond et al., 2019).

2.5 Mental Health Stigma Theory

In the following related work sections, I introduce and define *stigma* and variants of stigma called *self-stigma* and *public stigma*. I discuss highly utilized stigma reduction approaches from the literature: education, contact, paracontact, and disclosure.

2.5.1 The Life-limiting Impact of Stigma

Sociologist Erving Goffman defined stigma as a social process used to discredit an individual based on prejudicial thinking associated with a distinguishing characteristic or personal trait in his landmark text, *Stigma: Notes on the management of a spoiled identity*, published in 1963. The result of being discredited due to a discreditable attribute or label “results in social rejection” and forces an individual to conceal their identity meticulously (Goffman, 1963). Since 1963, sociologists and social psychologists have continued to situate this initial understanding of stigma into larger societal contexts and processes. Sociologist

Bernice Pescosolido extends Goffman's definition to emphasize the complex and interwoven societal system through which stigma operates. She states, "An individual is labeled by a difference. Then through a multi-level set of societal interactions with individuals, groups, organizations, the difference either advantages them or disadvantages them" (Pescosolido & Martin, 2015). In recent years, sociologists have further developed the sociology of stigma to demonstrate how stigmatization is used as a political and structural apparatus to keep certain people "in, down, and away" (Link & Phelan, 2014). This process uses prejudicial thinking to drive discriminatory practices, creating a loss of power and opportunity for those labeled. Stigmatization results in stigmatizers retaining their power and social affluence (Link & Phelan, 2014).

In this dissertation work, I focus on stigma related to mental health. In the following paragraphs, I highlight the life-limiting impact of mental health stigma. In addition, I focus on mental health stigma embedded within predominantly Black and African-descended communities. Though my work focuses on mental health stigma, Black and African American people have multiple stigmatized identities in the US and are subject to racial and ethnic discrimination *and* discrimination associated with mental illness (K. Conner, 2010). The impact of embodying multiple stigmatized identities can have a compounding effect. For instance, historically, Black and African American people have been burdened by racial bias embedded within the mental health treatment system, resulting in inadequate care and reduced access to health-enhancing resources (K. Conner, 2010).

In general, mental health stigma has been shown to directly undermine care-seeking behaviors (K. O. Conner et al., 2009) and disrupt access to an individual's critical social support networks (Corrigan, Druss, & Perlick, 2014). As a result, in communities where mental health stigma is relatively high, such as Black and African American communities (Mantovani et al., 2017), individuals living with mental illness are more likely to avoid treatment, lack mental health knowledge, and lack the required support to adequately facilitate care-seeking behaviors (e.g., accessing formal care services like medication or talk therapy). Such factors reinforce racial and ethnic disparities in the treatment (K. O. Conner et al., 2009; P. W. Corrigan et al., 2014; Shefer et al., 2013).

2.5.2 Intersectional Complexities of Race and Religiosity on Mental Health Stigma and Treatment Utilization

Research has demonstrated that religious involvement (Davenport & McClintock, 2021a), group identification (Kearns et al., 2018), and culture mediate the impact of stigma on mental healthcare service utilization and informal care-seeking (e.g., pastoral counseling, prayer, faith-centered support groups, etc.). Previous work in sociology, psychology, and population health indicates that actively belonging to a religious organization can reinforce stigma (Peteet, 2019). Religious involvement's impact on stigma depends on multiple factors; however, one critical factor is how a faith community and its leadership approach mental health promotion and awareness. For this review, I focus on the mediation of mental health stigma due to the following cultural factors: 1. religious involvement and 2. racial identity in the context of the Black church. Readers should be aware that though this review focuses on Christian tradition, mental health stigma exists in other faith traditions.

Social Standing and Status

In communities where social cohesion is high, such as is typically found in religious communities; stereotypes may have social and behavioral consequences for the individual, including loss of individual aspirations, increased internalized shame, loss of reputation, and increased social isolation and rejection (K. O. Conner et al., 2009). These consequences have a cascading effect: individuals concerned about such consequences expend strenuous effort meticulously hiding their mental illness and symptoms from their social network and often forgo formal and informal care-seeking to protect their social standing and reputation within the community. Some Black Americans believe that stigma related to mental illness is felt more intensely in Black communities and is more deleterious to a Black American's social standing when compared to a white person due, in part, to systematic racism and historical disenfranchisement (R. D. Campbell & Mowbray, 2016). As a result, Black Americans may fear adopting an additional stigmatized identity, such as mental illness, to avoid any risk of losing life opportunities (R. D. Campbell & Mowbray, 2016).

Beyond reasonable concerns regarding social status and standing, systematic racism and racial bias in the healthcare system also shape the experiences that African Americans have when engaged in formal treatment. African Americans are more likely than their white counterparts to be misdiagnosed, misunderstood, ignored, and assumed to be more dangerous than other racial and ethnic groups diagnosed with mental illness by both members of the public and clinical staff (R. D. Campbell & Mowbray, 2016). As a result, interventions should consider explicitly addressing treatment hesitancy and avoidance by focusing on reasons for such avoidance when tailoring anti-stigma and care-seeking interventions for this population.

Cultural Expectations that Decrease Mental Health Service Utilization

Some researchers attribute high mental health stigma to inaccurate beliefs stemming from commonly held cultural beliefs (Mantovani et al., 2017). A qualitative investigation of the sociocultural beliefs of African-descended faith communities found that values and beliefs affiliated with an individual's religious community or culture can promote or reduce stigma (P. W. Corrigan et al., 2014; Mantovani et al., 2017). In some religious communities, individuals experiencing a mental health condition are seen as spiritually unwell, weak, or morally corrupt (Mantovani et al., 2017). In addition, Mantovani et al.'s participants identified several social consequences of stigma embedded in faith communities, including the propagation of self-stigma (internalization of stereotypes), loss of aspirations reinforced by structural discrimination, feelings of shame, and increased social isolation (Mantovani et al., 2017).

Woods-Giscombe et al. (2016) conducted an exploratory qualitative study focused on Black experiences of depression in a series of focus groups with 48 participants. Woods-Giscombe et al. (2016) examined the impact of intersectionality on stigma and formal service utilization, stating that spirituality has historically been an intrinsic part of the African American experience and is closely associated with a cultural expectation of appearing strong under trying circumstances such as emotional

distress. Taking this concept one step further, some of their participants reported that the perceived cultural obligation to appear strong is closely tied to religion and that God should be the ‘first line of defense’ against depression (Ward & Mengesha, 2013).

Beyond cultural pressures and obligations, attitudes of the faith-based organization can reinforce stigma. In the work of, Woods-Giscombe et al.’s (2016) participants reported wanting to turn to clergy as an initial first step; however, some shared that the clergy they approached lacked compassion, encouraged the individual to rely more heavily on God, and did not actively encourage treatment utilization, resulting in the participant being “unable to open up.” This work reminds us that when deploying stigma reduction interventions in community-based organizations, one should consider intervening on multiple levels.

The Impact of Religious Involvement on Treatment Utilization and Symptom Reduction

Just as the Black experience of depression and stigma is multifaceted and diverse, so too are experiences of religion and religious involvement. Researchers in religious studies, sociology, and social work have identified three dimensions of religious involvement: subjective, organizational, and non-organizational religiosity (Lukachko et al., 2015). Subjective religiosity is an individual’s perception of religion’s importance and impact on their life. Organizational religiosity describes formal participation in organized activities such as worship, attending fellowship, or experience working with a ministry. Nonorganizational religiosity refers to private spiritual practices such as prayer or consuming religious media (e.g., listening to gospel music, engaging in Bible study, viewing previously recorded sermons) (Lukachko et al., 2015).

Researchers surveyed a sample of African American adults experiencing depression to investigate the relationship between dimensions of religious involvement on attitudes toward mental health treatment (Davenport & McClintock, 2021b). They found that African Americans with high subjective and organizational religiosity were less likely to utilize formal mental health services and instead rely on religious coping (e.g., prayer, meditation, Bible study, etc.) and informal care utilization (e.g., attending a ministry meeting, meeting with the pastor, etc.). In another that sought to investigate the mediation of depressive symptom reduction by type of religious involvement, Davenport et al. (2021a) found that organizational religiosity may be related to an individual-level reduction depressive symptom. The authors hypothesized that the correlational relationship between symptom reduction and organizational religiosity might result from the sophisticated social needs and caring infrastructures common in congregations seated in an African American Christian faith. Essentially, churches that offer social support programs to help mitigate life stressors (e.g., financial need, assistance with medical care, food delivery services, etc.) may indirectly benefit a care receiver’s mental well-being. The authors of both studies did not evaluate the relationship between religious involvement and stigma.

Considerations When Designing for Intersectional Identities

This work highlights several research gaps that should be explored when designing tech-enabled mental health promotion interventions with and for Black and African American populations. This related work demonstrates that interventions

prioritizing religious communities with a high prevalence of mental health stigma should address cultural factors specific to Black and African Americans, such as the racialization of mental illness and dimensions of religious involvement (Rivera et al., 2021). In addition, church leadership should be approached directly to partner with researchers when developing mental health and care-seeking interventions (Berkley-Patton et al., 2021). Doing so may reinforce with congregants that reducing mental health stigma and increasing care-seeking opportunities are missions vital to the organization (R. D. Campbell & Mowbray, 2016). Future work should consider how intersectional complexities should shape the development of culturally-responsive interventions to assist in bridging the racial and ethnic mental health service gap.

2.5.3 Self-stigma: a stigma variant

As previously described, stigma exacerbates several societal processes that ultimately lead to adverse health outcomes and increased racial and ethnic health disparities. Notably, there are also individual-level behavioral processes caused by stigma that impact an individual's life outcomes. Self-stigma is the process through which individuals perceive the stereotypes and prejudice imposed on them by society and apply these stereotypes to themselves (P. W. Corrigan, Bink et al., 2016; Harkness et al., 2016). People who experience high levels of self-stigma have worse health outcomes (P. W. Corrigan & Watson, 2002; Harkness et al., 2016). In their work, Corrigan and colleagues present the Stage Model of Self-stigma, which breaks down how prejudice and discrimination impact an individual's *behavioral response* to societal stigma (Corrigan et al., 2016).

In Corrigan's Stage Model of Self-stigma, individuals who experience self-stigma are intimately aware of the stereotypes related to mental illness, and they not only believe these stereotypes but apply these stereotypes to themselves (P. W. Corrigan, Bink, et al., 2016). This results in the individual avoiding potential life opportunities. Corrigan et al. (2016) enumerate the stages of the model and label these stages as *Aware*, *Agree*, and *Apply*. These stages have an outcome of both *Diminished Self-respect* and what Corrigan et al. (2016) label as the *Why Try Effect*. In practice, an individual is *aware* that society believes in certain stereotypes about people with mental illness. For example, one such stereotype is that "people with a mental illness are unreliable and cannot hold a job." Individuals experiencing self-stigma are not only *aware* of the stereotype but *agree* with it and subsequently *apply* it to themselves. The outcome of the aware, agree, apply process results in *diminished self-respect* and ultimately makes it less likely that they will apply to a new job, believing that their mental illness makes them a poor job candidate. In essence, individuals with self-stigma may ask themselves, "*Why* should I even *try* applying to that job? I will not get the job anyway; I am unreliable."

2.5.4 Public stigma: a stigma variant

Another stigma variant, *public stigma* is a set of negative attitudes and beliefs that motivate community members to reject, fear, and avoid individuals with mental illness (Pescosolido et al., 2008). Public stigma has been shown to decrease informal care-seeking opportunities (e.g., family support and support from informal healthcare organizations such as churches) (P. W. Corrigan,

2012). Pervasive perceived public stigma can further limit access to informal resources and horizontal social capital opportunities (e.g., leveraging one's peers to access informational and tangible resources within the community) (Pattyn et al., 2014). This is particularly problematic for Black and African Americans, who often rely on and/or prefer informal care networks to receive support compared to formal treatment (e.g., primary care, counseling, medication) (Blank et al., 2002; Bopp et al., 2013).

2.6 Stigma-Reduction Interventions

The following sections present an overview of stigma-reduction strategies and interventions. The focus of this work will be on Black and African-descended populations. However, there is limited work in the mental health stigma reduction literature that centers this population. Rivera and colleagues conducted a narrative review of mental health anti-stigma interventions for African American and Black people in the United States, and they observed that only *six studies* met the inclusion criteria (Rivera et al., 2021). Such a paucity of research is problematic when empirical investigations have suggested that mental health stigma is prevalent in African American communities (Rao et al., 2007). Therefore, this review will include work that addresses other concealable stigmatized identities (i.e., identities that can be hidden from others) in Black and African Americans.

Reducing self-stigma in communities where stigma is high, such as in religious and predominantly African American communities, is challenging. To increase care-seeking behaviors and minimize the effect of stigma, researchers have identified multiple strategies: (1) education (psychoeducation, mental health literacy), (2) contact (strategic face-to-face interactions with persons who identify as experiencing a mental illness), (3) paracontact (contact through testimonials or storytelling), and (4) disclosure (Corrigan et al., 2001). While the following sections are organized by approach, several of the interventions reviewed employ a combination of these strategies.

Psychoeducation has been shown to challenge false beliefs and prejudicial thinking by reviewing facts about mental health conditions and the effect of stigma (P. W. Corrigan et al., 2014). Advocates of psychoeducational approaches for reducing self-stigma believe that education or education combined with techniques used in Cognitive Behavioral Therapy (e.g., thought restructuring) can minimize feelings of internalized shame and hopelessness and increase self-efficacy and personal empowerment (P. W. Corrigan et al., 2014; Mittal et al., 2012; Yanos et al., 2015b). Psychoeducation approaches can include challenging cultural beliefs that promote myths about mental health conditions and providing culturally-tailored information about the realities of care-seeking and service utilization unique to this community. Typical topics used in anti-stigma include reviewing facts around causes, symptom manifestation, identification, prevalence, available treatment options, and self-care strategies (Mittal et al., 2012). For instance, Rusch et al. (2010) used deep-structure cultural tailoring to investigate the efficacy of an anti-stigma video developed for African American adults by comparing a contextual model of mental illness (i.e., life stressors lead to mental health challenges) to a biomedical model of mental illness (i.e., depression is the result of a genetic predisposition). In their randomized control trial, participants exposed to the biomedical model experienced increased stigmatizing thoughts and attitudes immediately following the session (Rusch et al., 2010).

Mittal et al. (2012) conducted a comprehensive review spanning 2001-2011, which described fourteen studies evaluating self-stigma reduction strategies. They found that psychoeducation was the most widely implemented and evaluated stigma reduction strategy. Of all fourteen studies reviewed, only one prioritized Black and African American people (Mittal et al., 2012). Alvidrez et al. (2009, 2010) evaluated a culturally-tailored educational brochure called “Getting Mental Health Treatment: Advice from people who’ve been there.” The booklet focused on professional service utilization, covering topics such as challenges in seeking mental health treatment and strategies for overcoming such challenges (Alvidrez et al., 2010). Alvidrez and colleagues partnered with community members to tailor the booklet using both surface-structure characteristics (e.g., images of Black and African Americans) and deep-structure factors (e.g., barriers to formal care-seeking and problem-solving strategies) that resonated most with their priority community (Alvidrez et al., 2009).

The booklet itself was co-designed by Black and African American community members interested in debunking myths about depression and treatment, sharing their experiences navigating mental health care as Black and African American people (“making treatment work better for them”), and discussing their experiences with stigma (Alvidrez et al., 2009, 2010). In a randomized pilot study, investigators found that the culturally-tailored psychoeducation booklet was more effective in reducing stigma in those participants who showed “greater uncertainty about treatment” than a generic pamphlet that identified community-based treatment and services (Alvidrez et al., 2009). This compelling finding emphasizes the importance of including culturally-tailored anti-stigma messaging in interventions instead of merely providing Black and African American consumers with information about healthcare service providers.

Contact is defined as a meaningful face-to-face interaction with an individual who self-identifies as someone living with a mental illness (Corrigan et al., 2014). Often used to minimize public stigma, a contact-based intervention uses planned interactions between a person with a mental health condition and a priority group (e.g., treatment providers, police officers, others experiencing a mental health condition). Contact-based interventions can take different forms and structures. Some contact-based interventions focus on a single storytelling session where individuals with mental illness describe the effects of mental health on their experiences receiving formal support (e.g., medication, therapy, support groups). This type of intervention is often used in large-scale mental health awareness campaigns to address public stigma (a variant focused on reducing popular myths and stereotypes) (Evans-Lacko et al., 2012).

Contact-based initiatives that intervene on self-stigma are typically multi-session group-based interventions where all participants self-identify as people experiencing a mental health condition (Yanos et al., 2015b). These interventions usually include a combination of psychoeducation and self-care strategies facilitated by either an expert or a trained peer mental health advisor (Phelan & Link, 2004; Yanos et al., 2015a). In one such contact-based and psychoeducation approach, Hansson and colleagues evaluated a manualized self-stigma reduction program called Narrative Enhancement Cognitive Therapy (NECT), a combination of thought restructuring and storytelling techniques used to promote personal empowerment and self-efficacy in a

group setting (Hansson et al., 2017). The authors found a reduction in self-stigma at a six-month follow-up for those who completed the NECT treatment. Racial and ethnic demographics of the study population were not characterized in this paper, yet findings still illustrate a thoughtful contact-based approach embedded in psychoeducation.

Young et al. (2020) evaluated the effectiveness of a culturally-tailored multi-session group-based intervention that also utilized techniques from CBT in a randomized control trial with Chinese participants. Self-stigma was assessed immediately following intervention delivery. The authors found that the treatment group showed a statistically significant decrease in self-stigma, demonstrating the effectiveness of a contact-based CBT intervention for their population.

Given the limitations of anti-stigma intervention research that centers Black and African-descended people, I widened my search to include papers that utilized standard anti-stigma approaches (e.g., psychoeducation, contact, etc.) on other types of concealable stigmatized identities such as persons living with HIV. The unity workshop was developed in 2012 by Rao and colleagues (Rao et al., 2012). Its focus was to reduce HIV self-stigma in African American women living with HIV infection. This workshop included both culturally-responsive and sensitive HIV education and a contact component by incorporating testimonials from Haitian and Afro-Caribbean people to further tailor their education intervention further (Lipira et al., 2019). This work was effective in reducing self-stigma.

Although some mental health stigma researchers state that contact-based approaches combined with education produce the most durable gains in stigma reduction (Dalky, 2012), other researchers present a more nuanced and complicated view of the effectiveness of contact-based approaches. Evans-Lacko et al. (2012) demonstrate that the effect of contact on mediating stigma may rely on the quality of the contact (i.e., positive or negative social relationship) and the degree of familiarity or closeness. In their work, they state that contact-based approaches should be carefully crafted so as not to increase self or public stigma.

Contact-based approaches are designed to reduce both public and self-stigma. However, these approaches are nuanced, and limitations exist. For example, in-person contact interventions have similar constraints to any in-person intervention where participation is limited to those who can access a physical location. Individuals experiencing self-stigma may avoid an in-person contact intervention to maintain privacy, which could reinforce self-stigma. In addition, contact-based approaches administered in community settings, where individuals rely on existing social relationships to maintain group membership and access health-enhancing resources, could present several unknown challenges that intimidate potential participants.

While most anti-stigma digital health interventions have mainly focused on interventions that include educational components, some researchers have used technology to facilitate *paracontacts* (Ferrari, 2016; Sebastian & Richards, 2017). A paracontact is an asynchronous technology-mediated contact where a person with a mental health condition is portrayed positively. For example, first-hand accounts shared by individuals experiencing mental health conditions presented through awareness campaign commercials are *paracontacts*. Tech-enabled paracontact interventions have typically included prerecorded video presentations of testimonials or text-based testimonials (Ferrari, 2016; Sebastian & Richards, 2017).

In addition to psychoeducation, contact, and paracontact, researchers have investigated how the process of *self-disclosure* may affect stigma. Disclosure is the process through which an individual (the discloser) tells another person (the confidant) about an otherwise concealable stigmatized identity (e.g., HIV infection status, mental health status, sexual orientation) (Chaudoir & Fisher, 2010). Disclosing one's stigmatized identity has been associated with a reduction in shame and decreased effects of negative consequences related to self-stigmatization (the process of applying stereotypes to oneself) (P. W. Corrigan & Rao, 2012). Corrigan & Rao (2012) explain that when individuals feel free from the impact of others' opinions on their identity, their concern about concealing their stigmatized identity is reduced, often decreasing their stress and improving their quality of life.

Some disclosure events have negative consequences, such as when the confidant does not respond positively but instead responds with prejudice (Chaudoir & Fisher, 2010). Depending on the degree of familiarity and closeness between the discloser and confidant, an adverse reaction can have personal, social, and professional ramifications. Though disclosure approaches have been used in some stigma reduction interventions (P. W. Corrigan, Kosyluk, et al., 2016; Henderson et al., 2017), I found no work examining the efficacy of mental health disclosure interventions in predominantly Black religious communities.

As an initial step towards investigating how a digital health tool might benefit self-disclosure processes in predominantly Black church communities, my dissertation work will educate participants on the benefits and consequences of self-disclosure and encourage users to consider disclosing their mental health status to a medical provider to facilitate treatment.

2.7 Digital Stigma-Reduction Interventions

The translation of anti-stigma approaches to digital health interventions is nascent. This review includes the small number of currently existing published works within the literature focused on African-descended people. Griffiths et al. conducted an RCT to evaluate the effectiveness of a depression literacy website compared to a website that focuses on reducing self-stigma explicitly (Griffiths et al., 2004). Both websites lowered self-stigma. Lipira and colleagues translated their Unity workshop to a multi-session digital approach featuring testimonials produced by individuals from a similar community and population as the target end-users. This approach was evaluated with Black and African American participants; however, only four participants were included in this pilot work (Lipira, 2018).

Rodriguez-Rivas et al. conducted a systematic review of experimental studies evaluating digital anti-stigma interventions. Their search resulted in nine papers that met their inclusion criteria. Technology solutions spanned from electronic contact using video conference software to video games and virtual reality (Rodríguez-Rivas et al., 2022). All studies included in the review prioritized young populations with non-Black users. The review recommends that future work prioritize older adults and communities of color, as well as investigate additional approaches applied to smartphone-based technologies. Lastly, the article calls upon researchers to “generate integrated technological systems not only to reduce stigma, but also to consider

preventive interventions in mental health,” such as incorporating resources, screening, and mental health literacy. This dissertation responds to this call by centering psychoeducation, screening, and resources into a stigma reduction digital health intervention for church members.

2.8 Takeaways from the Stigma-Reduction Intervention Literature

In summary, apart from studies that leveraged cultural-tailoring, most studies reviewed did not center African-descended populations explicitly. In a recent narrative review conducted by Rivera et al., only six mental health anti-stigma studies focused on Black and African-descended people, specifically (Rivera et al., 2021). Those six studies varied in the approaches utilized and their evaluation plans. Two of the six utilized church communities, and three included community-engaged principles such as intervention co-creation and reporting findings back to the communities (Rivera et al., 2021). Intervention times varied, with the shortest intervention consisting of a 10-minute contact and the longest intervention featuring a semester-long research participation period. Their review recommends (1) partnering with key community stakeholders to design and test interventions and (2) considering the cultural, social, and political contexts specific to Black and African-descended people by acknowledging issues of medical mistrust and how racial prejudice and discrimination interact with mental illness stigma.

2.9 Digital Interventions to Enhance Readiness for Therapy

As part of this dissertation work, intervention components to encourage care-seeking will be incorporated into the final system. As a result, I reviewed literature that specifically addresses enhancing motivation to seek professional treatment. In a recent systematic review conducted by Jardine et al., researchers characterized the state of digital interventions to enhance readiness for professional treatment (Jardine et al., 2022). Of the 48 interventions reviewed, the types of interventions spanned various technology formats, including but not limited to web-based interventions, video-based interventions, and automated emails or SMS messaging. The results of this systematic review indicate that intervention components, technology type, and intervention length were broad and varied. Of note, one barrier to intervention use was self-stigma. Researchers indicate that currently, most of the interventions analyzed as part of the review did not measure or respond to self-stigma. In addition, only 25% of the interventions included qualitative data. Authors recommend that future work consider using in-depth qualitative data collections to better understand “individual differences in emotional responses to readiness interventions and how constructs such as self-stigma affect motivation,” (Jardine et al., 2022).

This work, combined with the limited work addressing mental health promotion in church communities, suggests that mental health promotion interventions that use preventative strategies, combined with anti-stigma messaging, represent a unique research opportunity.

2.10 Embodied Conversational Agents in Mental Healthcare

Embodied Conversational Agents (ECAs) are animated characters that simulate face-to-face communication using speech and non-verbal behaviors (e.g., gaze, hand gestures, head nods, etc.) to interact with an end user (Cassell et al., 1999). Prior work has demonstrated the effectiveness of ECAs in delivering health promotion interventions and effectively using health counseling techniques (e.g., Cognitive behavior therapy, Motivational Interviewing, etc.) to facilitate behavior change on a myriad of health topics, including but not limited to physical activity (King et al., 2020), genetic counseling (Zhou & Bickmore, 2022), preconception care (Jack et al., 2020), substance use (T. Bickmore et al., 2020b), and depression (Laranjo et al., 2018; Ring et al., 2016a). ECAs have been shown to successfully develop a positive therapeutic working alliance with users presenting a somewhat straightforward translation from in-person screening and counseling to a digital intervention (T. Bickmore & O'Leary, 2023). Unlike text-heavy user interfaces, ECAs can provide support, empathy, and compassion (T. W. Bickmore & Picard, 2005). In addition, such interventions have resulted in positive behavior change outcomes in historically marginalized populations with higher rates of low health, numeracy, and technology literacy (T. W. Bickmore et al., 2010). Mental health screening and promotion has become a burgeoning application area of study for ECAs, with work spanning screening, skill building, and maladaptive behavior detection.

2.10.1 Screening

Bickmore et al. designed a tablet-based alcohol misuse screening ECA. They evaluated the system in two counterbalanced mixed methods validation studies with 57 participants, comparing the ECA to a human and an existing forms-based assessment (T. Bickmore et al., 2020a). Both studies demonstrated the validity of the ECA as a screening tool. In addition, participants disclosed significantly more to the ECA compared to the human, stating that the ECA was a nonjudgmental alternative to a human-administered screener. My proposed system adds to this body of work by providing a space where individuals can discuss symptoms of depression with the ECA versus having to share sensitive information with others in the church.

Philip et al. conducted a counterbalanced within-subject experiment comparing a clinical interview with a psychiatrist to an ECA-administered clinical assessment for major depressive disorder (Philip et al., 2017). In their evaluation, patients found the ECA acceptable, able to communicate empathy, elicit trust, reduce feelings of being judged by a human evaluator, and reduce barriers to disclosure of the user's mental health status and emotional state. Investigators found that the validity of the assessment performed by the ECA compared to a human clinician was satisfactory, with increased measure sensitivity in those with severe depression. Both studies demonstrate that ECAs are promising tools for conducting clinical interviews and screeners in general populations.

2.10.2 Psychoeducation and Skill-building.

Ring et al. developed an *affect-aware* ECA responsive to user affective state designed to help college students reduce symptoms of depression (Ring et al., 2016b). Components of their work included psychoeducation and skill-building using principles of CBT such as thought restructuring, behavioral activation homework, and self-monitoring. Results of their work indicated that participants had significantly higher levels of system engagement and a reduction in state anxiety when using the *affect-aware* ECA. SABORI is a Japanese ECA that promotes mental health using an automated behavioral activation intervention (Suganuma et al., 2018). SABORI asked users questions about their mood and physical condition and provided feedback based on CBT principles.

2.10.3 Detection

HelPath was a digital health tool designed to use sensing data from the patient and the patient's electronic health records to support the identification of risks associated with suicidal behavior (Martínez-Miranda et al., 2019). The ECA was designed to use this information and make suggestions based on principles from CBT to promote mental wellness. As a result, the ECA was the primary user interface. Results of this pilot work showed general acceptance of the ECA; however, experimenters hypothesized that individual differences such as symptom severity and preferences for interactive technologies (e.g., avatars) resulted in broad and varied use among the 12 pilot study participants.

2.10.4 Stigma Reduction

Sebastian et al. created a mental health literacy intervention for individuals with anorexia nervosa to address stigma (Sebastian & Richards, 2017). In a mixed experimental design, the authors compared video-based with ECA-based interactions using two stigma reduction strategies: education and paracontact. Both experimental conditions reduced stigma associated with anorexia nervosa. Importantly, this is the only ECA-facilitated intervention that explicitly focuses on stigma and stigma reduction.

2.10.5 Takeaways from the ECA Mental Health Promotion Literature

In summary, while much of this work has been evaluated using feasibility or small experimental studies, they demonstrate the potential for ECAs to provide access to preventative and life-saving services where logistical (cost, convenience, transportation, etc.) or societal issues (privacy, stigma, etc.) currently prevent or limit access. Researchers have compared user uptake and disposition towards activities when using an ECA to text on a screen (Egede et al., 2021). They found that user uptake and attitudes towards the tasks were positively impacted when interacting with an ECA, indicating that an ECA may be useful in maintaining user engagement. With the exception of one paper, no ECAs explicitly focused on mental health and stigma as a mechanism for encouraging care-seeking and improving user mental wellness. Given that this dissertation's topic of study is designing a digital health tool to address depression and mitigate social stigma, developing a relationship with the user may increase engagement. In addition, an ECA could signal community membership and simulate peer support in communities where peer support and fellowship are highly valued community dynamics and attributes.

Chapter 3: Phase I – Understanding Community Context

This dissertation is part of a larger ongoing research project funded by the National Science Foundation to design, develop, and evaluate an mHealth application with and for historically marginalized church communities in partnership with our community partner, the Black Ministerial Alliance Ten Point. This research used a socioecological approach where I spent time building rapport with church members and learning about their communities. This chapter presents Phase I of this dissertation work and is comprised of three discussion-based focus groups. Each focus group provided me with the necessary background to develop a deep understanding of how these communities operate, laying the groundwork for understanding church members' current practices, needs, and community assets. Furthermore, these focus groups played a critical role in building our research partnership with members. In this chapter, I first present common recruitment and analytic procedures and then the purpose, research questions, method, and findings of each focus group. Table 1 displays a brief summary of Phase I.

Table 1: Summary of Phase One's Procedure, Purpose, and Findings.

<i>Data Collection</i>	<i>Procedure</i>	<i>Purpose</i>	<i>Findings</i>
		Phase I: Understanding Community Context	
<i>Focus Group 1: Understanding Organizational Mission & History</i>	Discussion-Based Focus Group	RQ: How do participants describe their church communities and define their church's current goals and missions?	Church A & B have a rich histories and organizational missions that emphasize nurturing the spiritual health of their memberships. Participants shared their church's history and current in reach and outreach activities (i.e., ministries) at a high-level. The core value of fellowship was identified.
<i>Focus Group 2: Exploring Social Support within the Church</i>	Discussion-Based Focus Group	RQ: In what ways are parishioners currently providing social support to one another?	Social support is exchanged formally through ministries and informally member to member. Prayer support was the most common form of support exchanged. There is a desire to support matters of health, but fewer formal mechanisms currently exist.
<i>Focus Group 3: Identifying Personal & Organization Technology Utilization</i>	Focus Group & Inspiration Cards	RQ: What are current church and church member technology practices?	Church A & B participants use a variety of applications to nurture both their spiritual and physical well-being. Church A does not utilize technology during worship. Church B integrates tech into their worship services and uses technology to for the coordination of events and activities.

3.1 Participant Recruitment Phase I & Phase II

My team and I partnered with an organization that provides resources to over a hundred faith organizations serving predominantly Black churches near our city in the Northeast U.S. Our findings refer to this organization as the *partner organization*. This organization identified two churches to support the project's first year: Church A is a historical Baptist church

with approximately 600 members. Church B is a more recently founded African Methodist Episcopal (AME) church with about 500 members.

I worked closely with the executive director of the partner organization and the health ministry leaders from both churches to identify members of each church who could provide different perspectives and expertise regarding the strengths and needs of their respective church communities. Participants were eligible to participate in the formative work if they were 18 years or older, self-identified as a member of one of the two churches, were English-speaking, and owned a smartphone. I actively sought out participants with various levels of *embeddedness* in the church community to represent a cross-section of experiences reflected in the church membership. Participant consent was obtained at the start of formative work. Participants were paid an hourly rate of 24 dollars. These participants were consented as *key informants*: members of the community who act as primary sources offering deep insight into their respective communities (McKenna & Main, 2013). Key informants will be referred to as participants throughout this dissertation; however, please note that their level of involvement surpasses that of a one-time research participant. All key informants were part of Phase I (Chapter 3) and Phase II (Chapter 4) of this dissertation work. This research was conducted with my University's Institutional Review Board approval.

3.2 Phase I Participant Overview

Eleven church members participated as key informants during Phase I. Seven participants were from Church B and four participants from Church A. All participants self-identified as Black and/or African American. Three participants identified as men and eight as women. Participant age ranged from 26 to 76 (Mean = 57, SD = 17).

3.3 Qualitative Analytic Procedures Common to Phase I and Phase II

A standard set of grounded theory qualitative analytic procedures were utilized in Phase I and Phase II (Chapter 4). Qualitative research centers participant experiences and illuminates nuances of these experiences that, as a method, quantitative analysis does not typically capture (R. D. Campbell & Mowbray, 2016). In health research, qualitative approaches provide detailed descriptions of complex health experiences. This is particularly important for areas of study that concern minoritized community members who have been underrepresented in clinical research (Erves et al., 2017).

Phase I and Phase II data collections were analyzed using grounded theory analytic techniques, namely, initial coding, axial coding, selective coding, and memoing, using NVivo qualitative data analysis software v. 12 (Corbin & Strauss, 2014). The data was analyzed with the assistance of a second Ph.D. student who worked in partnership with me throughout year one. As part of our work, we met regularly to resolve coding discrepancies and discuss concepts and themes. The relevant focus group subsection will report analytic procedures unique to that data collection.

I took an assets-based approach to our work throughout the interviews, activities, and analysis. In recent years, both HCI researchers and community-engaged public health interventionists have made calls to use asset-based approaches in place of

deficit-based models when designing technology-facilitated health interventions (Chisholm et al., 2020; Cho et al., 2019; Dickinson et al., 2019; Harrington et al., 2019). This is especially important when partnering with minoritized communities. Well-meaning public health interventions have historically approached these communities using a deficit-focused lens. As a result, their work highlighted community needs and deprivation of resources at the direct expense of overlooking possible and important community strengths (e.g., a culture of communal support, existing services, etc.).

3.4 Focus Group 1: Understanding Organizational Mission & History – “People follow the Pastors lead”

3.4.1 Purpose

When working with historically marginalized community-based organizations, sociotechnologists should understand the organization's history as part of their exploratory analysis. This process not only assists in building rapport between researchers and community members (M. K. Campbell et al., 2007b), but it further defines the current perspectives and goals of the organization. In Focus Group 1, we learned about Church A and Church B's unique communities. This work addressed this research question:

RQ: How do participants describe their church communities and define their church's current goals and missions?

3.4.2 Method

I conducted a 90-minute facilitated discussion with Churches A and B separately, asking participants to describe their Church community from their perspective. Participants were encouraged to describe the activities they engage in, whom they engage with, an overview of the church's history, its organizational mission, and their perceptions of their church leaders. This session aimed to learn about the participants' church communities in detail.

3.4.3 Findings

Church A's History & Organizational Mission

Church A is an American Baptist Church. Members describe Church A as *traditional*, wherein participants define traditional as a slow adaptation to changes in how church business is conducted. In discussing how they would describe their church to others, participants from Church A felt strongly that their historic church would be well known to others in their state. Inside the walls of the church, members enjoy the benefits of preservation efforts that have maintained their historic worship hall. Fewer than five pastors have led this church community in its history, creating a continuity in leadership. The current pastor led the church for over 40 years. P3 described the powerful influence the current pastor has:

“Well, I will say again, people follow the pastor's [lead]. We don't really have that young people mentality, and I'm big on stuff like that [technology], but I look at what we do [have] and then...what a lot of churches have...Anyway, we can use a lot more of that that we don't do. A LOT more of it.”

Participants routinely stated that the Pastor's length of service, his emphasis on tradition, the conventional practices of their church, and the aging congregation all simultaneously impact the traditional mindset and feel of Church A. When asking participants to describe the various ministries that serve current church members, otherwise known as in-reach ministries, we were impressed by the number. In response to our surprise, one member exclaimed, *“Well, the church has been there for [200+] years,”* further emphasizing the influence of history and longevity on church culture. This focus on in-reach indicates Church A's mission to directly support the spiritual growth and development of the church members. The mission statement of Church A emphasizes the importance of maintaining the spiritual life of members through Scripture and church worship in the black religious tradition. Church A extends this mission to the broader community through activities that focus on bringing new people into the church in order to share in religious expression (e.g., community dinners, Bible study, etc.) as well as tangible support activities (e.g., back to school, clothing, and food drives, etc.).

Church B was established less than 40 years ago and has grown considerably. Church B is an African-Methodist Episcopal (AME) church, a denomination with a rich history of engagement in social and political action. Two co-pastors lead church B. When they founded Church B, they incorporated technology rapidly. Established in the pastors' homes, the church grew from just a few members to over 500 and now offers two services per Sunday to accommodate their membership. Participants describe the arc of this growth as starting from a table and evolving to a gymnasium then to an auditorium, with its latest growth culminating in the transformation of the church campus through the addition of a school.

Church B's History & Organizational Mission

The mission statement of Church B describes collaborations with public, nonprofit educational, community, governmental, and interfaith partners to initiate programs that serve thousands of people both locally and worldwide. While Church B indeed celebrates Scripture and nurtures the spiritual growth of its members, the mission statement indicates Church B's focus on outreach and social justice. Indeed, the AME church was born out of a response to injustice; this Protestant denomination was founded by Black churchgoers in 1787 in response to the racial discrimination they were subjected to (e.g., restrictions where they were allowed to pray in church) (Church, 2023). The AME church's mission statement is, “to minister to the social, spiritual, and physical development of all people.” In keeping with this denominational tradition, Church B's conceptualization of community is not constrained to a particular neighborhood, town, or even country. P8 passionately described her church's mission to serve all people by saying:

“I appreciate the fact that they both said that [Church B] has a connection to the community. It's not just about...ministry at [Church B] is not just about the members of the [Church B] congregation. It's about how we extend ourselves to heal and help

people outside of our immediate faith community. How do we serve as a witness to other people for what faith and belief in God and trust in God should be about? How do we help people overseas who are looking to build a ministry and connect people to God? It's all of those things, I think that makes it kinda special." Church B and its members are committed to addressing societal issues through community engagement. P7 explained, "I think social action and social justice is a big part of our identity as a church."

Conclusion

Findings from Focus Group 1 characterized the organizational identity and mission of both churches. Both churches have a rich history and current organizational mission statement that emphasizes taking care of the spiritual health of their congregants. This workshop helped illuminate these goals. Given the obvious and vital emphasis on togetherness, fellowship, and service to others within each institution, I conducted Focus Group 2 intending to depict the provision of social support within each church.

3.5 Focus Group 2: Exploring the Exchange of Social Support *"You can get whatever you need in the church."*

3.5.1 Purpose

Research has demonstrated that the provision of social support contributes positively to individual wellness and health behaviors. Predominantly Black churches have historically been sources of resources and social support for minoritized members and non-members alike and central gathering places for social networks (Eng et al., 1985; Kaziunas et al., 2019). Kaziunas et al. have argued that beyond social support, faith-based organizations provide critical healthcare services and resources to their populations in need, including those most vulnerable and socially marginalized. Beyond explicit community health promotion efforts, churches represent key cultural institutions where members connect through shared experiences to exchange social support, including spiritual and prayer support. This type of community support (Ferrell et al., 2020) has become even more critical during the COVID-19 pandemic, where Black and minoritized communities have experienced significantly higher rates of disease, disability, death, and economic disruptions than their white counterparts (Cyrus et al., 2020; Millett et al., 2020). The goal of Focus Group 2 was for community members to describe how types of social support are communicated, provided, and accepted by church members. Through their discussions, participants answered the following research question:

RQ: In what ways are church members currently providing social support to one another?

Social support is the provision of verbal and nonverbal communication between providers and receivers that helps the receiver know that they are cared for during times of uncertainty or challenge (Glanz et al., 2015). Social support theory elucidates dimensions of social support and categorizes these dimensions into four different types: emotional, instrumental, informational, and appraisal support (Glanz et al., 2015; Ko et al., 2013). See Table 2 for working definitions of social support. Literature has expanded these categories to include a fifth category called prayer support (Smith et al., 2020). In their study of how spiritual support is communicated in online health communities, Smith et al. (2020) identified another social support

dimension called spiritual support. However, Smith and her participants characterized spiritual support as a category of social support that underlies other social support types (Smith et al., 2021).

Table 2: Social Support Type Definitions

<i>Social Support Type</i>	<i>Definition</i>
<i>Emotional</i>	Demonstrations of compassion, affection, confidence, and concern.
<i>Instrumental</i>	Tangible assistance that can be physically provided
<i>Informational</i>	Advice, suggestions, and information that can be used for problem solving
<i>Appraisal</i>	Information that can be used for self-evaluation
<i>Prayer</i>	Support is offered by another through prayer, spiritual blessings, and warm thoughts.

3.5.2 Method

The interview guide was designed to leverage these categories to investigate what types of social support church members receive and give within these two church communities and the institutional and community processes by which this social support is proffered. Similar to the previous workshop, this workshop was conducted in two sessions, one session per church, and ran an average of 90 minutes. The discussion was recorded and transcribed for later analysis.

3.5.3 Findings

Support through Ministry

Social support across both churches is provided and received in various ways across numerous activities. While neither church has a health-specific ministry, both churches have a range of ministries with numerous social configurations (i.e., age/gender-specific life groups to promote the building and nurturing small group relationships). Participants reported the positive impact of informational support facilitated by sharing private health matters within their smaller ministry groupings. Specifically, one participant reported the helpfulness and positive impact of sharing his recent prostate cancer diagnosis with his men's Bible group. In both communities, ministries ranged in size from a membership of sixty to a small group of five. The larger ministries were designed to enrich the broader church community. For example, an arts ministry where members come together to beautify a shared space, a kitchen ministry to feed members, and events called "Dinner Church" where leaders emphasize that Dinner Church is the place for food, fellowship, and family. Other ministries with larger memberships enrich Sunday worship (e.g., a dance ministry, church choir, etc.). The primary purpose of some ministries is to promote spiritual growth (Bible study) of the laity; while others offer prayer support as their primary role and function (prayer ministry). The prayer ministries in both churches are highly active. Both ministries collect prayer requests submitted to church leadership using an online web form or are requested actively during service. Prayer ministries provide a foundation for a triadic spiritual experience of praying with others and with God (Smith et al., 2021). For Christians, praying together is an important spiritual practice.

Though neither church has a dedicated health ministry, both churches have ministries dedicated to ensuring that older adults and individuals with disabilities have access to the spiritual and prayer support they require. These ministries also identify needs and assist those members who are homebound. Both churches have a caring infrastructure to address both members' physical health crises, financial crises, and bereavement support. Services provided by these infrastructures include meal delivery, assistance getting to and from appointments, resource identification and sharing, financial assistance to help members through immediate financial strain, and spiritual and emotional support. One member, in particular, spoke about money being collected to help congregants with unexpected bills resulting from a medical crisis.

Members from Church B described unofficial social configurations created due to a shared medical experience, in this case, knee replacement. Other members pointed to the [geographic location redacted] walking group, a group of older women who formed a walking group during the spring. Though the initial list of different unofficial groupings around shared healthcare topics and experiences was impressive, many of these initiatives did not have staying power. They were virtually unknown to other members of the focus group.

Support Proffered by Leaders

Participants discussed the support offered by their Church's pastors and ministries as a powerful source of spiritual and emotional support, especially during times of critical need (e.g., a hospitalization). Stories of support included the pastor visiting members in the hospital, calling members to pray with them on the phone, or linking individuals to care opportunities and recommended services. Church leaders were described as well-connected individuals who make a point to develop inter community-based organization connections to both establish collaborative and synergistic outreach ministry opportunities and service-based connections to share with their laity. Participants clearly emphasized a feeling of being *seen* when receiving support directly from the pastors and their ministry staff.

Prayer Support

In Church B, support is provided in various ways similar to that of Church A. However, Church B spent most of the focus group discussing the importance and power of shared prayer. These discussions described a sophisticated infrastructure for matching members with prayer partners (individuals who desire a regular 1-1 prayer contact) and extra weekly services dedicated to group or communal prayer offered in-person and remotely. Such services are devoted to supporting church members using the power of communal prayer. It was clear that these prayer experiences were an integral and powerful aspect of belonging to the community, maintaining a sense of hope, and inspiring spiritual resilience. Members often leveraged such support during times of immense stress or fear.

Barriers to Receiving Support

Three barriers were identified by participants as preventing the exchange of support in their church community: 1) a collective sense of minimizing any perceived burden placed on the pastor or leadership with *small matters*, 2) a lack of awareness surrounding which ministries were active or defunct, and 3) volunteer burn-out. Members emphasized multiple times throughout both focus group sessions that while anyone in need *could* approach the pastor, there is a collective understanding that the Pastor is too busy to address the needs of all individuals within the congregation. Often, members stated that while they knew they *could* go to their pastor, they would not. P7 explained:

“Our congregation, it’s large...at least 600 people. He can’t possibly have one-on-one time with us all. No, I would rather find support somewhere else in the church and make sure that he’s helping those who have more pressing problems than I’m currently dealing with.”

Participants echoed this sentiment multiple times, insisting that while the pastor’s proverbial door is always open, they would only approach church leadership if absolutely necessary.

I observed participants swapping or informing one another of health resources during our focus group sessions on several occasions. Even the most well-connected church members were unaware of many resources. One participant indicated that she had wanted to refer someone from her personal life to a *caring ministry* (a ministry whose purpose is to address the tangible needs of those who may be experiencing a personal challenge or life crisis). This member had incorrectly assumed that the ministry was no longer active and was surprised to find herself sitting next to an active member of that ministry. This missed opportunity is a clear candidate for how a technology system could assist churches and their members in broadcasting currently offered services to the broader community.

Lastly and perhaps unsurprisingly, participants mentioned volunteer burnout as one central reason ministries and services come and go within the walls of their church. For instance, at the time of the focus group, Church B no longer offered most of its “life groups,” small groups designed to build social intimacy and relationship building. These life groups would meet within different geographic communities so that members not necessarily geographically close to the church could enjoy in-person meetings with a small group of church members. The life groups required their leader to develop the curricula, organize the groups, and coordinate life group events and meetup locations. Participants mentioned enjoying the groups but acknowledged that running these groups was no small matter. P11 stated:

“Sister [redacted] did a wonderful job coordinating these. But, it’s a lot to take on. And she just didn’t have enough room on her plate to do it anymore. I hope someone picks it up though.”

Unaddressed Needs

During Focus Group 2, participants from Church A mentioned that while they felt confident that if experiencing a physical health crisis, they could utilize community support for emotional, spiritual, and informational support, they had difficulty imagining

how they could leverage the same community connections for matters of mental or sexual health, stating that both areas are incredibly sensitive in the context of their church. Specifically, Church A members mentioned that even though they know of a few mental health practitioners within the community (other members who work professionally as mental health providers), they could not envision how they could gracefully ask for their help without others in the church finding out or overhearing. Church B members echoed a similar point of view. P12 shared:

“I mean, it’s not necessarily that I’m concerned about others seeing me walk up to Sister [redacted]. It’s that but, it’s also her Sunday too. She’s there to worship...she’s where she gets spiritually fed...she’s there with her family. She’s not necessarily there to hear my problems, even if she doesn’t mind, I mind.”

Here, we see a similar pause in leveraging a community asset, in this case, a church member instead of the pastor. Participants communicated strongly that there needed to be an appropriate *time and place* when asking certain members for assistance.

Conclusion

During Focus Group 2, participants identified how the church provides social support to its members. Much of this work is proffered via numerous social configurations (1-1, small group, large group, etc.), and support providers are either domain experts, peers, or leaders. Through these discussions, participants identified what social support activities currently exist, how those activities are communicated between members, and what barriers currently diminish the provision of care within their communities. See Table 3. for an overview of support exchanged within the church communities based on Focus Group 2.

Table 3: Examples of Social Support Exchanged

<i>Social Support Type</i>	<i>Examples</i>
<i>Emotional</i>	1-1 peer support, member check-ins, hospital visits from pastors or ministry staff
<i>Instrumental</i>	Rides to appointments, financial assistance, meal drop-off.
<i>Informational</i>	Resource sharing, personal recommendations.
<i>Appraisal</i>	Pastoral counseling, self-reflection offered during small group ministry meetings or life groups.
<i>Prayer</i>	Prayer partners, in-person prayer meetings, synchronous remote prayer meetings.

3.6 Focus Group 3: Describing Personal Use and Church Technology Utilization

“I was shocked. That’s one modern screen app. I couldn’t have imagined it.”

Focus Groups 1 and 2 provided the necessary context to understand the current organizational mission and aims, and the various social support activities facilitated by the church. Focus Group 3 was designed to explore what technology church members currently use and their perspectives on technology use within the walls of the church. Given that the overall research project aimed to design a community-engaged smartphone application to address health disparities, it was critical to understand church members' individual and community technology use patterns.

This focus group sought to answer this question:

RQ: What are current church and church member technology practices?

3.6.1 Method

During this two-hour facilitated discussion, participants described their current use of technology, potential benefits of such technology practices, and concerns. Inspiration cards were designed to assist participants in brainstorming the range of technologies they may or may not use. These cards enumerated 24 technologies. They included depictions of commercial software and consumer electronics such as calendar apps, health and wellness technologies, radios, smartphones, disembodied conversational assistants (i.e., Siri, Alexa, etc.), embodied conversational agents (i.e., fully embodied computer characters), smart watches, etc. The format of the cards included a short label and an image.

This workshop was not designed as an exhaustive probe into all the technologies. Instead, we focused on technologies that participants currently used to enrich their spiritual experiences and practices and technologies that promote, facilitate, and enriched their current health practices. These questions were framed on an individual, interpersonal, and community organization level. We ended the session by asking participants to describe what technology, if any, is used in their church, including technologies that enhance, support, and/or promote church services and programs. A review of faith-based technology used in the churches and technology used to support individual spiritual practices is summarized below. For a more detailed and thorough review, please refer to my published work in Human Factors in Computing Systems 2021 conference proceedings (O’Leary et al., 2021).

3.6.2 Findings

Church A and B participants described using various technologies to promote spiritual and bodily well-being. While some of these technologies served a dominant purpose (i.e., promote exercise), others promoted multiple facets of well-being and were leveraged by participants to accomplish various health and spiritually-focused goals. For example, participants reported using YouTube to consume both spiritual content in the form of online sermons *and* exercise content in the form of workout routines or cooking videos. A summarization of the personal use technologies participants identified is provided in Table 4.

Table 4: Technology Utilization by Church

<i>Technology</i>	<i>Church</i>	<i>Domain/Use</i>
<i>Physical activity trackers (e.g., smart watches, Fitbit)</i>	A & B	Physical well-being
<i>Nutrition and calorie trackers (e.g., WW, MyFitnessPal)</i>	B	Physical well-being
<i>Reproductive health (e.g., Flow)</i>	B	Physical well-being
<i>Medical Record Systems (MyChart)</i>	A	Physical well-being
<i>Social Media (e.g., Facebook, Instagram)</i>	A & B	Physical & Spiritual well-being
<i>Messengers (e.g., Facebook Messenger, WhatsApp, SMS)</i>	A & B	Physical & Spiritual well-being
<i>Team-based Messengers (Slack)</i>	B	Spiritual well-being

<i>Email</i>	A & B	Spiritual practices (communication with ministry members and church leadership)
<i>Alarm Clocks</i>	A & B	Physical well-being (reminders for doctors' appointments, medications, and events at the church)
<i>Calendar apps</i>	A & B	Physical & Spiritual well-being
<i>Bible Application, E-Bible Study, and Daily Devotional</i>	A & B	Spiritual well-being
<i>Conversational Assistants</i>	B	Physical well-being
<i>E-readers</i>	B	Physical and Spiritual well-being
<i>Collaboration tools (e.g., Google Drive)</i>	B	Spiritual practices (organizational capacity)

Most participants in Church A & B reported using a Bible application to complement their individual and interpersonal spiritual practices, such as supporting individual and group Bible study. Reviewing any one application is outside the focus of this workshop; however, participants shared a detailed description of the features and their use of the YouVersion Bible App (*YouVersion Bible App*, 2017).

The Bible application contains thousands of Bible translations, daily devotionals, and reading plans, encouraging users to “read, listen, watch, and share the Bible” daily as part of their mission statement. The application promotes daily practice by quantifying the user's spiritual practices, similar to fitness applications, by awarding badges for consecutive days of application use. Notifications are pushed to participants' phones, nudging them to continue their Bible study practice. Participants explained how quantifying their spiritual practice sometimes led to higher self-efficacy and a feeling of pride. However, missing a day and breaking their reading streak was disappointing and sometimes disruptive to the participants’ commitment to their spiritual discipline. One member lamented when she realized that she lost her 7-day Scripture reading streak. Another member responded, “*You read the word when you needed it,*” indicating that the participant should not let the application's attempts at nudging define her spiritual needs or progress.

Overall, Church A participants reported less frequent personal technology use than Church B and less experience with hardware outside of smartphone and computer usage. Church B reported more experience and daily use of tablets, e-readers, smartwatches, conversational assistants, and personal computers and smartphones. Notably, Church A’s discussion remained centered on individual technology use as their church, at the time of the focus group in 2019, did not utilize technologies apart from broadcasting Sunday worship by radio.

Though Church A participants seldom used technology within the church, they leveraged technologies for membership communication. Ministries and leadership utilize email and messaging for event coordination, organization, and support exchange. One member of Church A explained that the youth ministry uses WhatsApp to communicate between themselves and their co-pastor. Younger members from the focus groups described sharing Scripture and gospel music via their social media accounts (e.g., Facebook, Instagram, etc.). By doing this, participants leverage technology to connect to others in their larger community on the topics of faith and spirituality. When asked why sharing this content is essential, P2 explained, “If it helps me,

it could help someone else,” demonstrating that by sharing Scripture, she can offer spiritual support to her wider community (an important spiritual discipline).

Unlike Church A, Church B members utilized technology *within* and *outside* their church’s walls. When it comes to technology that supports physical well-being, Church B members reported using a wide variety of mobile phone fitness applications such as Noom, WW, MyFitnessPal, Apple’s native Health application, activity on Apple Watch, Fitbit activity application, and reproductive health apps such as Flow (a menstruation and ovulation tracker). Participants characterized technology integrations and eco-systems between their wearables, smart scales, and syncing their activity with friends and family. Members mentioned that their continued use of such wellness technologies often relies on a critical mass of total use by their social network. P3 stated, *“my steps don’t matter unless someone else can see it!”* Unlike Church A, Church B members use technology to connect to others with shared goals and experiences to promote their health.

Church B shared how their use of collaborative productivity applications like Google Drive and Slack assists ministry leadership with coordination efforts and ensures that each ministry receives the required resources. For instance, Church B uses a scheduling application to reserve rooms within the physical space of their church to hold ministry events. In Church B, Slack is a virtual platform for coordinating church-based activities, events, and ministry communications. A channel within Slack managed by the prayer ministry allows ministry members to post calls for prayer that were submitted by church members. The Slack channel is a database of sorts—tracking what prayer requests require ministry attention.

Members of Church B also described how messaging technologies support small-group communication. Church B described using video conference software, phone calls, text messaging, and social media platforms to interact with other church members regularly. For instance, one member explained that members had approached her via text message to discuss her experience with knee and hip surgery. She explained that several Church B members who have had knee surgery have gotten to know one another, and those connections were initiated using technology. In addition, Church B participants credited the technology they adopted with increasing their ministries’ capacity to support members and facilitate service utilization. P11 shared, *“Once our church grew, you sometimes only know the people in your small group, not necessarily everyone else. But technology helps us connect outside of Sunday and find other members who can help or have been through similar things.”*

3.6.3 Conclusion

I began Focus Group 3 with intentionally broad framings: asking participants to share what technologies they use daily that bring value. The focus group was framed in a positive light; we did not spend time assessing technologies at this point that did not serve the community or member interests. While participants were encouraged to tell us what problems they have encountered when using these technologies, our focus was on what these technologies augmented both for the individual and the broader communities. Participants shared that the most highly used and valued technologies they have adopted are those technologies that

enrich their spiritual values, promote social connection, and increase access to health-enhancing material and resources tailored to their community.

Chapter 4: Phase II Participatory Design Focus Groups

In Phase II of this dissertation work, we arranged and conducted four participatory design focus groups to explore concepts for an mHealth application using a combination of participatory design practices such as affinity diagramming, story completion, sketching and storytelling, and photo elicitation. Focus Group 1 of Phase II defined what health priorities church members most valued. In Focus Group 2, participants identified and characterized design guidelines for a culturally-tailored ECA. In Focus Group 3, participants used a novel story completion method called ‘modular storyboards’ to identify features and sources for in-app social support for a faith-centered mHealth app. Focus Group 4 examined the intersection of race and religion as it relates to designing a church-based mHealth application using photo elicitation. See Table 5 for a brief summary of Phase II.

Phase II’s four focus groups yielded critical findings that contributed to the identification of six guiding design principles that were subsequently leveraged in the final design of the mental health and stigma reduction intervention (presented in Chapter 7). As I describe Phase II’s focus groups in this chapter, and Chapters 5 and 6, I will formally establish each key design principle and its relevance to the final design.

Table 5: Phase II Data Collections.

<i>Data Collection</i>	<i>Procedure</i>	<i>Purpose</i>	<i>Key Finding(s)</i>
Phase II: Co-design Focus Groups			
<i>Focus Group 1: Health Priorities</i>	Focus Group & Affinity Diagramming	RQ: What health topics would church members identify as most essential for themselves and their community for a church-based mHealth app to address?	Church A & B participants identified as mental health as both their highest priority and least addressable health concern.
<i>Focus Group 2a: ECA design</i>	Focus Group & Drawing & Writing Activity	RQ: What role and persona should a community-based culturally-tailored ECA employ to support health and wellness in faith communities with predominantly Black and African American memberships?	The majority of ECAs created to support health and wellness were designed to support participant spiritual practices and experiences demonstrating the importance of integrating spirituality into the ECA design and functionality. (O’Leary et al., 2020)
<i>Focus Group 2b: ECA dialogue tailoring</i>	Focus Group, Demo, & Writing Activity	RQ: How might participants adapt a health counseling conversation to include matters related to their faith and spirituality?	A set of ECA design guidelines were established for creating culturally-tailored community-based ECAs for specialized populations. (O’Leary et al., 2020)
<i>Focus Group 3: Exploring Features to Deliver In-App Support</i>	Focus Group, Individual Interviews & Story Completion Activity	RQ: What resources would church members expect or seek-out in a church-based health app.	Participants preferred ECA support for sensitive health topics. App should increase access to community-based and national resources. (Stowell & O’Leary et al., 2020)
<i>Focus Group 4: Exploring the Intersection of Race & Religion in Health Technology Design</i>	Focus Group, Dyad Interviews & Photo Elicitation	RQ: How can we design digital health tools that are responsive to the intersectional identities of race and religion in multicultural church communities?	Church members were interested in community-building framings to engage members in health and wellness content that supports their church’s mission and goals. (O’Leary et al., 2021)

For an overview of the six key design principles that will be laid out in both this chapter and the next two, please refer to Table 6.

Table 6: Digital Mental Health Promotion Key Design Principles for Predominantly Black Church Communities

<i>Data Collections</i>	<i>Key Design Principles</i>
<i>Focus Group 1: Health Priorities</i>	Address stigma when designing for mental well-being.
<i>Focus Group 2a & b: ECA design</i>	Integrate spiritual elements and faith-based concepts into both ECA design and health promotion content.
<i>Focus Group 3: In-App Social Support</i>	Offer resources to increase church capacity for mental health support
<i>Focus Group 4: Race & Religion in Tech Design</i>	Use a community-building framing to encourage community members to interact with the mental well-being and anti-stigma content
<i>Church Connect Field deployment</i>	Balance tensions between the values of maintaining privacy & discretion and communicating care
<i>Mental Health Feature Identification</i>	Prioritize features that build mental health literacy <i>not</i> the provision of unstructured social

4.1 Methods Common to Phase II Focus Groups

Each data collection was run as two focus group sessions (one focus group per church) and incorporated a design activity (e.g., photo elicitation, affinity diagramming, story completion, etc.) to engage participants and encourage co-design. Focus group activities were intentionally designed to afford members several ways to participate. For instance, all design activities had optional worksheets allowing participants to describe their thoughts or ideas without feeling pressured to fully engage in the design activity if that activity did not feel comfortable.

4.2 Focus Group 1: Establishing Health Priorities – “If you don’t talk about mental health, you aren’t designing for our community.”

4.2.1 Purpose

This research took a community-based participatory research approach. As a result, what health topics the mHealth application would address was left for the community to decide. During Focus Group 1, participants identified health topics most relevant for themselves and their church.

As part of this work, we investigated the following research questions:

RQ1a: What health topics would church members identify as most essential for themselves and their community for a church-based mHealth app to address?

RQ1b: What types of church-based care networks, programming, and initiatives currently support these member-identified health topics? If at all?

4.2.2 Method

To answer these research questions, I facilitated an affinity diagramming activity. Participants were asked to brainstorm five to six health priorities for themselves individually and another five to six for their community collectively. Next, participants wrote each idea on a Post-it note (generating an average of 11 Post-it notes per participant). At the end of the 10-minute brainstorming period, participants posted the notes on the wall of the meeting room. This brainstorming activity resulted in 164 Post-it notes across Churches A and B.

Following traditional affinity mapping procedures (Hanington & Martin, 2019), participants were given fifteen minutes to review the post-it notes as a group. Next, participants were asked to group similar ideas and give each grouping a “few-word” name or label. We asked participants to work together to come up with these category names. This procedure resulted in 14 categories for Church B and 11 categories for Church A.

After creating the categories, participants ranked the categories from lowest to highest health priorities (Figure 2). Next, participants ranked the categories by least and most addressable given the current church’s capacity, social processes, and culture (Figure 3).

4.2.3 Findings

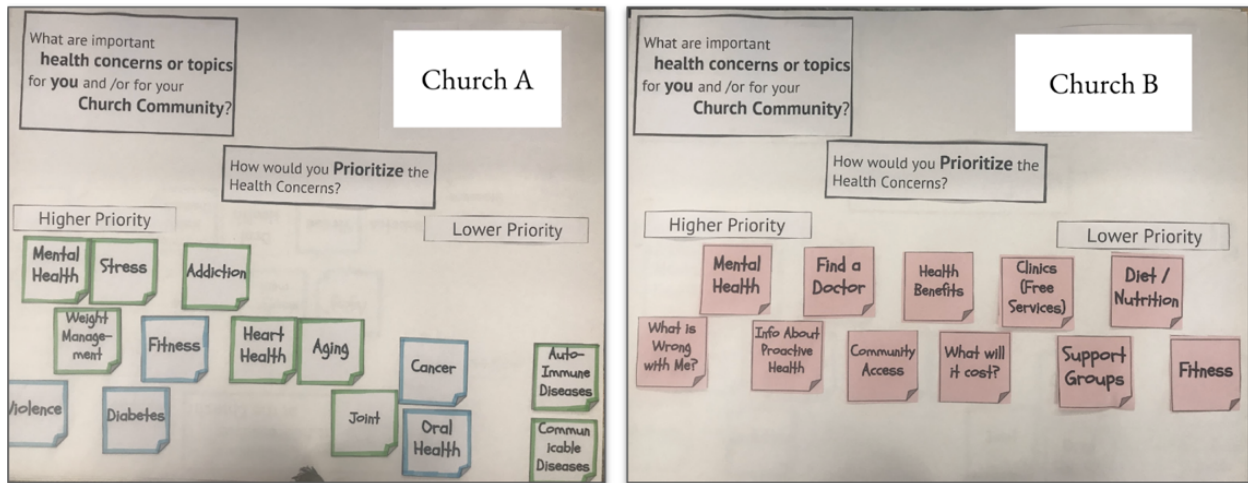
Participant Overview

Eleven church members participated in this data collection. Seven church members were from Church B and 4 from Church A. All participants self-identified as Black and/or African American. Three participants self-identified as men and eight as women. Participant age ranged from 26 to 76 (Mean = 57, SD = 17).

Highest Priority Health Concerns

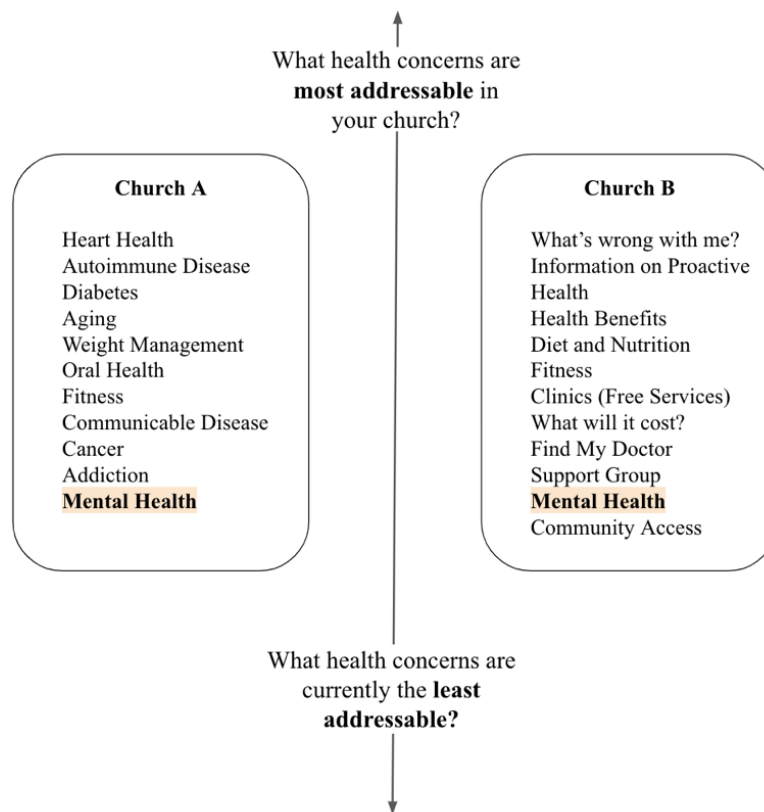
Participants provided categories for a range of health topics, such as heart disease, physical activity, and nutrition, as well as more sensitive topics, such as violence, sexual health, and mental health. Church A focused on uplifting and recognizing community health centers and resources. How the church members constructed the categories in Church A highlighted the need for informational health content in the form of linking participants to services or, at a minimum, promoting awareness of what services currently exist within the geographic region. Church B’s categories pinpointed topics of concern for the community. All participants came to a consensus that all issues listed in the affinity diagrams could be addressed within the confines of the church. However, they remained uncertain as to how the church could support mental wellness. See Figure 2 for an overview of the participant created affinity diagramming categories.

Figure 2: Ranked Health Priorities by Church



When asked to rank health priorities from highest health priority to lowest, participants from both Church A and B named mental health as one of their *highest-priority* community health concerns (Figure 2). As shown in Figure 3, participants simultaneously rated mental health as one of the *least addressable* health topics, stating that current church practices and initiatives cannot support mental health adequately.

Figure 3: Health Topics Ranked as Least to Most Addressable by Church A & B



Participants attributed this discrepancy (mental health ranked as one of the highest health priorities but currently the least addressable) to **mental health stigma**. Both discussions in Church A and B focus groups supported findings from the literature that talking about mental health and depression, especially in the context of the Black church, is not a widely accepted practice. Participants emphasized that mental health is “*just not talked about*” and is “*just taboo*” within the confines of the Church walls. Participants explained that because church members who are experiencing mental illness are not typically validated in their lived experience, there is a possibility that an individual who is currently engaged with the church community may avoid care-seeking. One participant shared her experience, stating that when she shared her struggles with depression, her Bible study ministry group told her to “*pray on it.*” A second focus group member said, “*But you need more than prayer.*” Importantly, participants stated that while spiritual support is integral to individual mental wellness, it is not a cure-all. Participants reasoned that the lack of community-wide mental health literacy contributed to mental health not being explicitly discussed within the community.

4.2.4 Implications for Design

Results from this focus group identified a current gap in care and a potential research opportunity: How can a smartphone-based ECA promote mental health and build community support for those experiencing mental illness in communities where the prevalence of stigma is perceived as high? This focus group and analysis established the first key design principle, “*address mental health stigma when designing for mental wellness in communities where the prevalence of stigma is high*” (Table 6).

4.3 Focus Group 2: Co-designing a Culturally-tailored ECA – “*She’s speaking my language!*”

Phase II Focus Group 1 established what types of intervention topics the ECA should promote. However, I had yet to explore how to culturally tailor the ECA for predominantly Black church members with participants. Many ECA applications are developed for specific user populations with unique beliefs, attitudes, language, and cultural practices. For these users, seemingly subtle mismatches between ECA behavior and community norms can lead to lower satisfaction and engagement and, in some cases, complete rejection if ECA behavior is deemed inappropriate or offensive. It can be challenging for ECA designers who are not community members to incorporate such subtleties into their ECA designs and behavior without exhaustive iterative prototyping and user testing.

While often difficult to achieve, cultural tailoring is vital to maximize the impact of ECAs. Several studies have demonstrated that culturally-tailored ECAs lead to better user attitudes and persuasion than non-tailored ECAs (Noar et al., 2007). Participatory design methods, in which members of a priority user group participate directly in system design decisions, represent a promising approach to developing culturally tailored virtual agents. Representative users can ensure, from the outset, that agent designs will be acceptable to and appropriate for community members by filtering out inappropriate designs at the earliest possible stage and guiding designers towards options that the community will value.

I developed the ECA design and dialogue focus groups and engaged participants in the design of community-based, culturally-tailored ECAs for a church-based mHealth application. During these workshops, we investigated opportunities for ECA-administered support. This research was published as a full paper in the Intelligent Virtual Agent’s 2020 conference proceedings (O’Leary et al., 2020). This work addressed the following research question:

RQ2: What role and persona should a community-based, culturally-tailored ECA employ to support health and wellness in faith communities with predominantly Black and African American memberships?

During this workshop, church members co-designed their aspirations for their ECAs. Through this exercise, they developed ECA back stories, roles, and functions for their idealized church-based health application.

4.3.1 Method

Embodied Conversational Agent Demonstration

Participants interacted with two different ECAs, one designed to assist people with atrial fibrillation and the second created to provide palliative care counseling to people with an advanced illness, including religious content for spiritual support. Both agents were introduced to participants to 1) ground participant conception of embodied conversational agents, including how a virtual agent could look or behave when deployed on a smartphone, 2) demonstrate how religious content could be included in an mHealth app, 3) and gauge participant reactions to a faith-based agent that is not tailored to them.

Design an Embodied Conversational Agent

Next, participants were instructed to design their own agent by being asked: “Think of a time in the past week or month that you needed help; design an agent who could have helped you. Give this agent a name and think about who they are.” To ensure the activity was accessible to participants with varying design backgrounds and skill levels, I created 13 agent templates (eight female, five male) as a starting point for participant design. The templates were based on people of color with various body sizes, hairstyles (including lack of hair to encourage participants to draw their own hair), and clothing styles. The agent templates were printed in black and white and appeared as unfinished coloring book pages (Figure 4a) to encourage participants to make their own designs (Hanington & Martin, 2019). Participants were encouraged to customize the agent template using the provided design materials. With each agent, participants completed a worksheet that prompted them for the agent's name, details about the types of support the agent could provide, and where and when they would interact with the agent (Figure 4c). Participants were encouraged to create three agents in the 60 minutes allocated to agent design. Participants then discussed differences and similarities in their designs, allowing them to endorse or challenge each other's agent designs. The focus group lasted 2.5 hours.

Figure 4: A) Blank Agent Templates. B) Created Agents. C) Completed Worksheet.

 A	 B	1. Who is your agent? What do they do, how do you know them, do they go to your church Agent's Name: <u>INER - SHE'S</u> A MEMBER OF THE CHURCH - A VERY FAITHFUL MEMBER; SHE'S ALWAYS HELPING OUT WITH OTHERS AND SPEAKS TO MOST OF THE MEMBERS IN THE CHURCH. 2. What do they talk to you about? <u>I SHE TALKS ABOUT SPIRITUAL HEALING - BEING SWAY WITH YOURSELF - NEVER GIVING UP ON GOD - BEING HOPEFUL HOW GOD NEVER FAILS!</u> 3. What do you talk to them about? <u>My relationship with GOD and myself. How I want to grow in the church. How I want to know the bible better. How happy that I am to know someone like her.</u> 4. What kind of help would you seek from them? <u>Definitely emotional support. Information, Advice, Reminders, Emotional Support, Companionship. She is easy to talk to and is a good listener. We always talk about spiritual up-lifting and the greatness of GOD. How GOD never fails.</u> 5. When and where do you interact with them? <u>Mostly at church - after the church, home, store, doctor's office, work, in a car ... Before bed, commuting to work, before after or during worship service, during Coalition. We talk about amongst other members about the pastors message for the day, how we are taking this message out into the world. Sometimes we will call each other at home before bedtime and read bible to each other. We will talk about strengthening ourselves. Digging deep into your soul to find your self and to stay steadfast.</u> C
---	---	--

4.3.2 Findings

Participant Overview

A total of 11 church members participated in the design and evaluation workshops. Eight participants self-identified as women and three identified as men. Participants ages ranged from 27 to 73. In terms of education, one participant had some high school education, three had some college, four had a BS/BA, and three had advanced degrees. Their health literacy was high, with all participants scoring 64 or above on the 66-word REALM health literacy test (Davis et al., 1993). Participants had been members of their respective churches for an average of nine years (SD=8.4, range: 2-27). Their involvement in church-affiliated groups or ministries ranged from *no involvement* to *serving multiple roles*, including attendee, ministry team leader, choice member, and trustee.

Reactions to the Embodied Conversational Agent Demonstration

Participant responses were focused on the palliative care agent that provided spiritual support. Participants suggested changes in the agent's appearance related to attire and perceived ethnic identity. Given the agent's spiritually-focused functions (e.g., discussions around end-of-life, religious upbringing, etc.), participants did not approve of the manner of dress. As P8 explained, *"She had on jeans. Well, you know, [it] wasn't that I didn't like them. It was just that she was a ... You know, for what she was doing, she was a bit casual."*

While the agent's clothing would have been acceptable in other contexts, such as a health coach, participants explained that casual clothing was incongruent with their preconception of someone they would turn to for guidance in their faith. Further, the white demo agent was described by participants as a *"a typical minister at a nursing home or hospital."* One participant challenged the agent design and explained that she would prefer a Black agent. When asked why, she explained, *"I think it's the idea of trying to have a character that reflects your, whoever your audience is. That is going to allow them to identify with the character, cause what... I think people are more likely to listen to and get mess-, and receive messages from people that they identify with."* Several other participants echoed this message of creating agents similar to themselves during the focus group.

Most of our participants indicated they attended church for most of their lives and engaged in Bible study. While they appreciated the attempt at incorporating spiritual discussions into the dialogue, the prayers, and Scripture included were from a different version of the Bible than is used by their church communities. Further, the dialogue was too basic and uninteresting to be helpful. P7 explained, *"So, I don't know if they're taking [the religious knowledge] into consideration as to the type of user. And um, the other thing...it's like, the tempo. It sort of seemed kind of, talking down? As opposed to I'm here with you."*

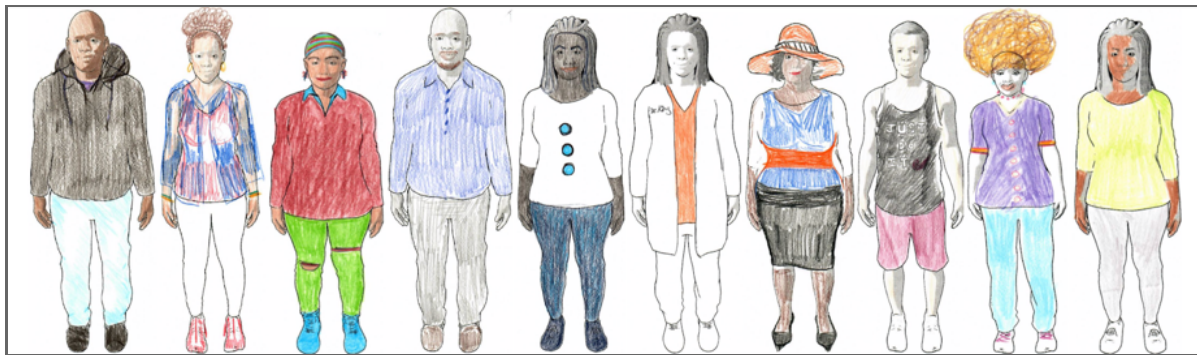
Participants also indicated that the religious knowledge demonstrated by an agent should be tailored to their level of religious knowledge and expertise. For example, when prompted by the agent to say another prayer, P3 anticipated the prayer that would be used, stating, *"That's why I said, no [to praying with the agent]. I know that's the default one [The Lord's Prayer]. So that's*

why I said, “No,” okay?” By accurately predicting the prayer the agent would suggest, the participant demonstrated how the agent loses credibility and functionality by presenting materials without considering the user’s religious knowledge.

Participant Created Embodied Conversational Agents

Participants designed 33 agents that could provide them with support in various aspects of their lives. Participants gave these agents detailed personas and often designed them with multiple functions to support the user. Fourteen of the agents are based on family or friends of the participants, suggesting that in creating agents, a strategy for grounding the exercise might be to ask about people in their lives who support them. See Figure 5 for a subset of created agents.

Figure 5: Examples of Participant-Created Embodied Conversational Agents



Agent Persona

There are a variety of personas that agents embodied in our findings (Table 7). Over half of the agents were described as Peers or Friends, where participants imagined engaging in social chat, spiritual discussions, and encouraging each other on their health goals. Three peer agents offered expertise and advice, sometimes on serious health concerns such as intimate partner violence, substance use, and mental wellness. Church Leader agents embodied the persona of a pastor, minister, and long-time church member, all a source of information, resources, and advice. Church support staff included agents labeled as special event coordinators and Bible study coordinators. Health and Wellness Experts included various roles such as a yoga instructor, a tennis coach, and a nutrition coach.

Table 7: Embodied Conversational Agent Personas by Type

<i>Persona</i>	<i>N</i>
<i>Friends/Peers</i>	17
<i>Church Leaders</i>	3
<i>Church Support Staff</i>	3
<i>Health & Wellness Expert</i>	3
<i>Personal Assistants</i>	5
<i>Church Newcomer</i>	1
<i>Role Model</i>	1

Agent Functions

Participants created agents to support them in various aspects of their lives, often assigning multiple functions to a single agent (e.g., encouragement, advice, nutrition support, check-ins, stress management). While the prompt for the types of support or functions was broad, participants created a striking number of agents that engaged with them on faith-based challenges or incorporated the participant's faith into discussions about health and wellness (24/33). We referred to agents who provided these functions as *Spiritual Agents*. Agents supported participant spirituality by encouraging prayer, teaching about the Bible, incorporating spiritual content as a source of encouragement, and sending reminders for church events. P12 shared how he would interact with his agent, *"I talk to [her] about my relationship with God and how I want to grow in the church, and how I want to know the Bible better, how she could help me, and how happy I am to know someone like her."*

Of the 33 agents, 13 (40%) focused only on the participants' spiritual development, and an additional 11 (33%) agents facilitated the participants' spiritual journey and provided other non-spiritual functionality (see Table 8). P4 described her interactions with the agent, saying, *"It sort of goes from one thing to another [nutrition], but we always come back to that personal path and that journey. That spirituality."* She demonstrated how spirituality and faith are not limited to Sundays, but are interwoven throughout all aspects of her life. As a result, P4 expected that her core beliefs would be included in the design as part of her interactions with the agent.

Fourteen agents were designed to support participants' health and wellness goals (42%) by guiding meditation, facilitating connections to health resources in the participants' community, and encouraging their self-care through preparation for a doctor's appointment.

Table 8: Breakdown of Spiritual and Non-Spiritual ECA Designs by Primary Function

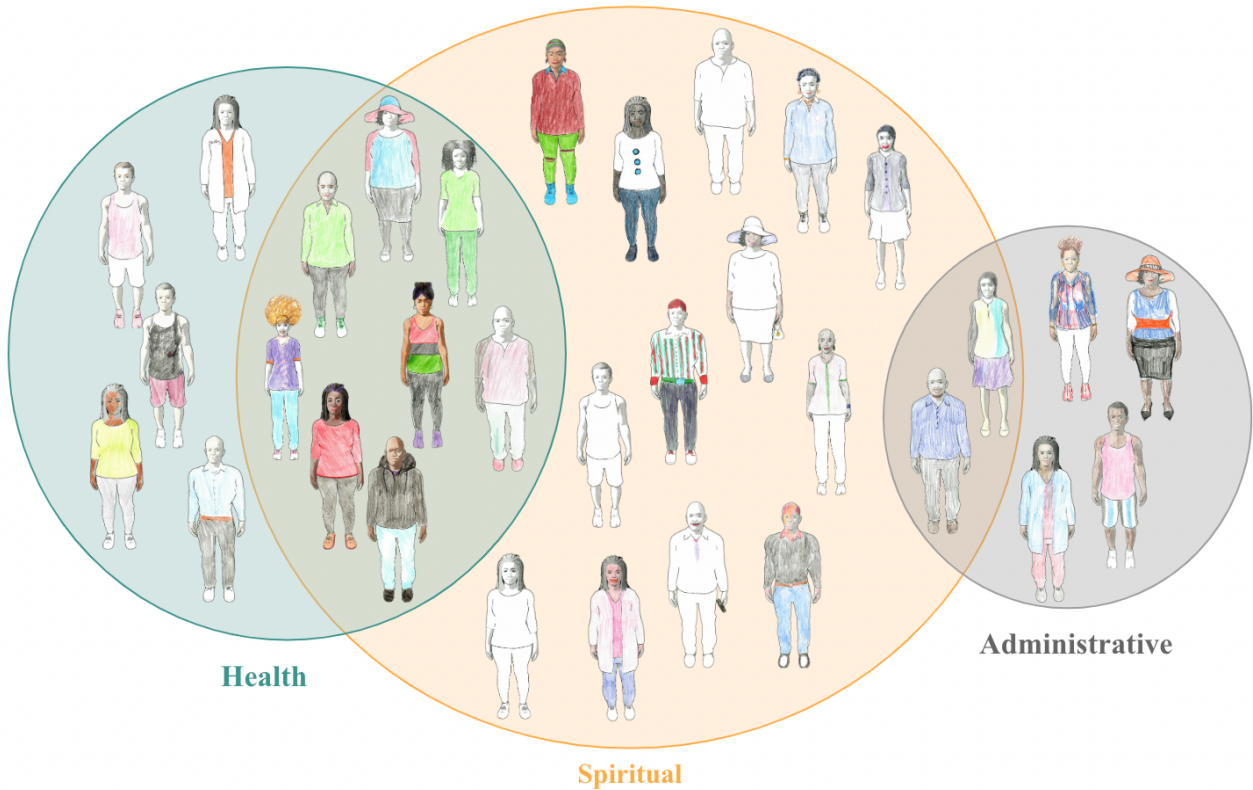
<i>Primary Function</i>	<i>Agent Type</i>	
	Spiritual (n=24)	Non-Spiritual (n=9)
<i>Spiritual-Only</i>	13	0
<i>Health</i>	9	5
<i>Personal Assistant</i>	1	3
<i>Work Assistant</i>	1	1

4.3.3 Design Implications

The Agent Design focus group gave us insight into participant preferences in agent appearance, personas, and desired functions. Of the 33 ECA designs, 24 were described as persons of faith and were developed to support a range of spiritual disciplines and experiences (e.g., prayer, Bible study, supporting church-based activities, sending reminders for worship, and incorporating spiritual content to support user health goals, etc.) (Figure 6). Even though participants knew that the ECA would be the central feature supplying health content, education, and counseling in the mHealth app's final version, they requested that

the ECA be spiritual, framed as a peer, and prepared to support multiple facets of well-being (i.e., spiritual, mental, and physical well-being).

Figure 6: All Embodied Conversational Agents Designed by Participants Grouped by Function(s)



4.4 Focus Group 2b: Tailoring Embodied Conversational Agent Dialogue

4.4.1 Purpose

Since participants communicated strongly that the ECA should incorporate spiritual content as part of the health dialogue with the end user, I conducted a dialogue-tailoring workshop to investigate the research space. The workshop allowed participants to adapt a secular agent application to incorporate language reflecting their communities' beliefs and values. This work answered the following question:

RQ2b: How might participants adapt a health counseling conversation to include matters related to their faith and spirituality?

4.4.2 Method

Authoring Worksheets

To scaffold participant responses, we created worksheets that decomposed the agent-user conversation into adjacency pairs (Figure 7). Each sheet contained one turn of dialogue: one agent utterance and a menu with possible user utterances. On the left-hand side of each worksheet, participants were presented with the conversational agent utterances and user menu options they saw in the demo. Participants could rewrite the utterances on the right-hand side and reimagine the user menu options. Participants were instructed to leave the right-hand side of the document blank if they endorsed the current conversational script. We intentionally created worksheets with lo-fi design attributes to encourage participants to adapt content, challenge the current design, and elicit additional feedback.

Authoring Session

Five participants took part in the focus group. To begin, participants were shown a live demonstration of the health promotion conversation that they would be adapting. Participants were then given authoring worksheets of the conversation. They were instructed to endorse agent utterances and user menu options that were representative of what they would say or want to hear when conversing with another member of their community. For utterances and user menu options that were not representative, participants were instructed to author new turns of dialogue guided by the following prompts: (a) If you were having a conversation with someone at your church, how would you phrase this? (b) How would you change what the agent says, using language or content related to your faith? (c) How do you use language or content related to your faith to encourage or congratulate another church member on their health-related progress? (d) What language or content related to your faith would you find encouraging, motivating, or meaningful to hear while you worked on a new health goal? These prompts were displayed for participants to refer to throughout the 60 minutes they worked on the packets. Directly following, an investigator-led 40-minute group discussion containing open-ended interview questions allowed participants to contextualize their script choices and provide overall feedback on the design session.

Figure 7: Embodied Conversational Agent Dialogue Tailoring Worksheet

Conversation 1		S8: Worksheet	Date: _____ ID: _____
Step 1 - Clara says: Have you had any other problems walking that you would like to talk about?	Step 2 - Write your own message from Clara. Clara should say: <u>WHAT CAN WE DO</u> <u>TO MAKE YOUR WALKING A</u> <u>MORE ENJOYABLE EXPERIENCE</u>	Step 4 - Write your own responses. You would respond: A. I'M LOOK TO WALK WITH B. WALK MY DATE ASSOCIATE TO RETURN C. THEIR INTEREST IN WALKING W/ME. D. AS MY WALKING BUDDY.	
Step 3 - Cross out responses you do not like. You can say: Yes. Nope. I THINK THAT'S A GOOD IDEA	Step 2 - Write your own message from Clara. Clara should say: _____ _____ _____ _____		

4.4.3 Findings

Participants authored a total of 71 agent utterances (Mean=14.2, SD=2.5) and 35 user menu options (Mean=7, SD=2.5) each (Table 9). All but two agent utterances and one user menu were modified, and three of the five participants contributed an additional turn of dialogue. The majority of the utterances were secular (e.g., “Clara should say: *Let's discuss the main problem that prevented you from hitting your walking objectives*.”) Language related to the user's religion was woven into both agent utterances (e.g., “Clara should say: *I know things can be hard at the beginning, but with perseverance and prayer we can get through this, if you need me, I am just a click away*”) and user menus (e.g., “You would respond: A) Blessings to you. B) May God continue to bless you. C) See you next week!”); however, scripture was used in agent utterances far more often than user menus (25.4% and 5.7%, respectively). Of the 20 utterances with scripture, 16 unique passages were chosen from eight different books of the Bible.

Table 9: N (%) of Agent Utterances by Category

	Secular	Faith	Scripture
Agent Utterance (n=71)	69% (49)	12.7% (9)	25.4% (18)
User Menu (n=35)	82.9% (29)	11.4% (4)	5.7% (2)

During the workshop, participants discussed considerations for spiritual content included in health conversations. Participants agreed that Scripture should be integrated into the dialogue, emphasizing that the agent should speak similarly to how their friends would speak. Further, participants indicated that the degree of religious tailoring may need to be dynamically tailored for individual users, even from the same church. For example, a newcomer to the church or religion may feel intimidated by too much faith-related language. P12 explains, “*I also think about newbies. People that are just coming into the faith? Sometime[s] they might feel a bit intimidated, to talk a lot of Scriptures... so that might be a barrier.*” Participants proposed two solutions to tailoring religious content to user preference: 1) explicitly asking the user if they would like the agent to share Scripture and 2) creating adaptive user menu options that dynamically alter the levels of faith-centered, that is, as users select more faith-centered responses (e.g., “*Praise the Lord*”), the agent would respond more frequently with similar language.

Through analyzing both the design and dialogue workshops, I derived a set of ECA design guidelines for creating culturally-tailored ECAs for church-affiliated users (Table 10) (O’Leary et al., 2020). In Chapter 7 (system design), I will walk readers through how these guidelines are operationalized in the final design of the mental health promotion and anti-stigma intervention.

Table 10. Embodied Conversational Agent Design Guidelines

Guidelines

Establish the role and cultural background of the agent early in conversation.
Establish common ground <i>and</i> solidarity through tailored social chat.
Leverage religious concepts and Scripture to motivate and persuade users.
Provide culturally-tailored problem solving
Adapt the degree of tailoring to the user’s degree of cultural involvement.

Agent Validation Study

Next, I conducted a counterbalanced within-subjects experiment to evaluate the effect of spiritual tailoring on attitudes toward health behavior change. The study had two conditions: SPIRITUAL, in which the agent used religious and spiritual language in its counseling dialogue based on the ECA design guidelines (Table 12), and SECULAR, in which the agent used secular dialogue of equivalent duration instead of religious content. Participants had conversations with two different characters during a study session. One promoted exercise, and the other promoted diet (fruit and vegetable consumption), with one of these conversations SPIRITUAL and the other SECULAR. The order and association of character, topic, and study condition were randomized. Following the interaction with both agents, I conducted a qualitative semi-structured interview. Participants were asked about their impressions of both agents and their attitudes towards and perceptions of the spiritual language. In addition, participants were asked to describe the impact of spiritual language in the context of this healthcare conversation with an agent.

During the interviews, participants stated that they recognized the incorporation of language and content related to their faith when interacting with the SPIRITUAL agent. They further endorsed the use of spiritual and religious content and described

it as appropriate, genuine, and motivating when used in health counseling. When asked which agent they would like to continue to talk with, all participants endorsed the SPIRITUAL agent.

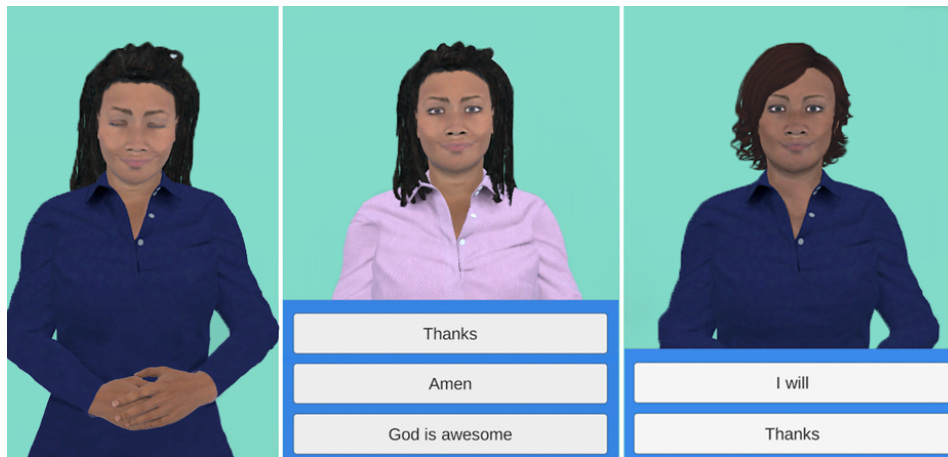
Participants characterized the positive and holistic impact that spiritual language had in agent dialogue. When asked to describe the impact of spiritual language in agent dialogue, P5 stated, *“She's talking my language. It reinforces what I believe, I never mind talking about prayer life, and never mind talking about...the good he [God] wants for you. It's healthy. It's spiritually healthy. You can do other things because the Word, it gives you strength.”* This demonstrates that spiritual language was not only welcomed, but it evokes participants' self-efficacy that their underlying core belief system provides, helping them to follow through with coping strategies. P1 echoed this sentiment when she expressed, *“I think it's helpful to hear a message when you are trying to change your life, or do something that is seriously difficult,”* reaffirming that incorporating spiritual support can be powerful when promoting behavior change.

All participants stated they felt more connected to the SPIRITUAL agent, who demonstrated membership in the church community. Not only was the agent's use of spiritual language perceived as genuine, it contributed to the perception that the agent was a member of the participant's faith community. P4 affirmed, *“What she [SPIRITUAL agent] did is very much akin to some of the people I talk to at church. I would have liked the first one [SECULAR agent] if some of that had been in there.”*

One participant specifically highlighted that the SPIRITUAL agent, when compared to the SECULAR, felt like a friend. In contrast, she identified the SECULAR agent as a clinician. She stated, *“Overall, [SECULAR agent] had a clinical feel. She wasn't a buddy. I don't need a clinician; I need help, a friend.”* P2's preference for an agent she identifies as a friend demonstrated the value she and her community place on peer support and how this translates to preferences for the role of a virtual agent.

Participants offered direct feedback on the tailored dialogue's authenticity, placement, and frequency. P2 voiced, *“I liked the way the Scripture was infused into the dialogue, it lifted my spirits. They were appropriate, they came at the right time, it gave us a better connection.”* All participants reported that the frequency of Scripture and religious language was “subtle” and “just right,” but that future iterations may consider fine-tuning the tailoring. P6 explained, *“When I'm super stressed, I need to pray more... I could use the spiritual language more when things are harder. Less when things are good,”* Other participants expressed that a user's age and current level of engagement within the church community may also impact the desired frequency and distribution of religious content.

Figure 8: Spiritual and Non-Spiritual Validation Study Embodied Conversational Agents



4.4.4 Design Implications

I developed a novel methodology for co-designing culturally-tailored ECAs with specialized populations. Through these methodologies, I distilled a set of ECA design guidelines for culturally-tailoring dialogue, and identified numerous design concepts for health promotion agents for two church communities. Following these ECA design guidelines, I developed two health promotion agents and qualitatively evaluated the effects of spiritual tailoring in a within-subjects experiment. Results demonstrated that spiritual tailoring positively influenced the relationship between the user and agent. Further, faith-centered content was the primary reason users gave to explain their choice to continue working with the SPIRITUAL agent. Although participants have a common shared value of spirituality, this was the first focus group where participants demonstrated to our team how spirituality could be meaningfully integrated into both ECA design and dialogue. This finding supports the second key design principle of this dissertation work: *“integrate spiritual and faith-centered concepts into both the ECA design and the educational content”* (Table 6).

4.5 Focus Group 3: Exploring Features for In-app Social Support Using Modular Storyboards

In Phase I, members identified what types of social support they currently give and receive from their church communities. We identified in the prior workshop how an agent could support members on matters related to health and spirituality. Now, I aimed to explore what resources members would seek from a church-based mHealth application. In addition, this workshop was designed to investigate how church members currently access health resources. This work was published in the CHI 2019 conference proceedings, received an honorable mention award, and was presented as a full paper.

This work investigated the following research question:

RQ3: What resources would church members expect or seek in a church-based health app?

4.5.1 Procedure

We developed a modular storyboard method. The modular storyboards combined elements of two traditional storyboarding methods (i.e., a set storyboard versus a blank canvas) by presenting participants with the same overarching storyline (Halko & Kientz, 2010) while also allowing them to customize, change, and remove parts of the board (Berry et al., 2019). The similar storylines facilitated participant comparison of each other's boards. Through the customizations, participants drove significant design decisions and outcomes within the scenarios, resulting in end products that reflected their values and experiences.

The storyboards featured a hypothetical church member experiencing three different health journeys: an acute health crisis, an ongoing experience of mental illness (Figure 9), and a wellness journey (Stowell et al., 2020). These storyboards were used to elicit church member feedback on the activities and features the application should include. Unlike a static storyboard, each contained opportunities for participant story completion (e.g., multiple choice options, fill-in-the-blanks, and matching support opportunities to support providers), creating a flexible story narrative. Storyboards were structured into three parts of the character's health journey: introduction to (Figure 9a), seeking social support from, and giving social support using the health app. The jumps in time probe the types of support a member would seek in the app. Participants were given four packets of blocks: social support and resources (e.g., free and discounted resources, preventive health information, and places to vent), people (e.g., health professionals, church leadership and laity, family), context (e.g., church service, downtime), and message blocks (e.g., health, inspirational, social) (Figure 9c). In the character's time of need, participants were encouraged to tape on blocks. Enabling participants to mix and match blocks allowed us to explore factors that might impact their preference for app-facilitated social support. Participants were encouraged to create their own blocks if the existing blocks did not fit their needs.

Fill in the Blank & Multiple Choice

At key plot points, the storyboards had multiple-choice and fill-in-the-blank sections to allow the participants to create a story that felt authentic to their experiences (Figure 9b). Participants were encouraged to customize the storyboards however they saw fit, including writing directly on them. At the end of each storyboard frame, participants were provided with smiley face feedback stickers to indicate their level of satisfaction with the storyboard frame (Figure 9d)

Participants were encouraged to choose storyboards where they felt they could personally relate to the health topic to illuminate how and why certain features would be required or valued when developing a health technology system for their community. The storyboards took approximately 90-minutes to complete. Participants presented their finished storyboards to the group and answered questions about their choices. At the end of the focus group, participants briefly discussed the differences and similarities they saw in the storyboards. These sessions were 3 hours each.

Interviews

Given the time constraint of the individual storyboard focus groups, participants were invited to a follow-up 1-on-1 interview where they walked through each decision they made in the storyboard. Participants were prompted to discuss whether the storyboard they created reflected their personal experiences and preferences for engaging with a health promotion application. Each interview lasted 1 hour.

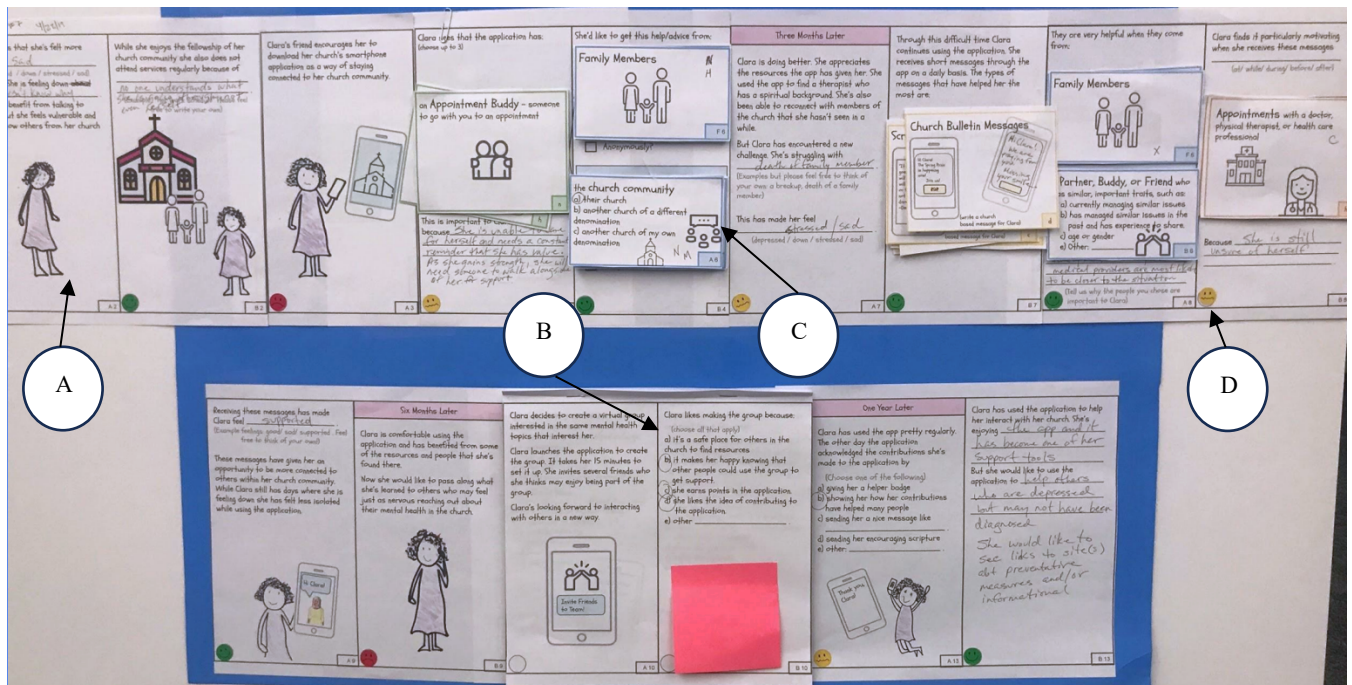
4.5.2 Findings

Nine church members participated in the individual storyboard session, four from Church A and five from Church B. One member of Church B dropped out of the study for personal reasons before individual interviews. The following subsections will highlight findings related to this dissertation work. Although not all participants chose to complete the mental wellness board during the focus group session, participants were able to comment on those boards during the discussion as a group.

Participant Overview

All church members self-identified as Black. Six self-identified as women and three self-identified as men. Participant ages ranged from 33 to 73. One participant had completed some high school, two had some college, three had a BS/BA, and three had advanced degrees.

Figure 9: Mental Health Modular Storyboard



Mental Wellness as a Priority

Similar to previous focus groups, all participants across both churches agreed that mental health treatment and diagnosis is a healthcare priority in their predominantly Black church communities. However, participants stated that mental wellness is challenging to promote and support within the church, due to cultural attitudes and perspectives. P2 from church A emphasized the importance of including mental wellness content by stating: *“if a faith-centered smartphone app doesn’t include mental health in it, it isn’t an application for the Black community,”* as it would not assist in one of the most prevalent yet stigmatized issues that she believed the church currently faces. In addition, participants explained that the mere existence and incorporation of mental well-being content presented within the church-based application might positively suggest to church members that the church (and its leaders) *endorse* mental health promotion. As a result, when it comes to matters of mental health, the church would be seen as more inclusive than members realized.

A Need for Privacy

While privacy was a concern for all participants, it was most prevalent when examining the mental health storyboard. Participants explained that fear of judgment from others prevents members from receiving mental health support, connecting with others within the church around topics of mental wellness, and obtaining mental health services. Unlike the other two health journey storyboards (i.e., acute crisis and physical wellness), participants stated that providing the user with alternative sources of support outside of peers and other church members was a priority. Participants in Church B had a rich discussion at the end of the storyboard session, stating that, like many religious organizations, stigma from mental illness results in church members avoiding treatment opportunities because participating in treatment is considered a sign of spiritual weakness and a lack of resilience or strength. P5 shared: *“If someone is needing to go outside the church for help, people might think she’s touched. She’s not doing what she should be. She needs to fix her relationship with God.”*

In 9/11 completed storyboards, members selected peer support and mentorship more than healthcare professionals, with the exception of the mental health storyboard. While buddy system approaches such as peer mentorship have shown to be effective for reducing both community and self-stigma (Yanos et al., 2015a), members stated that they do not necessarily feel comfortable or supported disclosing their experiences to other members in this context. This need for secrecy means that, unfortunately, other members experiencing mental illness do not benefit from sharing their personal experiences with others. While creating opportunities for enabling structured peer support could be a future step, participants stated firmly that a mental wellness intervention should start with a set of culturally-informed resources, a brief psychoeducation, and screening.

Embodied Conversational Agents Can Provide Support

To afford members an opportunity to receive informational support in a comfortable setting, participants selected the ECA in their storyboards to provide users with credible support anytime and anywhere. Participants explained that the ECA was an

acceptable source of support and information, especially when supporting users experiencing sensitive health topics (i.e., mental illness). During group discussion, participants stated that given the level of perceived stigma within the community, an ECA would allow for easy access to mental health education and necessary resources without having to *come out* to other members within the community—keeping member privacy intact while simultaneously providing content to reduce stigma. One participant stated that a person living with depression may not have the social support necessary to connect with other members of the church. Thus, the ECA could fill-in.

The Value of Spirituality in Maintaining Bodily Health

In addition, this focus group confirmed previous findings of church members' expectations for including spiritual practice as a means of maintaining their bodily and mental health. Participants expected this application to, at a minimum, situate their bodily and mental health and their health goals into the context of their religion and spirituality. Participants insisted that the smartphone-based health application should honor their religious beliefs and values in a way that conventional secular health-based applications currently do not.

Resource Identification

Church members further explored the utility of using the application to provide access to support offered by the church that church members otherwise may not know exists. For instance, one member spoke about wanting to leverage the services of a marriage counseling ministry (a service provided by the church for couples who wish to improve their communication) that she believed was no longer active. During the focus group, another participant chimed in, stating that the ministry is indeed active and that she herself has worked with them recently. Without the focus group, this first member would have remained completely unaware of a legitimate resource already offered within her church. Clearly, an application listing resources offered within the church could help increase service utilization and access for those experiencing stigmatized health conditions like mental illness. These findings helped establish the third key design principle: *"Include resources to increase church capacity for mental health support"* (Table 6).

4.6 Focus Group 4: Examining the Intersection of Race, Religion, and Technology

4.6.1 Purpose

Phase II Focus Group 4 engaged participants in discussing how to design health technologies for the intersectional identities of race and religion (O'Leary et al., 2021). Race is a socially constructed concept in which people are grouped based on physical traits (e.g., skin color and hair textures) (Harris et al., 2017; Winker, 2004). As a social construct, race reflects how societies have interpreted and defined the *meanings* of these people groupings and in turn, shaped experiences and access to resources and power (D. R. Williams & Rucker, 2000). Researchers have demonstrated how inattention to race when developing technology

has reinforced the resource-reducing power of systematic racism by leading to numerous consequences for different racial minorities (e.g., infrared sensors unresponsive to darker skin tones, policing software that predicts criminal recidivism using racially biased data sets) (Benjamin, 2020). Indeed, if researchers ignore an explicit focus on race even well-intentioned designs can lead to the development and implementation of racist socio-technical systems.

Recently, there has been a call to engage critically with race when designing, developing, and evaluating socio-technical systems (Ogbonnaya-ogburu et al., 2020). Such engagement requires using methods to democratize the design process (Harrington et al., 2019). In this focus group, and in the published work (O’Leary et al., 2021), I present a case study of such critical reflection on and engagement with the concept of race in community-engaged HCI research.

In Phase I Focus Group 3, respondents characterized their current technology use on an individual level and while participants described how their church uses technology at a high level, these descriptions were largely presented as a surface-level overview with a lack of detail about *how* and *why* such technologies are used or not used. In Phase I Focus Group 1, participants described their church’s history, leadership, events, and programming. Both discussion-based workshops from Phase I required participants to describe their communities and reflect on technology use without preparation or use of environmental cues. In this later stage of the project, I wanted to take a deeper look at how the intersectional identities of race and religion impacted the choice of technology and a community’s desires for technology. In order to engage with race and religious identity critically, I chose Photovoice, a research method where participants lead their own data collection and analysis using photographs. This work sought to answer the following research question:

RQ4: How can we design digital health tools responsive to the intersectional identities of race and religion in multicultural church communities?

4.6.2 Method

Photovoice is a flexible and accessible method that requires only that participants use a camera to explore “matters of relevance” (Gubrium, 2009; Hergenrather et al., 2009; Wang, 1997). Using community member photos to drive subsequent interviews, participants are not passive but active catalysts in identifying and addressing their most pressing concerns, as well as celebrating their community’s strengths (Catalani & Minkler, 2010; C. Wang, 1997; C. C. Wang et al., 2004). Importantly, these representations are captured by participants instead of the researcher. In recent years, HCI researchers have made a similar call to incorporate asset-based approaches into the design of technology, and have used similar approaches of selecting affirming workshop activities that address goals held by participants, and celebrate strengths (Chisholm et al., 2020; Cho et al., 2019; Dickinson et al., 2019; Harrington et al., 2019).

Photovoice Session 1

To begin the session, I asked participants to define the racial and ethnic makeup of their community to guide the racial and ethnic identity terminology I used in the subsequent design activity. While this project focuses on predominantly Black churches, previous sessions made it clear that participants used a range of descriptors to label the racial and ethnic identity of their church, while others did not. I felt it necessary that participants define the terminology that we would use later in the focus group to engage in a dialogue about race and their Church communities.

I conducted an affinity diagram activity in order to co-develop with participants the prompts for the photovoice activity, where the aim of the affinity diagram activity was to scaffold a participant-led exploration of the intersection of race, church, health, and technology. Participants generated five to six words or concepts related to the following prompts: 1) How does race relate to your church experience? 2) How does your [focus group selected ethnic identity] church experience relate to health? 3) How does your [focus group selected ethnic identity] church experience relate to technology? If asked to explain the prompts further, we provided two additional secondary prompts. 1) How does your [focus group selected ethnic identity] church experience impact/promote/impede health? 2) How does your [focus group selected ethnic identity] church experience impact/promote/impede technology use? Post-its were then displayed on the wall for all participants to review. As a group, participants then clustered post-its with similar concepts and labeled each cluster. The resulting categories (formed from groupings) were used as inspiration for the participant photo data collection (O’Leary et al., 2021).

Between Sessions

Next, participants were instructed to take photos during their typical day-to-day experiences over the next three weeks. Participants were provided with a list of the categories created in the affinity diagram activity and were asked to take at least one photo per category and to write a title and caption for each photograph.

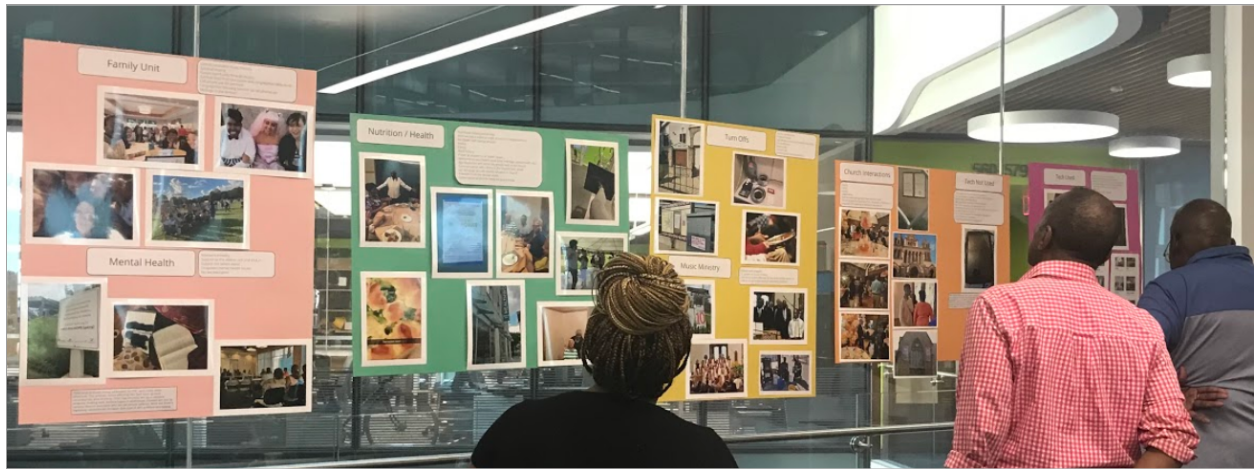
Session Two

In the second session, we displayed all photographs grouped into their respective categories. At the start of the focus group, participants were invited to view all the photographs (Figure 10). After 15 minutes, we asked participants for their overall impressions of the photographs as a whole. For both church focus groups, these discussions lasted about 20 minutes. Next, Participants were instructed to pick one photograph that they felt was most important and most representative of something they would like us to know about their church community or church experience before we developed an application. After selecting one photo, participants completed a worksheet about the selected photograph. Worksheet questions were adapted from the SHOWED photovoice methodology (Hergenrather et al., 2009).

The following are the worksheet questions: 1) What important persons, places, or objects are in this photograph? 2) What moment did you capture? OR What do you want to say about it? 3) How does this photograph relate to your community's lives?

Why is it important? 4) Why does this strength, situation, or problem exist? 5) What must we keep in mind when developing this health application?

Figure 10: Participants Viewing a Gallery Display of All Photos Taken



4.6.3 Findings

Using Photovoice, church members gave nuanced descriptions of the racial and ethnic identities of their communities, and how church history and aspirations for the future impacted implications for a smartphone-based health technology. In combination, these social factors provided important insights into designing race-informed technology that not only responds to the needs of community organizations but also celebrates their strengths.

Participant Overview

Ten church members participated in the first photovoice introduction session, and photovoice follow-up session, four from Church A and six from Church B. One member of Church B could not attend the follow-up focus group, and instead participated in an interview. All church-members self-identified as Black. Seven identified as women and three as men. Participant ages ranged from 33 to 73. One participant had completed some high school, one completed a GED, two had some college, three had a BS/BA, and three had advanced degrees.

Participants from Church A have been members for an average of 12 years ($SD = 11.5$, Range: 2-27), whereas participants from Church B have been members for an average of 4.5 years ($SD = 2.4$, Range: 2-7). They range from no involvement in any church affiliated group or ministries to serving multiple roles in the church. These roles include attendee, ministry team leader, choir member, and trustee.

Understanding Community Identity

Throughout the Photovoice focus group participants used various terms to refer to their church communities. In some cases, this included racial and ethnic descriptors. Defining the racial and ethnic makeup of the church could not be accomplished with a single term, and instead, a participant-led discussion focused on the multicultural makeup of their church communities.

Participants were hesitant to label the cultural makeup of their church for several reasons. P5 communicated his concern:

“Well, how our race relates to your church experience, and...you know, that's, you also have to think about...Who I am, and what I represent as a Black man. And how I perceive church and White expectations of the Black church. You know, from the stereotypes...Like you have to dress a certain way to be accepted. You have to, um, act a certain way for acceptance.”

P5 felt that his church lacked characteristics of a “stereotypical” Black church that I, a white researcher, might expect and that the label would lead to incorrect, and potentially harmful, assumptions about the organization. With further probing, participants enumerated various characteristics that come with the label “the Black church” such as a “fire and brimstone preacher” and certain formal attire. Participants tied this hesitancy around labeling their church to concerns for their broader community when interacting with academic research institutions. Given the history of abuse of BIPOC by white researchers in the United States, as well as current practices of broad-brush assumptions based on race (e.g., not all minority communities are low-income), participants reminded me that other church members may be wary of any technology system my team and I designed. P7 characterized this sentiment, *“People tend to lump us into a certain category.”* Participants further expressed disillusionment with a corrective approach that emphasizes deficits within their community. As P1 stated, *“A lot of time, organizations approach Black and Brown communities always with an under-served approach. Or low income,”* P1 goes on to state that as long as, *“grants for underserved groups get people to develop for the community, fine. But it can't be all about that.”* While P1 recognizes that the *underserved* label may come from funding opportunities, she speaks to recognizing strengths and variations within each community.

Due to the multicultural makeup of their churches, participants in Church B grappled with labeling their church community, and they did so through conversation with one another. Participants avoided a particular racial or ethnic label that made others in the community feel unwelcome and asked each other about their responses to various terms. For example, participants explored the nuances of the terms ‘Black,’ a broad descriptor, and ‘African American,’ a descriptor that emphasizes the United States as a person’s country of birth (Agyemang, 2005). It was clear that participants strongly believed that the label used to describe the racial and ethnic identity of their community should be broad enough to celebrate diversity within the Black community and promote inclusivity of people from other racial and ethnic backgrounds. P9 described the importance of inclusivity at Church B:

“Even though they say it's a Black church, it's not all Black...the ministers are Black... But we welcome everybody. It's an open-

arm church... and I think that's one of the core things about the church. We love everybody." Engaging new members is exceedingly important to both churches.

During the focus group, church members from each community reached a consensus. In Church A, participants agreed that the label "*predominantly Black Church*" was representative of their church community's diversity, while Church B chose to label their community as a "*Multicultural Church seated in the African American tradition.*" Both groups favored the term Black when describing the racial identity of most of the church members.

Having this conversation was foundationally important for this work. At a practical level, to work with communities, you have to understand how to refer to them. Language and words are core to establishing rapport and facilitating effective communication between the research team and participants. Beyond the words themselves, these discussions helped me better understand how community members frame their racial and cultural identity as an organization, this is especially important given that this is a community of which I am not part. This conversation gave church members an opportunity to check in with me and ensure that I and members of our research team were not thinking of these churches as monolithic organizations but instead understanding that each organization is unique and diverse. These perspectives on identity have direct implications for their expectations of technology, such as the expectation that these technologies reflect the diversity of the communities and that technology designers understand the heterogeneity of each organization. At a minimum, several participants stated that the application's iconography and agent imagery should represent shades of black and brown skin tones and different hairstyles and textures.

Church A Technology Use

Church A focus group members emphasized the implications of how church history and legacy inhibit the integration of technology within the church. Technologies commonly found in some sanctuary halls are nonexistent in Church A (Wyche, 2006). There are no digital displays, and instead of a video live stream, Church A opted to air their service using radio. This provides access to services for those who cannot attend while minimizing technologies in their historic sanctuary. The only display technology used on the church campus is an overhead projector system (Figure 11a). This system is used exclusively for Bible study and is located in a newer, adjoining building not bound by the same historical preservation efforts.

During worship, church members opt for physical Bibles and Hymnals. The use of the YouVersion Bible application during worship (mentioned in Phase I Focus Group 1) is met with mixed reactions. P1 explained, "*People may think you're doing something else [on your smartphone] instead of listening to the pastor*"

Figure 11: Technology Use in the Churches. A) Church A - a rare use of technology. B), C) & D) Church B - depictions of technology used during worship



Outside of the physical Church, leaders communicate with members mainly by phone. P3 stated, *“A lot of our people don't even have email”* At the start of our work, Church A had a public-facing website and Facebook group; however, some of the participants were notably surprised to find this out, exclaiming, *“There's a website?!”* Other focus group members were aware of the website to the extent that information had not been updated in over a year. When asked how people learn about events, P2 responded:

“I think it's a lot of word of mouth. If you're in the church, there's a section in the book [church bulletin] that has all the events. They do have Facebook. It gets a little bit of traffic. Yeah, they don't really update it as much as they should.”

Perhaps most indicative of participant perceptions of technology in Church A is that photographs representing the participant-identified category *Technology Used* did not depict technology used within the Church. Instead, participants took photographs of their personal technologies. In lieu of a photo, P3 submitted a paragraph describing his frustration with the lack of current technology practices within the church. Participants viewed technology use as *“modernizing”* and as a way to reach new members, implying that new members expect to see technology integrated into worship (e.g., recorded sermons, digital displays, or webcasts, etc.) and leveraged to promote events (e.g., calendar of events, events conducted by video conference, etc.).

P1 described being active in the church she grew up in, and partly attributed this to a smartphone app that her home church used to deliver content. By sharing the app with others, P1 introduced other participants to possible functions technology could provide. After looking through the app, P5 shared:

“It's a very modern thing...the...app blew me away. [Church A] is not there yet, it's very traditional. I would think of [Church A] as a traditional Black, historic Black church. You know, with the, with some of the...rules and uh, mission statements. Just they kept, keeping things traditional”

In describing her photograph (Figure 12), P1 further explored the types of *traditional* thinking that could constrain technology implementation in the church:

“This reminds me of old school or traditional ideals [points to sign in Figure 12] that are stuck in Black or Brown communities. This type of ideology [Sign reads: Jesus Christ is Lord of All. And he's coming back any minute. Get ready.] could be a hindrance to introducing new technology. Some people will be excited. Some people will view technology as the devil.”

For P1, the tone and message of the sign were judgmental and threatening. Her sentiment explained how mindsets towards technology can vary within any community-based organization. In the case of Church A, the participant felt that members would react to technology in various ways, ranging from enthusiasm to harsh judgment and skepticism. Understanding how ideologies vary within an organization is crucial when implementing technology interventions.

Figure 12: Old World Thinking



Participants in Church A were enthusiastic about the potential for future technology integration within their Church; however, these findings illuminated that Church A's incorporation of technology has been slow and, in some cases, deliberately restricted. Members of the Church emphasized that the goal of engaging new younger members sits in tension with current Church A technology practices done, in part, to preserve the worship hall of Church A as a historic landmark within the Black community.

Church B Technology Use

During the introduction to the Photovoice session, Church B created two technology categories *Technology Integration* and *Technology Outreach*. They returned to our second photovoice session with many images of types of technology that their church fully embraced.

Worship Technology

In contrast to Church A, Church B uses a number of technologies during church services. These technologies include digital displays to project scripture, worship lyrics, and multimedia auxiliary materials such as images or videos to support the sermon's

message. P10 explained, *“Sometimes whoever is preaching will integrate something from a modern video, music video...you can project that on the screen which enhances the sermon.”*

To increase engagement and appeal to younger audiences, Church B uses a pre-recorded cartoonish digital avatar to deliver announcements (Figure 13d). Unlike Church A, using personal technology is widely accepted during worship. P8 stated, *“I use the Bible app, yeah. Everybody seems to go on their phone on Sundays now. You see the phones and iPads coming [out] on Sunday.”* Unlike in Church A, in Church B it is assumed that church members on their personal devices, including both phones and tablets, are using the Bible app in coordination with service.

Church members who cannot attend church due to illness can view a live stream of the sermon (Figure 13b). While this option increases access, participants quickly pointed out that viewing from home reduces the number of people who attend in person. P8 explained: *“[Points to photograph of greeting during Worship Figure 13a] This one kind of speaks to what can happen... when you are in fellowship together. And this [Points to a photograph of a camera used to capture services Figure 13b.] facilitates, um, people having access to worship but not fellowship.”* These juxtaposing images (Figures 13a & 13b) showed the importance of fellowship and connection to church members, and that simply streaming church services does not address that core relational value.

As evidence of church members' rejection of a deficit lens when working with their communities, only six of the 83 photographs submitted during Photovoice depicted challenges or problems. Whereas 34 illustrated various positive facets of in-person interactions and fellowship. Participants spoke of the power of community and of the desire to facilitate community building around topics of interest.

Figure 13: Church B: A) Fellowship & B) Livestreaming the sermon



Communication and Coordination of Events

Church B's website includes an updated calendar of events, and a number of church-wide announcements (e.g., prayer concerns, birth announcements, etc.). Church B uses social network platforms such as Facebook, Instagram, and Twitter to communicate with church members and promote various outreach missions of interest to their organization. Participants spoke about coordinating ministry events using platforms such as Slack and YouTube. P7 described the numerous ways the dance ministry used technology:

“They post YouTube videos. They post instructional videos. They post, um, inspirational words, scriptures. Like when we’re going to minister [dance] on a particular Sunday, you don’t have time to rehearse collectively, So they will create a video and send it out so that we can do it at home.”

Technology-hosted Activities

Prior to the Covid-19 pandemic, the only interactive and synchronous activities that Church B hosted virtually were regular evening and early morning conference calls colloquially known as *prayer lines*. Such calls were coordinated to facilitate group prayer and meditation. These offerings created opportunities for group activities that otherwise would be missed if limited to events that occur only in person.

Summary of Technology for A & B

Church A was founded nearly two centuries before Church B. Our findings showed how considerations for types of technology integrated within Church A must honor its historical significance and legacy, as well as its attempts to grow its membership. In contrast, Church B has a shorter history and a more modern building with a focus on rapid technology integration. Differences in institutional, cultural, and political capital impact not only how Church A uses financial resources (e.g., money for preservation efforts versus technology integration in the worship hall)—it also impacts the normative digital practices of church members. We may have otherwise wrongly assumed that technological literacy causes these differences in digital practices when comparing both churches. See Table 11 for a comparison of technology use between Church A and B.

Table 11: Summarizing Technology Use in Church A and Church B

	<i>Church A</i>	<i>Church B</i>
<i>During Church Service</i>	No digital displays in sanctuary	Digital displays in sanctuary, use of multi-media content during worship and sermons (e.g., videos, pre-recorded announcements using an avatar, worship lyrics projected, images, etc.)
<i>Personal Devices in Sanctuary</i>	Limited use	Pervasive and regular use
<i>Private Worship Outside of Sanctuary</i>	Bible App	Bible App; Daily Bread; Daily Devotionals
<i>Communication between Church Members</i>	Primarily phone calls	E-mail; Slack; Facebook; Instagram; Twitter; Church Website
<i>Church Service Broadcast</i>	Synchronous Radio Streams	Synchronous Video Live Streams

Church A: Preservation V. Innovation

Engaging younger church members is a secondary goal of the church, and our participants posited that technology may be an answer for recruiting younger families and young adults. However, concern remained that technology integration could sit in tension with tradition. Utilizing technology to address such a challenge requires great sensitivity to values central to this particular organization. The extensive preservation efforts in Church A may speak to the value that the community places on not only a traditional space for worship but also a physical symbol and catalyst for social action within the Black community. Gaver et al. considered the importance of both materiality (e.g., aesthetics, cultural significance) and functionality when designing a device to support prayer among cloistered nuns (Gaver et al., 2010). Their design team was careful to understand the cultural significance of the space and of the religious practice (e.g., the need for ambience in both physical form and sound). Similarly, recommendations to embed technology in sacred church spaces may not be desirable for all predominantly Black church communities, and may not address some of their most valued and prescient missions. There instead is a need to understand these historic buildings as assets despite the potential constraints they may place on technology implementation.

Church B: Augmenting Social Action Efforts

The Black Church is an important site for past and present social justice efforts (Taylor, 1987). In examining the core missions and identities of both churches, it became clear that Church B's identity was rooted firmly in social action perspectives and activities. P10 demonstrated her church's commitment to social action when describing one of her photographs (Figure 14):

"This photograph is important because social justice is a significant part of our responsibility and our reputation at Church B."

Indeed, this photograph (Figure 14) depicts one of many active ministries at Church B that address various social justice issues.

Of note, these ministries are sustained entirely by volunteers. When asked directly how technology could augment their Church mission, Church B participants were interested in using technology to support the work of these ministries. P10 asked our team,

"How can the [mHealth] app improve access to resources or identify gaps in potential healthcare availability?" P10 explained

that the application should expand current social justice efforts of their health ministries, by assisting in collecting data on racial health inequities and providing access to needed education and resources. While not all churches with predominantly Black memberships have a social action orientation, those that do may be potential sites for technology innovations that amplify community-building efforts paramount to the community. Specifically, members indicated that technologies that amplify the health goals and topics that align with the church's current mission and goals would be valuable. Given church members' interest in mental health promotion, a digital health tool that assists with community resource identification and building community capacity may be highly valued. This work formalized the fourth key design principle, *"use a community-building framing to encourage community members to interact with the mental well-being and anti-stigma content"* (Table 6).

Figure 14: Church B Mission



4.6.4 Design Implications

This work investigated the intersection of racial identity and religion in the context of designing technology for two churches. These findings speak to the intricacies of community identity, namely how members balance honoring historical legacy with current culture and future organizational goals. While the findings from this work consist of a descriptive case study of technology use between two predominantly Black churches, these findings highlight how technology integration and implementation are the result of organizational identity, legacy, and current mission. For example, community-building around sensitive health topics, assisting with the identification of resources to augment institutional capacity, or limiting technology use within the walls of a historic building. These findings were integral to establishing the fourth key design principle, “*Use a community-building framing to encourage community members to interact with the mental well-being and anti-stigma content*” (Table 6).

Chapter 5: Phase III Church Connect Field Deployment

5.1 Field Deployment

Phases I and II culminated in Phase III a quasi-experimental 28-day field deployment, where 25 church members from Churches A and B used the first version of a smartphone-based ECA system, designed for their church (O’Leary et al., 2022). Phases I and II revealed that our stakeholders (church members and leaders) viewed health and well-being as interconnected concepts inextricably linked to spirituality. As such, we developed a smartphone application—*Church Connect*—that supports bodily health in a manner that facilitates community support and spiritual practices. This system included both spiritual and health-related content presented by the ECA. For the health-related content, users could discuss nutrition and physical activity and participate in ECA-guided meditations. For the spiritual content, church members had access to two social features: pastor announcements (a one-way messaging interface where Pastors could message all members using the app), use of a prayer request church-wide group forum, and two ECA-led features: Scriptural meditations and a daily Bible story.

I conducted this field deployment to get preliminary feedback from participants on the culturally-tailored aspects of the design and content, the overall ECA design (e.g., character appearance and culturally-tailored dialogue), and the ECA-facilitated physical wellness content (i.e., nutrition and physical activity). This pilot work did not include sensitive health content and topic areas (i.e., mental health). This work examined participants’ experiences using Church Connect for four weeks, after which I conducted individual semi-structured interviews with 17 of the 25 participants.

The findings of this work characterized a set of community values and *community dynamics*—processes through which communities change and develop—that shaped the way participants wanted to use the health, community, and spirituality features featured in the Church Connect application. These findings demonstrated how these values and dynamics can, in turn, be shaped through engagement with the app.

This work investigated the following research question:

RQ: How can digital health tools be designed to support bodily, spiritual, and social well-being within predominantly Black church communities?

The findings presented in this dissertation work include participant characterizations of how participants used the *Church Connect* spiritual and community features, why they used them, and the impact of their use. Specifically, findings subsections will cover the themes of *communicating care* and *maintaining privacy and discretion*. Such themes provided a more nuanced picture of participants’ perspectives on how a faith-centered community-based health technology should support multiple facets of well-being, including mental wellness. These findings established the fifth key design principle, “*balance tensions between maintaining privacy and communicating care*,” for the mental health promotion intervention (Table 6). For a complete enumeration of themes related to this work, please refer to our published paper in the Conference on Human Factors in

Computing Systems 2022 proceedings. Table 12 contains an overview of the purpose, procedure, and key findings of this data collection.

Table 12: Phase III Summary of Field Deployment Procedure, Purpose, Findings.

<i>Data Collection</i>	<i>Procedure</i>	<i>Purpose</i>	<i>Key Finding(s)</i>
	Phase III: Field Deployment		
<i>Field Deployment</i>	28-day Field Study & Semi-structured Follow-up Interview	RQ: How can digital health tools be designed to support bodily, spiritual, and social well-being within predominantly Black church communities?	A set of five community dynamics were identified. Certain dynamics sit in tension with one another depending on the sensitivity of the health topic. (O’Leary et al., 2022)

5.1.1 Church Connect Field Deployment System Design

This version of the Church Connect application contained health education modules that addressed physical activity, nutrition, healthy hydration, and stress reduction. The application featured an ECA named Clara, designed as a health counselor (Figure 15a). Clara introduced each health topic, nutrition, and physical activity and explained the benefits of engaging in each health behavior. Health content for nutrition and physical activity was tailored based on the user’s age, sex, and baseline behaviors. Clara used counseling techniques from an established person-centered counseling method called motivational interviewing (Miller & Rollnick, 2012). These sessions were structured using principles of motivational interviewing (e.g., expressing empathy, supporting self-efficacy, etc.), where Clara plays the part of a motivational interviewer who assists users with defining and setting incremental behavior change goals. All goals are set every week. During weekly follow-up conversations, Clara employed ipsative health behavior change strategies to influence behavior change positively (e.g., setting a new goal, providing feedback or problem-solving, etc.) based on current behavior change best practices (Han & Lee, 2018; Triantafyllidis et al., 2019).

Outside of the physical activity and nutrition counseling sessions, Clara led in-app guided meditation sessions (Figure 16a & b). For these sessions, users could choose an ambient background scene (e.g., snow falling, stars twinkling, etc.) and a selection of ambient background music (e.g., piano, crashing waves, church bells, etc.). Users could also specify the length of the meditation session (e.g., 5, 10, 15, or 20 minutes) and the type of meditation they would like to perform (e.g., body awareness, breathing exercises, scripture-based reflections).

All health dialogue administered by Clara was spiritually tailored to include content related to the Christian faith using principles leveraged from Phase II of this dissertation work. Clara also offered a Bible story of the day. If selected, Clara would select a Bible story from the project database and read the story aloud. Participants requested this feature throughout Phase II as they wanted a feature that exclusively served a *spiritual purpose*.

We designed two features to help users connect with their church community. The first is a prayer center, which has a community prayer wall (Figure 15b & c) and a private prayer list. All prayer center feature requirements were specifications that focus group participants from Phase II requested and evaluated favorably. In a similar manner to sequential group posts on contemporary social media platforms, users could post prayer requests and praise reports on a church-wide digital prayer wall. Prayer requests can be characterized as concerns, crises, or challenges (e.g., a family death, news of a terminal illness, anxiety over an upcoming job interview) that an individual would not only like to share with others in their community but through their request, ask for communal support. Sharing praise reports allows people to share expressions of joy or gratitude (e.g., getting a new job, recovering from an illness, engagements, births, etc.). Posting to the prayer center could be done anonymously or in an identified manner. Posting to the prayer wall could be deeply personal and may place the poster in a place of vulnerability. Accordingly, the poster may not feel ready to post the information associated with their name. Allowing for anonymous public-facing posts gave members flexibility regarding self-identification while asking for communal support.

The second community feature was also based on findings from Phase II, to “include features for building church capacity.” In response to this requirement, we developed Pastor Announcements, a one-way communication channel from the pastor to message congregants through the app. This allowed church leaders to share content directly with their church community, such as information about events, words of encouragement, Scripture, or church announcements.

Figure 15: Images from the Church Connect App a) the ECA Clara waiting for the user to respond using the menu, b) the church prayer wall, c) prayer request submission form, and d) prayer center push notification. The prayers shown are archetypal.

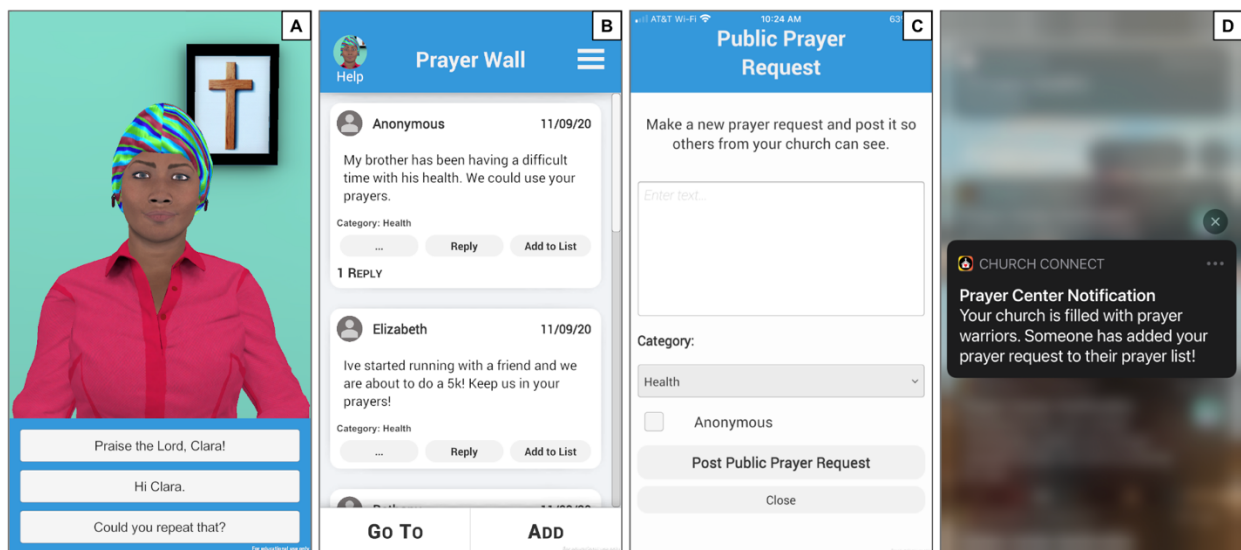
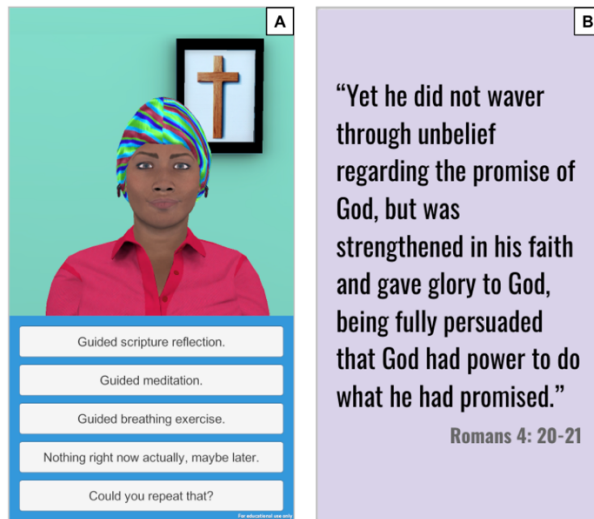


Figure 16: Clara-led Guided Meditation



5.1.2 Procedure

Participants were enrolled during an initial study appointment using video conference software. After obtaining informed consent, participants completed pre-study measures such as demographics and a technology use survey. Next, participants downloaded the application, logged in, and completed the first ECA-led app introduction. Participants were asked to use the app “a few minutes per day” for the next 28-day period. After 28 days, participants were invited to participate in a final study appointment with a semi-structured interview. To be eligible, church members had to be 18 years or older, English-speaking, and have access to a computer, tablet, or phone where they could connect using video conference software. In coordination with church ministry leaders (i.e., leaders who represent different church-affiliated groups) from both church communities, I advertised the field study in both church e-bulletins and by word of mouth via church ministry liaisons actively partnering with our research team. Prior to recruitment, the study protocol was approved by our university Institutional Review Board. Participants were compensated at a pay rate of \$24 an hour for the initial study session, an hour of application use over 28 days, and the follow-up semi-structured interviews. Total compensation was \$84 per study participant for 3.5 hours. The semi-structured interviews averaged 75 minutes.

During interviews, participants were asked for their overall impressions of using the system and probed component-by-component to elicit their attitudes, preferences, and motivations for using or not using different areas of the system (e.g., explicit health features versus techno-spiritual features). In addition, participants were asked the following questions: 1) To what extent did they see Church Connect as making use of or strengthening their community? 2) In what ways did Church Connect reflect or fall short of their community values?

5.1.3 Findings

These findings characterized participants' engagement with and reaction to Church Connect and their perspectives on how future tools can be designed to effectively promote well-being in church communities. Participants reflected on their desires for an app that builds on and enhances their community values to strengthen not only the health of their bodies but also the church community and their ability to engage in valued spiritual practices. Drawing upon their experiences with the Church Connect app, participants elaborated on how technology reinforced, enhanced, or disrupted specific community values.

Participant Overview

Twenty-five church members used the Church Connect app during the study period. Fourteen were from Church A and eleven from Church B. Nineteen church members self-identified as Black or African American, two self-identified as Black and American Indian, two self-identified as Black and Latinx, one self-identified as Latinx, and one self-identified as white. Eighteen self-identified as women, five as men, and two as non-binary. Participants' ages ranged from 19 to 75. For education, one participant reported having a high school degree, three reported having some college, three had an associate's degree, nine had a bachelor's degree, five had a master's degree, three had a professional degree, and one had a doctorate. Participants have been members for an average of 14 years ($SD = 8.9$, Range: 4-31). They range from no involvement in any church-affiliated groups to serving multiple roles in the church.

Church Connect App Usage

During the four-week field study, 25 participants used the app 560 times, with each participant logging in to the app an average of 22 times (Min=4, Max=54, $SD=13.3$). Participants posted 82 prayer requests, added 43 to private prayer lists, and wrote 45 replies on the prayer wall. Participants interacted with Clara for 36.6% of their sessions with the app. Participants used the meditation feature 84 times, Bible Story of the Day 50 times, Physical Activity 26 times, and Nutrition 25 times. Given that Clara is designed as a central component of the application, she delivers the majority of content outside of Pastor Announcements and the Prayer Center. As a result, participants interacted with Clara more compared to any other app component or community feature (e.g., the Prayer Center). It should also be noted that I did not capture passive usage of the Prayer Center, and as a result, I do not know the frequency by which church members may have checked for new content.

These usage statistics revealed that the spiritually-focused and community-focused features were used more than the physical activity and nutrition counseling sessions with Clara. It is important to highlight how each of these features were framed. The Bible Story of the Day feature provided new content every single day in the form of a new Bible story, setting the stage for daily usage. In addition, there were three types of guided meditations and five types of Scripture-based meditations, all of which could be tailored based on user music preferences, background scene selections, and time-based preferences. Clara encouraged users to join her multiple times a week to meditate and to partake in meditation as frequently as they desired. In contrast, the physical activity and nutrition sessions were framed as weekly interactions. Clara would offer a short tip if users returned before

the 7-day follow-up. Therefore, how the nutrition and physical activity content was framed compared to the other components may be partly why the physical activity and nutrition sessions were used less than other features that inspire daily usage.

Participants explained that at the time of the field deployment (March-May 2021), the continuation of the COVID-19 pandemic was causing immense strain on their mental well-being. Participants reported that their social connections with friends, family, and other church members were eroding and that the effects of long-term isolation were taking a toll.

Beyond the trauma and strain of the pandemic, participants explained that the collective racial trauma of George Floyd's murder also impacted their use of Church Connect. George Floyd was an African American man residing in Minneapolis, Minnesota (Times, 2022). While in police custody, he was murdered by the arresting police officer, Derek Chauvin, on May 25, 2020. George Floyd's murder was filmed and shortly after the murder went viral. After his murder, protests erupted against systemic police brutality sweeping across the United States and the global community. At the time of the participant interviews, Derek Chauvin's murder trial was ongoing. Participants acknowledged the importance of physical activity and nutrition overall. However, they explained that the combination of the pandemic and continued police brutality inflicted on Black people diminished their interest in the wellness content. Participants noted that devoting time and their limited energy to what seemed like a less acute problem was not high on their priority lists.

Instead, Bible stories of the day, daily meditations, and the prayer wall offered comfort and an identity-affirming way to address the stress of the times through spiritual support. Several of the pilot study participants expressed the desire for the application to include content to support community members' mental health and wellness during these times of increased stress. P9 stated: *"I honestly feel like, ya know, what would have helped me more would have been something about mental health and stress. There's so much heavy stuff is pressing down on us. And I honestly don't even know where to start. But you know the pandemic has made it easier to talk about mental health."*

The turmoil of the pandemic and current events compounded the need to provide increased access to mental health support in predominantly African American communities. This finding was not the focus of the field deployment or the interviews. However, participants provided unprompted confirmation that the addition of a mental wellness intervention was highly desired.

Community Dynamics: Communicating Care

Members identified *caring* as an essential community value driving church services and activities. Throughout the Phase I and II, concepts surrounding the value of community service, both in-reach (e.g., serving church members directly) and out-reach (e.g., supporting local, regional, or global community initiatives), were a consistent priority and one of the primary missions of both churches. Participants explained that since the COVID-19 pandemic, their typical volunteer activities had been disrupted. Members expressed a desire to serve their community directly and found that the Prayer Center was an actionable and meaningful way to give and receive prayer support. Church members explained that in their communities, *"we make sure everyone is covered."* When asked how it felt to respond to a posted prayer request, P10 stated:

“It felt good. I felt like there was something significant and meaningful that I could do for someone's life.”

The caring value is an intrinsic part of both communities, closely associated with the phenomena of reciprocity (i.e., the provision of support based on the notion that others from the community will respond in kind) (Pan et al., 2017). However, in religious communities, caring for others is another type of spiritual practice that brings the care recipient and caregiver closer to God (Smith et al., 2021). Church Connect's Prayer Center created additional mechanisms for demonstrating caring (e.g., the reply feature).

Providing Immediate and Intentional Prayer Support

Currently, outside of the Church Connect application, church members from both communities can submit a request for prayer publicly during their video conference worship or during synchronous remote-based prayer meetings called prayer conferences. However, church members pointed out that through the Church Connect Prayer Center, they could offer prayers, encouragement, and support with greater immediacy. P16 shared the benefits of the wall in terms of caring for members of the community:

“It's definitely a place for people to, to say, you know, can you please pray for me in this situation? Otherwise, we have the bulletin but the prayer is down at the bottom or at the back, you know, this, [Church Connect] is more intentional. And, it will be like a quick format. Like you [the poster] wouldn't have to wait till Tuesday to call, to ask for it, to be put in the bulletin you can just go there [to the prayer wall] and just do this... in the moment. When you need it.”

Multiple participants pointed out that reading the prayer posts as posted and written by the requester themselves felt even more personal than requests at the back of the church bulletin. The practice of being able to post a reply in the public thread created opportunities to respond to the poster with “*intention*.” The prayer wall created a front-and-center prayer request environment, allowing participants to access prayers, act on prayers, and respond directly and immediately.

Supporting a Wider Range of Health Challenges

Members stated that the addition of the prayer wall in the Church Connect app was not duplicative of their current prayer conferences or prayer requests printed in the church e-bulletin. Participants described how, using the mobile application, they felt that prayers could be added to the wall for milder health situations. P8 explained: *“A lot of times, you know, in the past we announced certain things but those are worst-case scenario situations. Someone is in hospice, or someone is on their way out, but sometimes, some people are just going through mild situations or challenges, or maybe it's a, you know, friend, right.”*

Calls for prayer to the entire church community by church leaders are typically reserved for health crises or extreme situations. Thus, participants discussed how Church Connect offered an additional mechanism for receiving support around health challenges that are not necessarily dire. Participants explained that the app enhances the communicating care dynamic by creating a platform to support everyday health challenges and stressors through communal spiritual support and can do so anonymously if needed. When discussing the prayer wall, members often spoke about utilizing the prayer wall to support church members facing challenges and life stressors in a more immediate and personal way.

Identifying Member Needs for Focused Outreach

Five of the 17 participants self-identified as ministry leaders and spoke about how they could envision using Church Connect to support members of their community and their ministry directly. For instance, one participant leads a Bible study group and states that the Prayer Center would be part of the application that he would enjoy using to ensure that everyone in his ministry receives the spiritual support they require. Similarly, a health ministry leader indicated wanting to use the application to get a better sense of what members of her ministry needed in a timelier manner: P4: *“Well, the alerts that we get [alerts sent by Pastors], we usually have to wait till Sunday. Um, somebody could post those daily [using the app]. Um, if somebody is sick, or if there has been a death in a family. And, um, then we could get on it 'cause we send cards and we make calls and texts and things like that...it would make the information that we receive as ministry members, uh, even more timely. And, um, it could be specific.”* By alerting members in the Church Connect community that prayer support is needed, participants who occupied leadership positions imagined how Church Connect could create opportunities for more focused and rapid service allocation to those community members in need.

Disrupting the Communicating Care Dynamic

Church members illustrated how posting a prayer concern places the poster in a vulnerable position, and responders feel a strong sense of responsibility to those members. While the phenomenon of the social pressure to respond on social media is not new, this phenomenon embedded in the practice of spiritual and prayer support makes it complex. Participants explained that unrealistic expectations for posted replies could lead to feelings of isolation for the prayer poster. While members appreciated the ability to reply to the individual who posted, some mentioned how the public nature of the reply made them worry that others in the community who posted a prayer concern would wonder why they had not replied directly to other posts on the wall. P17 shared: *“I guess maybe this is it. You have these people on, and maybe some prayer requests you wanna respond to, and then some you don't, but somehow, when you did, it was there for everybody to see, and it was kind of like, whoa, she didn't, she didn't, she didn't reply to mine. That made me pause a bit.”*

In a separate interview, a different member from a different church mentioned that they were surprised that more people from the church community did *not* make an effort to reply to other prayer concerns. P10 lamented: *“Certain person didn't respond, that kind of hit me.”* Similarly, one participant shared his disappointment that members of his ministry hadn't responded to his post: *“It was confusing once I wrote it, and then I came back, I didn't get the level of response from certain people. Like the guys from my Men's group. That felt bad.”*

In summary, while the social features of the Church Connect app enhanced the communicating care dynamic in some ways (e.g., offering opportunities to receive more immediate prayer support), it also challenged the communicating care dynamic by creating new expectations for digital prayer support (e.g., the burden of responding immediately). While receiving support in a

timelier manner gave participants an emotional boost, the absence of timely support can potentially damage an individual's sense of social connectedness if they do not receive the response they expected.

Community Dynamics: Adjusting Community Scope to Maintain Privacy

In addition to communicating care, another community dynamic revolved around the topic of privacy. Participants use of Church Connect sparked conversations around the importance of maintaining church member privacy when using aspects of the church-wide application, mainly the Church Connect prayer wall. While all members endorsed the prayer wall as a valuable aspect of the application, some shared concerns over how privacy could and should be maintained when posting prayer concerns. P13 shared this concern: *"I had to be mindful about how much health information I was putting on the prayer wall because I realized that it's more widely shared with others in my church."* Church members emphasized wanting to keep the space safe and continued to use discretion throughout the study period by curating posts to limit personal information and specifics. Participants stated that the subject matter of a prayer concern often contains private information about another church member. For instance, a post asking for prayers about a family member's chronic condition or mental health posted without the permission of the family member could create issues and would have to be managed appropriately by community members (e.g., establishing procedures for taking down the post, establishing guidelines for protecting other's privacy when posting). P17 stated: *"What if somebody put up a prayer request and somebody else knew that person and, you know, they kind of got into a verbal battle of what was actually the need for a prayer? Or, even question why the first person put the prayer up in the first place."* P17 explained that prayer posts made about others could be inaccurate. Moreover, she described how tensions over whether or not the prayer should have been posted is also a concern.

Participants offered solutions for the perceived privacy problem. One such solution was altering the audience size of the prayer wall to allow for varying levels of detail in the prayer concern posts. For instance, one church member described wanting to post a prayer for his Men's Bible study but felt that posting the prayer church-wide would feel both impersonal and reveal too much private information. P3 shared: *"I wanted to share a prayer request about Men's Bible study, something specific that only those guys know about. But..But..I couldn't find a way to limit the post. But, that would have been a nice way to really ask for prayers around something close to my heart."* Other church members echoed this desire, sharing that sometimes prayer support from a curated small group reduces feelings of vulnerability and may be more responsive in that those prayer supporters are privy to the context surrounding the post. On the opposite end of the spectrum, another participant spoke of broadening the scope of the post to lessen the feeling of vulnerability through anonymity. P5 reported: *"Well, I think being able to post to more than just your church that's a good idea. Um, but just because I just, you know, it's a bigger community. Someone might find, um, you know, they may find, um, a bigger community less intimidating. There's a lot of people who don't wanna share, about personal stuff you know, where they get [spiritually] fed."*

Our findings illuminated a contrast between the idea of sharing prayer concerns with a specific subset of people within the church versus a large group of people beyond the church. Both sentiments stemmed from a desire to ameliorate feelings of vulnerability when asking for prayer support. On one hand, participants hoped to accomplish this by shrinking their audience to those they feel closest to. In contrast, others desired an audience so large that their post is one of many; P5's quote suggested that this desire may stem from feelings of intimidation when sharing with a small community. While our data does not directly explain where these feelings stemmed from, possible explanations included not feeling a sense of connection to the larger community that makes sharing feel comfortable or having sensitive health concerns that one does not want those in their immediate network to be privy to.

5.1.4 Mental Health Design Implications

Phase III uncovered existing community dynamics defined as underlying social processes that establish community norms, roles, practices, and shared goals that characterize a particular social group—such as churches. Such processes can mediate the impact of a health intervention on a social group. Church members explained how existing dynamics affected their expectations for and engagement with the smartphone application and two dynamics critical to developing and implementing a digital mental health and stigma reduction intervention. Such community dynamics are as important as a community's physical assets (e.g., financial resources, space for in-person programming, etc.) in that they provide a lens for understanding how digital health tools could be designed to support community members as active agents in improving their community's health and well-being.

Across Phase I-III data collections, participants explained that maintaining privacy within the church was crucial and, at times, challenging. Similarly, those who have experienced sensitive health issues like mental illness indicated a desire to more effectively and efficiently give and receive care. It has been well-established that participants requested mental wellness be one focus of Church Connects content. During the Church Connect field deployment, the Prayer Center, an interpersonal social support feature, became the focus of much participant feedback. However, participants confirmed previous findings that mental health could and likely should be addressed using the ECA instead of social features like the Prayer Center. Indeed, if issues of mental illness were posted on the prayer wall, issues like responsiveness, privacy, and best practices would have to be developed by the church and its leadership to respond appropriately to members' mental health concerns. Using an ECA, participants indicated that members could receive mental wellness education and resources while maintaining anonymity. Participants explored the importance of balancing the values of *privacy* and *caring for one another* as it relates to sensitive health topics and severe health challenges. While these dynamics seemingly sit in tension, developing an ECA-facilitated system that honors both values may be crucial for maintaining community engagement when addressing mental health.

Thus, one goal of the community-engaged mental health promotion intervention will be to introduce mental health content in a way that sparks discussion while simultaneously providing members with the requisite knowledge and resources to adequately and sensitively respond to church members experiencing mental illness, in this case, depression. At a minimum, ECA-

administered content will protect the identities of those who do not yet feel comfortable sharing their mental health status with others while also providing community members with the necessary skills and strategies to become connectors to care and support. These findings confirmed the fourth key design principle, “*use a community-building framing to encourage community members to interact with the mental well-being and anti-stigma content*” and the fifth, “*balance tensions between the values of maintaining privacy and communicating care*” for the final system presented in Chapter 7 (Table 6).

Chapter 6: Phase IV Mental Health Feature Selection

6.1 Purpose

Given the sensitive nature of mental wellness and the prevalence of stigma within the communities of interest, I conducted a participatory design activity focused on depression to identify a set of features for the digital mental health and stigma reduction intervention. None of the previous phases of this research focused exclusively on or delved deeply into the topic domain of mental health. This data collection episode was designed to address this gap uncovering leader and church member perspectives on and expectations for features that would be most desirable when addressing church member experiences of depression (Table 13). Leaders were included as stakeholders in the research processes, given the critical role Black and African American pastors play in championing health promotion initiatives as part of a holistic approach to leading their spiritual communities (L. F. Williams & Cousin, 2021).

This data collection addressed the following research questions:

RQ1: How do predominantly Black churches and their members currently support members experiencing mental illness?

RQ2: What features would church members affiliated with predominately Black churches find most meaningful for supporting individuals with depression?

Table 13: Overview of Phase IV Procedure Purpose and Key Findings.

<i>Data Collection</i>	<i>Procedure</i>	<i>Purpose</i>	<i>Key Finding(s)</i>
	Phase IV: Mental Health Participatory Design		
<i>Identifying Mental Health Features with Key Stakeholders</i>	Individual Interviews and Storyboard	RQ: How do predominantly Black churches and their members currently support members experiencing mental illness? RQ: What features would church members affiliated with predominately Black churches find most meaningful for supporting individuals living with depression?	A set of highly desired features were identified: 1. Psychoeducation, 2. Resource Referral, 3. Scriptural Content & Framings.

6.2 Procedure

I developed a set of storyboard screens featuring a fictional church member's experience of depression. The fictional church member was depicted as a Black Indigenous Woman of Color struggling to find mental health support within her church (Figure 17). The story arc included a brief introduction of the character's difficulties with depression (Figure 17), her desire to receive support, and the support that she receives using a faith-based mental health app through the introduction of app features. The features depicted in the storyboard are listed (Table 14). The fictional ECA administered all of these features.

Eight church leaders (two pastors, two co-pastors, and four ministry leaders) were each invited to participate in individual semi-structured interviews facilitated using video conference software. One week prior, I sent the storyboards to each participant for their review. During each individual interview, I shared the storyboards using the video conference ‘share’ feature. Doing so allowed participants to refer to specific storyboard frames and helped anchor our discussions.

Figure 17: Depression Storyboard Excerpt

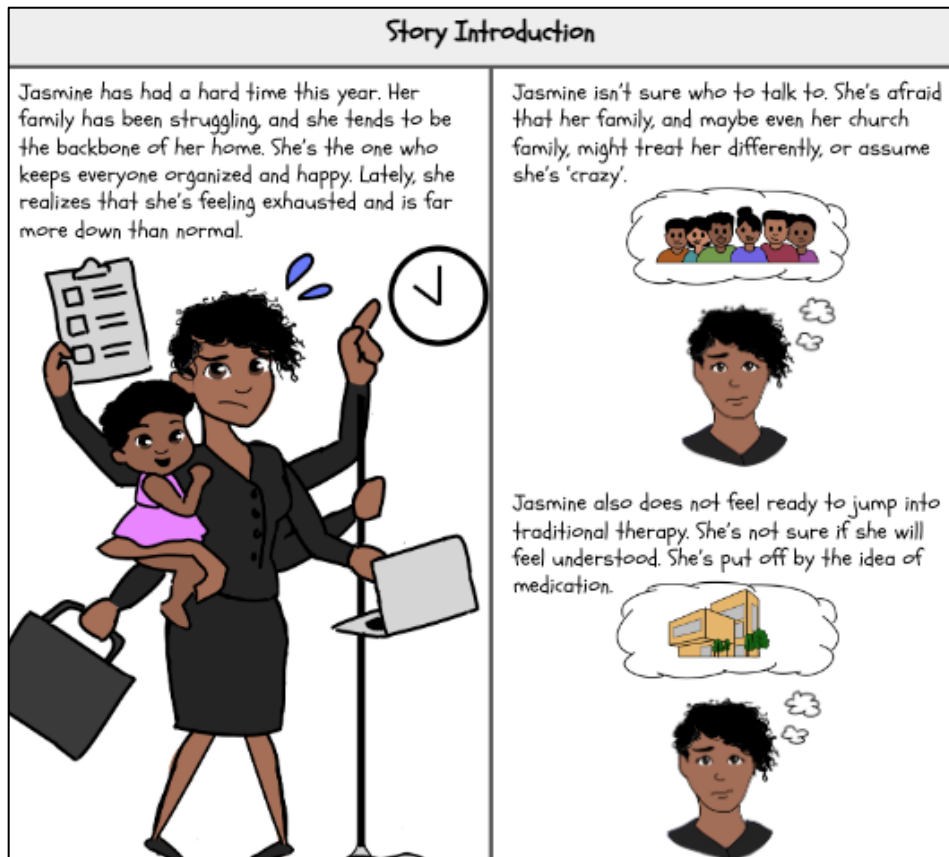


Table 14: Features Depicted in the Depression Storyboard

Features	Description
Psychoeducation	A series of sessions focused on learning about depression (signs, symptoms, treatment, recovery).
Check-ins	A mid conversation check-in to ensure that the participant is okay to continue to the conversation.
Screening	A conversational administration of the PHQ-8. In the storyboard, its depicted as agent-initiated as a means of tailoring content.
Resource Referral	A list of ‘communitysourced’ resources and descriptions available to the user via the ECA recommended by other church members.
Cognitive Behavioral Therapy	Mention of thought restructuring, activity scheduling, and breathing exercises
Self-management	The encouraging of self-care activities by the ECA.

<i>Role Playing</i>	A series of sessions where the ECA teaches users how to talk and support those with a mental health condition.
<i>Scriptural content</i>	Mention of the ECA connecting mental health content to Biblical teachings.

Each interview ranged from 60-90 minutes, with a median length of 73 minutes. Interviews were recorded and transcribed for data analysis. I used a qualitative template-based analysis technique called rapid qualitative data analysis (Gale et al., 2019) to identify a limited set of highly desirable features. As part of their interviews, participants described the presence of mental health stigma within their communities and how the combination of stigma, racial bias, and discrimination prevents church members from seeking care from both informal (e.g., pastoral counseling, church-run support groups) and formal treatment (e.g., therapy, psychiatry, etc.).

6.3 Findings

The following sections discuss findings from the analysis of participant transcripts. Findings describe leader's observations of how church members currently discuss and support those experiencing mental illness, participant feature selections, and participant descriptions of the perceived potential impact of the intervention if the application were developed and deployed.

6.3.1 Participant Overview

Eight church leaders from both Church A and B participated in this work. Two self-identified as men and six as women. Participant ages ranged from 58 to 76. Four leaders were from Church A and four from Church B. No other demographic information was collected.

6.3.2 Current Church-based Practices for Supporting Mental Well-being

Participants emphasized that their church communities struggle to effectively support those with mental health conditions, stating that broad community support for those experiencing mental illness is not normalized. Ministry leaders pointed out that individuals with mental illness do not receive the same support as those experiencing a physical illness (e.g., cancer, stroke, heart attack, etc.). P1 stated: *"No one shows up at your door with a casserole when you have depression. They just stop asking about you."* One leader surmised that those who have a mental health condition may choose to leave the church or choose not to access church-based resources due to fear of disclosure. Doing so means that some members who once felt connected to their community are further isolated. P4 shared: *"Over the years, I know we have lost members because they don't feel totally comfortable just being themselves."* Church B has an active caring ministry called Stephen's Ministry, which is supported by a national faith-based organization that funds training and supplies churches with manualized curricula. Cornerstones of the ministry include one-to-one faith-centered peer case management and care coordination. Currently, one pastor in Church B identifies *caregivers* and *care receivers*. Caregivers are trained by Stephen's ministry staff and carry a caseload of care receivers.

Care receivers are church members who have expressed concerns that typically present alongside an untreated mental health condition (e.g., difficulty accessing appointments, loss of financial stability, social isolation, etc.).

Though the presence of the caring ministry in Church B is undoubtedly beneficial, leaders hypothesized that even with a ministry-mandated confidentiality policy, a significant number of church members do not engage with the ministry due to fears of being exposed. In addition, the ministry does not focus on linkages to care or formal care. Nor do they focus on screening symptoms, promoting awareness, or teaching mental health literacy. Instead, they focus on resolving life stressors using peer-to-peer confidential support. Leaders of this ministry explained that the emphasis on resolving life stressors instead of mental health literacy, screening, and linkages-to-care may unintentionally reinforce stigma within the church.

In Church A, the co-pastor and ministry staff use a direct referral process to local clinicians who identify as Black or African American. This informal referral process requires that church members disclose their mental health status or symptoms directly to a pastor or church leader. In addition, the co-pastor explained that although they have personal relationships with some clinicians, the clinician's availability is not guaranteed. One leader lamented that having additional resources or knowledge about mental health would be helpful when assisting church members. Unlike Church B, in Church A there is no consistent mechanism for supporting members who have a mental health concern, with the co-pastor stating: *"It's luck of the draw who you talk to that day. They may know how to help you; they may make things worse."*

Across all interviews, participants hinted at the presence of stigma by describing how church members may avoid traditional and nontraditional treatment options to avoid perceived shame and discrimination. As the interviews progressed, participants who self-identified as members who experienced depression stated that, even as leaders, they had felt the impact of stigma from those in church. Participants described how being labeled as mentally ill can impact their livelihoods, reputation, and family's reputation. From participant perspectives, there is very little community-wide disclosure. However, participants pointed out that on rare occasions when others have disclosed, their disclosure was powerful. P5, a health ministry leader, shared: *"There's just something about hearing another person's story. Especially someone you can relate to. When people do actually open up, it makes me feel seen. But it's just not the norm. And I don't think it will be until there's more education on it."*

When asked how the proposed ECA system should address mental health and specifically depression, ministry leaders and pastors suggested challenging myths common in their communities (i.e., *"having a mental health condition is not a mortal sin"*). In addition, they stated that in-depth ECA-led psychoeducation would be crucial. One ministry leader brainstormed a potential mental well-being curriculum stating that the ECA should help reduce prejudicial thinking among community members and lead community member training by 1) defining depression and describing symptoms and 2) teaching church members how to ask people experiencing depression what they need.

6.3.3 Feature Ranking

During the interviews, participants were asked to rank the prospective features from the storyboard and add any additional features or changes to the proposed features as needed. Table 15 shows the top-ranked features with feedback from participants.

Table 15: Feature Ranking, Feedback and Endorsement

Feature	Feedback	N Endorsement
Psychoeducation	Participants stated that not only is psychoeducation one of their top features but that the session should emphasize normalization and include a stigma education component	8/8
Resource referral	Participants believed that links to credible resources provided by the agent is crucial for providing access to community members who may perceive stigma. However, they did not endorse communitysourcing these resources.	8/8
Scriptural content	Participants explained that a culturally-responsive mental health promotion intervention should tie teachings from the Bible and Scripture to mental health education and encourage engagement.	8/8
ECA Roleplaying	Participants believed that the ECA should educate users on how to help others within the community engage in formal or informal treatment/support. They felt content such as this could be directed towards caregivers or confidants.	6/8

6.3.4 Actionable Strategies for Coping

While most proposed features received positive feedback, participants were uncomfortable with the ECA using Cognitive Behavioral Therapy or any talk therapy with church members. When asked why, participants explained that doing so would be inappropriate, expressing concerns that church members should be encouraged to engage with formal care. Participants were dissatisfied with any scenario where technology could potentially replace good clinical care and wanted to avoid using the ECA as a stand-in for a trained professional. However, participants were comfortable with the ECA providing specific actionable support strategies (e.g., breathing exercises, meditation, prompts to take a walk, etc.). They stated that actionable strategies delivered privately (within the application by an ECA) could immediately increase access to good clinical advice for those not ready, unable, or on a waitlist to participate in formal treatment.

6.3.5 Crowdsourcing Structured Informational Social Support

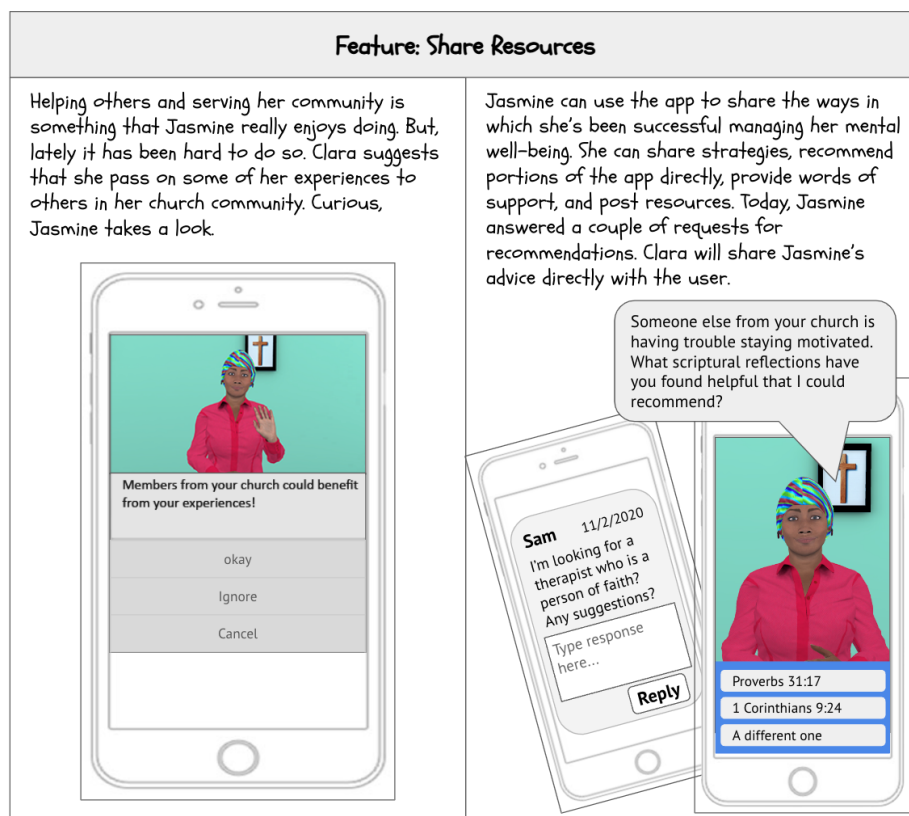
Crowdsourcing is the process of leveraging remote workers' collective intelligence to perform tasks for an individual or an organization (Howe, 2006). Effective Crowdsourcing techniques have been applied to various health domains, including mental health and well-being (Brabham et al., 2014; Wazny, 2018). In some instances, crowdsourcing requires knowledge that comes from within a specific community's social network, a practice called *communitysourcing* (Heimerl et al., 2012).

Given the importance of producing a culturally-informed application and the value church members placed on supporting one another, I hypothesized that communitysourcing certain aspects of the health content (e.g., supportive messages, words of encouragement, resources, etc.) could be one way to achieve a more culturally-relevant and responsive application experience. As a result, I included one highly structured communitysourced feature in the storyboard to serve as a conceptual

probe. In the storyboard, the agent asks the main character to share resources that have worked well for her in the past. The character is informed that the agent will share these resources with other app users (Figure 18).

Both church communities viewed this communitysourced feature as problematic and unrealistic. Leaders pointed out that member contributions being available on the app could increase the dissemination of misinformation. Church leaders and lay leaders alike stated that *any* communitysourced information would need to be evaluated by staff to discern the credibility and safety of the submitted resource. Given current church staffing and resources, such a review process would be complicated and nearly impossible. As a follow-up, I asked leaders if words of encouragement or supportive messages to be shared through the application might be preferred; all leaders and lay leaders alike stated that this feature was not currently welcomed as there is a potential for even well-intentioned messages to be harmful. As a result, none of the interviewees endorsed any feature with a communitysourcing component. These findings demonstrate how I arrived at the sixth key design principle, “*Prioritize features that build mental health literacy, not the provision of unstructured social support*” (Table 6).

Figure 18: Storyboard Communitysourcing Feature



6.3.6 Potential Future Impact

During the interviews, participants discussed their future aspirations for the application and its potential impact if used by church members broadly. Participants stated that the application could help normalize discussions around mental health. Given that the

application is being co-developed with church members, participants stated that its content would appear more relevant and credible to members. Such an app promoted by the church would reinforce that the church and its leaders endorse this topic (e.g., formal care-seeking). Members stated that the app could help pastors promote mental health awareness while providing lay leaders and pastors with standardized and credible information. Leaders believed that church members might gravitate towards the application as a first step towards addressing their mental health to maintain their privacy and anonymity. However, they hoped such use could catalyze future discussions within the community. Throughout the interviews, all lay leaders and pastors welcomed the development of a church-based digital mental health tool believing that such a tool would help begin a conversation about mental wellness in a more robust way than what is currently practiced.

6.4 Synthesis of Phases I-IV

Along with the related literature laid out in Chapter 2, I leveraged findings across Phases I-IV of my dissertation work to establish six key design principles (Table 6) for developing a technology-enabled mental health and stigma reduction intervention for predominantly Black church communities. Phase I provided the necessary background to understand these communities and drove the subsequent research questions and co-design activities in Phase II. In Phase II, participants communicated their desires and expectations for a community-based culturally-tailored ECA who could “*speak their language*.” Participants also identified values that were integral to their communities and health topics of relevance. The evaluation of Church Connect in Phase III demonstrated the feasibility and acceptability of working with an ECA that takes a holistic approach to health and wellness by incorporating content and features that support bodily, spiritual, and social well-being. In Phase IV, participants selected what features should be prioritized in the design and implementation of an ECA-facilitated mental health promotion intervention and dismissed any communitysourced features. The next chapter will focus on the technical design of the final ECA system and the development and implementation of the intervention curriculum.

Chapter 7: Mental Health and Stigma Reduction System Design

In this chapter, I present the overall system design and implementation of the final ECA-based mental health promotion system. First, I will provide a justification for the two user types, intervention framing, the ECA design and system architecture, and then a detailed presentation of the mental health and stigma-reduction intervention curricula and features. This chapter will demonstrate how I acted on learnings from my prior work to develop the final system evaluated in Chapter 8.

The design of the mental health and stigma-reduction system was developed using the *Key Mental Health Promotion Intervention Design Principles* based on findings from Phases I-IV of this dissertation work and the related work presented in Chapter 2. As demonstrated in Chapters 3-6 these principles were conceived using an iterative and incremental design process. Table 16 provides a summary of how these principles were operationalized and implemented into the final design of the system.

Table 16: Key Intervention Design Principles and their Operationalization in the Mental Health & Stigma Reduction System

Key Intervention Design Principles	Intervention Operationalization
Address stigma when promoting mental wellness	Three sessions were created to address stigma explicitly. One focused on general stigma education, the other on self-disclosure processes, and the third on discussing myths and facts associated with depression.
Integrate spiritual elements and faith-based concepts into both ECA design and health promotion content	See below Table 17 for how the ECA Community-based Cultural Tailoring Design Principles were utilized.
Offer resources to increase church member capacity for mental health support	Resources were included in the application and could be texted directly to user's phones at the user's request.
Use a community building framing to encourage members to interact with content	Developed in partnership with our community partner, the framing for the intervention was to <i>support others in your community</i> and <i>build community around mental health promotion</i> .
Balance tensions between the values of maintaining privacy & discretion and communicating care	Due to the smartphone implementation, users were able to access resources and health education while <i>maintaining privacy</i> . To assist members with <i>communicating care</i> , the intervention included recommendations for how to respond to members experiencing depression, how to assist with facilitating the care-seeking intentions of others, and <i>big and small ways</i> to reduce stigma in the community.
Prioritize features that build mental health literacy, as a first step, not the provision of unstructured social support.	This intervention prioritized psychoeducation, early screening, support strategies, and resource identification. This intervention did not include any communitysourced features.

7.1 Justification for Two User Types

The intervention targeted two groups of primary stakeholders: individuals with depression (PwD) and other community members (CM). While the two interventions shared significant elements (e.g., psychoeducation on depression) they also had unique elements, and were tailored differently. Prior research indicates that an individual's social network mediates the impact of stigma

on individual care-seeking intentions (P. Corrigan, 2004; P. W. Corrigan et al., 2014; Pescosolido et al., 2013; Pescosolido & Olafsdottir, 2013). The impact of stigma in a person's social network on care-seeking is twofold. First, if an individual decides they would like to access either informal or formal treatment their social network can act as facilitators to receiving care either through emotional or tangible support (e.g., driving a friend to an appointment, watching your friend's children while they speak with the pastor, etc.). If the social network remains unsupportive due to stigma, then the individual is unable to capitalize on their social connections. Second, if an individual experiencing symptoms of depression engages in treatment but has an unsupportive social network then they are forced to conceal their symptoms. Research demonstrates that such concealment of a stigmatized identity comes at a psychological cost to the individual's overall mental well-being (P. W. Corrigan & Rao, 2012). This twofold impact of stigma on individual mental wellness and the aforementioned community assets of prioritizing collective well-being and social interconnectedness present within these churches provides the reasoning and justification for creating an intervention that targets both persons with depression and community members.

For PwD the goal of the intervention was to provide psychoeducation, reduce self-stigma, and promote care-seeking intentions. For CM's the goal was to promote community-wide mental health awareness and reduce stigma using a community training model that emphasizes education and actionable strategies for stigma reduction within the church.

7.2 User Type Intervention Framing

Responding to the key intervention design principle *"use a community building framing to encourage community members to interact with the mental well-being and stigma content"* (Table 16), Clara framed and encouraged intervention engagement as a means of creating a more inclusive and welcoming church community, where people experiencing symptoms of mental illness could find community support. Clara assured church members that participation was confidential (information would only be accessed by members of the project team), aligning with the key intervention design principle of *"balancing tensions between maintaining privacy and communicating care"* (Table 16).

7.3 User Type Assignment

User type assignments were handled in-application by the ECA and depended on how users answered a series of questions during the introductory session and during a follow-up session. This approach was intentionally liberal in how depression is defined to capture how individuals from these communities who experience depression may present or describe their experiences.

Initial User Type Assignment

Users had two opportunities during the introductory conversation to identify as a PwD or a CM. First, the ECA administered a two-item depression screener, the Patient Health Questionnaire-2 (Kroenke et al., 2003), and asked users if they had a history of

depression, were in treatment for depression, or self-identified as depressed. These questions were asked to determine if the user was experiencing depression or had ever experienced depression.

Given the level of stigma in the community and concerns over privacy and confidentiality, I decided to include an additional goal-oriented question in an effort to ensure that those who may not have felt comfortable or ready to endorse a symptom of depression, depression treatment, or history had access to PwD content. Thus, directly following the depression questions, the ECA asked what the user's primary goal for using the intervention was and provided two pre-scripted user-response options: *"to get help managing my mental wellness"* or *"to support others in my church."*

If the user screened as depressed, self-identified as depressed or endorsed a history of depression or depression treatment, they were assigned to the PwD user type. If the user did not screen as depressed nor endorsed any of the depression history or identity questions but selected the goal *"to get help managing my mental wellness,"* they were assigned to the PwD user type. If the user did not screen as depressed and set the goal *"to support others in my church,"* they were assigned to the CM user type.

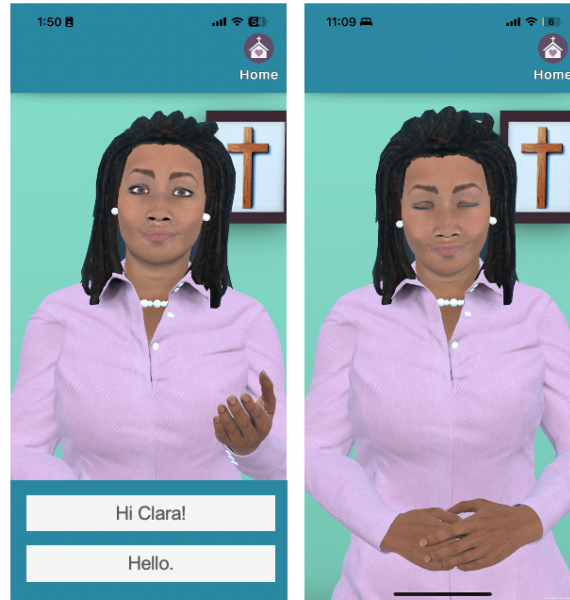
Updating User Type Post Introduction Session

Users had a second opportunity during the intervention to update their user type assignment. Given that the introduction session would be completed during the first interaction with the character, I wanted to allow users another opportunity at intervention assignment once the ECA had had more time to establish rapport with the user. During a follow-up session with all user types, the ECA administered a longer depression screener called the Patient Health Questionnaire-8 (PHQ-8) (Kroenke et al., 2009). If users screened as depressed using the PHQ-8 but had been previously assigned to the CM user type during the introduction, they were automatically reassigned to the PwD user type and had access to all PwD content. If the user did not screen as depressed but had been assigned to the PwD user type during the introduction session, they remained in the PwD curriculum. Again, these design decisions were made to ensure that anyone who may benefit from the PwD content could access it.

7.4 Embodied Conversational Agent Design

This digital mental health promotion intervention featured a smartphone-based ECA named Clara (Figure 19).

Figure 19: Screen Shot of Clara an Embodied Conversational Agent



To design the ECA, I leveraged the ECA Community-based Cultural Tailoring Design Principles (O’Leary et al., 2020) that were based on findings from the ECA design and dialogue tailoring focus groups to create an identity-affirming intervention responsive to aspects of participant faith and racial and ethnic identity. Table 17 provides an overview of how these design principles were operationalized, including how participant values of spirituality and religious concepts were reflected in the final mental health and stigma-reduction ECA system.

Table 17: ECA Community-based Cultural Tailoring Design Principles Mapped to the Final ECA Implementation

ECA Design Principles	ECA Intervention Implementation
Establish the role and cultural background of the agent early in conversation.	Clara’s role was defined as a peer health advisor of a caring ministry early in the initial introduction interaction.
Establish common ground <i>and</i> solidarity through tailored social chat.	Clara includes a set of ‘ <i>getting to know you</i> ’ scripts. For example, Clara asks users about the role faith plays in relation to their health and shares how her faith shapes how she thinks about mental well-being. These scripts were triggered at the start of a new study day. Seven unique faith-based <i>getting to know you</i> scripts were written for this dissertation work.
Leverage religious concepts and Scripture to motivate and persuade users.	Scripture and religious concepts were interweaved into the sessions to motivate users to interact with the system and provide encouragement to sustain intervention engagement.
Provide culturally-tailored problem solving	Clara addressed religious and racial and ethnic barriers to engaging in formal treatment.

7.5 System Architecture

Clara is an animated ECA developed and rendered using Unity3D game engine. The ECA's dialogue is scripted using a custom scripting markup language based on a Hierarchical Transition Network Model. ECA dialogue is designed as a series of conversational states, converted into executable code, and input into a state-based dialogue manager. The dialogue manager decides on state transitions based on user responses and predefined user attributes. For example, if users indicated they experienced depression, the dialogue manager transitions to states where the ECA offers different content and scripts than if they indicated they are not experiencing depression. ECA nonverbal behavior is automatically generated using BEAT (Cassell et al., 2001). Nonverbal behaviors included facial displays, head nods, gaze shifts, body posture to signal topic boundaries, and hand gestures for conversational emphasis. User variables were stored in a persistent remote database. The ECA spoke using an off-the-shelf text-to-speech synthesizer. ECA speech was synchronized with nonverbal behavior. Users were able to converse with the ECA by selecting user utterance options from a multiple-choice menu displayed on the screen, updated at each turn of conversation.

7.6 Stigma-Reduction and Depression Intervention

In following sections, I present the mental health and stigma reduction intervention. The intervention was designed as a series of *sessions* each comprising of a 10 to 15-minute conversation with Clara, intended to be experienced over a series of days. All sessions were based on my review of the related work (Berkley-Patton et al., 2021; Hays, 2018; Rivera et al., 2021), a self-stigma manualized treatment for the stigma reduction curricula (Hansson et al., 2017), and the *Key Mental Health Promotion Intervention Design Principles* from phases I-IV of this dissertation work.

The mental health and stigma reduction interventions were interweaved. The design choice to include stigma reduction and depression content as part of the overall mental health promotion intervention was to be 1) culturally responsive (K. Conner, 2010; Rao et al., 2007), and 2) increase the effectiveness of the mental health promotion intervention by exploring stigma as a barrier to care-seeking (P. W. Corrigan et al., 2014; Pattyn et al., 2014; Shefer et al., 2013).

Two domain experts reviewed all sessions: a pastor from one of the two churches to validate the integration of spiritual elements and a clinical psychologist to determine that the mental health and well-being content met good clinical standards. See Table 18. for the curricula developed for each user type.

Table 18. Sessions by User Type

*note sessions are not in sequence but rather users can choose the order.

<i>Sessions</i>	<i>Tailoring by User Type</i>
<i>Psychoeducation</i>	Tailored by mental health status
<i>Stigma and self-stigma education</i>	Tailored by self-stigma status

<i>Myths and facts quiz</i>	Tailored by mental health status
<i>PHQ-8 baseline</i>	Tailored by mental health status
<i>Encourage care-seeking</i>	Person who experiences depression only
<i>How to support others</i>	Community member only
<i>Disclosure Education</i>	Person who experiences depression only
<i>Resources</i>	Tailored by mental health status
<i>Frequently Asked Questions</i>	Tailored by mental health status

In the following subsections, I provide a detailed overview of first the stigma-reduction intervention and then the depression intervention components.

7.6.1 Stigma-Reduction Intervention

In response to the key intervention design principle, “*address stigma when promoting mental wellness*,” This system incorporated three stigma reduction strategies from the literature. They are: 1) education, 2) paracontact, and 3) self-disclosure planning.

Education Approach

The intervention included both psychoeducation and a social stigma education facilitated by Clara. The stigma reduction education aimed to explicitly discuss stigma as a harmful societal process that negatively impacts people with concealable stigmatized identities such as depression. The purpose of discussing stigma explicitly was to label stigma as a resource-reducing social process that diminishes an individual’s life opportunities. Stigma disrupts typical care-seeking pathways, such as capitalizing on one’s social network for informational resources or leveraging social relationships to facilitate care. I hypothesized that by explicitly labeling stigma, individuals experiencing depression may be less likely to hide their symptoms and more likely to seek mental health care and support.

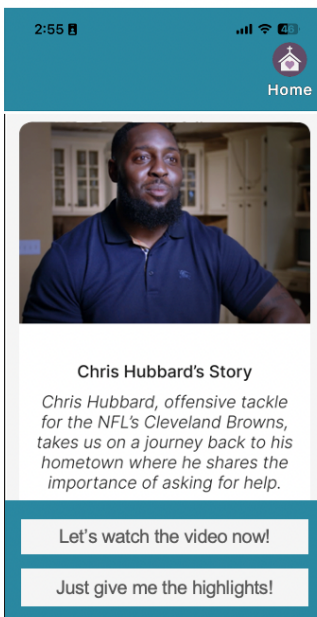
The ECA began this session by defining stigma and self-stigma. Next, the ECA asked PwD users if they have ever experienced self-stigma. If yes, users received a tailored version of the script explicitly addressing self-stigma. CMs received a mental health impact exercise where they reflected on the types of support activities offered to church members experiencing a mental illness compared to those experiencing an acute health crisis such as surgery.

Paracontact Approach

While contact-based approaches have demonstrated significant reductions in public and self-stigma (Evans-Lacko et al., 2012), creating opportunities for computer-mediated social contact between members in a small community setting could negatively impact cohesion and social well-being. This was confirmed during the Mental Health Feature Participatory Design Session (Chapter 6). As a result, a paracontact approach was adopted. As a first step, culturally tailored testimonials were selected for inclusion. These stories were adopted directly from the following resources: the National Alliance for Mental Illness *Stigma Free* campaign and McLean Hospital's *Deconstructing Stigma* campaign.

The stories are auto-biographical in nature and featured African American people experiencing depression. Each story is told through the eyes of the narrator with the aim of discussing misconceptions of mental health. These stories each provided an overview for how symptoms presented for each narrator, the barriers they encountered while trying to access treatment, and their recovery process. One story featured a United States National Football League player, Chris Hubbard, and his experience of depression (NAMI, 2020) (Figure 20). His story spoke directly to the consequences of stigma on his life and relayed the importance of asking for help. The second story featured an African American woman named Dominique (McLean Hospital, 2018). Her story described how her symptoms presented and the impact they had on key moments in her life. Her story ends by encouraging readers to ask for help sooner in order to take care of themselves. These testimonials were shared in dialogue by the ECA. Users could read the stories, or opt to have the character summarize them.

Figure 20: Paracontact



At the end of the story, users were invited to discuss story takeaways with the character. While users received the same paracontact testimonials, the story reflection discussion with Clara was tailored. For instance, PwD were asked how they related

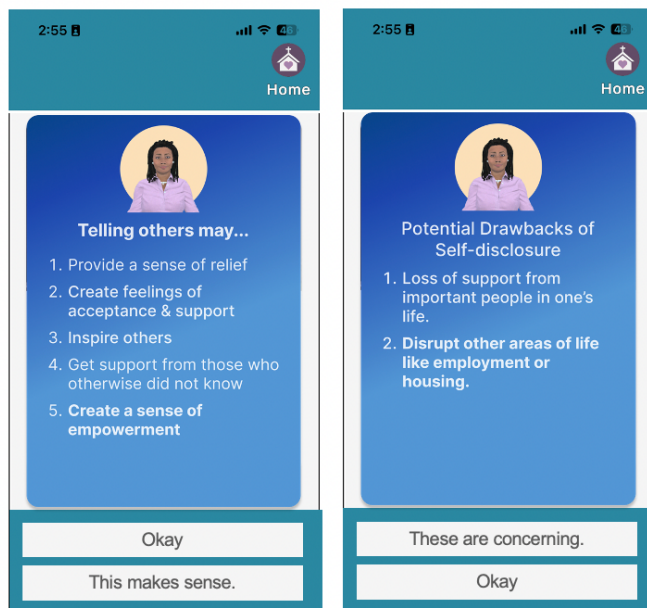
to the experiences relayed in each story. CMs were asked if they knew anyone who had experienced the situations described in the paracontact testimonials. Clara ended the story reflection discussions by sharing her favorite takeaways with the user.

Disclosure Approach

As discussed in Chapter 2, self-disclosure (a strategic decision to let others know about one's experience and stigmatized identity) is another approach used to reduce stigma (P. W. Corrigan & Rao, 2012). The final system included an introduction to self-disclosure planning session facilitated by the ECA for the PwD user type. The aim of this session was to introduce the concept of *planning* self-disclosures so as to maximize the benefits of self-disclosure outcomes. During this session, the ECA provided a brief introduction to disclosure where users learned about the potential benefits and drawbacks of disclosures (Figure 21), possible goals for disclosures (e.g., receive work accommodations, explain behavior, create closeness in a social relationship) and potential audiences for disclosures (e.g., friends, family, co-workers, employers, healthcare providers, social media etc.). The disclosure planning education was based on the Disclosure Processes Model (Chaudoir & Fisher, 2010) and the disclosure section of the Narrative Enhancement Cognitive Therapy facilitator's manual (Hansson et al., 2017).

This conversation allowed users to consider how and why they might want to self-disclose their mental health status to others. The ECA intentionally took a neutral stance to self-disclosure and did neither encourage nor discourage users from self-disclosing. However, the ECA did recommend if interested in self-disclosure that users first self-disclose to a medical care provider as a pathway to seeking care

Figure 21: Featured Self-disclosure Planning Session Screens



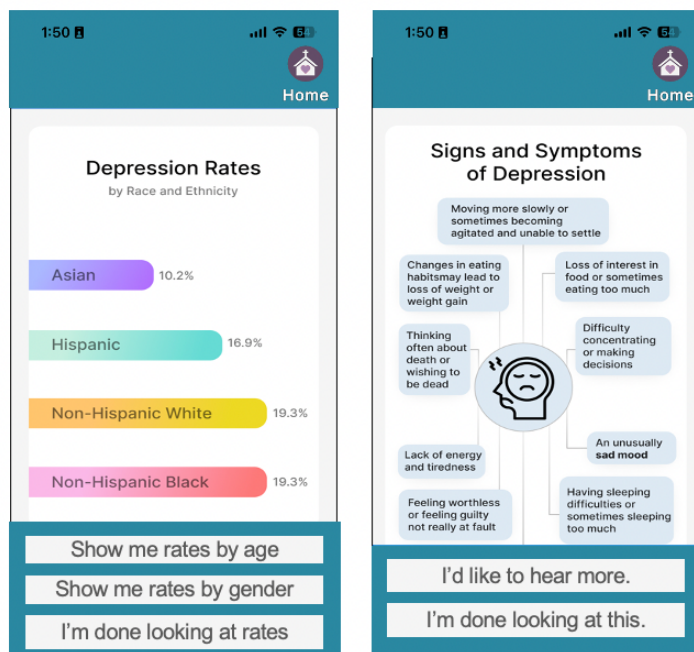
7.6.2 Depression Intervention

In response to the key intervention design principle, “prioritize features that build mental health literacy, as a first step, not the provision of unstructured social support,” this intervention took a mental health awareness approach. In this section, I summarize the intervention components: 1) psychoeducation, 2) screening, 3) resource lists, and two additional components focused on enhancing and increasing support for church member care-seeking intentions. For individuals with depression, Clara engaged users in a counseling conversation to enhance readiness for professional treatment. For users in the CM user type, Clara discussed both how to support PwD generally and how to effectively support the care-seeking process.

Psychoeducation

The depression psychoeducation is an interactive education session led by Clara. During the education session, Clara oriented users to the topic of depression, defined depression, and presented a culturally-informed description of symptoms. The session included a set of population-level statistics where the user could select depression prevalence by demographic characteristics (e.g., gender, race and ethnicity, or age etc.).

Figure 22: Featured Psychoeducation Screens



Symptom Screening

In this session, Clara administered the eight-item Patient Health Questionnaire depression scale (PHQ-8) in dialogue for all user types. The PHQ-8 is a standardized and validated questionnaire that measures symptom severity (Kroenke et al., 2009). For the

PwD, Clara framed the conversation as a way to better understand how depression impacts the user. For the CM Clara framed the screener has a way to explore symptoms of depression.

Symptom Check-ins

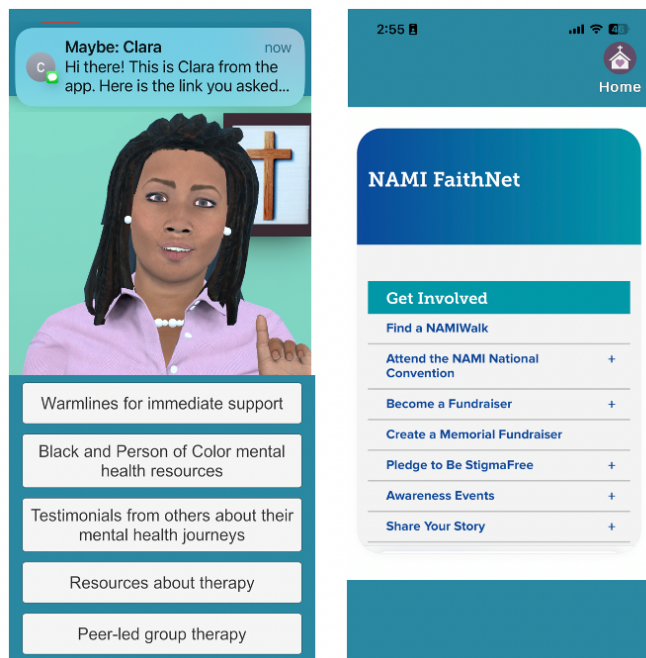
After establishing a PHQ-8 baseline score, Clara conducted a brief check-in with users once a day for every positive symptom (where a positive symptom is defined as any symptom where the participant endorsed the symptom frequency of *some of the days* or greater). This symptom check-in occurred as part of the initial *start of the session* mental well-being interaction protocol, where Clara greeted the user. Then, if any positive symptoms existed, Clara intervened on the most severe reported symptom by asking the user if they were still experiencing the symptom, providing an empathic response, and offering suggestions.

Suggestions were authored by my clinical collaborator.

Resources

In response to the intervention key design principle, “*offer resources to increase church member capacity for mental health support*,” users had access to a culturally-informed list of resources, including resources that featured non-white service providers. Administered through dialogue, Clara introduced the resources, offered a description of the resource, and mentioned why and how the resource might be helpful to the participant. Resources included both local and national service organizations. Users were given the option of having Clara send the resource via SMS to the user’s phone (Figure 23). This way users could refer back to the resource and share resources with others outside of the study at their convenience. Resources were available for both PwD and CM user types and were tailored accordingly.

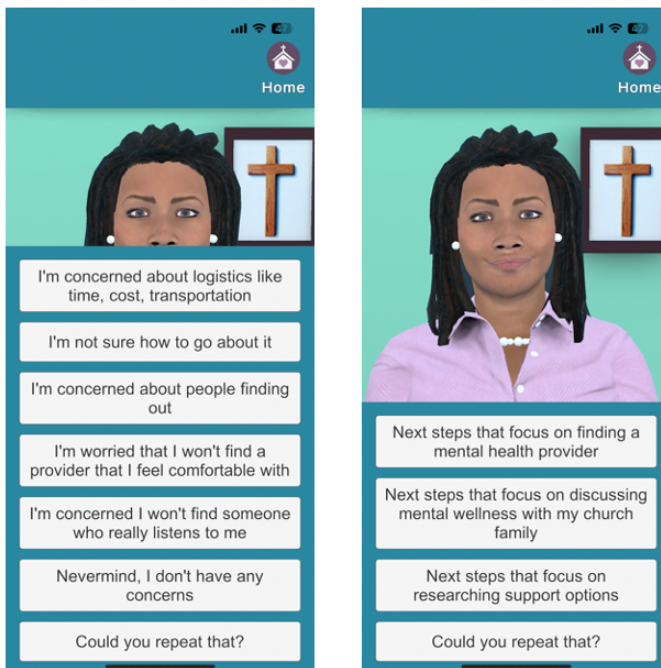
Figure 23: Featured Resource Screens



Enhance Readiness for Care-seeking (PwD only)

PwD users were encouraged to seek care from formal and/or informal mental health support options. The ECA asked users what types of support they were interested in, discussed potential barriers or concerns (Figure 24), and presented them with options for their *next steps* (e.g., discuss mental wellness with others in the church, research providers, etc.) (Figure 24). All *next steps* were selected to either 1. increase mental health support or 2. normalize discussions around mental wellness. For those engaging with both professional and alternative forms of mental health support (i.e., therapy and a church-sponsored grief group), Clara asked users if they had received conflicting advice (conflicting secular versus religious perspectives) and empathized with users if this was the case. This dialogue was constructed to be sensitive to support opportunities outside the traditional and often racialized lens of mental healthcare. For instance, if an individual is already engaged in pastoral counseling, the ECA offered information for formal mental health care but did not undermine the positive benefits of engaging in informal care.

Figure 24: Encourage Care-seeking Addressing Barriers (left) and Next Step Activity (right) Screens



Enhancing Community Member Capacity for Facilitating Care-seeking (CM Only)

The system offered CMs a community training intervention focused on increasing support and empathy for treatment-seeking persons with depression to complement the content that the PwD user type received. This intervention component first educated users on treatment, addressed concerns that users may have regarding treatment, and then covered how CMs could help PwD establish services. In addition to persuading CMs to empathize with PwD, Clara includes advice for assisting PwD that is both actionable and tangible.

Chapter 8: Summative Evaluation

I evaluated the efficacy of the mental health and stigma reduction intervention in a quasi-experimental two-week field deployment. This work investigated participant perspectives on what additional content should be included in a digital mental health and stigma-reduction tool tailored for their community, as well as collected their current perceptions of the culturally-tailored content incorporated in the intervention.

Specifically, this work addressed the following research questions:

RQ1: Is it feasible and acceptable to provide a stigma reduction and care-seeking encouragement intervention using a smartphone-based ECA modeled after a peer health advisor?

RQ2: What effect does a smartphone-based ECA that utilizes stigma reduction strategies to enhance readiness for care-seeking have on African American and Black church-affiliated users?

RQ3: What impact does introducing the topic of disclosure have on Black church member disclosure intentions?

RQ4: How does a smartphone-based ECA that utilizes stigma reduction strategies increase confidence in discussing the topic of mental illness and mental health?

8.1 Participant Sampling

I recruited twelve adults from historically Black churches in Boston using network and stakeholder connections from my preliminary work in Churches A and B. To be eligible to participate, prospective participants were required to be 1) 18 years or older, 2) speak and read English, 3) own and use a smartphone, 4) self-identity as being a member of a historically Black church, and 5) have access to video conference software. All prospective participants were pre-screened for depression and/or depressive symptoms prior to enrollment using the PHQ-8 (Kroenke et al., 2009) and four questions that asked participants if they are currently in specialty treatment for depression, are taking medication for depression, or self-describe as experiencing depression. Participants were excluded from study participation if they were screened as *severely depressed* using the PHQ-8. I stratified the study sample to include at least six participants who screened as depressed.

I worked directly with church health liaisons to identify potential participants and advertise the study. Participants were invited to contact me directly using email, text message, or phone call. All participants elected to speak directly by telephone. During this initial communication period, I screened participants for eligibility, introduced the project and study, and answered any questions regarding participation. If eligible and interested, I scheduled the first study appointment.

8.2 Measures

Participants completed several measures pre and post study period. Refer to Table 19 for a measurement schedule.

Table 19: Measure Schedule

<i>Measure Name</i>	<i>Pre-interaction</i>	<i>Post-interaction</i>
Demographics		
Multidimensional Measure of Religious Involvement in African Americans (MMRI-AA)		
PHQ-8		
Mental Help Seeking Intention Scale (MHSIS)		
Confidence in Discussing Mental Health Topics with Community Members		
Depression Stigma Scale (DSS)		
Inventory of Attitudes Toward Seeking Mental Health Services (IASMHS)		
Agent and System Satisfaction		

All measures except *Confidence in Discussing Mental Health* and *Agent and System Usability* are standardized and validated measures. The Confidence measure was modeled after primary outcome measures designed to assess *community support* in community-engaged mental health first aid interventions (Evans et al., 2021; Mohatt et al., 2017).

8.2.1 Multidimensional Measure of Religious Involvement in African Americans (MMRI-AA)

The MMRI-AA measures is comprised of three different subscales designed to measure three dimensions of religiosity: 1) organizational religiosity, namely public participation in organization-based roles and practices such as leadership roles, attending worship, volunteering at a church-sponsored event etc., 2) non-organizational religiosity, i.e., private participation in religious practices such as prayer, reading the Bible, etc., and 3) subjective religiosity, i.e., the importance of religion to the respondent (Frazier et al., 2005). Higher scores on the composite measure indicates a greater degree of participant religiosity. This measure was chosen because of its specificity in characterizing different dimensions of religiosity important to African American respondents (Frazier et al., 2005).

8.2.2 Patient Health Questionnaire-8 (PHQ-8)

The PHQ-8 is a standard and validated self-report measure of depressive symptoms (Kroenke et al., 2009). The measure consists of 8 Likert-type scale items. Each item corresponds to a symptom of depression. Each item's Likert-type scale response options indicates the frequency by which a symptom occurs over the past two weeks (0 = *Not at all*, 1 = *several of the days*, 2 = *more than half the days*, 3 = *nearly every day*). Total score ranges on the PHQ-8 are from 0-24 with higher scores indicating a greater severity of depressive symptoms. Scores between 0 and 4 indicate *no depression*. Scores between 5 and 9 are coded as *mild*

depression. Scores between 10 and 15 are coded as *moderate depression*. Scores between 16 and 20 indicate *moderately severe depression*. Scores greater than 20 indicate *severe depression*.

8.2.3 Mental Help Seeking Intention Scale (MHSIS)

The MHSIS is a 3-item 7-point Likert scale used to measure participants' intention to seek help from a mental health professional if they have a mental health concern. A higher score indicates a greater intention to seek care.

8.2.4 Inventory of Attitudes Toward Seeking Mental Health Services (IASMHS)

The IASMHS assesses respondents' attitudes towards professional mental health services and care. The IASMHS consists of 24 Likert-type scale items that contribute to a total score and three subscales: 1) psychological openness (i.e., the extent to which an individual is open to acknowledging a mental health condition), 2) indifference to stigma (i.e., the extent to which individuals are impacted by stigmatizing thoughts and beliefs of others) and 3) propensity for help-seeking (i.e., the extent that individuals believe they can get help, if they wanted to). Scores on the IASMHS range from 0-96. Higher scores indicate more positive attitudes towards professional mental health care. Fifteen of the 24-items were reverse coded prior to data analysis.

8.2.5 Confidence in Discussing Mental Health Topics with Church Members

The Confidence in Discussing Mental Health Topics with Church Members has been used in studies assessing mental health first aid and community trainings. The scale was adapted to reference *church members* in lieu of the more general term *community members* and *your church* instead of the more general term of *community*. The measure uses five 5-point Likert-type scale items (0 = *not at all confident* and 4 = *very confident*) to assess the respondents' confidence in discussing mental health with others in their church. A higher score indicates a higher degree of confidence.

8.2.6 Depression Stigma Scale (DSS)

The DSS is an 18-item measure that contains two stigma subscales: 1. personal stigma and 2. perceived stigma. Personal stigma is defined as stigmatizing attitudes that an individual personally believes. Survey respondents were asked to choose how strongly they agree (0 = *strongly disagree* to 4 = *strongly agree*) with each statement. Higher scores on this subscale indicate higher levels of personal stigma or stigmatizing attitudes towards individuals experiencing depression. The perceived stigma subscale measures the participants' perceptions of the attitudes of others towards individuals experiencing depression. Similar to the personal stigma subscale, higher scores on the perceived stigma subscale indicate higher levels of perceived depression stigma. Higher scores on the total DSS indicates higher levels of depression stigma.

8.2.7 Agent and System Usability

The Agent and System usability measure is an 11-item Likert-type scale questionnaire designed to assess concepts related to longitudinal engagement with an ECA. Concepts included: ease-of-use, likelihood of continued use, satisfaction, likelihood of recommending the ECA to others, and relational qualities, i.e., degree of familiarity with agent, how knowledgeable the agent is, how trustworthy the agent is, how much the user believes the agent cares about them. Item response options were on a 7-point Likert-type scale (1 = *not satisfied at all* and 7 = *very satisfied*). Higher scores on items indicate higher levels of satisfaction with system use.

8.3 Procedure

8.3.1 Initial Study Appointment

This study was conducted in-person. Following informed consent, participants completed pre-interaction study measures (Table 19). Next, I conducted a short semi-structured interview where I asked participants to explore their current perceptions of mental health and well-being within their church communities as it relates to community support and stigma. Participants were invited to explore how their current experience of depression or symptoms of depression impact their lives, and in what ways they wish their churches would respond. Directly following the interview, Participants downloaded and installed the intervention smartphone app with my assistance. Next, participants completed the eight-minute mental well-being introduction session with the ECA.

8.3.2 Fourteen-Day Field Study

Following the initial interaction with Clara, participants were instructed to use the application at home for a period of 14-days. A paper journal was provided to participants. User journals consisted of a prompt per day asking users to reflect on the session in 2-5 sentences. The layout and expectations for journal writing were covered during the initial study appointment. Paper journals and diaries have been used in digital health and HCI research to capture daily thoughts when interacting with digital health interventions (R. N. Brewer et al., 2017; Claisse et al., 2022). My aim for using paper journals was to capture participant thoughts in a way that does not require additional technology.

8.3.3 Follow-up Study Appointment and Interview

At the end of the 14-day field study, users were invited to a remote follow-up study appointment with myself where they completed the post-intervention study measures (Table 19), and were interviewed about their experience using the system. The semi-structured interview explored users' satisfaction and expectations for the mental health promotion and stigma-reduction intervention. In addition, participants were invited to respond to cultural-tailoring throughout the intervention including both

content (e.g., barriers to treatment, spiritual framing for the intervention) and agent role and persona (ministry member, person-of-faith, Black woman, peer mental health advisor).

8.4 Results

Sixteen church members responded to the recruitment flyer and announcements made at their respective churches. While all prospective participants were eligible, four were unable to participate due to various medical and personal reasons. This left 12 participants who successfully completed the study, with six in each user type condition. One participant of the twelve completed the first two post-study questionnaires (MHSIS and Confidence), but due to technical difficulties was unable to complete the remaining post-study questionnaires. A second participant elected not to complete the post-study questionnaires due to a medical crisis. All participants completed the pre-interaction measures. Thus, when reporting sample descriptive statistics an $n = 12$ will be used. When reporting on pre and post study measures an $n = 11$ or an $n = 10$ will be reported. All twelve participants completed the post-intervention qualitative interview.

8.5 Quantitative Findings

8.5.1 Participant Overview

For an overview of descriptive statistics refer to Table 20. Six church members were from Church A, six from Church B. All participants self-identified as Black or African American. All participants self-identified as women. Participants' ages ranged from 31 to 83 ($M = 58.2$, $SD = 18.68$). For education, one participant reported having a high school degree, four reported having some college, three had a bachelor's degree, and four had a master's degree. All participants were members of their respective churches.

Participant scores on the PHQ-8 ranged from zero to 16 with a median of 5 ($IQR = 6$). Six participants did not meet the minimum score (< 5) to be considered as meeting a clinically significant threshold of depression and thus were considered *not depressed*. Of the remaining six participants, five scored as having *mild depression* and one participant scored as having *moderately severe depression*.

The minimum score on the organizational religiosity subscale is 2 to 12 with the upper limit varying depending on one open-ended numerical response item ("*how many church clubs or organizations do you belong to?*"). Participant organizational religiosity ranged from 7 to 16 with a mean of 11.5 ($SD = 3.26$) indicating a high degree of religious involvement across the study sample. Given that participants were recruited directly from their places of worship this is expected.

Non-organizational religiosity was measured using four Likert-type scale questions ranging from 1 = *never* to 5 = *nearly every day*. Scores on the non-organizational religiosity subscale can range from 5 to 20. In this study, participant non-organizational religiosity ranged from 7 to 20 with a mean of 13.58 ($SD = 4.12$).

The subjective religiosity scale consists of three Likert-type scale questions with response items ranging from 1 = *not important at all* to 5 = *very important*. Scores can range from 3 to 15. For this study, participant subjective religiosity ranged from 3 to 12 with a mean of 9.75 (SD = 2.7).

Table 20: Descriptive Statistics

	n = 12
Age – Mean (SD)	58.2 (18.68)
Race (non-exclusive):	
Black or African American	12 (100%)
Ethnicity (non-exclusive):	
Non-Hispanic	12 (100%)
Education (non-exclusive) (count):	
High School Degree	1 (8.33%)
Some College	4 (33.33%)
Bachelor’s Degree	3 (25%)
Master’s Degree	4 (33.33%)
PHQ-8 (count):	
No depression	6 (50%)
Mild depression	5 (41.66%)
Moderate depression	0 (0%)
Moderately severe depression	1 (8.33%)
Severe depression	0 (0%)
Organizational Religiosity – Mean (SD)	11.5 (3.26)
Non-organizational Religiosity – Mean (SD)	13.58 (4.12)
Subjective Religiosity – Mean (SD)	9.75 (2.7)

8.5.2 System Use

Eleven of the 12 participants completed the intervention curriculum during the two-week study period. Seven of the 11 curriculum completers chose to revisit completed content. As previously mentioned, participants were able to complete their mental wellness sessions with the ECA in any order they desired. As a result, the path that participants took through the system varied considerably (Table 21). Out of six possible core session choices, five participants selected the psychoeducation session first. Though each education session with the agent was designed to be five to twelve minutes in length, the intervention was flexible and allowed participants to leave education sessions early and complete them at a later time. Over the course of the study, the number of participant conversations with the agent ranged from 3 to 29 with a median of 10 and inter-quartile range of 10.25. The time of day that participants used the system varied considerably with system use spanning from 5am to midnight.

Table 21: Path Through the Core Curriculum for all Study Participants

<i>PID</i>	<i>Type</i>	<i>Choice 1</i>	<i>Choice 2</i>	<i>Choice 3</i>	<i>Choice 4</i>	<i>Choice 5</i>	<i>Choice 6</i>
1	CM	Support Others	Screeners	Psychoeducation	Social Stigma	Myths & Facts	NA
2	CM	Psychoeducation	Social Stigma	Screeners	Myths & Facts	Support Others	NA
3	PwD	Psychoeducation	Social Stigma	Screeners	Self-disclosure	Myths & Facts	Care-seeking
4	PwD	Social Stigma	Psychoeducation	Screeners	Myths & Facts	Care-seeking	Self-disclosure
5	CM	Psychoeducation	Social Stigma	Screeners			NA
6	PwD	Myths & Facts	Screeners	Social Stigma	Care-seeking	Psychoeducation	Self-disclosure
7	CM	Screeners	Social Stigma	Myths & Facts	Support Others	Psychoeducation	NA
8	PwD	Self-disclosure	Psychoeducation	Social Stigma	Care-seeking	Screeners	Myths & Facts
9	PwD	Psychoeducation	Social Stigma	Screeners	Support Others	Myths & Facts	Self-disclosure
10	PwD	Myths & Facts	Psychoeducation	Care-seeking	Screeners	Social Stigma	Self-disclosure
11	CM	Psychoeducation	Screeners	Social Stigma	Myths & Facts	Support Others	NA
12	CM	PHQ-8	Social Stigma	Myths & Facts	Psychoeducation	Support Others	NA

* note: Persons with Depression (PwD) had 6 core sessions in their curriculum. Community Members (CMs) had five core sessions. P5 did not complete the curriculum.

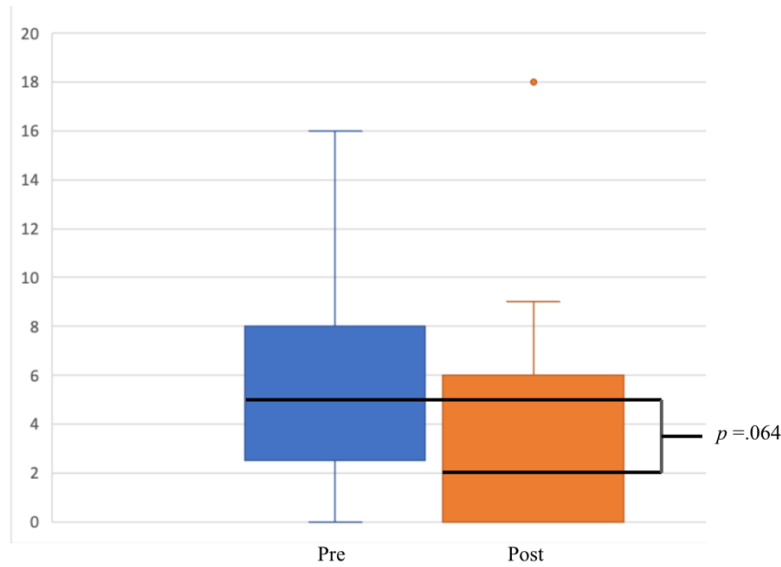
8.5.3 Depression Symptoms

For this dissertation study, the PHQ-8 was administered as a pre/post study measure to examine what impact, if any, the intervention had on users' depressive symptoms. Given the non-normal distribution of both the pre/post measures, a paired samples Wilcoxon signed-rank test was performed to assess statistical significance. Though not statistically significant, the Wilcoxon signed-rank test revealed that depressive symptoms were lower post intervention (Med = 2, IQR = 6, n = 10), $W = 7$, $p = .064$, $r = .585$ compared to before (Med = 5, IQR = 6, n = 10) (Table 22). Next, I conducted a sub analysis of treatment effects between groups. A Mann-Whitney U test showed that PHQ-8 change scores were not significantly different between PwD (Med = -2, IQR = 3.5, n = 5) and CMs (Med = -1, IQR = 4.5, n = 5), $U = 12.5$, $p = 1$, $r = 0$.

Table 22: Differences in Depressive Symptoms, measured before and after the intervention, and the direction of the change.

Measure	Scale	Before Med (IQR)		After Med (IQR)	Statistical Test
PHQ-8	0-4	5 (6)	>	2 (6)	$W=7$, $p=.064$, $r=.585$

Figure 25: PHQ-8 scores were higher before the intervention compared to after the intervention.



8.5.4 Formal Help-seeking Intentions and Attitudes towards Formal Treatment Providers

The measure distribution of the MHSIS at both time points was non-normal. A Wilcoxon signed-rank test was conducted to assess statistical significance. Help-seeking intention scores before the intervention (Med =18.5, IQR = 10, n = 11) compared to after the intervention (Med =18, IQR = 6, n = 11), were not significantly different $W = 15$, $p = .344$, $r = .285$ (Table 23). A Mann-Whitney U test revealed that changes in PwD (Med = 0, IQR = 11.5, n = 5) help-seeking scores compared to CMs (Med = 0, IQR = 4.5, n = 6) were not significantly different between groups $U = 12.5$, $p = .662$, $r = .167$.

Table 23: Differences in Help-seeking Intentions, measured before and after the intervention, and the direction of the change.

Measure	Scale	Before Med (IQR)		After Med (IQR)	Statistical Test
MHSIS	1-7	18.5 (10)	>	18 (6)	$W=15$, $p=.344$, $r=.285$

The distribution of pre/post IASMHS scores was non-normal, therefore a paired samples Wilcoxon signed-rank test was chosen to assess statistical significance. The test showed that there was not a significant difference ($W = 44$, $p = .092$, $r = .533$). The median score pre-study intervention was Med = 75, IQR = 14, n = 10 compared to Med = 81.5, IQR = 14, n = 10 post-study intervention (Table 24). A Mann-Whitney U test demonstrated no significant difference in IASMHS change scores ($U = 10$, $p = .69$, $r = .2$) between the PwD group (Med = 4, IQR = 2.5, n = 5) compared to CMs (Med = 10, IQR = 24, n = 5).

Table 24: Differences in Attitudes Toward Seeking Mental Health Services, measured before and after the intervention, and the direction of the change.

Measure	Scale	Before Med (IQR)		After Med (IQR)	Statistical Test
IASMHS	0-4	75 (14)	<	81.5 (14)	$W=44$, $p=.092$, $r=.533$

8.5.5 Confidence

As with previous measures, the measure distribution of both pre and post measures were non-normal. A Wilcoxon signed-rank test was not significant $W = 35.5$, $p = .122$, $r = .467$ when comparing post-intervention scores (Med = 14, IQR = 4, $n = 11$) compared to pre-intervention scores (Med = 12.5, IQR = 3, $n = 11$) (Table 25). A Mann-Whitney U test demonstrated no significant difference in confidence change scores ($U = 13$, $p = .792$, $r = .133$) between the PwD group (Med = 1, IQR = 5, $n = 5$) and CMs (Med = 1, IQR = 4.5, $n = 6$).

Table 25: Differences in Confidence in Discussing Mental Health Topics with Church Members, measured before and after the intervention, and the direction of the change.

Measure	Scale	Before Med (IQR)		After Med (IQR)	Statistical Test
Confidence	0-4	12.5 (3)	<	14 (4)	$W=35.5$, $p=.122$, $r=.467$

8.5.6 Depression Stigma

A Wilcoxon signed-rank test was chosen due to the non-normal distribution of both DSS pre/post measures. The test demonstrated a non-significant finding ($W = 13$, $p = .483$, $r = .222$) when comparing participant scores before the intervention (Med = 27.5, IQR = 18, $n = 10$) to after (Med = 24.5, IQR = 17, $n = 10$) (Table 26). In a between-group analysis, A Mann-Whitney U test showed that changes in depression stigma scores were not significantly different ($U = 19$, $p = .222$, $r = -.52$) between PwD (Med = 0, IQR = 6.5, $n = 5$) and CMs (Med = -5, IQR = 29, $n = 5$).

Table 26: Differences in Depression Stigma, measured before and after the intervention, and the direction of the change.

Measure	Scale	Before Med (IQR)		After Med (IQR)	Statistical Test
DSS	0-4	27.5 (18)	>	24.5 (17)	$W=13$, $p=.483$, $r=.222$

8.5.7 Agent and System Usability

Non-parametric inferential statistics were used to determine statistical significance on each of the ordinal items that comprise the 11-item usability measure. The median and inter-quartile range for each item is reported and a comparison of those ratings with a neutral score of 4 is displayed in Table 27.

Table 27: Participant average ratings of single-scale items for Agent and System Usability and results of a one-sample Wilcoxon signed-rank test.

Item	Anchor 1	Anchor 2	Scale	Med (IQR)		Neutral Score	Comparison to Neutral
How satisfied are you with the overall app?	Not at all	Very much	1-7	6 (2)	>	4	W=45, p=.007, r=.854
How much would you like to continue using the app?	Not at all	Very much	1-7	5 (3)	>	4	W=21, p=.026, r=.706
How likely are you to recommend the app to other members in your church?	Not at all likely	Very likely	1-7	6 (2)	>	4	W=36, p=.011, r=.809
How easy was it to use the app?	Very difficult	Very easy	1-7	7 (0)	>	4	W=55, p=.002, r=.999
How satisfied are you with the animated character?	Not at all satisfied	Very satisfied	1-7	5.5 (3)	>	4	W=21, p=.026, r=.706
How much would you like to continue working with the animated character?	Not at all	Very much	1-7	5 (2)	>	4	W=41, p=.020, r=.736
How much do you trust the animated character?	Not at all	Very much	1-7	6.5 (3)	>	4	W=28, p=.014, r=.778
How much do you like the animated character?	Not at all	Very much	1-7	6.5 (2)	>	4	W=50, p=.017, r=.753
How would you characterize your relationship with the animated character?	Complete stranger	Close friend	1-7	5 (4)	>	4	W=25, n.s., r=.246
How much do you feel the character cares about you?	Not at all	Very much	1-7	5 (3)	>	4	W=30, n.s., r=.028
How much do you feel that you and the character understand each other?	Not at all	Very much	1-7	5 (3)	>	4	W=29, n.s., r=.492

All usability ratings for the overall application (items 1-4) and agent (items 5-8) were rated significantly higher than neutral (Table 27). While the median score for *how much the character cares about you* 5(3) was greater than a neutral rating (4) the comparison was non-significant ($W = 29, p > .05, n = 10$). Participants median ratings for *how much do you feel you and the character understand each other* and *how would you characterize your relationship* were 5(3) and 5(4) respectively. Scores were not significantly higher than neutral.

8.6 Qualitative Findings

All participants completed pre and post intervention interviews. Pre interviews and post interviews totaled 774 minutes (13 hours) of recorded data. Interviews were transcribed by a professional transcription service. Transcripts and participant journal entries were analyzed using rapid qualitative analysis an approach that utilizes matrix summaries to condense interview data (Gale et al., 2019). This method instructs researchers to leverage templated summaries as a way to persevere vertical data segments (individual interviews) while also encouraging researchers to interpret patterns cutting across data segments using an analytic tool called display matrices. This approach is useful when evaluating innovative health initiatives where the research questions and purpose for the data collection are well-defined. Given the research questions for this summative evaluation were well-defined, I decided to leverage this method to guide my analysis.

As part of the interviews, participants were asked their overall impressions of the ECA, the ECA's purpose and goal, and for their reactions to the mental health promotion and stigma-reduction curriculum. The last half of the interviews explored

participants' thoughts on how the application could be used to spark community discussion about mental wellness and assist church members experiencing depressive symptoms, if at all. Four themes were identified. Participant quotations will be annotated with the participant's ID number and their user type: Person with depression (PwD) or Community member (CM).

8.6.1 Theme 1: Clara as a Partner

Participants described Clara as a partner, meaning that participants recognized Clara as a peer who could offer informational and emotional support while simultaneously respecting participant beliefs and autonomy. This was understood through participants' perspectives on Clara's purpose and value—including her ease of use, personalization, and normalization of discussing mental wellness with others.

Clara's Purpose

The ECA, Clara, was described by participants as a “partner,” “peer,” “guide,” “church lady,” and “calming spirit” who was interested in “the cause”—supporting member mental health in their predominantly Black church communities. When asked to describe Clara's purpose further, participants stated that she was designed “as an entry point” for mental health promotion and awareness. Participants went on to note how the application provided a thorough starting place for members to “dip their toe in” to learn about depression while also having “immediate” access to resources. P2(CM) described Clara's role as “Clara is a persona and a mental wellness app that walks you through mental well-being topics, and that by going through them, you can learn about mental health and how to get support and information about it and learning about the stigma and also getting a lot of resources.”

The Value of an ECA when Designing Digital Mental Health Promotion Interventions

Participants identified several affordances of the ECA that complimented the mental health promotion and stigma reduction intervention. The ECA 1) promoted ease of use, 2) created a more “*personalized*,” “*personable*,” and culturally-tailored user experience, and 3) normalized the practice of discussing mental health with others.

The System is Easy to Use

Leveraging an ECA or conversational assistant as the main interaction mechanism for guiding participants through the curriculum was appreciated by all participants, with the exception of one participant who described the character's embodiment as “*off-putting*.” Eleven of 12 participants explained that the character made the application “*easy to use*” and intuitive. While nearly all participants had recommendations for improving the ECA's voice (e.g., creating a less robotic voice), non-verbal behavior (e.g., creating more fluid movements), or gender presentation (e.g., creating a male-presenting agent for men, or an agent that appears more feminine), all participants stated that the information was pleasant and well-presented. Some participants compared the app to other smartphone apps that they had tried. P2(CM): “*I don't get lost in it. It's very easy to talk to her. Other*

stuff, I stopped using it. This [Clara] is clear.” For these participants, unlike other smartphone apps, Clara reduced their technology hesitancy because of how intuitively she presented the content through conversation.

Clara is Personalized and Personable

Participants compared Clara to different modalities like a video of a *real person*, a disembodied voice assistant, or a text-based chatbot. While several of the participants reported that a video featuring a person from *their community* may be more relatable and authentic, they noted the caveat that the intervention may, as a result, be less interactive and responsive. Utilizing the embodied character and user-menu options helped users feel actively engaged instead of passively consuming information from a webpage or educational video. P9(PwD): *“She gave me choices. I could say I wasn’t ready to talk about certain things or that I was ready to hear a suggestion.”* Participants appreciated and understood that their user-menu selections steered the conversation between themselves and Clara.

When asked how an embodied character benefited the intervention, P10(PwD) stated, *“Clara’s face makes her more personable. It doesn’t just feel like AI.”* Unlike P10, four participants commented that, at first, Clara’s physical embodiment distracted them from the content she presented. P6 laughed as she shared, *“I mean at first. It’s a robot. It’s AI. So, I have to just live with that. But I was able to get into it...I’d be like oh yeah, this is really great, Clara, yay.”* Most participants who aired similar concerns stated, like P6, that after a few interactions, they *“got into it”* and *“enjoyed”* talking with Clara daily.

Clara is Culturally Tailored and Credible.

Participants were asked in the interview and diary how they would characterize their relationship with Clara. Almost all participants explained that while they do not necessarily feel *“close”* to *“an avatar,”* they did believe she was a trustworthy information source culturally tailored for their community. P5(CM) commented on Clara as a peer, *“I like that Clara reads the Bible and shares the verse.”* P2(CM) was impressed with the variety of resources Clara offered and her use of Scripture, which seemed to add to the character’s credibility. While most participants appreciated the intervention’s incorporating spiritual and religious content, one disagreed. P12(CM) *“She just seemed too churchy. She could have had less of that.”* P12(CM)’s perspective, though in the minority, provides a valid perspective—spiritual tailoring should not necessarily be the same for all participants.

Beyond spiritual tailoring, participants appreciated the culturally relevant content in the psychoeducation; P8 said, *“It was nice seeing that depression rates aren’t actually higher necessarily for Black people compared to others.”* P12(CM) also noted the cultural tailoring. She stated, *“Because I feel like oftentimes there’s a lot [of apps] put out in terms of catering to mainstream....And so for the minority...the fact that that was taken into consideration and it was like, okay, some of these things are applicable to the mainstream, but also it she [the app and Clara] is really tailored for somebody who has a lived experience [Black experience].”* The tailored content and agent persona signaled that the intervention was created with the predominantly African-American communities in mind.

A Non-threatening Tool for Discussing Sensitive Topics

Several participants described The ECA in the interviews and journals as *non-threatening* and *non-judgmental*, two essential qualities for a digital mental health promotion intervention prioritizing communities where the prevalence of stigma is high. Participants further explained that individuals interested in learning about the topic may not otherwise talk to another person about it and, as a result, lose the opportunity to obtain health-enhancing resources and services. Participants felt strongly that using the embodied character assisted in “*getting people closer to a human connection.*” P6(PwD) echoed this finding, “*People who are dealing with a mental illness tend not to want to talk with other people because it causes them to have to be confronted. [Clara] can help you realize that “I’m not alone. There are other people in the world.” ... It [Clara] gets them closer to a human connection. And I also think people find it probably much more evocative. And what I mean by that is that they have more feelings when it’s a person’s face than when they are reading [information]. It makes it more connecting.*” Several participants confirmed this finding, stating that simulating conversation about mental wellness using Clara helped them imagine discussing the content with others. One participant took this a step further, saying that she hoped future iterations would include a role-playing activity with Clara so that she could practice skills such as approaching another member of her community whom she believes may be experiencing mental illness. Another member stated they would have enjoyed practicing questions they should ask a prospective therapist to ensure that their needs are being met and that they aren’t “*being offensive.*” These findings demonstrate how an ECA simulating human-to-human conversation can help reduce participant anxiety surrounding conversations about sensitive health topics that church members identify as potentially uncomfortable or awkward.

P6 encapsulated the intervention benefits of the ECA-facilitated intervention when she said, “*Clara is an automated mental health professional who holds space for you to learn and to be heard and to be supported, and who helps you get used to the idea of what a real mental health support system might look like.*” Her words described the ECA as a supportive mental health professional, responsive to her needs, who provided aspirational examples of mental health support.

This theme included participant descriptions of the ECA, her intended purpose and characterized what perceived benefits an ECA provides when delivering a digital mental health promotion intervention to Black and African American church members. The next theme will demonstrate how the intervention shifted participant perspectives of mental illness, primed treatment intentions, and led to the practice of supporting others during the two-week study period.

8.6.2 Theme 2: Starting to Consider Stigma

This theme characterizes participant shifts in *attitudes* towards mental illness during the study period. These shifts notably resulted in *actions* that participants took for themselves and others, namely: sharing mental wellness resources, discussing mental wellness, and care-seeking. This theme thus covers two subthemes: *attitudes toward mental illness* and *actions for the self and others*.

Attitudes Towards Mental Illness

Through both the interviews and journals, participants described how the application prompted them to *start to consider stigma*. These shifts in attitudes occurred for nine of the 12 participants. These shifts included: 1. changes in how they plan to respond to members with mental illness (i.e., being a good listener, paying more attention to others, and refraining from using stigmatizing language); 2. changes to their sense of confidence when it comes to helping others, and 3. a higher degree of self-awareness when it comes to examining their own experiences of depressive symptoms.

Responding to Members with Depression

Church members discussed how the ECA-facilitated intervention assisted them in building empathy towards individuals experiencing depression by having them consider stigma explicitly. One member explored how the intervention made her question her beliefs: *“Wait, am I no better than the rest of the population? Have I been doing this [stigmatizing people]?”* P4 described how the intervention helped her realize that she may have been avoiding another member, partly because of the member’s mental health challenges. Before the study, P4 stated that she had convinced herself that she had not checked in on her friend because she didn’t want to make her friend uncomfortable. After working with Clara, P4 stated, *“There’s the stigma right there because sometimes you want to call, but then you’re like, “Oh God, should I call?” Because everything is a rough time and something you don’t feel like dealing with. And that’s another stigma...Clara just really set the fire under my feet.”* In stigma theory and reduction literature, social distance is defined as the act of avoiding others who have a mental illness. In this case, the intervention has helped P4 reconsider her avoidance of another member experiencing mental illness.

The application helped users challenge their assumptions about people with mental illnesses. For example, P4(PwD) stated she *“assumed things about people. I’m grateful for Clara because she’s made me more compassionate.”* Other participants reported how the application increased their sensitivity towards others. P5(CM) stated, *“I realize now that I just haven’t paid much attention to what others are going through...I haven’t thought about stigma.”* When asked how P5 intends to use the information she’s learned, she responded, *“I’m going to really listen.”* Listening was one of the most commonly referred to skills that participants repeated during the interviews. P2(CM), P11(CM), and P5(CM) all stated that they plan to be better listeners. P11(CM) shared her point of view: *“yeah, because sometimes we think we are listening. We are hearing them talk, but we aren’t listening to what they are saying.”* P2(CM) explained, *“In the past, I’ve been quick to talk. I’m a talker. Now I realize I need to listen.”* Another participant mentioned using humor (e.g., *“we’re all crazy.”*) in the past as a response to others disclosing information regarding their mental health, believing it would make the other person feel more comfortable. Instead, she realized that even a well-intentioned strategy to mitigate social discomfort may have been dismissive.

Participants also shared that apart from impacting how well they listen to others; the application has also made them reconsider how they talk around others. P11(CM) stated that the experience of using the system has made her rethink the *“callous language”* she may have used in the past. P11 noted, *“Think that, say for instance when you were in a group and people started*

listening to people talk and some of the conversations are really, really off the wall. And you may say that person is delusional or that person...is crazy. We sometimes do that. And it is very harmful, but at that time, we don't think it's very harmful." Apart from using this language to characterize the actions of others, seven participants credited the application with encouraging them to reconsider how they use language such as, *"that's crazy" "she's nuts" "that's psychotic" "she's touched"* to refer to surprising or adverse events. P1 shared how the application encouraged her to change how she speaks about mental illness. P1 explained, *"One of the things in the app talked about how our language can change. Instead of saying crazy or using those kinds of negative terminology, we can use something different. I said, 'Oh, yeah. I never thought of it that way.' ...I can use it [the app] to change the language in which other people speak as well as myself."* Five CM users and one PwD user all stated that they would reconsider the language they use when discussing mental illness and depressive symptoms when at the church.

Building Confidence

Beyond building empathy for those experiencing depressive symptoms, the application helped participants begin to build confidence in broaching the topic of mental wellness with others. P8 stated, *"You know it gives you facts. And the more facts you have, the more confident you feel when responding to other people in conversation. Sometimes you know something doesn't sound right, but if you don't know the fact, you can't speak about it with authority."* P9 shared a similar point of view.

P9: *"Now, when someone asks for help or needs something, I can share resources or say more than 'you should see a therapist.'"* Following the intervention, P12 reported a boost in confidence in responding to members with mental illness, stating that she felt less prepared to discuss mental illness before the intervention. She said, *"There's awkwardness. I mean, if somebody physically sick, it's okay. You're like, oh, how are you doing? But mentally, if mental health is an issue for the individual, it's a little more challenging; how do you approach them? Clara gave two simple questions you can ask...So I felt like there was value in that piece for me through the app."* This quote demonstrates how the system assisted members in identifying strategies for approaching and responding to members experiencing depression.

When starting the study, I asked all participants what interested them in this work. P1, assigned as a CM and with a history of clinical depression, shared that she was interested in seeing if the application could help the youth in her church struggling with mental illness post-pandemic. During the study period, the youth ministry leader approached P1, asking if she would like to be involved. P1 declined, stating, *"They [the young adults in youth ministry] want young people. They don't want old cronies like myself."* The self-stigma education linked to NAMI's stigma-free campaign, where an NFL football player connects with teenagers to reduce mental illness stigma and increase care-seeking among teens. Viewing the video was meaningful to P1, she explained, *"I felt I didn't have anything that I could bring to the table. But after seeing that video, I felt different."* For P1, the video changed her self-assessment of her capabilities.

Facilitating Self-Awareness

Participants described instances of self-discovery and greater self-awareness while using the application. One CM told how using the application made her much more aware of her symptoms, particularly her sleep and appetite difficulties. P2, in recent months, had lost her spouse and reported having not properly grieved his absence from her life. P2(CM) stated, *“So I check myself in listening to Clara and to say, “Is that me?” And I know one thing that can impact me is my being too tired. The app puts that on your mind. The app could help me stay ahead of it.”* P7(CM) echoed this finding when she reported that she wondered about some symptoms she endorsed during the PHQ-8 session with the ECA once she graduated from the curriculum. Though this participant initially rejected identifying as depressed and believed that the symptoms she endorsed were not an issue, she later wondered if she, too, could benefit from more support. She questioned if the PwD content could help her, especially the symptom check-in. Both participants stated that though they believed their system-assigned user type was accurate, they had hoped to have access to the entire curriculum regardless of user type by the end.

Three of the six participants assigned to the PwD user type reported never engaging in formal mental health treatment. Two of the six users were actively involved in therapy or psychiatry, and one participant had a treatment history. Of the three participants who have never engaged in formal care, two reported shifts in their self-assessments of their mental wellness. P9 stated, *“Clara helped open my eyes... How did you feel half the time? All of the days or most of the days? [referring to the mental health screener Clara administers] When I thought about those questions, I thought about how many days when I was feeling down, what did I get to do? Did I get to go out or do something all of the days, some of the days, most of the days, or none of the days? I really got a chance to focus on my own mental health...Now I realize depression is a mental illness.”* When asked if this type of self-discovery was helpful or unhelpful, P4 explained that, at first, it was difficult to digest. However, it also assisted her in realizing that both memory and sleep could be related to an untreated mental illness. It propelled her to consider discussing these symptoms with a doctor. In addition, the intervention reframed how she thinks about herself when she’s depressed by normalizing depression, *“I shouldn’t be sad that “I’m depressed.” Now I’m okay with saying, “You know what? I’m going through about of depression right now. I’m going to make it through, but right now, this is what I need.”* P9 shared a similar point regarding the discomfort she experienced while using the intervention during her interview, *“even though being honest with Clara [about mental health] was uncomfortable at times, I was ready for it.”* And though P9 stated that she’s not ready to commit to the label of *depression*, she is ready to address some of the symptoms that *“Clara had her take a look at.”*

Actions for the Self and Others

In addition to shifts in attitudes, participants reported taking new actions they had not considered before the intervention. Participants described having done the following during the two-week intervention: 1. sharing mental health content and resources with others, 2. discussing mental wellness with others, and 3. facilitating care-seeking for themselves. Several users shared application content directly with family members and friends. Two participants shared the application with their spouses

and explored the content together. Two additional participants reported discussing the application with their primary care physicians to facilitate formal mental health treatment.

Sharing Content

Participants described how they shared resources and application content with others during the study period. P2(CM) was supporting an out-of-state family friend whose mother passed away last month. The family friend was a middle-aged Black man who had been the primary caretaker of his aging-in-place mother. P2(CM) explained that she worried he was isolating himself and having difficulty transitioning from a full-time caretaker to living independently. P2(CM) texted two resources directly from the application to him. Though specific resources were tailored for the state the intervention was deployed, P2(CM) remarked that Clara gave her enough information to realize that some of the resources were national, prompting her to find the state equivalent resource for the family friend and send it to him. Several members shared stories of sharing the application with others during the study period. P1 shared the application with a member who has not attended church lately due to a mental health crisis. P6 and P9 shared content with their adult sons, and P6 asked during the follow-up interview how she could share the app directly with her son to “*start a conversation.*”

Discussing mental wellness with others

Participants described how using the application led to discussions with others regarding mental illness. As previously mentioned, P1 experienced a boost in confidence after exploring content from the social stigma session of the app. P1(CM) used this confidence to volunteer for her youth ministry, a sub-community of her church that she cares deeply about. She planned on sharing her experiences with mental illness directly with ministry attendees. P4(PwD) shared the application with her husband during the two-week study. Before starting the study, P4 explained that her husband could “use help in this area,” but he wasn’t ready to participate. During the study period, she described several instances of using the app to talk with Clara “together.” She shared, “*But when she [CLARA] was talking, and there were many links that we could go to, I would ask him, “Do you want to hear this?...that was really good that I was able to listen and write my little thoughts down. He was able to also listen.”*” P3(PwD) and P4(PwD) revealed that the application primed them to discuss mental wellness openly during Sunday dinner with their immediate family members, “*But the fact that we were having a little family gathering and it wasn’t about mental health. For her to pull out... mentioning Clara...[redacted: P3’s spouse name] wants to talk to Clara. [redacted: P4’s spouse name] already hung out with Clara.*” When asked if P3 or P4 had ever had these discussions during mixed non-private family gatherings, they said they had not.

Preparing for Care-Seeking

Participants shared stories of care-seeking and self-disclosure during the interviews. Nearly all of these narratives were shared by participants assigned to the PwD user type except one. At the time of the interview, several participants shared new research-based activities (e.g., researching providers, asking trusted friends or family for recommendations, etc.) they were currently

engaged in that were helping them identify mental health providers. P8(PwD) explained that she had been in treatment. However, she stopped attending therapy due to a lack of progress with her therapist. Since using the application, P8(PwD) stated that she is recommitted to finding a provider, crediting Clara's advice and reminders. P8 shared, *"Although depression is tough, you do have to work to get yourself out of it, and that there are so many options. I like that it gives you a call to action, and it forces you, well, it doesn't force you, but it kind of provokes you to do more for yourself."* P8 used various resources recommended by Clara for researching therapists and support groups both online and in the area. P10(PwD) reported a similar objective, stating that she has been contemplating if and how she might benefit from speaking with a counselor since having a child. P10(PwD) also leveraged app-recommended resources for finding a culturally responsive therapist and reviewed questions she may want to ask a prospective clinician to ascertain if they match her needs.

Care-Seeking

Beyond facilitating preparation for care-seeking, participants reported active steps they took toward treatment and increasing their support network. Three PwD participants reported sharing the application with their treating primary care providers during the study period. Two participants attended appointments during the study period where they leveraged content Clara had shared in the application to address a medication change (P3) and ask to be referred to a counselor (P9). One participant described an emotionally charged experience where she asked her PCP for a memory exam due to concerns about her memory. P4(PwD) *"You have professional help, but they're NOT hearing you. I shared with her [about my forgetfulness] with my PCP, and she said, 'Your labs look good. I'm not worried about your mind or your forgetfulness.' And I said, you are not, BUT I AM."*

P4's story emphasizes a new sense of confidence and empowerment that pushed her to advocate for the treatment she had been interested in, even though her provider had initially dismissed her concerns. P4 shared how using the app impacted her life, *"Honestly, I don't want to cry, but I'm really grateful to the app. It may seem like a small thing for somebody else. It may be a small thing...But I definitely am grateful."*

Participants who received the PwD content were encouraged to self-disclose their experiences to professionals as a mechanism for facilitating treatment. The CM content focused on mental health awareness and strategies for helping others who may be experiencing depressive symptoms and are looking for support. Near the end of the interview, P2(CM) described how the application prompted her to reflect on how she may benefit from additional support advice that has led to her decision to disclose her challenges to both her adult daughter and a trusted church member and friend.

8.6.3 Theme 3: Intervention Approaches for Engaging Church Members with Depression (meeting them where they are)

In this theme, participants characterized how the current application assisted users with depression and additional features and functionality that they believed would be helpful.

Managing Mental Health Using Self-evaluation Activities

For the PwD user type, the current intervention had a largely health education orientation coupled with a counseling component designed to encourage users to consider how working with a provider or reaching out to other supportive relationships may benefit them. During the interview, I asked participants to brainstorm recommendations for how else they would like a digital mental health promotion intervention with Clara to support them. Participants expressed various ideas; however, most focused on managing symptoms through self-re-evaluation activities.

In the current system, Clara screens participants using the PHQ-8. Clara takes any symptoms that participants endorse, ranks them by severity, and then intervenes on each symptom once per day at the start of the next session; I refer to this as Clara's symptom check-in. This check-in asked participants if they were still having trouble with the symptom and then offered a suggestion if the symptom was still present. Participants appreciated the symptom check-in but indicated that if using the application for a more extended period, they would like to be able to rescreen. Three participants imagined a daily wellness check with Clara. During which Clara would offer suggestions and a brief intervention (e.g., five-minute breathing exercise, suggest a walk, Scripture, reminders for self-care activities, etc.) tailored to their responses. One participant took this a step further and suggested therapeutic journaling with Clara, *"But if Clara could be something that creates some reflection time, in other words, and every so often outside of teaching me or talking to me about stuff, give me a chance to write about something that I'm dealing with or record my thoughts about something that I'm dealing with that could help."* P6 explains how journaling with Clara could offer time for self-reflection and self-reevaluation.

Introducing Self-Disclosure and Encouraging Supportive Helping Relationships

The self-disclosure planning session was intentionally designed to take a neutral stance on self-disclosure, neither encouraging nor discouraging the practice, but instead, educating users on how to plan a successful self-disclosure using concepts from the Disclosure Process Model (Chaudoir & Fisher, 2010). Reactions to the session varied slightly. All participants welcomed the session, and none believed it should be removed from the intervention's curriculum. However, participant reactions to the session's content and neutral stance varied. On one end of the spectrum, participants endorsed aspects of the self-disclosure session that promoted autonomy and confidence. Indeed, P4 credited the self-disclosure session with giving her the confidence to approach her PCP and discuss her symptoms.

P8 appreciated the messaging around carefully selecting the audience for a disclosure. P8 stated, *"So, it's nice that she's not like, 'Yeah, just go for it.' You never know. I have friends that I've told them, and it's just it turns around and bites you in the ass."* Similar to P8, P3 appreciated the agent's explicit care to encourage users to make a plan, *"Yeah. I thought it was a good reminder about how to go about talking to someone and then also emphasizing that it's good to be in as good of a place you can be to share that so that it's not a more stressful situation. I thought that was good advice. And yeah, I just thought it [self-disclosure] was good for people to think about. Yeah, I thought that was just good."*

On the other end of the spectrum, some participants believed that Clara's stance should take a more prescriptive approach by adapting to how impacted the user is by concealing their mental health status. This participant explained what she believed Clara's stance should be, *"if something is being really negatively impacted, maybe it would be helpful for you, or it could be beneficial for you to tell someone so that because really with depression and stuff, you can really lose friends and loved ones and jobs and stuff if you're not communicating stuff when you should be, even if you're not at a good place."* She believed Clara's stance towards disclosure should shift if an individual's mental health results in resource-reducing challenges like losing friends or a job. This same participant hoped the intervention would also provide a more thorough understanding of one's rights post-disclosure (e.g., accommodations and advice about protections when disclosing mental illness in the workplace).

Creating Flexible Intervention Entry-points (a way in)

To ensure that participants had enough time over the two weeks to finish the curriculum, they were told to prioritize the mental wellness content over other topics presented within the application (e.g., meditation, nutrition, healthy relationships, meditation). However, a quarter of the study participants *did* investigate other intervention topics within the application. In interviews, participants proposed more finely integrating these wellness interventions as a mechanism for providing users who otherwise do not believe that the mental wellness intervention applies to them with alternative interventions focused on bodily or spiritual health. Participants viewed these pathways connecting the interventions as a *"way in"* to the mental health content and a supplement to formal treatment options.

P6 exclaimed, *"We have to think about when you are going to put together any kind of support about mental health; you've got to be holistic about it. This is not just therapy. Because not everyone is going to feel that that's right for them. There are places around nutrition and mental health as a partner of the app. And maybe even a module on the Bible and mental health...there is certainly exercise...there's meditation...And so maybe even saying at the beginning, here are some other things [approaches other than 1-1 therapy] that you can do research on to look at to see what's right for you. This list of things that they could do [with Clara]."*

P6 and others believed that by approaching mental wellness through multiple facets of wellness (e.g., interpersonal relationships, bodily well-being, or spiritual well-being), users can take incremental steps towards mental wellness in immediate and realistic ways that make them most comfortable. P6 believed such an approach would lead to more sustainable intervention engagement by church members.

Participants provided feedback on the intervention's framing explicitly stated by the ECA in the introduction, highlighting it as another approach for creating an *entry point* for hesitant users who may be unaware or in denial of their mental health challenges. Early in the introduction, and later referred to in subsequent modules, the ECA states that the intervention has been designed to help create a more loving and supportive community and consistently emphasizes the importance of building a culture of support within the church walls for members experiencing mental illness. Two CMs who elected to use the intervention

to support others in their church learned skills when using the system that they then leveraged for themselves. For instance, P2 used the screener to self-reflect on her difficulties with sleeping and changed her sleep hygiene by turning off the television at night.

Before the last interview, participants were unaware that the intervention was tailored by user type, nor did they realize they had slightly different curriculums. Once asked how they felt about the other user types, participants all believed this level of tailoring was valuable as it made the intervention feel responsive to their goals for use early on. However, CMs and PwD wanted to access the companion user type's content post-intervention. P10(PwD) stated, *"I wish there was...you could have access to both because I feel like, as a church member, you could be both. You could be a person experiencing a mental illness, and then you could be someone who supports someone experiencing a mental illness. Both things could be true."* P10's point was shared by multiple participants who believed that while the tailoring made the intervention feel relevant from the start, as they progressed, they wondered if other material from the companion curriculum could have benefited them.

8.6.4 Theme 4: "Feeding the flock," Community Approaches for Mental Health Promotion in the Church.

The last third of the interview prompted participants for their thoughts on how the application, if deployed widely, could promote mental health and decrease stigma in their churches. All participants agreed that for the application to be more broadly adopted and catalyze a culture of open discussion surrounding mental wellness, the application would have to be introduced by way of a church-sponsored event and backed by church leadership. P3(PwD) put it aptly when she stated, *"Leadership should always be paving the way."* Throughout the interviews, participants described leadership's vital role when starting any new initiative in the church. Participants explained that leadership actively promoting the application accomplishes two goals. First, it gives the app credibility *and* demonstrates to the broader church membership the importance of supporting mental wellness. Participants added that leadership buy-in and in-person programming (e.g., church-based small groups or special events) would be the most successful method for attracting members to the application.

A Holistic Approach to Ministry

One participant described how her church provides spiritual ministry to its members but is reluctant to minister to other facets of their members' lives. In her opinion, this reluctance diminishes the church's ability to attract new members and maintain its current level of membership and engagement. P7 explained that using the app and integrating its mental health content into sermons and church-based activities could help *"shepherd your flock, and not just spiritually but...holistically...this is how I see this. I see this as outreach holistically...as feeding the flock."* In her eyes, the application could be used to move her church into a more *"contemporary way of ministering"* and, as a result, assist in maintaining the current membership and reaching new members.

Imagined Hybrid Mental Health Promotion Approaches

Participants discussed various ways the app could be used in concert with church-sponsored activities and events. I identified two broad approaches described across interviews: 1) *embedding the application in the church* and 2) *reflecting community in the application*. Table 28 lists participant ideas for how to introduce the application to their church and ideas for synergistic in-person community events. Participants believed that these brainstormed events would motivate and sustain application use and augment the effectiveness of the intervention.

Table 28: Hybrid Approaches to Community-based Mental Health Promotion

	<i>Place-based Programming</i>	<i>Description</i>	<i>Hybrid Approach</i>
<i>Approaches</i>	Invited Speaker Series	The Church sponsors an in-person expert speaker or clinician focused on raising mental health awareness.	The Kick-off event introduces the application. If the event is a series and continues, the app content is used in tandem.
	Panel Discussion	The Church hosts a panel discussion with volunteer church members who have experienced mental illness.	The contact-based in-person discussion compliments and responds to the social stigma and psychoeducation sessions.
	Small Discussion Group	Leadership advertises a small interest group centered on supporting members with mental illness. Interested members join.	Small discussion groups are used to kick-off app use. Members use the app and come together to discuss Clara's sessions.
	Sermons	The pastor and/or co-pastor(s) speak on the topic of mental wellness and stigma during multiple sermons.	Leadership introduces the application and its content to the broader church membership. Leadership has to demonstrate application use.
	Mental Wellness Committee	A mental wellness committee is instantiated. This committee focuses on gathering community-based resources for members and functions as a touchpoint for concerned members.	The app encourages users to approach the mental wellness committee as another avenue for obtaining informational resources and communal support.
	Resource Pamphlets	Tangible pamphlets that match the resources Clara discusses in-app are available in the Church.	The pamphlets represent a way to bring informational aspects of the application into the walls of the church. Pamphlets can also assist in advertising the app.
<i>Advertising</i>	Church Bulletin	The church bulletin is used to advertise any of the above events.	The app could be advertised within the bulletin.
<i>Timing</i>	Mental Health Awareness Month	The Church launches events during this month.	Church uses the month as a way to broach the topic and introduce the app to their wider membership.

I asked participants, if such in-person or synchronous remote events existed, would there be a need for the application? All participants stated emphatically that the ECA-facilitated mental health promotion intervention would be critical. First, from the participant's perspectives, users believed that the app provides access to content that, even with in-person programming, may still

be considered stigmatizing. P5(CM) highlighted this point when she said, *“Oh yeah, we need the app because this would be a gateway. A gateway to the stigma education to all this stuff. It is important. And even to the person that has mental illness. They might not want to become full; they feel there's a stigma on, I don't want people in church to know, and stuff like that. And to let them know that your feelings are real, that you feel like this, and that shouldn't stop you from getting help.”*

Thus, the app can be used privately to help church members who otherwise would avoid in-person programming receive informational resources. Users highlighted additional affordances of the app: the app is 1) self-paced and thus adapts to the user's time constraints and 2) provides a *“bird's eye view”* of the depression and mental health stigma content. An educational overview ensures that church members are *“on the same page”* by establishing an educational foundation. The in-person events, panels, or small groups allow members to share their experiences of mental illness and their reactions to the content as they use the application. In addition, participants believed that a small group or a group facilitated by an expert could provide a certain level of community *personalization*. Such personalization could go beyond the intervention's pre-scripted content to more robustly address the group's interests and respond to current events promptly.

For Church A, brainstormed in-person opportunities fulfill an additional need—assistance with the technology itself. In my previous work from 2019, we found that Church A used and leveraged less personal use technology (O'Leary et al., 2021). Although these interviews now took place years later, post-pandemic participants reported that technology use amongst older members has not changed. Participants from Church A advocated for other members they believed would find using their smartphone to access the application challenging and anxiety-producing. Participants stated that some members might feel embarrassed, hesitant, anxious, or ill-equipped when it comes to downloading the app and *“getting started.”* Thus, in-person initiatives could serve these members by assigning others as *“tech helpers”* to assist.

Reflecting Community in the Application

My prior work indicated that the value of fellowship is intrinsic to both Church A and B (O'Leary et al., 2022). Several participants communicated the importance of incorporating community into the intervention. P3 emphasized the value of community when talking about mental health promotion, *“at the end of the day, everything we're doing, everything is about society. So, it doesn't make sense that we would be doing this work siloed by ourselves...and community is really important for healing and for growth.”* For most participants, the desire for the application to incorporate community was crucial in establishing intervention relevancy and creating a more effective intervention. For example, P1(CM) described how an in-person event could generate personalized content for her church specifically. P1(CM) imagined a church-sponsored event panel inviting members to *tell their stories* about mental wellness, stigma, and treatment. Stories from the panel could be recorded and then featured in the application. P1(CM) proposed, *“If you had video clips of actual people...It'd be personal to your church...it could be part of the app. [Pretending to be Clara]: Did you know there are others in your church that have gone through mental challenges?”* Like P1, all participants brainstormed various ways their community could be reflected at key points within the

application. Table 29 presents the participants' brainstormed ideas paired with the participants' hypothetical impact of each feature and any concerns participants have about the idea.

Table 29: Reflecting Community in Digital Mental Health Promotion Applications

Community Feature	Description	Impact	Concerns
<i>Church Member Mental Wellness Stories</i>	Recorded stories from church members describing their experience of mental illness and/or treatment on their lives. Each story could be tailored to the session's topical content (e.g., a description of symptom presentation, a story about researching or interviewing providers, etc.)	Decreasing stigma through credible, targeted, and local self-disclosures.	Recruiting members to share their stories could be challenging. Technical aspects of story collection could also be challenging.
<i>Structured Support (Peer Matching)</i>	The application matches users to peers who can answer specific resource questions (e.g., recommending service providers, or swapping support strategies). The application reviews content submitted for safety or is implemented in such a way to restrict the use of open-ended responses.	Support coming directly from others within the community may feel more relevant, evocative and credible to users. Structured support may mitigate disinformation.	There is a potential for misinformation to be spread if content is not appropriately mediated.
<i>Unstructured Support (Group Forums)</i>	The group forum affords public-facing unstructured member support. Participants imagined that this feature could help users answer specific questions or share experiences.	A mechanism for crowdsourcing resources, suggestions, and stories from the wider cross-section of the community.	This option brought up concerns regarding misinformation or well-intentioned but stigmatizing responses from users.
<i>Mental Health Community Quiz Results</i>	Aggregated results from the myth and facts quiz could be posted. If results are positive, meaning a decrease in community myth endorsement, users could be notified.	Higher scores demonstrate a reduction in stigmatizing beliefs across the community.	Scores could be de-motivating if low scores are shared.
<i>Numerical Representation of App usage</i>	A counter representing total # of users displayed in the application user interface.	A way for members experiencing stigma or mental illness to see that others in the community are committed to supporting member experiences of mental illness and mental health promotion.	De-motivating if participants perceive the number is low in comparison to the overall church membership.
<i>Listing ministries / events that are church-sponsored</i>	The ECA includes a list of resources offered by the user's church.	A list would signal the church's commitment to mental health promotion and give users access to local mental wellness opportunities within their own communities.	None mentioned by participants.

8.7 Discussion

This study evaluated an ECA-facilitated mental health promotion and stigma reduction intervention designed for individuals experiencing depressive symptoms and community members who were not. Church members were recruited to participate in a two-week field deployment. They were assessed on their care-seeking intentions, confidence in discussing mental wellness with others, attitudes towards individuals experiencing depression, and their overall satisfaction with the app and ECA.

This application was not designed as a mental health treatment intervention, but rather, a mental health promotion intervention, where individuals with mental illness could explore the ways in which stigma has negatively impacted their ability to receive assistance and support from both the community and the mental health treatment system. Through their interviews and journals, participants characterized how the mental health promotion application facilitated shifts in attitudes towards mental illness, primed care-seeking behaviors, and could be used to support members experiencing mental illness. Participants provided insight into how the application should be introduced to their respective communities to ensure community uptake, sustain intervention engagement, and drive intervention outcomes more robustly. Although the literature supports individual-level mental health promotion interventions that center health education, screening, and resource identification for predominantly Black church communities, participants emphasized the importance of incorporating community features to increase communal support and intervention relevancy.

In the following subsections, I will discuss the limitations of this work, review each of the research questions and present opportunities for future work, offer method reflections, and discuss the critical and synergistic role of anchoring digital mental health promotion interventions in community settings to promote use and engagement.

8.7.1 Limitations

The evaluation study reported in this chapter has several limitations. First, a small convenience sample was used, and participants hailed from the two church communities where I conducted the formative work. As a result, the generalizability of these findings is limited. The intervention length of two weeks was short; thus, we do not know how participant attitudes toward depression or their care-seeking intentions will change over time. Nor is it possible to evaluate long-term knowledge retention. On a similar note, given the short study period, we also cannot determine whether participants who intended to seek help were able to execute their plans. Lastly, the study participants all self-identified as women. Thus, we do not know how other genders may respond to the intervention, nor can we claim the efficacy of the intervention for different genders.

8.7.2 Feasibility and Acceptability of an ECA-administered Mental Health Promotion Intervention

Findings from the summative evaluation revealed that participants from each treatment group were highly satisfied with the ECA-facilitated mental health promotion intervention. Participant interviews provided nuanced descriptions of how the ECA

promoted intervention accessibility, added credibility to the intervention, and, for some participants, the ECA’s embodiment was credited with normalizing the discussion of mental illness.

Multi-level ECA-based Intervention Approaches

Although the agent-based education sessions were designed for two stakeholders, persons with depression and community members, the ECA intervened on the individual level and focused on individual outcomes. The ECA supported members experiencing stigma by promoting self-acceptance, encouraging care-seeking, and educating users on essential matters to consider before disclosing to others. In addition, findings demonstrated that the intervention activated lay community members to be more empathetic toward those experiencing mental illness; however, given that stigma is a societal process, decreasing community stigma measurably may require a community-level component.

In their interviews, participants explored expanding stakeholder user types to include a tailored intervention for church leaders, purporting that doing so could help increase church leaders’ mental health literacy, provide leaders with actionable strategies for responding to member mental illness, encourage leaders to create a clear and consistent organizational message regarding mental health stigma and brainstorm strategies for promoting a culture of community support surrounding these issues. For instance, a future version of the intervention could include educational content facilitated by Clara to encourage leaders to include messages of support for those experiencing mental illness in their community’s prayer intentions list or prepare a sermon focused on mental health. Clara could provide leaders with pre-written, templated mental health awareness messages for leaders to personalize and disseminate using push notifications. Leaders could use the same messaging interface to share their own mental wellness stories and testimonials. Any of these approaches may be helpful in reducing stigma in the community by demonstrating leadership’s support for discussing these topics more openly and solidifying the church’s organizational message regarding mental illness. In addition, the intervention could assist ministry leaders in identifying how their ministry’s resources and current initiatives could be leveraged to explicitly promote mental wellness in the community (e.g., discussing treatment seeking as part of the bereavement ministry). The ECA could assist senior leadership in brainstorming how current community-level events or programs could be utilized to reduce stigma in the community and promote the exchange of social support.

Culturally-tailored ECAs for Mental Health Promotion

The intervention contained both surface and deep structure tailoring (Table 30).

Table 30: Examples of Surface and Deep Structure Tailoring for the Intersectional Identities of Race and Religion

Surface Structure Tailoring	Agent was designed to physically represent an African American woman who identifies herself as a church member.
Deep Structure Tailoring	The agent references Scriptures and incorporates culturally-relevant health content: 1. Stigma-reduction education, 2. education addressing stigma in religious communities, 3. racial and ethnic barriers to effective medical care including racial and ethnic medical bias.

The agent frames the intervention with a focus on collective goals and building collective-efficacy.

The agent encourages both formal and informal care-seeking and does not undermine resources and services provided within religious settings.

Core values of fellowship and social connectedness are mentioned in rapport building dialogues.

The agent was designed as a peer mentor and health liaison *not* explicitly a health professional.

Such tailoring efforts were based on prior work where I evaluated agent modalities (O’Leary et al., 2020). This summative study confirms my previous work, demonstrating that a community-based culturally-tailored ECA that employs surface and deep structure tailoring helps establish intervention credibility and relevance when creating automated health promotion initiatives for diverse populations. Participant interviews revealed that participants not only recognized but *valued* these tailoring efforts. However, while the tailoring integrated into this module was met with positivity and was viewed by most participants as a genuine reflection of the participants’ community, we cannot ignore that the implementation of the cultural and spiritual tailoring used a *one-size-fits-all* approach.

Future work should investigate technical innovations for adaptive spiritual tailoring when designing digital health promotion initiatives for religious communities and people. In prior work, participants mentioned that spiritual or religious content preferences vary between members and, at times, change for the individual. For an individual, spiritual needs for Scripture or religious content can shift depending on the person’s emotional state, life stressors, and experiences, or relationships (O’Leary et al., 2020). In this study, participants were recruited directly from their churches. Findings on participant MMRI-AA scores confirm a range of religiosity amongst the participants, demonstrating that, indeed, within one church community, there is a diverse range of spiritual needs.

Similarly, while my prior work showed that these churches label the predominant racial and ethnic identity of their memberships as Black, the membership is not monolithic but heterogeneous and diverse (O’Leary et al., 2021). In addition, more work exploring how to design these applications and modalities for intersectional identities (race, religion, gender, education, income, etc.) to create more effective interventions that resonate with diverse populations is needed. Thus, researchers interested in creating culturally tailored health promotion initiatives using ECAs should consider investigating opportunities for designing deeper reflections and more diverse representations of Black and African American intersectional identities.

8.7.3 Impact on Care-seeking

The second research question focused on determining the effectiveness of a smartphone-based ECA that utilizes stigma reduction strategies to enhance readiness for care-seeking in African American and Black church-affiliated users. I hypothesized that the intervention would increase care-seeking intentions. Participant interviews captured several descriptions of how the intervention

supported care-seeking intentions in all but one PwD user. Seven participant interviews (5 PwD and 2 CMs) focused on changes in attitudes and actions associated with care-seeking.

Behavior change theories are often helpful for describing and contextualizing participant behaviors and characterizations. Thus, to better situate and discuss these qualitative findings, I'll briefly review the Transtheoretical Model of Behavior Change (TTM). The TTM states that changing one's behavior is a process and that where an individual is in that process can be mapped to five stages of change: pre-contemplation, contemplation, preparation, action, and maintenance (Hashemzadeh et al., 2019). Table 31 defines each stage (Prochaska et al., 2013).

Table 31: Stage of change definitions

<i>Stage of change</i>	<i>Definition</i>
Precontemplation	Not currently considering or recognizing a need to change
Contemplation	Ambivalent to change
Preparation	Getting ready to change
Action	Actively changing behavior
Maintenance	Continued commitment to sustaining change

The TTM has been applied to numerous behaviors and populations (e.g., eating more fruits and vegetables, increasing physical activity, adopting vaccines, quitting smoking, etc.) (Hashemzadeh et al., 2019). Edwards et al. (2006) leveraged the TTM to conceptualize descriptions of help-seeking attempts by African American women who have experienced interpersonal violence. Similarly, the TTM can also help conceptualize the care-seeking intentions and experiences characterized by the interviewees in this summative study.

Participants reported engaging in self-assessment activities that helped them recognize symptoms they are currently experiencing, prompting them to seek help from close social relationships or a physician. Based on participant interviews, the intervention successfully empowered *contemplators*, individuals interested in change who had previously lacked confidence in requesting assistance from their medical provider. The intervention also seemed to support those in *preparation*, individuals who were determined to engage in formal treatment but who otherwise were unsure how, by encouraging and supplying resources for researching mental health services. Two users who opted into the CM user type reported that, before the intervention, they did not believe they had or had experienced depression or depressive symptoms. As mentioned in the findings post-intervention, both participants described having a newfound interest in investigating some of the symptoms that they reported to Clara during the conversationally administered depression screener. Notably, participants met the threshold for clinically significant depression on their self-reported pre-interaction depression screener but opted into the CM user type. It is possible that the community member framing, which allowed participants to enter the intervention without having to already self-identify as experiencing depression, was able to support *pre-contemplators*. These qualitative findings demonstrated that the intervention successfully primed participants' care-seeking behaviors across several stages of change.

Future work opportunities in digital mental health promotion for predominantly African American religious populations are vast. Such efforts could include a more formal evaluation of participant staging through pre/post measurement and leveraging the TTM to create an intervention to assist individuals through these stages. At this point, an analysis of the qualitative interviews does not suggest that the action stage was supported using this intervention. Thus, reconceptualizing the intervention to incorporate features and functionality that intentionally leverage change processes suitable for the action stage could be a worthwhile extension of this work.

Ethics of Priming Care-seeking

There is another critical component to consider when developing interventions that debunk treatment myths and encourage treatment-seeking. This dissertation focused on activating participant care-seeking intentions by including education surrounding formal treatment utilization such as information on how to initiate care and the importance of initiating treatment earlier to minimize symptom severity. The current intervention acknowledged long wait-times for care, a lack of providers of color, and recommended leveraging support groups and individual support strategies while waiting to begin treatment. However, as researchers we must ask ourselves, what does it mean to create interventions designed to activate treatment-seeking when the United States mental healthcare system is currently unable to meet exceedingly high treatment demands?

In the United States, there has been a shortage of mental health providers for years; now, post-pandemic, the treatment demand has soared from 11% pre-pandemic to 33% of adults reporting symptoms of anxiety and depression (Weiner, 2022). Unfortunately, this unprecedented demand for providers is co-occurring with an even more significant physician shortage, making accessing the mental healthcare system challenging (Nguyen et al., 2023). Too often, this shortage results in an individual who is ready for treatment but unable to efficiently and effectively access care, which can result in that individual's decompensation or an increase in symptom severity. Creative and collaborative solutions are required to increase treatment options while priming care-seeking intentions. This is especially true for medically underserved racial and ethnic groups such as African Americans (Nguyen et al., 2023). Thus, we must question the ethical implications of galvanizing individual care-seeking intentions when the path to receiving such care remains challenging to traverse and, in some instances, entirely unresponsive to an individual's acute needs. Thus, collaborations with local mental health clinics, training programs, or larger community-based healthcare organizations to increase access to providers before enhancing participant motivations to pursue treatment are essential next steps when disseminating these care-seeking encouragement interventions. In addition, such community-based collaborations could significantly strengthen the impact of the care-seeking intervention while also addressing the ethical implications of priming motivation to seek treatment without addressing underlying system-based inequities in treatment availability.

Future Work: Engaging Providers as Intervention Stakeholders

Collaborations with healthcare organizations such as local community health centers or clinics could create additional intervention opportunities, such as extending the intervention to train healthcare providers to be receptive to patient engagement during treatment. As discussed in the findings, P4 leveraged language related to advocating for one's care to request a memory exam from her primary care provider. This language was part of both the self-disclosure and care-seeking encouragement education. Initially, P4's provider dismissed her concerns. P4 described how she reiterated her goals for the appointment to the provider and requested the desired exam again. This resulted in the provider acknowledging and honoring the participant's request—shifting the tone of the appointment from prescriptive to one of mutual collaboration and partnership.

Fortunately, the provider chose to acknowledge the participant's concerns, which resulted in the provider changing the participant's treatment. From the participant's point of view, the appointment was an empowering experience where she successfully leveraged concepts shared by Clara to advocate for a particular type of treatment. Historically, the patient-provider relationship included a significant power imbalance where patients relied on their physician's expertise to access medical information (Kilbride & Joffe, 2018). The birth of the internet, social media, and digital health transformed this culture of care into a model where patients can more fully participate (Kilbride & Joffe, 2018). In the case of P4, the intervention was successful. However, the provider could have just as likely refused to accommodate the participant, creating tension in the patient-provider relationship. Researchers must recognize that when implementing individual-level interventions into complex social systems, unintended consequences may arise, like eroding the relationship between patients and their providers. As a result, there is an opportunity for future work to extend ECA-facilitated interventions to educate and train treating providers on how to be prepared for increased patient engagement to better support patient autonomy and maintain a positive patient-provider relationship.

8.7.4 Responses to Self-disclosure Planning

The third research question explored the impact of self-disclosure planning on Black church members' disclosure intentions. Given the limited research in this area, the disclosure planning session was intentionally designed as a high-level introduction to self-disclosure and thus was a single-session intervention. The disclosure planning education was incorporated into the overall mental health promotion and stigma reduction intervention to collect participant feedback on the ECA's neutral stance on self-disclosure and reactions to the concept of explicitly *planning* a self-disclosure through thoughtful preparation. Although all participants endorsed the session as a critical component of the overall intervention curriculum, the qualitative findings found no evidence that the self-disclosure planning education session impacted self-disclosure intentions in the study sample except for one participant (P4). As noted in the findings, P4's interview emphasized empowerment after the ECA self-disclosure session. P4's interview credited the part of the session where the ECA encouraged self-disclosing to a medical professional to facilitate specialized care and treatment. All remaining participants stated that while their self-disclosure intentions had not necessarily

changed, they did appreciate the session's emphasis on participant control over their self-disclosure process and narrative (e.g., deciding carefully who to tell or how much information to divulge, etc.). These findings resonate with similar related works that focus on empowering individuals with mental illness using therapeutic narrative enhancement techniques (Hansson et al., 2017; Yanos et al., 2015a).

The interviews identified topics ripe for future research. The self-disclosure planning session was not sensitive to prior participant experiences with self-disclosure, beliefs surrounding self-disclosure, nor tailored to self-disclosure contexts or participant goals for self-disclosure. For instance, self-disclosing disability status to an employer is an inherently different process with different desired outcomes than disclosure for psychological (i.e., self-disclosing status to reduce self-stigma) or psychosocial benefits (i.e., self-disclosing status to minimize strain on interpersonal relationships). Participants mentioned interest in both contexts (i.e., work and interpersonal relationships).

As previously noted in the findings, these same participants questioned the ECA's neutral stance on self-disclosure. Instead, they proposed that the ECA adapt how strongly she encourages self-disclosure based on how negatively the participant is impacted by concealing their mental health status. For instance, if the user was at risk of losing their job and disclosing their mental health status could afford them specific accommodations and rights, participants advocated for an ECA who would encourage them to self-disclose at work and then prepare them with resources surrounding their rights before the disclosure.

An adaptive approach to self-disclosure encouragement may be ethically problematic and technically challenging. For the ECA to adapt her stance, a method for determining how detrimental concealing mental health status is for the individual would have to be implemented. In addition to this challenge, further research would have to explore how to mitigate risk and ensure that participants are not worse off post-disclosure than concealment. In the current system, I mitigated risk by encouraging users interested in self-disclosing to first self-disclose to a medical professional. While I understand participant perspectives, I remain wary of any fully automated solutions that encourage users to self-disclose, as there is a lack of empirical evidence evaluating the impact of self-disclosure on individuals.

8.7.5 Impact on Confidence in Discussing Mental Wellness in the Church

The fourth research question examined how a smartphone-based ECA that utilizes stigma reduction strategies impacts confidence in discussing the topics of mental illness and mental health. The qualitative analysis of participant interviews and diaries contained several stories of participants sharing information from the intervention and discussing the intervention with others in their immediate social circles. In addition, participants mentioned both awareness and adoption of new conversational strategies (e.g., being a better listener, asking how they can help) for discussing these topics with other members. During their interviews, participants described powerful moments of initiating contact and supporting individuals from their churches *and* broader social circles.

While the impact of the intervention on confidence was not supported by the quantitative findings, the qualitative findings suggested a few reasons why we did not observe a significant change in confidence scores post-intervention. Interview findings indicate that while participants did act on sharing information, they still felt fairly hesitant and uncomfortable starting conversations. Though participants recognized and adopted new strategies for discussing these sensitive topics with others, they did not necessarily *feel confident* executing these skills. Indeed, this intervention was education-based and did not explicitly assist users in developing or practicing their newfound skills.

Thus, future work in this area should explore tech-enabled strategies for increasing user confidence surrounding these challenging topics in community settings. One such related work evaluated the use of a pedagogical and role-playing agent to train laypeople to promote influenza vaccination. In a single-use study, authors found that participants could execute these counseling techniques reasonably well during a conversation with an unvaccinated participant (Murali et al., 2022). These results indicate that using a role-playing agent to train laypeople to promote desirable health behaviors within their close social networks is promising. Incorporating such an approach into digital mental health promotion interventions for place-based communities may help improve participant confidence in discussing mental wellness and supporting members with mental illness.

8.7.6 Method Reflections

In the following subsections, I offer four reflections on the research methods and processes used for readers to consider when building on this work.

Exploring Intervention Length When Fielding Church-based Stigma Reduction Applications

Given my extensive experience partnering with these church communities, I'd like to offer my perspective on an ideal intervention duration as a next step if deploying the intervention more broadly to the entire church. When reviewing the digital care-seeking encouragement intervention literature, I found that intervention durations varied widely from 15-minute sessions to four-week-plus interventions with no set best practice recommendations for intervention duration (Jardine et al., 2022). For this study, I chose a study duration of two weeks. Given the exploratory nature of this work, I designed the evaluation to include a post-interaction interview where participants would reflect on session-specific details from the study period. Two weeks was an appropriate amount of time for ensuring: 1. participants could complete about two hours of ECA total content (the approximate intervention curriculum total time), and 2. participants would be able to remember content-specific details for the interviews.

For a fielded intervention, I recommend that the intervention be timed to match an awareness month or current community development goal. In both church communities, it is customary to introduce health initiatives and programming in coordination with a topically relevant awareness month. For instance, pre-pandemic, both churches sponsored events (e.g., a farmers market, workshops with a nutritionist, CPR training, etc.) to promote cardiovascular health and awareness during February's *American Heart Month*. A similar approach leveraging the digital mental health promotion intervention during a

mental health awareness month could be as effective in encouraging a coordinated community-wide intervention start date and communicating a clear time commitment to congregants.

For interventions intended to exceed four weeks, I recommend incorporating community features to create an intervention that evolves with and responds to the community's interests and supports the community's changing goals. Features that reflect community resources (e.g., a calendar of church-sponsored events), mediate community member interactions (e.g., resource exchanges), or augment the community's capacity to deliver caring initiatives for individuals experiencing mental illness (e.g., coordinating prayer intentions, meal support, or check-ins) are worthwhile approaches for sustaining intervention engagement for longer-term persistent initiatives.

Comparing Stigma-reduction Approaches

This dissertation leveraged multiple stigma approaches—education, paracontacts, and self-disclosure planning. Participants responded well to the mix of approaches and indicated that all aspects of the stigma-reduction curriculum were helpful. Although the experimental design of the evaluation makes it impossible to declare which stigma approach unequivocally worked best, the culturally informed paracontacts remained one of the most favored and valued aspects of the stigma-reduction intervention by study participants. The paracontacts supplemented the educational content presented by Clara by providing relevant and relatable examples for participants to reflect on and examine. I recommend that future work designing and evaluating digital mental health promotion interventions randomize the inclusion of approaches to investigate which approach is most effective for this population. In addition, an exciting and valuable extension of this work could be to examine the impact of including a more diverse sample of paracontacts (e.g., race, religiosity, gender, age) to understand better the impact of representing intersectional experiences of stigma and mental illness in these interventions.

Reflecting on the Strengths and Limitations of Community-engaged Research

This research drew heavily on principles of Community-Based Participatory Research (CBPR) (Holkup et al., 2004), specifically, treating the community as a unit of identity, exploring community strengths and resources, and leveraging data collections where community members drove research priorities and goals. As a result, the institutional support proffered by these two communities was both an inherent strength and limitation. Such institutional support means that the relationships between the participants and me were deeper than those created during a typical one-off research interaction. Over the course of the four-year project, I became a pseudo-member of their community, known to the individuals I interacted with and the wider church community to which they belonged. As a result, my identity evolved from an academic researcher to a partner and community ally. The relationships I established within these communities eventually mitigated the historical mistrust often found when large research-based institutions work in marginalized communities. This intentional relationship-building afforded me access to community members and established a sense of mutual respect where participants felt comfortable sharing their insights. Since this work

addressed a stigmatized health concern, I needed to build these relationships with members to recruit participants and maintain long-term research engagement.

Even still, one must acknowledge that although the community relationships established during the project were critical for supporting an iterative design process that democratizes research participation, my presence in these communities limits the generalizability of these findings. As a result, the effect of my involvement with the communities and my 1-1 interactions with participants cannot be separated from the impact of the intervention alone. One must ask, how would the intervention perform if I, the researcher, had not become the trusted community ally that I was? Unfortunately, given the history of research abuse between researchers and marginalized communities, it's unlikely that church members would feel comfortable using the application without a mechanism for establishing the credibility and source of the intervention. As a next step, I encourage researchers to consider intervention implementation approaches where researchers train community members to champion and disseminate the digital intervention to limit this confound. Without interested community liaisons and leaders, promising health promotion efforts may die on the vine if not bolstered by community member support.

Examining the Impact of Leadership Buy-In

It is worth highlighting that senior leadership from each church not only accepted the intervention but also assisted in the design of the application and provided support throughout the project. Members trusted their leaders to protect their community by scrutinizing new initiatives *before* endorsing them. This is especially true of initiatives developed and sponsored by individuals outside the community, such as academic researchers or healthcare provider organizations. Leadership buy-in was both a strength and a limitation of this work, and thus, it's valuable to reflect on how their support may have impacted the progression of this research and study outcomes.

Given regional differences and the inherent heterogeneity of predominantly Black churches, we must acknowledge the possibility that senior church leadership at other institutions may not be as accepting of the dissemination of a digital mental health promotion application. Indeed, related work examining the role of Black church leaders as connectors to outside mental health treatment has revealed notable differences in whether or not church leaders support and refer members to formal mental healthcare (Allen et al., 2010). Such variations existed between churches and varied between leaders of the same church. As a result, we should assume that support for a digital mental health promotion intervention may differ between and within other churches, which, in turn, may impact how the intervention performs if disseminated within a community where church leadership believes that engaging on matters of mental health is outside the scope of their churches purpose and mission.

8.7.7 A Case for Community Embeddedness

Since the early 2000s, the Department of Health and Human Services in the United States has provided federal funding for faith-based community health initiatives to address racial and ethnic health disparities in medical care (Peterson et al., 2002). Mobile

health interventions are often framed as solutions for the barriers presented by community-based initiatives delivered primarily in-person, such as a lack of privacy when addressing sensitive health topics or other barriers like scheduling, transportation, child or eldercare obligations, etc., (L. P. C. Brewer et al., 2020). Some digital health researchers have proposed that tech-enabled health interventions for historically marginalized communities should consider the community's integral role in individual health and, thus, look for ways to embed these interventions into community infrastructures (Parker et al., 2018). Although my research evaluated a digital tool focused on a sensitive health topic instead of physical wellness, based on these findings, there is a similarly strong case for embedding digital mental health promotion interventions into place-based communities. This may be especially true for communities that value togetherness and collective well-being, such as predominantly African American churches.

Participant interviews emphatically expressed expectations for the system to be embedded within their community. Participant findings explored ideas for how the intervention, as is, could be leveraged by their communities. Members also expressed a strong desire for features that facilitate the exchange of social support (e.g., swapping informational resources through tech-enabled peer matching, sharing support strategies through a group forum), reflect community assets (e.g., a digital calendar that displays church-run support groups), and signal community progress toward a culture of support (e.g., in-app display of myths and facts quiz results to build collective-efficacy).

Supporting Intervention Diffusion

Participants reasoning for these hybrid approaches to digital mental health was focused on how to effectively create awareness and adoption of the intervention in their communities. To better contextualize the interview data, I will utilize concepts from the Diffusion of Innovations theory (Kaminski, 2011), which describes how innovations like new technologies spread or diffuse through a community or population. By incorporating community in the design of these digital health interventions, there is potential to normalize intervention use amongst the community and increase the priority populations' sense of intervention compatibility (i.e., how the intervention is consistent with community values (Kaminski, 2011)). The most obvious way this was done in the current intervention was through the intervention's framing, which highlighted the importance of caring for others within the church. In addition, participant ideas for kick-off events and recurring small discussion groups may assist in establishing intervention awareness, adoption, and continued use. Participant in-app community feature ideas (e.g., in-app structured social support or pre-recorded church member stories of living with depression) were mentioned as additional mechanisms for promoting intervention engagement and continued use. Given this research, I recommend that future technology-enabled stigma reduction and digital health work consider consciously and carefully incorporating in-app community features in concert with community programming when partnering with community-based organizations.

Increasing Intervention Effectiveness

In terms of intervention effectiveness, Social Action Theory (SAT) helps conceptualize participant preferences and expectations of hybrid programming if the application is brought to scale. SAT presupposes that a person's social network and relationships are crucial in catalyzing individual-level behavior change (Hays, 2018). Related work suggests that because SAT goes beyond an individual focus and prioritizes the goals and missions of the individual's community organization, the individual's social network, *and* the individual, it may be more effective than behavior change models that focus solely on the individual or interpersonal levels (Hays, 2018). The use of SAT has been suggested for in-person community-based mental health initiatives anchored in predominantly African American churches as it may improve intervention outcomes by delivering an intervention consistent with religious values and cultural norms to effect change at the community level.

In the interviews, participants noted that while they found the intervention helpful in changing their intentions and attitudes, they desired an intervention that could *effect change*, like stigma reduction on the congregation level. In addition, the churches that were part of this dissertation work spoke to their preferences for fellowship and togetherness throughout our prior work (O'Leary et al., 2022; Stowell et al., 2020). SAT considers not just self-change processes, such as self-efficacy, but the impact of the individual's environment and social relationships (Hays, 2018). As Hays et al. (2018) point out, the church is an organizational structure with unique affordances, culture, institutional norms, community dynamics, and history, all of which may influence individual goals and behavioral processes. Church leaders are often considered role models by their congregants. Thus, their buy-in and championing of any initiative bears tremendous influence on whether or not individuals within the congregation will adopt an intervention. Our participants emphasized this point, stating that not only should the application be championed by leadership, but it must be *used* by leadership. SAT includes a concept called *social interdependence*, which suggests that the actions of others from within the individuals' community-based organization will invariably influence their actions (Hays, 2018). Findings from participant interviews align with this concept as participants prioritized brainstormed community features or community programming that emphasized taking action *together*.

SAT remains an underutilized theoretical framework when designing and evaluating health promotion interventions for predominantly Black churches (Hays, 2018). While SAT was helpful in this dissertation work for contextualizing participant findings, I recommend that future work explore how explicitly using SAT to guide the development and implementation of a church-based digital mental health initiative may improve intervention relevance and effectiveness.

Researchers should consider leveraging culturally relevant theoretical frameworks to guide the development of interventions designed for historically marginalized communities. A recent systematic review focused on examining the use of digital health tools for historically marginalized populations found that a limited number of interventions leveraged theoretical frameworks in both the design and evaluation of the system (Stowell et al., 2018). This dissertation work was guided by stigma theory and models of cultural tailoring.

8.8 Conclusion

This 14-day field deployment evaluated an ECA-facilitated mental health and stigma reduction intervention among twelve church members from two communities. Results included acceptance and feasibility of the ECA as demonstrated by interviews and post-intervention measures. Participant interviews highlighted how the intervention positively shifted participant attitudes toward mental illness, facilitated discussions of mental wellness amongst community members, and primed care-seeking intentions. This work adds to a subset of literature advocating for the thoughtful integration of digital health interventions into community settings using hybrid approaches. This work contributes to the fields of human-centered computing, digital mental health research, intelligent virtual agents, and digital faith-based health interventions for historically marginalized communities. To the best of my knowledge, this study adds to the limited literature on evaluating mental health promotion initiatives in predominantly African American church communities and is the first of its kind to deploy a digital mental health promotion intervention in these settings. This work serves as an example of how to successfully frame mental health promotion interventions for religious communities to benefit both individuals experiencing mental illness and community members without mental illness.

Chapter 9: Conclusions

In this dissertation, I reported on a 4-year community-engaged research project which aimed to create an mHealth intervention with and for predominantly Black church communities. Through an iterative and incremental design process, I reported on the efficacy of a digital mental health and stigma-reduction intervention with the primary aims of 1) increasing care-seeking intentions of church members experiencing depression and 2) reducing community stigma by activating lay church members as sources of mental health support. As a result, the intervention was tailored to two unique user groups: individuals with depressive symptoms interested in managing their mental wellness and community members interested in supporting others within the church.

9.1 Phase I: Discussion-based Focus Groups

In Phase I of this work, I engaged community members in discussion-based focus groups to understand these communities and their assets before engaging in participatory design activities. Phase I answered these research questions:

RQ1: How do participants describe their church communities and define their church's current goals and missions?

RQ2: In what ways are parishioners currently providing social support to one another?

RQ3: What are current church and church member technology practices?

This work provided me with a rich understanding of each church community. Participants characterized both churches' organizational history and missions. We learned that both organizations have robust care infrastructures responsible for the provision of social support for members and non-members alike. At a high level, we gained an understanding of the technology use practices of both communities.

9.1.1 Phase I Contributions

Phase I highlighted the assets of each community, from its institutional and cultural resources to its social capital shared between members. These three discussion-based focus groups helped build rapport between me and the members. Early-on, these focus groups signaled the importance of mental wellness as a topic of interest for members. Most importantly, Phase I provided a lens of understanding these communities that rejected a deficit and solution-focused model, an approach too often taken by researchers when partnering with historically marginalized communities.

9.2 Phase II: Participatory Co-design

Phase II of the preliminary work set forth the participatory co-design phase. During this phase, we focused on exploring health topics of interest to community members, and preliminary designs and modalities for the final system. This phase included four participatory sessions and investigated these research questions:

RQ1a: What health topics would church members identify as most essential for themselves and their community for a church-based mHealth app to address?

RQ1b: What types of church-based care networks, programming, and initiatives currently support these member-identified health topics? If at all?

RQ2a: What role and persona should a community-based, culturally-tailored ECA employ to support health and wellness in faith communities with predominantly Black and African American memberships?

RQ2b: How might participants adapt a health counseling conversation to include matters related to their faith and spirituality?

RQ4: What resources would church members expect or seek in a church-based health app?

RQ5: How can we design digital health tools responsive to the intersectional identities of race and religion in multicultural church communities?

In Phase II, I facilitated an *affinity diagramming* session with church members to identify health topics of relevance and importance for their communities. Participants identified mental health as one of their churches' highest health priorities but currently least addressable due, in part, to mental health stigma.

During the *design an agent* and *dialogue tailoring* participatory design sessions, participants highlighted the importance of designing a culturally-tailored agent with surface and deep-structure tailoring. Highlights from this work included that the agent should be designed as a church member and peer who utilizes spiritual language and content to support and encourage members.

Next, I conducted a *modular storyboard* session where participants finished storyboard narratives, selected mechanisms of support for the imagined app, and prioritized app-based features that they most desired. Using their storyboard designs, participants confirmed that ECA's would be a preferable modality when addressing sensitive health topics like mental health and illness. In addition, including resources within the application was listed as a necessary feature so as to increase organizational capacity for health-based resource identification.

Finally, a photo elicitation activity known as *Photovoice* helped us examine the intersectional identities of race and religion when designing health technologies for predominantly Black church communities. This work provided important insights for reflecting on how cultural and organizational identities influence organizational technology priorities. This work offered recommendations for conducting photovoice activities with this population.

9.2.1 Phase II Contributions

Contributions of this phase included:

- A description of the health priorities and intervention opportunities for predominantly Black churches. Mental health was identified as one of the highest health priorities.

- Preliminary design concepts for how ECAs can be used for culturally-tailored, responsive, and acceptable health interventions in predominantly Black churches.
- ECA design guidelines for tailoring community-based ECAs.
- A unique story completion design methodology called *Modular Storyboards*. A methodology that democratizes participant co-design work by providing participants with semi-structured design materials. Their use resulted in end products that reflected participant values and experiences while also leveraging behavior change concepts, strategies, and theories.

9.3 Phase III: Church Connect Field Deployment

Based on the work of Phase II we designed the first version of the *Church Connect* application. The app featured the first version of the co-designed ECA and a suite of health, spiritual, and social features and interventions. The application was deployed and evaluated in a 4-week field deployment with twenty-five church members from Church A and B. This study addressed this research question:

RQ: How can digital health tools be designed to support bodily, spiritual, and social well-being within predominantly Black church communities?

Results of this research uncovered a set of *community dynamics* defined as social processes, group norms, and goals that effect community change. These dynamics demonstrated how tech-enabled health interventions may enhance or disrupt the social dynamics of the communities that host them. This work emphasized the value of building mHealth applications that take a holistic or whole person approach to addressing well-being in religious communities. Such framings are more consistent with the church member values, making the intervention more compatible and relatable.

9.3.1 Phase III Contributions

- Demonstration of feasibility and efficacy of an ECA-administered health intervention that takes a trifold focus on well-being—supporting bodily, spiritual, and social well-being in predominantly Black churches.

9.4 Phase IV: Designing for Mental Wellness

At the end of the field deployment, I sought to develop the mental health promotion intervention, however, I needed additional input from church members. This resulted in my final participatory design session focused on identifying preferred features and concepts for a smartphone-based mental health promotion intervention. This work confirmed the recommendations from the albeit limited literature on mental health promotion in predominantly Black churches which recommended that interventions prioritize education, screening, and resource identification. Leaders did not support including features that included a communitysourced component as they feared that such approaches may reinforce stigma due to a lack of mental health literacy.

Participants selected psychoeducation, check-ins, screening, and resource lists as their most desired features for the mental health promotion intervention.

9.4.1 Phase IV Contributions

- A description of digital mental health promotion intervention opportunities for predominantly African American church communities.

9.5 Phase I–IV Design Implications

A synthesis of findings across Phases I–IV provided me with a set of six key design principles used to develop my final digital mental health and stigma-reduction system. They are:

- Address stigma when designing for mental wellness
- Integrate spiritual elements and faith-based concepts into both ECA design and health promotion content
- Offer resources to increase church capacity for mental health support
- Use a community-building framing to encourage members to interact with mental wellness and anti-stigma content
- Balance tensions between the values of maintaining privacy and discretion while also communicating care
- Prioritize features that build mental health literacy

9.6 Summative Evaluation

Finally, in my last study, I implemented an ECA-based mental health and stigma reduction smartphone application responsive to two user personas. Persona one included dialogue and content tailored for individuals with depression or depressive symptoms. Persona two featured dialogue and content tailored for individuals interested in supporting others in their community who do not self-identify or screen as depressed. This evaluation answered these research questions:

RQ1: Is it feasible and acceptable to provide a stigma reduction and care-seeking encouragement intervention using a smartphone-based ECA modeled after a peer health advisor?

RQ2: What effect does a smartphone-based ECA that utilizes stigma reduction strategies to enhance readiness for care-seeking have on African American and Black church-affiliated users?

RQ3: What impact does introducing the topic of disclosure have on Black church member disclosure intentions?

RQ4: How does a smartphone-based ECA that utilizes stigma reduction strategies increase confidence in discussing the topic of mental illness and mental health?

Overall, participants were satisfied with the system and found the ECA pleasant, non-threatening, and reflective of their community priorities and values. Participants endorsed the curriculum in its entirety. While all users believed that the user

type, they were assigned in the introduction session with the ECA was favorable and responsive to their intentions, they recommended that future versions adapt to changes and shifts in their attitudes towards their own mental wellness.

While intentions for care-seeking were not statistically significant, interviews captured detailed narratives that demonstrated how the intervention facilitated and catalyzed help-seeking intentions across user types.

Post-intervention disclosure intentions remained the same across the sample. However, all users endorsed the introduction to disclosure planning session as an importance aspect of the mental health promotion intervention. Participants desired a more complete version that included more content and guidance for different types of disclosure goals and audiences

Though we did not see confidence in discussing the topic of mental illness with others in the community increase, this intervention was successful in catalyzing the exchange of mental health support between participants and others within their close personal networks.

9.6.1 Summative Evaluation Contributions

- Design of an ECA-based intervention for mental health promotion and stigma reduction that takes a multi-level approach by supporting church members as active agents in improving their community's mental health and well-being.
- Demonstration of efficacy of an ECA-based mental health promotion and stigma-reduction intervention.

9.7 Concluding remarks

The mental health and stigma reduction intervention included intervention components for activating communal support, encouraging care-seeking, and stigma reduction. There are many exciting directions one could take to build on and extend this work. This research offers design implications and future directions for digital mental health and stigma-reduction interventions for predominantly African American church communities. This work recommends that digital health researchers interested in designing for these communities consider how to better incorporate synergistic community-based initiatives in partnership with community stakeholders, including how to engage church leadership. Although such efforts do not represent a technical innovation, they represent an essential program development and implementation step. Supporting community assets such as the community values of togetherness, fellowship, and community-building may lead to higher digital intervention awareness, adoption, engagement, and effectiveness.

The final intervention incorporated content related to participant faith and spiritual practice. Overall, participants appreciated these efforts and responded well to including this content. However, results from the summative evaluation confirmed previous findings that participants desired spiritual tailoring responsive to changes in individual preferences and

differing preferences among members. I recommend that future work consider developing adaptive spiritual-tailoring mechanisms to create more individually responsive and identity-affirming interventions.

The final evaluation of this system was short and included a small sample of participants. An immediate next step would be to evaluate the intervention using a larger sample of church members and compare it to a non-tailored system version. Doing so would provide us with an understanding of how tailoring the intervention impacts intervention effectiveness.

In addition, this intervention incorporated a community support framing that served to minimize barriers to intervention entry. However, participants emphasized a desire to move more fluidly between user types as they progressed through the curricula. Future research should consider how to create longitudinal interventions with content that adapts to shifts in participant goals and priorities when utilizing digital mental health promotion interventions.

Finally, findings from this work revealed that while community members learned new skills for responding and supporting members with mental illness, they did not necessarily feel confident using these skills. Future work developing community mental health literacy and mental health promotion interventions may consider developing technical approaches that support confidence building to increase church members' self-efficacy in responding to and supporting members with mental illness.

References

- Agyemang, C. (2005). Negro, Black, Black African, African Caribbean, African American or what? Labelling African origin populations in the health arena in the 21st century. *Journal of Epidemiology & Community Health*, 59(12), 1014–1018. <https://doi.org/10.1136/jech.2005.035964>
- Allen, A. J., Davey, M. P., & Davey, A. (2010). Being examples to the flock: The role of church leaders and African American families seeking mental health care services. *Contemporary Family Therapy*, 32(2), 117–134. <https://doi.org/10.1007/s10591-009-9108-4>
- Alvidrez, J., Snowden, L. R., & Kaiser, D. M. (2010). Involving Consumers in the Development of a Psychoeducational Booklet About Stigma for Black Mental Health Clients. *Health Promotion Practice*, 11(2), 249–258. <https://doi.org/10.1177/1524839908318286>
- Alvidrez, J., Snowden, L. R., Rao, S. M., & Boccellari, A. (2009). Psychoeducation to Address Stigma in Black Adults Referred for Mental Health Treatment: A Randomized Pilot Study. *Community Mental Health Journal*, 45(2), 127–136. <https://doi.org/10.1007/s10597-008-9169-0>
- Bell, G. (2006). No more SMS from Jesus: Ubicomp, religion and techno-spiritual practices. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 4206 LNCS, 141–158. https://doi.org/10.1007/11853565_9
- Benjamin, R. (2020). *Race after technology: Abolitionist tools for the new Jim code*. Oxford University Press.
- Berkley-Patton, J., Thompson, C. B., Williams, J., Christensen, K., Wainwright, C., Williams, E., Bradley-Ewing, A., Bauer, A., & Allsworth, J. (2021). Engaging the Faith Community in Designing a Church-Based Mental Health Screening and Linkage to Care Intervention. *Metropolitan Universities*, 32(1), 104–123. <https://doi.org/10.18060/24059>
- Berry, A. B. L., Lim, C. Y., Hirsch, T., Hartzler, A. L., Kiel, L. M., Bermet, Z. A., & Ralston, J. D. (2019). Supporting communication about values between people with multiple chronic conditions and their providers. *Conference on Human Factors in Computing Systems - Proceedings*, 10, 1–14. <https://doi.org/10.1145/3290605.3300700>
- Bickmore, T., & O’Leary, T. (2023). Chapter 9—Conversational agents on smartphones and the web. In N. Jacobson, T. Kowatsch, & L. Marsch (Eds.), *Digital Therapeutics for Mental Health and Addiction* (pp. 99–112). Academic Press. <https://doi.org/10.1016/B978-0-323-90045-4.00010-1>
- Bickmore, T., Rubin, A., & Simon, S. (2020a). *Substance Use Screening using Virtual Agents*. 1–7. <https://doi.org/10.1145/3383652.3423869>
- Bickmore, T., Rubin, A., & Simon, S. (2020b). Substance Use Screening using Virtual Agents: Towards Automated Screening, Brief Intervention, and Referral to Treatment (SBIRT). *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents*, 1–7. <https://doi.org/10.1145/3383652.3423869>

- Bickmore, T. W., Pfeifer, L. M., Byron, D., Forsythe, S., Henault, L. E., Jack, B. W., Silliman, R., & Paasche-Orlow, M. K. (2010). Usability of conversational agents by patients with inadequate health literacy: Evidence from two clinical trials. *Journal of Health Communication, 15*(S2), 197–210.
- Bickmore, T. W., & Picard, R. W. (2005). Establishing and maintaining long-term human-computer relationships. *ACM Transactions on Computer-Human Interaction (TOCHI), 12*(2), 293–327.
- Blank, M. B., Mahmood, M., Fox, J. C., & Guterbock, T. (2002). Alternative mental health services: The role of the Black church in the South. *American Journal of Public Health, 92*(10), 1668–1672. <https://doi.org/10.2105/AJPH.92.10.1668>
- Bopp, M., Baruth, M., Peterson, J. A., & Webb, B. L. (2013). Leading their flocks to health? Clergy health and the role of clergy in faith-based health promotion interventions. *Family and Community Health, 36*(3), 182–192. <https://doi.org/10.1097/FCH.0b013e31828e671c>
- Brabham, D. C., Ribisl, K. M., Kirchner, T. R., & Bernhardt, J. M. (2014). Crowdsourcing applications for public health. *American Journal of Preventive Medicine, 46*(2), 179–187. <https://doi.org/10.1016/j.amepre.2013.10.016>
- Brewer, L. C., Jenkins, S., Lackore, K., Johnson, J., Jones, C., Cooper, L. A., Radecki Breitkopf, C., Hayes, S. N., & Patten, C. (2018). mHealth Intervention Promoting Cardiovascular Health Among African-Americans: Recruitment and Baseline Characteristics of a Pilot Study. *JMIR Research Protocols, 7*(1), e31. <https://doi.org/10.2196/resprot.8842>
- Brewer, L. C., & Williams, D. R. (2019). We've Come This Far by Faith: The Role of the Black Church in Public Health. *American Journal of Public Health, 109*(3), 385–386. <https://doi.org/10.2105/AJPH.2018.304939>
- Brewer, L. P. C., Hayes, S. N., Jenkins, S. M., Lackore, K. A., Breitkopf, C. R., Cooper, L. A., & Patten, C. A. (2019). Improving Cardiovascular Health Among African-Americans Through Mobile Health: The FAITH! App Pilot Study. *Journal of General Internal Medicine, 34*(8), 1376–1378. <https://doi.org/10.1007/s11606-019-04936-5>
- Brewer, R. N., Morris, M. R., & Lindley, S. E. (2017). How to remember what to remember: Exploring possibilities for digital reminder systems. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 1*(3), 1–20.
- Bryant-Moore, K., Haynes, T., Davis, A. H., Turner, J., & Smith, J. (2019). Engaging the Faith Community in Mental Health Research: Lessons Learned from the REJOICE Project. *APHA's 2019 Annual Meeting and Expo (Nov. 2-Nov. 6)*. https://apha.confex.com/apha/2019/meetingapi.cgi/Paper/451131?filename=2019_Abstract451131.html&template=Word
- Campbell, M. K., Hudson, M. A., Resnicow, K., Blakeney, N., Paxton, A., & Baskin, M. (2007a). Church-based health promotion interventions: Evidence and lessons learned. *Annual Review of Public Health, 28*, 213–234. <https://doi.org/10.1146/annurev.publhealth.28.021406.144016>

- Campbell, M. K., Hudson, M. A., Resnicow, K., Blakeney, N., Paxton, A., & Baskin, M. (2007b). Church-based health promotion interventions: Evidence and lessons learned. *Annual Review of Public Health, 28*, 213–234. <https://doi.org/10.1146/annurev.publhealth.28.021406.144016>
- Campbell, R. D., & Mowbray, O. (2016). The Stigma of Depression: Black American Experiences. *Journal of Ethnic and Cultural Diversity in Social Work, 25*(4), 253–269. <https://doi.org/10.1080/15313204.2016.1187101>
- Cassell, J., Bickmore, T., Billinghamurst, M., Campbell, L., Chang, K., Vilhjálmsón, H., & Yan, H. (1999). Embodiment in conversational interfaces: Rea. *Conference on Human Factors in Computing Systems - Proceedings, May*, 520–527. <https://doi.org/10.1145/302979.303150>
- Cassell, J., Vilhjálmsón, H. H., & Bickmore, T. (2001). Beat: The behavior expression animation toolkit. *Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques*, 477–486.
- Catalani, C., & Minkler, M. (2010). Photovoice: A review of the literature in health and public health. *Health Education and Behavior, 37*(3), 424–451. <https://doi.org/10.1177/1090198109342084>
- Chaudoir, S. R., & Fisher, J. D. (2010). The disclosure processes model: Understanding disclosure decision making and postdisclosure outcomes among people living with a concealable stigmatized identity. *Psychological Bulletin, 136*(2), 236–256. <https://doi.org/10.1037/a0018193>
- Chisholm, C. J., Falk, C. A., & Kozak, L. E. (2020). Sticks, Ropes, Land: Confronting Colonial Practices in Public Space Design. *ACM International Conference Proceeding Series, 2*, 63–67. <https://doi.org/10.1145/3384772.3385140>
- Cho, A., Herrera, R. G., Chaidez, L., & Uriostegui, A. (2019). The “COMADRE” project: An asset-based design approach to connecting low-income Latinx families to out-of-school learning opportunities. *Conference on Human Factors in Computing Systems - Proceedings*, 1–14. <https://doi.org/10.1145/3290605.3300837>
- Church, A. M. E. (2023, May 22). *Church Mission*. AME Church. <https://www.ame-church.com/our-church/our-mission/>
- Claisse, C., Kasadha, B., Stumpf, S., & Durrant, A. C. (2022). Investigating Daily Practices of Self-care to Inform the Design of Supportive Health Technologies for Living and Ageing Well with HIV. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 1–19.
- Conner, K. (2010). Mental Health Treatment Seeking Among Older Adults with Depression: The Impact of Stigma and Race. *Am J Geriatr Psychiatry, 23*(1), 1–7. <https://doi.org/10.1038/jid.2014.371>
- Conner, K. O., Koeske, G., & Brown, C. (2009). Racial differences in attitudes toward professional mental health treatment: The mediating effect of Stigma. *Journal of Gerontological Social Work, 52*(7), 695–712. <https://doi.org/10.1080/01634370902914372>
- Corbin, J., & Strauss, A. (2014). *Basics of qualitative research: Techniques and procedures for developing grounded theory*. Sage publications.

- Corrigan, P. (2004). How stigma interferes with mental health care. *American Psychologist*, 59(7), 614–625.
<https://doi.org/10.1037/0003-066X.59.7.614>
- Corrigan, P. W. (2012). Challenging the Public Stigma of Mental Illness: A Meta-Analysis of Outcome Studies. *Psychiatric Services*, 63(10). <https://doi.org/10.1176/appi.ps.005292011>
- Corrigan, P. W., Bink, A. B., Schmidt, A., Jones, N., & Rüsch, N. (2016). What is the impact of self-stigma? Loss of self-respect and the “why try” effect. *Journal of Mental Health*, 25(1), 10–15. <https://doi.org/10.3109/09638237.2015.1021902>
- Corrigan, P. W., Druss, B. G., & Perlick, D. A. (2014). The impact of mental illness stigma on seeking and participating in mental health care. *Psychological Science in the Public Interest, Supplement*, 15(2), 37–70.
<https://doi.org/10.1177/1529100614531398>
- Corrigan, P. W., Kosyluk, K. A., Markowitz, F., Brown, R. L., Conlon, B., Rees, J., Rosenberg, J., Ellefson, S., & Al-Khouja, M. (2016). Mental illness stigma and disclosure in college students. *Journal of Mental Health*, 25(3), 224–230.
<https://doi.org/10.3109/09638237.2015.1101056>
- Corrigan, P. W., & Rao, D. (2012). On the self-stigma of mental illness: Stages, disclosure, and strategies for change. *Canadian Journal of Psychiatry*, 57(8), 464–469. <https://doi.org/10.1177/070674371205700804>
- Corrigan, P. W., & Watson, A. C. (2002). The paradox of self-stigma and mental illness. *Clinical Psychology: Science and Practice*, 9(1), 35–53. <https://doi.org/10.1093/clipsy/9.1.35>
- Costanza-Chock, S. (2020). *Design justice: Community-led practices to build the worlds we need*. The MIT Press.
- Cyrus, E., Clarke, R., Hadley, D., Bursac, Z., Trepka, M. J., Dévieux, J. G., Bagci, U., Furr-Holden, D., Coudray, M., Mariano, Y., Kiplagat, S., Noel, I., Ravelo, G., Paley, M., & Wagner, E. F. (2020). The Impact of COVID-19 on African American Communities in the United States. *Health Equity*, 4(1), 476–483. <https://doi.org/10.1089/heq.2020.0030>
- Dalky, H. F. (2012). Mental Illness Stigma Reduction Interventions: Review of Intervention Trials. *Western Journal of Nursing Research*, 34(4), 520–547. <https://doi.org/10.1177/0193945911400638>
- Davenport, A. D., & McClintock, H. F. (2021a). Let Go and Let God: A Study of Religiosity and Depressive Symptoms in the Black Church. *Community Mental Health Journal*, 57(7), 1340–1347. <https://doi.org/10.1007/s10597-020-00757-7>
- Davenport, A. D., & McClintock, H. F. (2021b). Religiosity and Attitudes Toward Treatment for Mental Health in the Black Church. *Race and Social Problems*, 13(3), 226–233. <https://doi.org/10.1007/s12552-020-09311-2>
- Davis, T. C., Long, S. W., Jackson, R. H., Mayeaux, E. J., George, R. B., Murphy, P. W., & Crouch, M. A. (1993). Rapid estimate of adult literacy in medicine: A shortened screening instrument. *Family Medicine*, 25(6), 391–395.
- Derose, K. P., Fox, S. A., Reigadas, E., & Hawes-Dawson, J. (2000). Church-based telephone mammography counseling with peer counselors. *Journal of Health Communication*, 5(2), 175–188. <https://doi.org/10.1080/108107300406884>

- Dickinson, J., Diaz, M., Le Dantec, C. A., & Erete, S. (2019). The cavalry ain't coming in to save us. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW), 1–21. <https://doi.org/10.1145/3359225>
- Egede, J. O., Price, D., Krishnan, D. B., Jaiswal, S., Elliott, N., Morriss, R., Trigo, M. J. G., Nixon, N., Liddle, P., Greenhalgh, C., & Valstar, M. (2021). Design and Evaluation of Virtual Human Mediated Tasks for Assessment of Depression and Anxiety. *Proceedings of the 21th ACM International Conference on Intelligent Virtual Agents*, 52–59. <https://doi.org/10.1145/3472306.3478361>
- Eng, E., Hatch, J., & Callan, A. (1985). Institutionalizing Social Support Through the Church and into the Community. *Health Education Quarterly*, 12(1), 81–92. <https://doi.org/10.1177/109019818501200107>
- Erves, J. C., Mayo-Gamble, T. L., Malin-Fair, A., Boyer, A., Joosten, Y., Vaughn, Y. C., Sherden, L., Luther, P., Miller, S., & Wilkins, C. H. (2017). Needs, Priorities, and Recommendations for Engaging Underrepresented Populations in Clinical Research: A Community Perspective. *Journal of Community Health*, 42(3), 472–480. <https://doi.org/10.1007/s10900-016-0279-2>
- Evans, J., Romaniuk, M., & Theal, R. (2021). Evaluation of mental health first aid training for family members of military veterans with a mental health condition. *BMC Psychiatry*, 21(1), 1–12.
- Evans-Lacko, S., London, J., Japhet, S., Rüsch, N., Flach, C., Corker, E., Henderson, C., & Thornicroft, G. (2012). Mass social contact interventions and their effect on mental health related stigma and intended discrimination. *BMC Public Health*, 12(1), 489. <https://doi.org/10.1186/1471-2458-12-489>
- Ferrell, B. R., Handzo, G., Picchi, T., Puchalski, C., & Rosa, W. E. (2020). The Urgency of Spiritual Care: COVID-19 and the Critical Need for Whole-Person Palliation. *Journal of Pain and Symptom Management*, 60(3), e7–e11. <https://doi.org/10.1016/j.jpainsymman.2020.06.034>
- Frazier, C., Mintz, L. B., & Mobley, M. (2005). A Multidimensional Look at Religious Involvement and Psychological Well-Being Among Urban Elderly African Americans. *Journal of Counseling Psychology*, 52(4), 583–590. <https://doi.org/10.1037/0022-0167.52.4.583>
- Gale, R. C., Wu, J., Erhardt, T., Bounthavong, M., Reardon, C. M., Damschroder, L. J., & Midboe, A. M. (2019). Comparison of rapid vs in-depth qualitative analytic methods from a process evaluation of academic detailing in the Veterans Health Administration. *Implementation Science*, 14(1), 1–12. <https://doi.org/10.1186/s13012-019-0853-y>
- Gaver, W., Blythe, M., Boucher, A., Jarvis, N., Bowers, J., & Wright, P. (2010). The prayer companion: Openness and specificity, materiality and spirituality. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2055–2064. <https://doi.org/10.1145/1753326.1753640>
- Glanz, K., Rimer, B. K., & Viswanath, K. (2015). *Health behavior: Theory, research, and practice*. John Wiley & Sons.

- Griffiths, K. M., Christensen, H., Jorm, A. F., Evans, K., & Groves, C. (2004). Effect of web-based depression literacy and cognitive-behavioural therapy interventions on stigmatising attitudes to depression: Randomised controlled trial. *British Journal of Psychiatry*, 185(OCT.), 342–349. <https://doi.org/10.1192/bjp.185.4.342>
- Gubrium, A. (2009). Digital Storytelling: An Emergent Method for Health Promotion Research and Practice. *Health Promotion Practice*, 10(2), 186–191. <https://doi.org/10.1177/1524839909332600>
- Gutierrez, J., Devia, C., Weiss, L., Chantarat, T., Ruddock, C., Linnell, J., Golub, M., Godfrey, L., Rosen, R., & Calman, N. (2014). Health, Community, and Spirituality: Evaluation of a Multicultural Faith-Based Diabetes Prevention Program. *The Diabetes Educator*, 40(2), 214–222. <https://doi.org/10.1177/0145721714521872>
- Halko, S., & Kientz, J. A. (2010). Personality and Persuasive Technology: An Exploratory Study on Health-Promoting Mobile Applications. In T. Ploug, P. Hasle, & H. Oinas-Kukkonen (Eds.), *Persuasive Technology* (Vol. 6137, pp. 150–161). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-13226-1_16
- Han, M., & Lee, E. (2018). Effectiveness of Mobile Health Application Use to Improve Health Behavior Changes: A Systematic Review of Randomized Controlled Trials. *Healthcare Informatics Research*, 24(3), 207. <https://doi.org/10.4258/hir.2018.24.3.207>
- Hanington, B., & Martin, B. (2019). *Universal methods of design expanded and revised: 125 Ways to research complex problems, develop innovative ideas, and design effective solutions*. Rockport publishers.
- Hansson, L., Lexén, A., & Holmén, J. (2017). The effectiveness of narrative enhancement and cognitive therapy: A randomized controlled study of a self-stigma intervention. *Social Psychiatry and Psychiatric Epidemiology*, 52(11), 1415–1423. <https://doi.org/10.1007/s00127-017-1385-x>
- Harkness, S. K., Kroska, A., & Pescosolido, B. A. (2016). The Self-Stigma of Psychiatric Patients: Implications for Identities, Emotions, and the Life Course. In *50 Years After Deinstitutionalization: Mental Illness in Contemporary Communities* (Vol. 17, pp. 207–233). Emerald Group Publishing Limited. <https://doi.org/10.1108/S1057-629020160000017008>
- Harrington, C. N., Erete, S., & Piper, A. M. (2019). Deconstructing community-based collaborative design: Towards more equitable participatory design engagements. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW). <https://doi.org/10.1145/3359318>
- Harris, A., Stefancic, J., & Delgado, R. (2017). *Critical race theory: An introduction*.
- Hashemzadeh, M., Rahimi, A., Zare-Farashbandi, F., Alavi-Naeini, A., & Daei, A. (2019). Transtheoretical model of health behavioral change: A systematic review. *Iranian Journal of Nursing and Midwifery Research*, 24(2), 83. https://doi.org/10.4103/ijnmr.IJNMR_94_17
- Hawkins, R. P., Kreuter, M., Resnicow, K., Fishbein, M., & Dijkstra, A. (2008a). Understanding tailoring in communicating about health. *Health Education Research*, 23(3), 454–466. <https://doi.org/10.1093/her/cyn004>

- Hawkins, R. P., Kreuter, M., Resnicow, K., Fishbein, M., & Dijkstra, A. (2008b). Understanding tailoring in communicating about health. *Health Education Research*, 23(3), 454–466. <https://doi.org/10.1093/her/cyn004>
- Haynes, T., Turner, J., Smith, J., Curran, G., Bryant-Moore, K., Ounpraseuth, S. T., Kramer, T., Harris, K., Hutchins, E., & Yeary, K. H. K. (2018). Reducing depressive symptoms through behavioral activation in churches: A Hybrid-2 randomized effectiveness-implementation design. *Contemporary Clinical Trials*, 64, 22–29. <https://doi.org/10.1016/j.cct.2017.11.010>
- Hays, K. (2018). Reconceptualizing church-based mental health promotion with African Americans: A social action theory approach. *Journal of Religion & Spirituality in Social Work: Social Thought*, 37(4), 351–372. <https://doi.org/10.1080/15426432.2018.1502643>
- Hays, K., & Aranda, M. P. (2016). Faith-Based Mental Health Interventions With African Americans: A Review. *Research on Social Work Practice*, 26(7), 777–789. <https://doi.org/10.1177/1049731515569356>
- Heimerl, K., Gawalt, B., Chen, K., Parikh, T. S., & Hartmann, B. (2012). Communitysourcing: Engaging local crowds to perform expert work via physical kiosks. *Conference on Human Factors in Computing Systems - Proceedings*, 1539–1548. <https://doi.org/10.1145/2207676.2208619>
- Henderson, C., Robinson, E., Evans-Lacko, S., & Thornicroft, G. (2017). Relationships between anti-stigma programme awareness, disclosure comfort and intended help-seeking regarding a mental health problem. *British Journal of Psychiatry*, 211(5), 316–322. <https://doi.org/10.1192/bjp.bp.116.195867>
- Hendricks Sloan, D., Peters, T., Johnson, K. S., Bowie, J. V., Ting, Y., & Aslakson, R. (2016). Church-Based Health Promotion Focused on Advance Care Planning and End-of-Life Care at Black Baptist Churches: A Cross-Sectional Survey. *Journal of Palliative Medicine*, 19(2), 190–194. <https://doi.org/10.1089/jpm.2015.0319>
- Hergenrather PhD, MEd, MRC, CRC, K. C., Rhodes PhD, MPH, CHES, S. D., Cowan MDiv, C. A., Bardhoshi MEd, G., & Pula MAEd, S. (2009). Photovoice as Community-Based Participatory Research: A Qualitative Review. *American Journal of Health Behavior*, 33(6), 686–698.
- Holkup, P. A., Tripp-Reimer, T., Salois, E. M., & Weinert, C. (2004). Community-based Participatory Research. *ANS. Advances in Nursing Science*, 27(3), 162–175.
- House, T. W. (2022, February 14). *FACT SHEET: Biden-Harris Administration Celebrates First Anniversary of the Reestablishment of the White House Office of Faith-Based and Neighborhood Partnerships*. The White House. <https://www.whitehouse.gov/briefing-room/statements-releases/2022/02/14/fact-sheet-biden-harris-administration-celebrates-first-anniversary-of-the-reestablishment-of-the-white-house-office-of-faith-based-and-neighborhood-partnerships/>
- Howe, J. (2006). The rise of crowdsourcing. *Wired Magazine*, 14(6), 176–183.

- Imber-Black, E. (2020). Rituals in the Time of COVID-19: Imagination, Responsiveness, and the Human Spirit. *Family Process*, 59(3), 912–921. <https://doi.org/10.1111/famp.12581>
- Jack, B. W., Bickmore, T., Yinusa-Nyahkoon, L., Reichert, M., Julce, C., Sidduri, N., Martin-Howard, J., Zhang, Z., Woodhams, E., Fernandez, J., Loafman, M., & Cabral, H. J. (2020). Improving the health of young African American women in the preconception period using health information technology: A randomised controlled trial. *The Lancet Digital Health*, 2(9), e475–e485. [https://doi.org/10.1016/S2589-7500\(20\)30189-8](https://doi.org/10.1016/S2589-7500(20)30189-8)
- James, D. C., Harville, C., Sears, C., Efunbumi, O., & Bondoc, I. (2017). Participation of African Americans in e-Health and m-Health Studies: A Systematic Review. *Telemedicine and E-Health*, 23(5), 351–364. <https://doi.org/10.1089/tmj.2016.0067>
- James, D. C. S., Harville, C., Efunbumi, O., Babazadeh, I., & Ali, S. (2017). “You Have to Approach Us Right”: A Qualitative Framework Analysis for Recruiting African Americans Into mHealth Research. *Health Education and Behavior*, 44(5), 781–790. <https://doi.org/10.1177/1090198117727324>
- Jardine, J., Bowman, R., & Doherty, G. (2022). Digital Interventions to Enhance Readiness for Psychological Therapy: Scoping Review. *Journal of Medical Internet Research*, 24(8), e37851. <https://doi.org/10.2196/37851>
- Jordan, A., Babuscio, T., Nich, C., & Carroll, K. M. (2021). A feasibility study providing substance use treatment in the Black church. *Journal of Substance Abuse Treatment*, 124, 108218.
- Joseph, R. P., Ainsworth, B. E., Hollingshead, K., Todd, M., & Keller, C. (2021). Results of a Culturally Tailored Smartphone-Delivered Physical Activity Intervention Among Midlife African American Women: Feasibility Trial. *JMIR mHealth and uHealth*, 9(4), e27383. <https://doi.org/10.2196/27383>
- Kaziunas, E., Klinkman, M. S., & Ackerman, M. S. (2019). Precarious interventions: Designing for ecologies of care. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW). <https://doi.org/10.1145/3359215>
- Kearns, M., Muldoon, O. T., Msetfi, R. M., & Surgenor, P. W. G. (2018). Identification Reduces Stigma of Mental Ill-Health: A Community-Based Study. *American Journal of Community Psychology*, 61(1–2), 229–239. <https://doi.org/10.1002/ajcp.12220>
- Kilbride, M. K., & Joffe, S. (2018). The New Age of Patient Autonomy. *JAMA*, 320(19), 1973–1974. <https://doi.org/10.1001/jama.2018.14382>
- King, A. C., Campero, I., Sheats, J. L., Sweet, C. M. C., Hauser, M. E., Banda, J. A., Ahn, D. K., & Bickmore, T. (2020). PHYSICAL ACTIVITY ADVISING BY HUMANS VS. COMPUTERS IN UNDERSERVED POPULATIONS: THE COMPASS2 TRIAL MAJOR RESULTS. *ANNALS OF BEHAVIORAL MEDICINE*, 54, S526–S526.

- Ko, H.-C., Wang, L.-L., & Xu, Y.-T. (2013). Understanding the Different Types of Social Support Offered by Audience to A-List Diary-Like and Informative Bloggers. *Cyberpsychology, Behavior, and Social Networking*, 16(3), 194–199. <https://doi.org/10.1089/cyber.2012.0297>
- Kroenke, K., Spitzer, R. L., & Williams, J. B. (2003). The Patient Health Questionnaire-2: Validity of a two-item depression screener. *Medical Care*, 1284–1292.
- Kroenke, K., Strine, T. W., Spitzer, R. L., Williams, J. B. W., Berry, J. T., & Mokdad, A. H. (2009). The PHQ-8 as a measure of current depression in the general population. *Journal of Affective Disorders*, 114(1–3), 163–173. <https://doi.org/10.1016/j.jad.2008.06.026>
- Lahijani, A. Y., King, A. R., Gullatte, M. M., Hennink, M., & Bednarczyk, R. A. (2021). HPV Vaccine Promotion: The church as an agent of change. *Social Science & Medicine*, 268, 113375. <https://doi.org/10.1016/j.socscimed.2020.113375>
- Lancaster, K. J., Schoenthaler, A. M., Midberry, S. A., Watts, S. O., Nulty, M. R., Cole, H. V., Ige, E., Chaplin, W., & Ogedegbe, G. (2014). Rationale and design of Faith-based Approaches in the Treatment of Hypertension (FAITH), a lifestyle intervention targeting blood pressure control among black church members. *American Heart Journal*, 167(3), 301–307. <https://doi.org/10.1016/j.ahj.2013.10.026>
- Laranjo, L., Dunn, A. G., Tong, H. L., Kocaballi, A. B., Chen, J., Bashir, R., Surian, D., Gallego, B., Magrabi, F., Lau, A. Y. S., & Coiera, E. (2018). Conversational agents in healthcare: A systematic review. *Journal of the American Medical Informatics Association*, 25(9), 1248–1258. <https://doi.org/10.1093/jamia/ocy072>
- Link, B. G., & Phelan, J. (2014). Stigma power. *Social Science and Medicine*, 103, 24–32. <https://doi.org/10.1016/j.socscimed.2013.07.035>
- Lipira, L., Nevin, P. E., Frey, S., Velonjara, J., Endeshaw, M., Kumar, S., Mohanraj, R., Kerani, R. P., Simoni, J. M., & Rao, D. (2019). The Positive Living Program: Development and Pilot Evaluation of a Multimedia Behavioral Intervention to Address HIV-Related Stigma and Depression Among African-Immigrant People Living With HIV in a Large, Northwestern U.S. Metropolitan Area. *Journal of the Association of Nurses in AIDS Care*, 30(2), 224–231. <https://doi.org/10.1097/JNC.0000000000000037>
- Lukachko, A., Myer, I., & Hankerson, S. (2015). Religiosity and Mental Health Service Utilization Among African-Americans. *Journal of Nervous & Mental Disease*, 203(8), 578–582. <https://doi.org/10.1097/NMD.0000000000000334>
- Mantovani, N., Pizzolati, M., & Edge, D. (2017). Exploring the relationship between stigma and help-seeking for mental illness in African-descended faith communities in the UK. *Health Expectations*, 20(3), 373–384. <https://doi.org/10.1111/hex.12464>

- Martínez-Miranda, J., Martínez, A., Ramos, R., Aguilar, H., Jiménez, L., Arias, H., Rosales, G., & Valencia, E. (2019). Assessment of users' acceptability of a mobile-based embodied conversational agent for the prevention and detection of suicidal behaviour. *Journal of Medical Systems*, 43(8), 246. <https://doi.org/10.1007/s10916-019-1387-1>
- McKenna, S. A., & Main, D. S. (2013). The role and influence of key informants in community-engaged research: A critical perspective. *Action Research*, 11(2), 113–124. <https://doi.org/10.1177/1476750312473342>
- McLean Hospital. (2018, January 5). *Stories | Deconstructing Stigma*. <https://deconstructingstigma.org/stories>
- McNabb, W., Quinn, M., Kerver, J., Cook, S., & Karrison, T. (1997). The PATHWAYS church-based weight loss program for urban African-American women at risk for diabetes. *Diabetes Care*, 20(10), 1518–1523. <https://doi.org/10.2337/diacare.20.10.1518>
- Miller, W. R., & Rollnick, S. (2012). *Motivational interviewing: Helping people change*. Guilford press.
- Millett, G. A., Jones, A. T., Benkeser, D., Baral, S., Mercer, L., Beyrer, C., Honermann, B., Lankiewicz, E., Mena, L., Crowley, J. S., Sherwood, J., & Sullivan, P. S. (2020). Assessing differential impacts of COVID-19 on black communities. *Annals of Epidemiology*, 47, 37–44. <https://doi.org/10.1016/j.annepidem.2020.05.003>
- Mitchell, T. (2021, February 16). Faith Among Black Americans. *Pew Research Center's Religion & Public Life Project*. <https://www.pewresearch.org/religion/2021/02/16/faith-among-black-americans/>
- Mittal, D., Sullivan, G., Chekuri, L., Allee, E., & Corrigan, P. W. (2012). Empirical studies of self-stigma reduction strategies: A critical review of the literature. *Psychiatric Services*, 63(10), 974–981. <https://doi.org/10.1176/appi.ps.201100459>
- Mohamed, B. (n.d.). 10 new findings about faith among Black Americans. *Pew Research Center*. Retrieved July 12, 2023, from <https://www.pewresearch.org/short-reads/2021/02/16/10-new-findings-about-faith-among-black-americans/>
- Mohatt, N. V., Boeckmann, R., Winkel, N., Mohatt, D. F., & Shore, J. (2017). Military Mental Health First Aid: Development and Preliminary Efficacy of a Community Training for Improving Knowledge, Attitudes, and Helping Behaviors. *Military Medicine*, 182(1), e1576–e1583. <https://doi.org/10.7205/MILMED-D-16-00033>
- Murphy, L. (2000). *Down by the riverside: Readings in African American Religion* (Vol. 8). NYU Press.
- NAMI. (2020, July 14). *Chris Hubbard discusses stigma around mental health for pro athletes | NAMI: National Alliance on Mental Illness*. <https://www.nami.org/Press-Media/In-The-News/2020/Chris-Hubbard-discusses-stigma-around-mental-health-for-pro-athletes>
- Neely-Fairbanks, S. Y., Rojas-Guyler, L., Nabors, L., & Banjo, O. (2018). Mental illness knowledge, stigma, help-seeking behaviors, spirituality and the African American church. *American Journal of Health Studies*, 33(4), 162–174.
- Newton, R. L., Carter, L. A., Johnson, W., Zhang, D., Larrivee, S., Kennedy, B. M., Harris, M., & Hsia, D. S. (2018). A church-based weight loss intervention in African American adults using text messages (LEAN Study): Cluster randomized controlled trial. *Journal of Medical Internet Research*, 20(8), 1–14. <https://doi.org/10.2196/jmir.9816>

- Nguyen, A. M., Schaler-Haynes, M., Chou, J., Nowels, M., Llaneza, D. H., & Kozlov, E. (2023). Increasing access to a diverse mental health workforce through emergency reciprocity licensure. *Journal of Medical Regulation*, 109(1), 5–21.
- Noar, S., Selby, C., & Harris, M. (2007). Does Tailoring Matter? Meta-Analytic Review of Tailored Print Health Behavior Change Interventions. *Psychological Bulletin*, 133, 673–693. <https://doi.org/10.1037/0033-2909.133.4.673>
- NW, 1615 L. St, Washington, S. 800, & Inquiries, D. 20036 U.-419-4300 | M.-857-8562 | F.-419-4372 | M. (n.d.-a). Mobile Fact Sheet. *Pew Research Center: Internet, Science & Tech*. Retrieved October 23, 2023, from <https://www.pewresearch.org/internet/fact-sheet/mobile/>
- NW, 1615 L. St, Washington, S. 800, & Inquiries, D. 20036 U.-419-4300 | M.-857-8562 | F.-419-4372 | M. (n.d.-b). Religious Landscape Study. *Pew Research Center's Religion & Public Life Project*. Retrieved October 23, 2023, from <https://www.pewresearch.org/religion/religious-landscape-study/>
- Ogbonnaya-ogburu, I. F., Smith, A. D. R., To, A., & Toyama, K. (2020). *Critical Race Theory for HCI*. 1–16.
- O'Leary, T. K., Parmar, D., Olafsson, S., Paasche-Orlow, M., Bickmore, T., & Parker, A. G. (2022). Community Dynamics in Technospiritual Interventions: Lessons Learned from a Church-based mHealth Pilot. *CHI Conference on Human Factors in Computing Systems*, 1–23. <https://doi.org/10.1145/3491102.3517700>
- O'Leary, T. K., Stowell, E., Hoffman, J. A., Paasche-Orlow, M., Bickmore, T., & Parker, A. G. (2021). Examining the Intersections of Race, Religion & Community Technologies: A Photovoice Study. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–19. <https://doi.org/10.1145/3411764.3445418>
- O'Leary, T. K., Stowell, E., Kimani, E., Parmar, D., Olafsson, S., Hoffman, J., Parker, A. G., Paasche-Orlow, M. K., & Bickmore, T. (2020). Community-Based Cultural Tailoring of Virtual Agents. *Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents, IVA 2020*. <https://doi.org/10.1145/3383652.3423875>
- Ormond, A., Barbour, R., Lewis, R. K., Montgomery, A., & Ponds, A. (2019). Results from a community mental health assessment in a historically black church: A call for action. *Journal of Prevention and Intervention in the Community*, 0(0), 1–16. <https://doi.org/10.1080/10852352.2019.1654257>
- Pan, W., Shen, C., & Feng, B. (2017). You Get What You Give: Understanding Reply Reciprocity and Social Capital in Online Health Support Forums. *Journal of Health Communication*, 22(1), 45–52. <https://doi.org/10.1080/10810730.2016.1250845>
- Pattyn, E., Verhaeghe, M., Sercu, C., & Bracke, P. (2014). Public Stigma and Self-Stigma: Differential Association With Attitudes Toward Formal and Informal Help Seeking. *Psychiatric Services*, 65(2), 232–238. <https://doi.org/10.1176/appi.ps.201200561>

- Pescosolido, B. A., Boyer, C. A., & Medina, T. R. (2013). The Social Dynamics of Responding to Mental Health Problems. In C. S. Aneshensel, J. C. Phelan, & A. Bierman (Eds.), *Handbook of the Sociology of Mental Health* (pp. 505–524). Springer Netherlands. https://doi.org/10.1007/978-94-007-4276-5_24
- Pescosolido, B. A., & Martin, J. K. (2015). The Stigma Complex. *Annual Review of Sociology*, 41, 87–116. <https://doi.org/10.1146/annurev-soc-071312-145702>
- Pescosolido, B. A., Martin, J. K., Lang, A., & Olafsdottir, S. (2008). Rethinking theoretical approaches to stigma: A Framework Integrating Normative Influences on Stigma (FINIS). *Social Science and Medicine*, 67(3), 431–440. <https://doi.org/10.1016/j.socscimed.2008.03.018>
- Pescosolido, B. A., & Olafsdottir, S. (2013). Beyond dichotomies: Confronting the complexity of how and why individuals come or do not come to mental health care. *World Psychiatry*, 12(3), 269–271. <https://doi.org/10.1002/wps.20072>
- Peteet, J. R. (2019). Approaching Religiously Reinforced Mental Health Stigma: A Conceptual Framework. *Psychiatric Services*, 70(9), 846–848. <https://doi.org/10.1176/appi.ps.201900005>
- Peterson, J., Atwood, J. R., & Yates, B. (2002). Key elements for church-based health promotion programs: Outcome-based literature review. *Public Health Nursing*, 19(6), 401–411. <https://doi.org/10.1046/j.1525-1446.2002.19602.x>
- Phelan, J. C., & Link, B. G. (2004). Fear of people with mental illnesses: The role of personal and impersonal contact and exposure to threat or harm. *Journal of Health and Social Behavior*, 45(1), 68–80. <https://doi.org/10.1177/002214650404500105>
- Philip, P., Micoulaud-Franchi, J.-A., Sagaspe, P., Sevin, E. D., Olive, J., Bioulac, S., & Sauteraud, A. (2017). Virtual human as a new diagnostic tool, a proof of concept study in the field of major depressive disorders. *Scientific Reports*, 7(1), 42656. <https://doi.org/10.1038/srep42656>
- Prochaska, J. O., Norcross, J. C., & DiClemente, C. C. (2013). Applying the Stages of Change. In G. P. Koocher, J. C. Norcross, & B. A. Greene (Eds.), *Psychologists' Desk Reference* (pp. 177–181). Oxford University Press. <https://doi.org/10.1093/med:psych/9780199845491.003.0034>
- Rao, D., Desmond, M., Andrasik, M., Rasberry, T., Lambert, N., Cohn, S. E., & Simoni, J. (2012). Feasibility, acceptability, and preliminary efficacy of the unity workshop: An internalized stigma reduction intervention for african american women living with hiv. *AIDS Patient Care and STDs*, 26(10), 614–620. <https://doi.org/10.1089/apc.2012.0106>
- Rao, D., Feinglass, J., & Corrigan, P. (2007). Racial and ethnic disparities in mental illness stigma. *Journal of Nervous and Mental Disease*, 195(12), 1020–1023. <https://doi.org/10.1097/NMD.0b013e31815c046e>
- Resnicow, K., Baranowski, T., Ahluwalia, J. S., & Braithwaite, R. L. (1999). Cultural sensitivity in public health: Defined and demystified. *Ethnicity & Disease*, 9(1), 10–21.

- Resnicow, K., Wallace, D. C., Jackson, A., Digirolamo, A., Odom, E., Wang, T., Dudley, W. N., Davis, M., Mitchell, D., & Baranowski, T. (2000). Dietary change through African American churches: Baseline results and program description of the eat for life trial. *Journal of Cancer Education : The Official Journal of the American Association for Cancer Education*, 15(3), 156–163. <https://doi.org/10.1080/08858190009528685>
- Ring, L., Bickmore, T., & Pedrelli, P. (2016a). An affectively aware virtual therapist for depression counseling. *ACM SIGCHI Conference on Human Factors in Computing Systems (CHI) Workshop on Computing and Mental Health*.
- Ring, L., Bickmore, T., & Pedrelli, P. (2016b). Real-Time Tailoring of Depression Counseling by Conversational Agent. *Iproceedings*, 2(1), e27. <https://doi.org/10.2196/iproc.6065>
- Rivera, K. J., Zhang, J. Y., Mohr, D. C., Wescott, A. B., & Bamgbose Pederson, A. (2021). A Narrative Review of Mental Illness Stigma Reduction Interventions Among African Americans in The United States. *Journal of Mental Health & Clinical Psychology*, 5(2), 20–31. <https://doi.org/10.29245/2578-2959/2021/2.1235>
- Rodríguez-Rivas, M. E., Cangas, A. J., Cariola, L. A., Varela, J. J., & Valdebenito, S. (2022). Innovative Technology–Based Interventions to Reduce Stigma Toward People With Mental Illness: Systematic Review and Meta-analysis. *JMIR Serious Games*, 10(2), e35099. <https://doi.org/10.2196/35099>
- Rusch, L. C., Kanter, J. W., Brondino, M. J., Weeks, C. E., & Bowe, W. M. (2010). Biomedical stigma reduction programs produce negative but transient effects on a depressed low-income community sample. *Journal of Social and Clinical Psychology*, 29(9), 1020–1030. <https://doi.org/10.1521/jscp.2010.29.9.1020>
- Samuel-Hodge, C. D., Keyserling, T. C., France, R., Ingram, A. F., Johnston, L. F., Davis, L. P., Davis, G., & Cole, A. S. (2006). A church-based diabetes self-management education program for African Americans with type 2 diabetes. *Preventing Chronic Disease*, 3(3), 1–16. <https://doi.org/10.13016/wg5d-64bf>
- Schorling, J. B., Roach, J., Siegel, M., Baturka, N., Hunt, D. E., Guterbock, T. M., & Stewart, H. L. (1997). A trial of church-based smoking cessation interventions for rural African Americans. *Preventive Medicine*, 26(1), 92–101. <https://doi.org/10.1006/pmed.1996.9988>
- Schueller, S. M., Hunter, J. F., Figueroa, C., & Aguilera, A. (2019). Use of Digital Mental Health for Marginalized and Underserved Populations. *Current Treatment Options in Psychiatry*, 6(3), 243–255. <https://doi.org/10.1007/s40501-019-00181-z>
- Shefer, G., Rose, D., Nellums, L., Thornicroft, G., Henderson, C., & Evans-Lacko, S. (2013). Our community is the worst: The influence of cultural beliefs on stigma, relationships with family and help-seeking in three ethnic communities in London. *International Journal of Social Psychiatry*, 59(6), 535–544. <https://doi.org/10.1177/0020764012443759>

- Shirley M, M., Elizabeth, W., Garlandria, J., Amy, Z., Latina, B., Cathy, B., & Tanya, S. (2022). Cultural Tailoring of Electronic Behavior Change Interventions for African Americans. *Journal of Nursing and Practice*, 5(1).
<https://doi.org/10.36959/545/407>
- Smith, C. E., Kaur, A., Gach, K. Z., Terveen, L., Kreitzer, M. J., & O'Conner-Von, S. (2021). What is Spiritual Support and How Might It Impact the Design of Online Communities? *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), 1–42. <https://doi.org/10.1145/3449117>
- Smith, C. E., Levonian, Z., Ma, H., Giaquinto, R., Lein-Mcdonough, G., Li, Z., O'conner-Von, S., & Yarosh, S. (2020). “I Cannot Do All of This Alone”: Exploring Instrumental and Prayer Support in Online Health Communities. *ACM Transactions on Computer-Human Interaction*, 27(5), 1–41. <https://doi.org/10.1145/3402855>
- Stowell, E., O'Leary, T. K., Kimani, E., Paasche-Orlow, M. K., Bickmore, T., & Parker, A. G. (2020). *Investigating Opportunities for Crowdsourcing in Church-Based Health Interventions*. 1–12.
<https://doi.org/10.1145/3313831.3376833>
- Suganuma, S., Sakamoto, D., & Shimoyama, H. (2018). An Embodied Conversational Agent for Unguided Internet-Based Cognitive Behavior Therapy in Preventative Mental Health: Feasibility and Acceptability Pilot Trial. *JMIR Mental Health*, 5(3), e10454. <https://doi.org/10.2196/10454>
- The Black Church | American Experience | PBS*. (n.d.). Retrieved July 12, 2023, from
<https://www.pbs.org/wgbh/americanexperience/features/godinamerica-black-church/>
- Times, T. N. Y. (2022, July 29). How George Floyd Died, and What Happened Next. *The New York Times*.
<https://www.nytimes.com/article/george-floyd.html>
- Triantafyllidis, A., Kondylakis, H., Votis, K., Tzovaras, D., Maglaveras, N., & Rahimi, K. (2019). Features, outcomes, and challenges in mobile health interventions for patients living with chronic diseases: A review of systematic reviews. *International Journal of Medical Informatics*, 132, 103984. <https://doi.org/10.1016/j.ijmedinf.2019.103984>
- Van Der Veen, Y. J. J., De Zwart, O., MacKenbach, J., & Richardus, J. H. (2010). Cultural tailoring for the promotion of Hepatitis B screening in Turkish Dutch: A protocol for a randomized controlled trial. *BMC Public Health*, 10.
<https://doi.org/10.1186/1471-2458-10-674>
- Wang, C. (1997). Photovoice Ethics: Perspectives from Flint Photovoice. *Health Education and Behaviour*, 37(October), 560–572.
- Wang, C. C., Morrel-samuels, S., Hutchison, P. M., Bell, L., & Pestronk, R. M. (2004). *Flint Photovoice: Community Building Among Youths, Adults, and Policymakers*. 94(6).
- Ward, E., & Mengesha, M. (2013). Depression in African American men: A review of what we know and where we need to go from here. *American Journal of Orthopsychiatry*, 83(2 PART 3), 386–397. <https://doi.org/10.1111/ajop.12015>

- Wazny, K. (2018). Applications of crowdsourcing in health: An overview. *Journal of Global Health*, 8(1), 1–20.
<https://doi.org/10.7189/jogh.08.010502>
- We Need to Tell the Story of the Black Church* | Time. (n.d.). Retrieved July 12, 2023, from <https://time.com/5939921/henry-lous-gates-american-history-black-church/>
- Weiner, S. (2022). *A growing psychiatrist shortage and an enormous demand for mental health services*. AAMC.
<https://www.aamc.org/news/growing-psychiatrist-shortage-enormous-demand-mental-health-services>
- Wilcox, S., Laken, M., Anderson, T., Bopp, M., Bryant, D., Carter, R., Gethers, O., Jordan, J., McClorin, L., O’rourke, K., Parrott, A. W., Swinton, R., & Yancey, A. (2007). The Health-e-AME Faith-Based Physical Activity Initiative: Description and Baseline Findings. *Health Promotion Practice*, 8(1), 69–78.
<https://doi.org/10.1177/1524839905278902>
- Williams, D. R., & Rucker, T. D. (2000). Understanding and addressing racial disparities in health care. *Health Care Financing Review*, 21(4), 75.
- Williams, L. F., & Cousin, L. (2021). “A Charge to Keep I Have”: Black Pastors’ Perceptions of Their Influence on Health Behaviors and Outcomes in Their Churches and Communities. *Journal of Religion and Health*, 60(2), 1069–1082.
<https://doi.org/10.1007/s10943-021-01190-0>
- Winker, M. A. (2004). Measuring Race and Ethnicity: Why and How? *JAMA*, 292(13), 1612.
<https://doi.org/10.1001/jama.292.13.1612>
- Wyche, S. P. (2006). Technology in S-iritual 2ormation4 An 67-loratory Study of Com-uter Mediated Religious Communications. *CSCW*.
- Yanos, P. T., Lucksted, A., Drapalski, A. L., Roe, D., & Lysaker, P. (2015a). Interventions targeting mental health self-stigma. *Psychiatric Rehabilitation Journal*, 38(1), 171–178. <https://doi.org/10.1037/prj0000100>. Interventions
- Yanos, P. T., Lucksted, A., Drapalski, A. L., Roe, D., & Lysaker, P. (2015b). Interventions targeting mental health self-stigma: A review and comparison. *Psychiatric Rehabilitation Journal*, 38(2), 171–178. <https://doi.org/10.1037/prj0000100>
- YouVersion Bible App*. (2017, August 24). YouVersion. <https://www.youversion.com/the-bible-app/>
- Zhou, S., & Bickmore, T. (2022). A Virtual Counselor for Breast Cancer Genetic Counseling: Adaptive Pedagogy Leads to Greater Knowledge Gain. *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, 1–17.
<https://doi.org/10.1145/3491102.3517553>

Appendix A – Summative Evaluation Measures

PHQ-8

How often during the past 2 weeks were you bothered by...

	Not at all (1)	Several days (2)	More than half the days (3)	Nearly every day (4)
Little interest or pleasure in doing things	☐	☐	☐	☐
Feeling down, depressed, or hopeless (	☐	☐	☐	☐
Trouble falling or staying asleep, or sleeping too much	☐	☐	☐	☐
Feeling tired or having little energy (	☐	☐	☐	☐
Poor appetite or overeating	☐	☐	☐	☐
Feeling bad about yourself, or that you are a failure, or have let yourself or your family down	☐	☐	☐	☐
Trouble concentrating on things, such as reading the newspaper or watching television	☐	☐	☐	☐
Moving or speaking so slowly that other people could have noticed. Or the opposite - being so fidgety or restless that you have been moving around a lot more than usual	☐	☐	☐	☐

Additional Depression Screener Questions

Depression is characterized by the following symptoms: Sleeping too much or too little, eating too much or too little, having difficulty concentrating, feeling restless, difficulty sleeping, a lack of enjoyment in activities that you used to enjoy, a persistent feeling of sadness or hopelessness, tiredness or feeling slowed down, feelings of guilt like you've let others down, chronic physical symptoms that don't respond to treatment (e.g., headaches, digestive issues, chronic pain).

Given these symptoms please answer the remaining questions.

Q1 Do you currently self-identify as a person who experiences symptoms of depression?

Yes

No

Q2 Have you ever experienced symptoms of depression?

Yes

No

Q3 Are you currently in treatment for depression?

Yes

No

Q4 When you experience symptoms of depression, are the symptoms persistent and ongoing or do they go away after a short period of time? In this case, a short period of time is two weeks.

My symptoms are ongoing.

My symptoms usually last less than two weeks.

I don't have symptoms of depression.

Demographics

Q1 What category(ies) best describe your racial/ethnic identity(ies)?

Note: you may select more than one category and/or enter multiple specific identifiers in the box at the bottom.

- ☐ American Indian or Alaska Native—for example, Navajo Nation, Blackfeet Tribe, Muscogee (Creek) Nation, Mayan, Doyon, Native Village of Barrow Inupiat Traditional Government, etc.
- ☐ Asian or Asian American—for example, Chinese, Filipino, Asian Indian, Vietnamese, Korean, Japanese, etc.
- ☐ Black or African American—for example, African American, Jamaican, Haitian, Nigerian, Ethiopian, Ghanaian, etc.
- ☐ Hispanic or Latinx—for example, Mexican or Mexican American, Puerto Rican, Cuban, Dominican, Salvadoran, Colombian, etc.
- ☐ Middle Eastern or North African—for example, Lebanese, Iranian, Egyptian, Syrian, Moroccan, Algerian, etc.
- ☐ Native Hawaiian or other Pacific Islander—for example, Native Hawaiian, Samoan, Guamanian or Chamorro, Tongan, Fijian, Marshallese, etc.
- ☐ White—for example, German, Irish, English, Italian, Polish, French, etc.
- ☐ Biracial
- ☐ Other race or ethnicity—can specify below
- ☐ Prefer not to answer
- ☐ Specific racial/ethnic Identifier(s) from above categories:

Q2 What category(ies) best describe your ethnic identity(ies)?

- ☐ African
- ☐ African American
- ☐ Albanian
- ☐ American
- ☐ Armenian
- ☐ Asian Indian (From-Family from India)
- ☐ Brazilian
- ☐ Cambodian/Khmer
- ☐ Cape Verdean
- ☐ Caribbean Islander
- ☐ Chinese

- ☐ Colombian
- ☐ Cuban
- ☐ Dominican
- ☐ English
- ☐ European (other than already in list)
- ☐ Filipino
- ☐ French
- ☐ German
- ☐ Greek
- ☐ Guatemalan
- ☐ Haitian
- ☐ Honduran
- ☐ Irish
- ☐ Italian
- ☐ Japanese
- ☐ Jewish
- ☐ Korean
- ☐ Laotian
- ☐ Mexican/ Mexican American/ Chicano
- ☐ Middle Eastern
- ☐ Native American/First Nation
- ☐ Peruvian
- ☐ Polish
- ☐ Portuguese
- ☐ Puerto Rican

- ☐ Other _____
- ☐ Prefer not to Answer
- ☐ Specific Ethnic Identifier(s) from above categories: _____

Q4 What category(ies) best describe your gender identity(ies)?

- ☐ Genderqueer
- ☐ Man
- ☐ Non-binary / third gender
- ☐ Transgender
- ☐ Woman
- ☐ Prefer not to answer
- ☐ Prefer to self-describe as _____

Q5 What is the highest level of education you have completed?

- ☐ Less than a high school degree or equivalent
- ☐ High school degree or equivalent (e.g. GED)
- ☐ Some college, no degree
- ☐ Associate degree (e.g. AA, AS)
- ☐ Bachelor's degree (e.g. BA, BS)
- ☐ Master's degree (e.g. MA, MS, MEd)
- ☐ Professional degree (e.g. MD, DDS, DVM)
- ☐ Doctorate (e.g. PhD, EdD)
- ☐ Prefer not to answer
- ☐ Other (please specify) _____

Q6 What is your date of birth? Please use the format mm/dd/yyyy.

Q7 What is your occupation? If you are retired or otherwise not working please let us know.

Q8 How often do you read books?

- ☐ Never
- ☐ Less than once a week
- ☐ Once a week
- ☐ A few times a week
- ☐ Every day

Q9 How often do you watch TV shows and movies?

- ☐ Never
- ☐ Less than once a week
- ☐ Once a week
- ☐ A few times a week
- ☐ Every day

Q10 How much experience do you have with computers?

- ☐ I've never used one
- ☐ I've tried one a few times
- ☐ I use one regularly
- ☐ I'm an expert

MMRI-AA

Answer the following question using a scale of 1-5 where 1 is never and 5 is nearly every day.

	1 - Never	2	3	4	5 - Nearly Everyday
How often do you usually attend religious services?	☐	☐	☐	☐	☐

Are you an official member of a church or other place of worship?

☐ Yes

☐ No

How many church clubs or organizations do you belong to or participate in?

Answer the following question on a scale of 1 to 5 where 1 is never and 5 is nearly every day.

	1 - Never	2	3	4	5 - Nearly Every day
Besides regular worship, how often do you take part in other activities at your place of worship?	☐	☐	☐	☐	☐

Do you hold any positions or offices in your church or place of worship?

☐ Yes

☐ No

How often do you participate in any of these activities?

	1 - Never	2	3	4	5 - Nearly Everyday
How often do you read religious books or other religious materials?	☐	☐	☐	☐	☐
How often do you watch or listen to religious programs on TV, radio, or podcasts?	☐	☐	☐	☐	☐
How often do you pray?	☐	☐	☐	☐	☐
How often do you ask someone to pray for you?	☐	☐	☐	☐	☐

Answer the following question on a scale of 1-4, where 1 is not religious at all and 4 is very religious.

	1 - Not religious at all	2	3	4 - Very religious
How religious would you say you are?	☐	☐	☐	☐

Answer the following question using a scale of 1 to 4, where 1 is not important at all and 4 is very important.

	1 - Not important at all	2	3	4 - Very important
How important was religion in your home when you were growing up?	☐	☐	☐	☐
How important is it for parents to send or take their children to religious services?	☐	☐	☐	☐

DSS

Please read each statement and respond with how much you strongly disagree or strongly agree using the scale.

	Strongly disagree (1)	Somewhat disagree (2)	Neither agree nor disagree (3)	Somewhat agree (4)	Strongly agree (5)
People with depression could snap out of it if they wanted.	☐	☐	☐	☐	☐
Depression is a sign of personal weakness.	☐	☐	☐	☐	☐
Depression is not a real medical illness.	☐	☐	☐	☐	☐
People with depression are dangerous.	☐	☐	☐	☐	☐
It is best to avoid people with depression so you don't become depressed yourself.	☐	☐	☐	☐	☐

Please read each statement and respond with how much you strongly disagree or strongly agree using the scale.

	Strongly disagree (1)	Somewhat disagree (2)	Neither agree nor disagree (3)	Somewhat agree (4)	Strongly agree (5)
People with depression are unpredictable.	☐	☐	☐	☐	☐
If I had depression I would not tell anyone.	☐	☐	☐	☐	☐
I would not employ someone if I knew they had been depressed.	☐	☐	☐	☐	☐
I would not vote for a politician if I knew they had been depressed.	☐	☐	☐	☐	☐
Most people believe that people with depression could snap out of it if they wanted.	☐	☐	☐	☐	☐

	Strongly disagree (1)	Somewhat disagree (2)	Neither agree nor disagree (3)	Somewhat agree (4)	Strongly agree (5)
Most people believe that depression is a sign of personal weakness.	☐	☐	☐	☐	☐
Most people believe that depression is not a medical illness.	☐	☐	☐	☐	☐
Most people believe that people with depression are dangerous.	☐	☐	☐	☐	☐
Most people believe that it is best to avoid people with depression so that you don't become depressed yourself.	☐	☐	☐	☐	☐
Most people believe that people with depression are unpredictable.	☐	☐	☐	☐	☐

Please read each statement and respond with how much you strongly disagree or strongly agree using the scale.

	Strongly disagree (1)	Somewhat disagree (2)	Neither agree nor disagree (3)	Somewhat agree (4)	Strongly agree (5)
If they had depression, most people would not tell anyone.	☐	☐	☐	☐	☐
Most people would not employ someone they knew had been depressed.	☐	☐	☐	☐	☐
Most people would not vote for a politician they knew had been depressed.	☐	☐	☐	☐	☐

MHSIS

Please mark the box that best represents your opinion.

	(Extremely unlikely)	2	3	4	5	6	(Extremely likely)
If I had a mental health concern, I would intend to seek help from a mental health professional.	☐	☐	☐	☐	☐	☐	☐
If I had a mental health concern, I would try to seek help from a mental health professional.	☐	☐	☐	☐	☐	☐	☐
If I had a mental health concern, I would plan to seek help from a mental health professional.	☐	☐	☐	☐	☐	☐	☐

IASMHS

Please rate the following statements using a scale of 1 to 5, where 1 is 'disagree' and 5 is 'agree'. For questions that refer to a professional, a professional is anyone who is trained to work with individuals who are experiencing mental illness. This could include a primary care doctor, family physician, pastor, therapist, counselor, or psychiatrist.

	Disagree 1	Somewhat disagree 2	undecided 3	Somewhat agree 4	Agree 5
There are certain problems which should not be discussed outside of one's immediate family	☐	☐	☐	☐	☐
I would have a very good idea of what to do and who to talk to if I was concerned about my mental health and decided to seek professional help.	☐	☐	☐	☐	☐
I would not want my best friend to know if I were experiencing a mental illness.	☐	☐	☐	☐	☐
Keeping one's mind on a job is a good solution for avoiding personal worries and concerns.	☐	☐	☐	☐	☐
If good friends asked my advice about their mental illness, I might recommend that they see a professional.	☐	☐	☐	☐	☐

Please rate the following statements using a scale of 1 to 5, where 1 is 'disagree' and 5 is 'agree'.

	Disagree 1	Somewhat disagree 2	undecided 3	Somewhat agree 4	Agree 5
Being labeled as mentally ill carries with it a burden of shame	☐	☐	☐	☐	☐
It is probably best not to know everything about oneself	☐	☐	☐	☐	☐
If I were experiencing a mental illness at this point in my life, I would be confident that I could find relief in therapy.	☐	☐	☐	☐	☐
People should work out their own problems; getting professional help should be a last resort.	☐	☐	☐	☐	☐
If I were experiencing mental distress, I could get professional help if I wanted to.	☐	☐	☐	☐	☐

Please rate the following statements using a scale of 1 to 5, where 1 is 'disagree' and 5 is 'agree'.

	Disagree 1	Somewhat disagree 2	undecided 3	Somewhat agree 4	Agree 5
If I were experiencing a mental illness and Important people in my life found out they would think less of me.	☐	☐	☐	☐	☐
mental health problems, like many things, tend to work out by themselves	☐	☐	☐	☐	☐
It would be relatively easy for me to find the time to see a professional if I were experiencing a mental illness.	☐	☐	☐	☐	☐
There are experiences in my life and past that I would not discuss with anyone	☐	☐	☐	☐	☐
I would be interested in getting professional help if I were worried or upset for a long period of time	☐	☐	☐	☐	☐

Please rate the following statements using a scale of 1 to 5, where 1 is 'disagree' and 5 is 'agree'.

	Disagree 1	Somewhat disagree 2	undecided 3	Somewhat agree 4	Agree 5
I would be uncomfortable seeking professional help for a mental illness because people in my social circles might find out about it	☐	☐	☐	☐	☐
Being diagnosed with a mental illness would be a blot on a person's life.	☐	☐	☐	☐	☐
There is something admirable in the attitude of people who are willing to cope with their conflicts and fears without resorting to professional help	☐	☐	☐	☐	☐
If I believed I were having a mental breakdown, my first inclination would be to get professional attention	☐	☐	☐	☐	☐
I would feel uneasy going to a professional because of what some people would think.	☐	☐	☐	☐	☐

Please rate the following statements using a scale of 1 to 5, where 1 is 'disagree' and 5 is 'agree'.

	Disagree 1	Somewhat disagree 2	undecided 3	Somewhat agree 4	Agree 5
People with strong characters can recover from their mental illness by themselves and would have little need for professional help	☐	☐	☐	☐	☐
I would willingly confide intimate matters to an appropriate person if I thought it might help me or a member of my family.	☐	☐	☐	☐	☐
Had I received treatment for a mental illness, I would not feel that it ought to be "covered up"	☐	☐	☐	☐	☐

Confidence in Discussing Mental Health with Church Members

Please rate the following statements on a scale of 1-4 where 1 is you are not confident at all and 4 is very confident. How confident are you...

	Not Confidant at All - 1	Somewhat Confidant - 2	Confidant - 3	Very Confidant - 4
How confident are you in recognizing symptoms of depression.	☐	☐	☐	☐
How confident are you in knowing what to say to a church member who is experiencing depressive symptoms.	☐	☐	☐	☐
How confident are you in knowing what to do for a church member experiencing depressive symptoms.	☐	☐	☐	☐
How confident are you in your knowledge of depression.	☐	☐	☐	☐
How confident are you in discussing mental well-being with other church members.	☐	☐	☐	☐

Agent and System Usability

This last set of questions are about your satisfaction with the animated character named Clara.

Q1 How **satisfied** are you with the overall **app**?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very satisfied (7)

Q2 How much would you like to **continue** using the **app**?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very much (7)

Q3 How likely are you to **recommend the app** to other members of your church?

- ☐ not at all likely (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very likely (7)

Q4 How **easy** was it to use the app?

- ☐ very difficult (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very easy (7)

Q5 How **satisfied** are you with the animated **character**?

- ☐ not at all satisfied (1)
- ☐ (2)

- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very satisfied (7)

Q6 How much would you **like to continue working** with character?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very much (7)

Q7 How much do you **trust** the character?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very much (7)

Q8 How much do you **like** the character?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)

- ☐ (6)
- ☐ very much (7)

Q9 How would characterize your relationship with the character?

- ☐ complete stranger (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ close friend (7)

Q10 How much do you feel that the character cares about you?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very much (7)

Q11 How much do you feel that you and the character understand each other?

- ☐ not at all (1)
- ☐ (2)
- ☐ (3)
- ☐ (4)
- ☐ (5)
- ☐ (6)
- ☐ very much (7)

Appendix B - Summative Evaluation Journal Questions

Participants were instructed to complete a daily journal entry.

*Questions 1, 4, 7, 10, and 13 are the same question. This is intentional and was pointed out to participants.

1. What did you talk to Clara about today? What did you like or dislike?
2. How does discussing the topic of mental wellness with Clara make you feel?
3. Clara doesn't say that she has experienced symptoms of depression. How would Clara sharing her own story about living with a mental illness impact your experience?
4. What did you talk to Clara about today? What did you like or dislike?
5. Clara uses Scripture and Biblical references in parts of the education. How does this impact the mental wellness content? If at all?
6. Let's talk about trust. When it comes to matters of mental wellbeing, do you trust Clara? Why or why not?
7. What did you talk to Clara about today? What did you like or dislike?
8. When it comes to mental wellbeing, how might a system like this help you receive support from people in your community? What about the opposite, how might a system like this help you provide support to others?
9. When it comes to mental wellness, was there something you wanted to talk with Clara about that she has yet to address? Tell us about that.
10. What did you talk to Clara about today? What did you like or dislike?
11. Share a brief story about learning about mental wellness with Clara. This could be based on a real experience you had with her or a fictional one.
12. Share a brief story about telling someone else in your church about the mental wellness content. This could be based on a real experience or a fictional one.
13. What did you talk to Clara about today? What did you like or dislike?

Appendix C - Summative Evaluation Interview Guides

Interview Guide 1

Not all questions were asked of each participant. Rather the interview was guided by participants experiences with the system (the sessions they completed vs. didn't complete) as well as their responses.

Interview Guide 1 Start

1. Tell me a little bit about why you decided to volunteer for this research project.
(warm-up question)

Action Contexts

Establishing how the church supports the individual and collective health goals/missions.

2. Can you tell me about any health-related goals or missions that your church has? (*setting/organizational structure*)
3. Imagine that you have a friend with a mental illness. What current programs or people exist in the church who can help?
(*setting and organizational structures*)
Reframe: **How does your church currently address member experiences of mental illness, if at all?** (*setting and organizational structure*)
 - What **qualities** of your church help **support** these members?
 - What **qualities prevent wider support**?
 - What are the **benefits** of these programs?
 - What are the **challenges**?
 - What resources outside of the church do they currently direct members to, if any?
4. How would you **like to see mental health promoted** within your church? (*action state - social interdependence*)
 - Are there models that you've seen that work well?
 - Are there wishes you have regarding how members with a mental illness should be supported, but for whatever reason they haven't been put into place in your church?
5. Are there **areas of your physical health** that you feel impact your mental health? (*biological structures*)
Or prevent you from focusing on your mental health?

Self-change Processes

Individual experiences of mental illness/wellbeing

6. **When you think of someone with depression**, what do you think of? Can you describe them? (*Declarative knowledge*)

7. Have you or someone you know ever experienced symptoms of depression, anxiety, or mental illness? Can you tell me about that? (*mood & arousal*)***

*If YES to MI: Social influences and care-seeking/support****

8. What do your close family, friends, and church family think about mental healthcare?

- Do they think it is effective?

9. As things currently are, would you feel comfortable going to your church for support? Why/why not. (*social interaction processes*)

- How might telling people change things for you at church?

10. Would you feel comfortable telling other people in your church about your mental health? Why/Why not?

- What about a healthcare provider?
- What about other people in your life, like friends or family?

(self-disclosure)

*If No to MI: Social influences and care-seeking/support**

8. What do your close family, friends, and church family think about mental healthcare?

- Do they think it is effective?

9. If a member of your church told others in the church that they are depressed, what would happen? What wouldn't happen? (*social interaction processes*)

- How might telling people this change their experience in the church?
- Do you think members feel comfortable telling others in the church about their experience of mental illness? (*stigma*).

10. When it comes to mental health support, how effective do you think therapy is in managing mental health/ improving things like fatigue, stress, changes in appetite, or sleeplessness? (*outcome expectancies*)

Scale of 0 to 10. Where 0 is not effective at all and 10 is very effective.

- How does the healthcare system, in your experience, promote or support mental health and wellbeing? (*community structures*)

Action States

Current Health Protective Strategies

11. How do you currently support your own mental wellness?

Reframe: what do you do? What strategies do you use?

12. Who in your life currently supports these strategies? (*social interdependence*)

- How does your church/ church family fit into that, if at all? (*social interdependence*)

Extra Action Contexts

Perceptions of the healthcare system & care-seeking.

13. If you needed to, would you feel comfortable seeing a mental health professional? Why/Why not.

14. Would you feel comfortable sending someone you care about to see a mental health professional?

- Friends, family, church family?

15. From your experience and point of view: Do you believe the mental healthcare system can adequately address the needs of Black and African Americans? (*organizational structures*)

Interview Guide 2

Preamble

Over the last two weeks, you've interacted with a character named Clara on the phone and discussed matters of mental wellness with her. Now, it's time for us to learn about your experience.

I want to remind you; you are the expert here. So please don't worry about what your feedback means, just tell me your honest thoughts. Lastly, while you can share with me what you think others might think or feel, it's also important that you speak from your own experience.

Overall impressions (Clara and Clara's goal)

1. Imagine I'm a friend or family member. Can you describe Clara to me?
 - Who is she? Who is she to you? (*probing agent role and perceived relationship to user*)
2. When it comes to mental wellness, what was her purpose? What was she designed to do?
 - Did she fulfill this purpose?
3. When it comes to working with Clara on the topic of mental wellness, what did you like most? what did you dislike most?

Clara's Purpose - ask what doesn't come up in question 2.

4. Clara was designed to **educate church members about depression**. How did the application do?
 - What did you like?
 - What did you dislike?
5. Clara was also designed to **help church members feel more comfortable** talking about mental wellbeing. Did she meet this goal?
If Yes: How?
If No: Could she? How? What would it take?
6. Clara was also designed to **encourage church members to seek-out mental health support**. Did Clara meet this goal?
*Seeking help is defined as formal and informal.
If Yes: How?
If No: Could she? How? What would it take?

Impact of app on self-change processes & health protective actions (SAT concepts and constructs).

only PwD

7. How confident do you feel managing your mental health? (*Self-efficacy*) Has this changed?
 - How confident do you feel talking to others or telling others about your mental health?
 - How confident do you feel talking to others in your church about their mental health?
8. What are your goals when it comes to mental health and wellness? Have they changed? (*Individual goals*)

Session specific feedback

Stigma-reduction content Feedback

9. During one of the sessions with Clara, she shared rates of depression by race, gender, and age as well as Dominic's story about depression and anxiety. Overall, what did you think about the content? (*psychoeducation session*)
10. Clara also had content about mental health stigma. What was that like? (*Declarative knowledge*)(*social stigma session*)
 - helpful/not helpful?
 - Has it changed how you feel about yourself? Or think about yourself?
11. Clara talks about the potential benefits and drawbacks of disclosing mental illness to another person. How was the session? (*disclosure education*)
 - Was it useful? Why or why not.
 - Has it changed your feelings around telling others? If no, why not if yes why?
12. Clara had you complete a myths and facts quiz about mental health treatment and depression. (*myths and facts session*)
 - What did you think of that?
 - What myth surprised you the most?
 - Was there a myth or fact that you did not agree with?
 - Did you trust the information she shares?
13. Since you started using this app, has it changed your perspective on how members may think about mental illness?
 - If yes, why?

- If no, why not?
- Did it make the stigma you might feel worse? *related to the stigma special issue.

Encourage Care-seeking Session Intervention Questions - only PwD

14. There was a session called ‘learn about mental health support options’. Where Clara covered all the different kinds of support out there. (*EHS session*)

- What did you think of this session?
- How did this session impact your feelings around getting more support?

15. In this session, Clara asked if you had any concerns about professional mental healthcare and then listed some common concerns. What did you think of them? (*EHS session*)

- Do you feel like the concerns listed were accurate?
- What concerns did she leave out (if at all)?

16. Clara talked about different types of mental health care including formal things like therapy or medication as well as informal services like speaking with a pastor or attending a prayer conference.

- What did you think about this? (*EHS session*)
- Was it appropriate/sensitive/relevant?

17. Clara assigned some homework at the end of this session. These home works were things like ‘reaching out to a provider’, ‘researching a provider’, or ‘sharing what you learned with someone else’.

What did you think about these home works? (*EHS session*)

EHS - CM Session Questions - only CM

18. In this session, Clara provided some education about professional mental healthcare. She also shared some suggestions for how you might support a friend or family member. What did you think? (*How to Support Others Session*)

- Do you believe this was important?
- Would you be able to do this?
- What worked? What didn’t?

19. In this session, Clara asked if you had any concerns about professional mental healthcare and then listed some common concerns. What did you think of them? (*How to Support Others Session*)

- Do you feel like the concerns listed were accurate?
- What concerns did she leave out (if any)?

GRADUATION SESSION - only PwD

20. Clara shared a few support/self-care strategies for managing symptoms. These were things like meditation, eating well, maintaining friendships, or taking a walk. What did you think about these? (*support strategies – graduation session*)

- Were they helpful? Why/why not.
- Were there things that you wished were there?

GRADUATION SESSION - only CM

21. Clara shared a few strategies for creating a loving and supportive church (reducing stigma). What did you think of these strategies? (*anti-stigma messaging – graduation session*).

Exploring opportunities for app Community Impact

22. How could Clara help **promote mental wellness in the church**, (where that means getting people talking about it more openly about mental wellness)?

- What should Clara actually be addressing?
- How could this be used at your Church, if at all?
- In your opinion, how could Clara help people feel **more comfortable responding** to members who are experiencing depression? what should Clara be doing?
- How could the app compliment what’s currently happening in your church? How might it detract?
- Could an app like this decrease mental health stigma in your church? If yes, why, if no why not?

13. When it comes to supporting member mental wellness in the church, what else would you like to see from an app like this?

- What should Clara do to better support your own mental wellness?
- When it comes to encouraging people in your church to seek professional support, what else should the Clara app do? How else could we encourage folks?

24. Do you think an app like this one could make church members feel more comfortable reaching out to each other for support around mental wellness?

25. This content and curriculum was delivered by Clara, a character designed for a smartphone. What value, if any, does the character add? Should it be a character? Why or why not.

26. Outside of information and resources, what else should Clara be doing?

27. If you were to create your own curriculum more responsive to the landscape of your church what would it be? Are there other topics (outside of depression, that would be more useful)

28. How could it be introduced in your church?

Cultural Relevance

29. Did the app feel relevant and genuine to your church? To your cultural identity?

30. Did it seem culturally appropriate? When I say culturally appropriate, what I mean is did it address concerns in a way that felt responsive to your cultural identity?

Tailoring by User Type

Not all participants received the same curriculum. For instance, you received the [USER TYPE] curriculum. However, other participants received [OTHER USER TYPE].

The focus of the other curriculum was: to increase community support and knowledge about mental wellness.

The focus of the other curriculum was: to reduce self-stigma, the shame one might feel because of the discrimination and stigma associated with depression.

31. What do you think about this? How does it make you feel?

Appendix E – Summative Evaluation Protocol

Study Appointment 1

PRESENT STUDY & STUDY CONSENT

“You volunteered to help test a research prototype of an app designed to promote mental wellness in Black and African-American churches. You will be asked to test the app for a two-week period of time for about 10-15 minutes per day. In addition, I’ll be providing you with a notebook where you can take notes on your experience and answer one question per day. After testing the app for two weeks you and I will meet again by zoom where I get to hear all your thoughts! The things you liked and the things you didn’t like. You’ll receive a \$30 e gift card today and a second \$30 e gift card after our second appointment. Sounds okay so far?”

Consent is voluntary

“You should know your participation at every step of the study is completely voluntary. Meaning, if you feel uncomfortable answering certain questions that is fine. If you suddenly feel like you do not want to continue the study. That is also okay. Just let me know.”

Information is confidential

“All of your study information that we collect today will not be linked to any of your personally identifiable information like your name. Instead, you will be given a study ID which will be linked to your survey data. All your data will be kept on a secure server and will be encrypted.

Risks/Benefits

There is minimal risk associated with participating in this study. Your participation and feedback will benefit others and help us improve the overall app experience as well as drive our design choices for future features.

Compensation

You will be paid \$30 dollars for your participation today and another \$30 gift card at our next study appointment.

Time

The session today will take about 90 minutes (start to finish). During this session I will ask you to fill out a survey which will include a number of questions around your socio-demographics, and your feelings about mental health and wellness. After you complete that survey, I will help you download the app to your phone. You'll have your first conversation with the computer character while I'm on the line. I'll mute your video and mine to give you privacy. After you do the intro conversation, I will ask you some questions. And then you'll be all set and finished for today.

At the end of the 2 weeks, you and I will meet just like this again. You'll do another questionnaire and then we will chat about your experience.

If Participant signs consent form continue. Otherwise thank them for their time.

ICEBREAKER activity!

1. Why did you decide to volunteer for this study? Tell me a bit about that?

ADMINISTER PRE-INTERACTION QUESTIONNAIRES

Now, I'm going to have you fill out a questionnaire. The questionnaire will ask you questions about mental wellness, your feelings about mental health care, some demographic questions, and your feelings about religion and religious practices.

DOWNLOAD THE APPLICATION ONTO PARTICIPANT PHONE

OK get your smartphone out, I will be texting you the link you need to tap on your phone to download the app. Let me know when you are ready. Assist participant with downloading and registering the app.

FIRST AGENT INTERACTION

Now, let's have you do your first session with the character. During the conversation, the computer character will speak to you. In order for you to interact with the character you will see a menu of buttons pop up with things that you can say back to the character. Use these buttons and click on what you would like to say back to the character. I'll be outside the lab the entire time if you need me.

CONDUCT QUAL INTERVIEW USING INTERVIEW GUIDE 1

Now, I'm going to ask you some questions about mental wellness and your church community. If at any point you'd rather not continue the interview, or if you'd like to skip a question, please let me know. Every step of the study is voluntary.

THANK PARTICIPANT

We thank you so much for your time today, your expertise and point of view is important and will shape what we do next with this smartphone app. We couldn't do this without you. If you have any questions or concerns, you can contact the number on the consent form we provided.

EXPLAIN PAYMENT AGAIN & SCHEDULE APPOINTMENT 2

You will be paid \$30 in the next day or so. And then the final \$30 at the follow-up appointment with me! Let's schedule that now.

Study Appointment 2

GREET PARTICIPANT

Thank you so much for meeting with me again. For today, I'll have you complete your questionnaires first which will ask you questions about your experience over the last two weeks.

COLLECT JOURNAL FROM PARTICIPANT

I'd like to collect your journal from you. Thanks so much for taking the time to answer the questions.

ADMINISTER POST-INTERACTION QUESTIONNAIRES

Now, I'm going to have you fill out your last set of questionnaires. The questions are very similar to the ones you filled out last time. However, there are some new questions like questions about the character. This should take you about 15 minutes to complete.

CONDUCT QUAL INTERVIEW USING INTERVIEW GUIDE 2

Now, we are going to chat about your experience using the system and speaking with the character. This interview should take about 45 minutes. If at any point you would like to stop, please let me know. As a reminder all of these questionnaires are voluntary, so if I ask you a questionnaire that you would prefer to skip, please let me know and we'll move on to something else.

THANK THEM FOR THEIR TIME AND EXPLAIN STUDY PAYMENT

"I thank you so much for your time today, your expertise and point of view is important and will shape what we do next with this smartphone app. We couldn't do this without you. If you have any questions or concerns, you can contact the number on the consent form we provided. As a reminder, your study payment will be sent by email and will be a \$30 amazon giftcard."

Appendix E - Depression Feature Storyboard

Mental Wellbeing Part 1: Introduction

The goal of this story is to get your feedback on potential features to add to a health ministry app. These features aim to support mental wellbeing, specifically focusing on anxiety and depression. All feedback and suggestions for changes will help guide the design of these features.

Story Introduction

Jasmine has had a hard time this year. Her family has been struggling, and she tends to be the backbone of her home. She's the one who keeps everyone organized and happy. Lately, she realizes that she's feeling exhausted and is far more down than normal.



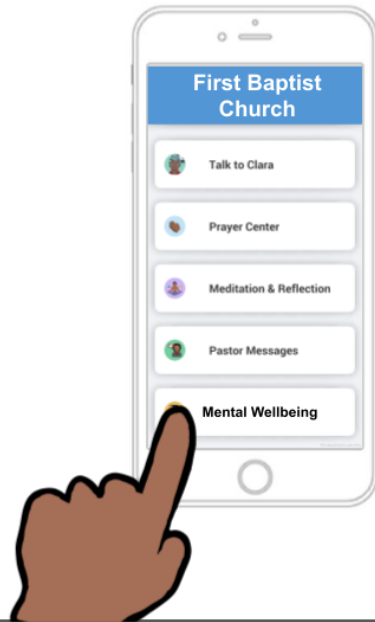
Jasmine isn't sure who to talk to. She's afraid that her family, and maybe even her church family, might treat her differently, or assume she's 'crazy'.



Jasmine also does not feel ready to jump into traditional therapy. She's not sure if she will feel understood. She's put off by the idea of medication.



Jasmine remembers that the Church App has some health and wellness resources on it. Jasmine opens the app to check out the Mental Well-being section.



Jasmine meets Clara, a computer character, who guides Jasmine through the app. Clara reminds Jasmine of a church member. Conversations with Clara can be short and fit in whenever Jasmine has time. To start, Clara keeps the conversation light. As Jasmine becomes comfortable with Clara, Clara talks more directly about depression and anxiety.

People can experience anxiety, depression, sadness, and stress in different ways. So, I would like to hear how you are feeling from your perspective.



Part 2: App Features

The next pages provide examples of each of the features Jasmine could use.

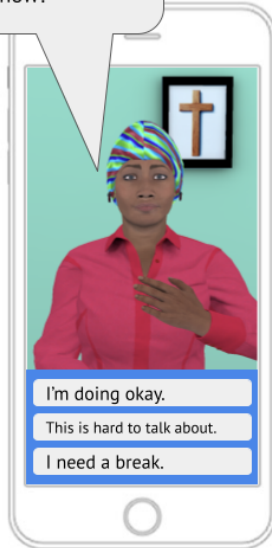
Which features would be the most helpful for someone like Jasmine?

How would you change the features?

Feature: Mid Conversation Check-In

Jasmine appreciates that Clara continues to check-in with her during conversations where the topic is more serious.

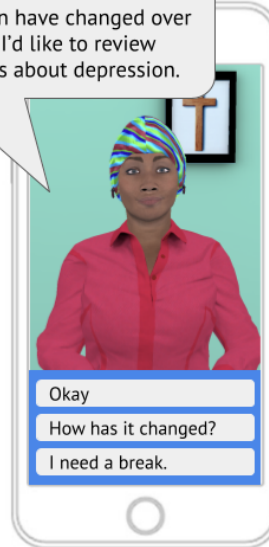
I know talking about these topics can be tough. I was just wondering how you are feeling right now?



Feature: Psychoeducation

During one of their conversations, Clara suggests some of the app's educational content. This includes "recognizing depression" and "depression myths." This helps her understand that depression is more common than she thought.

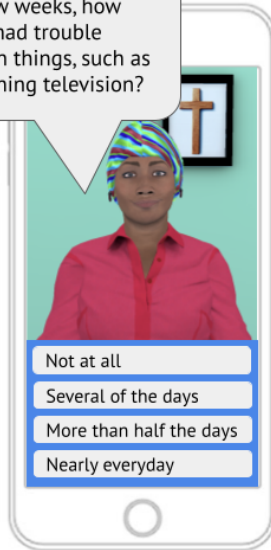
People's ideas about depression have changed over the years. I'd like to review some facts about depression.



Feature: Cognitive Behavioral Therapy

Jasmine is curious about the Cognitive Behavioral Therapy sessions with Clara. To start, Clara asks Jasmine a number of screening questions. Based on the screening questions, Clara recommends that Jasmine start by learning more about deep breathing exercises.

Over the last few weeks, how often have you had trouble concentrating on things, such as reading or watching television?



Over time, Clara helps Jasmine put into practice a number of CBT tools for coping such as: daily breathing exercises, thought restructuring, and activity scheduling. Jasmine finds these strategies to be helpful during challenging weeks.

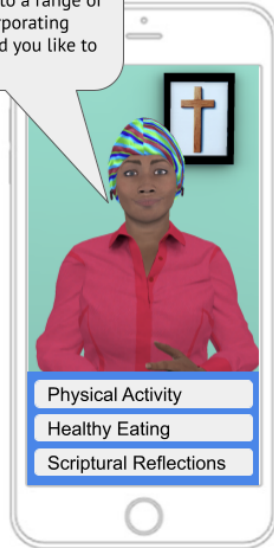
Let's start by finding a comfortable seated position. You can sit on a chair or on the floor. Allow yourself to settle. You can also put the phone down and just listen



Feature: Additional Coping Strategies

In addition to CBT strategies, Clara offers many other ideas for coping strategies. These include increasing physical activity, encouraging healthy eating practices, and scriptural reflections.

Finding the right strategies and resources that make you feel comfortable is important. I'll introduce you to a range of options, like incorporating faith. Which would you like to check out first?

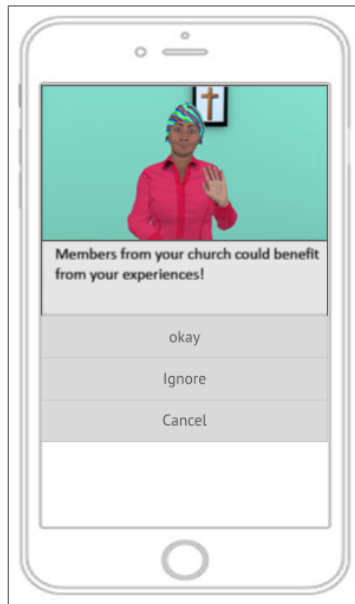


Jasmine is surprised to see many different options for coping strategies. She appreciates that Clara supports her spiritual growth alongside her mental health.

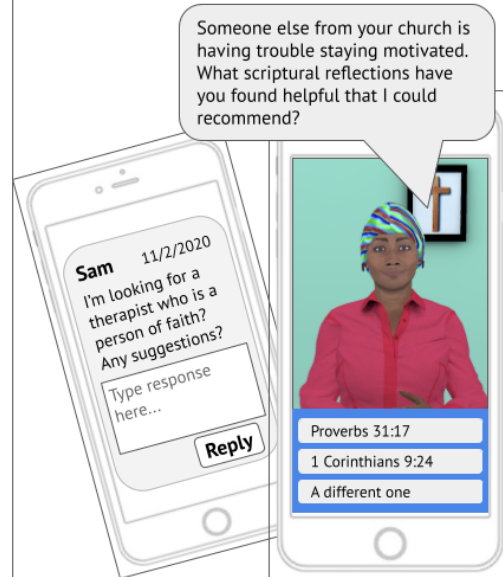


Feature: Share Resources

Helping others and serving her community is something that Jasmine really enjoys doing. But, lately it has been hard to do so. Clara suggests that she pass on some of her experiences to others in her church community. Curious, Jasmine takes a look.

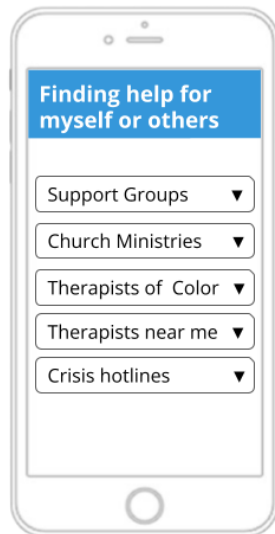


Jasmine can use the app to share the ways in which she's been successful managing her mental well-being. She can share strategies, recommend portions of the app directly, provide words of support, and post resources. Today, Jasmine answered a couple of requests for recommendations. Clara will share Jasmine's advice directly with the user.



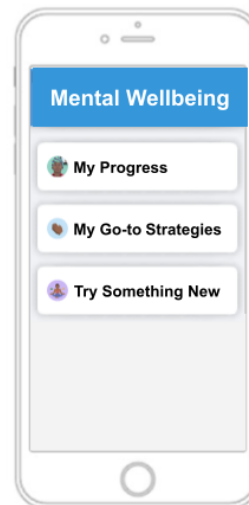
Feature: Find Resources

Throughout her time using the church app Jasmine uses the resource page. Today, She sees that her Church Health Ministry posted a link to a virtual event. She's found that content posted in the app feels relevant.



Feature: Self-Monitoring

After using the app regularly, Jasmine is able to look back on her progress, and can see what app-based coping strategies have helped her the most and when.



Appendix F – Depression Feature Interview Guide

CHURCH CONTEXT

1. Is depression a problem in your church community?
 - Tell me a bit more about that?
 - In what way?
2. How does your church currently address church member depression?
 - How has this worked?
 - What have been the benefits?
 - What hasn't worked?
 - Why?
3. How would you like to see church member mental health addressed in your church?
 - Are there models you've seen at other churches?
 - Are there wishes you have but for whatever reason aren't put into place?

MH STIGMA

4. How is mental illness or depression perceived in your church?
5. How does your church currently help church members address matters of mental health?
 - What resources does your church currently send people to?
 - When it comes to supporting members with mental illness, what barriers currently exist?
6. Do you feel that church members openly and freely disclose their experiences of mental illness to other members of your church?
 - What do you think plays into this decision at your church?
7. Have you, or someone else you know, ever experienced stigma related to mental health from other church members? *Reframe:*
8. Have you witnessed MH stigma at your church?
 - Can you describe what that stigma looked/felt like?
 - What were the consequences of that experience?

STORYBOARD OVERALL

9. Tell me your overall thoughts about the stories that you reviewed?
10. What were the strengths of the story?
11. Did the story feel realistic to you?
12. Did it feel realistic for your Church?
13. Did the story feel problematic to you?
14. Did you feel the story was problematic for your Church?
15. Imagine that this application exists with these features.
16. How would an app like this impact your community, if at all?
17. How would your community react?
18. What features would be most accepted?
20. What would be met with a healthy dose of skepticism?
21. How could an app like this help or hurt your work that you are already doing for your church? Does the app match your ministry goals?

FEATURES

22. Which features did you find most useful and why?
23. Which features did you find the least useful? Tell me why?
24. Now thinking about others at your church who are experiencing depression.
25. Which features do you think someone at your church might find the most useful? Why?
26. Which features do you think someone at your church might find the least useful? Why?
27. Did you find that any features were problematic?
 - Reframe: Dangerous? Harmful? Offensive?
28. Were there any features that seemed confusing?
 - Reframe: Was anything difficult to understand?
29. Were there features that you'd like more information on/ would like to discuss?

CBT-LED FEATURE

30. The computer character incorporated a feature called character-led CBT. Can you tell me your reactions to that? Problematic? Fine?

RESOURCE SHARING

31. We incorporated a way for church members to share resources using the application. Can you tell me your thoughts regarding this feature?

AGENT ROLE PLAY

32. The computer character incorporated a feature called role-play. Can you tell me your reactions to that?
Problematic? Fine?

MOTIVATION

33. We are interested in understanding what might make church members interested in using the depression content. Thinking of both the app overview and the content presented in the storyboards, what do you think would motivate someone like [character] to use the app content?
What would make a church member motivated to share resources in the app?

ProQuest Number: 30811592

INFORMATION TO ALL USERS

The quality and completeness of this reproduction is dependent on the quality and completeness of the copy made available to ProQuest.



Distributed by ProQuest LLC (2023).

Copyright of the Dissertation is held by the Author unless otherwise noted.

This work may be used in accordance with the terms of the Creative Commons license or other rights statement, as indicated in the copyright statement or in the metadata associated with this work. Unless otherwise specified in the copyright statement or the metadata, all rights are reserved by the copyright holder.

This work is protected against unauthorized copying under Title 17,
United States Code and other applicable copyright laws.

Microform Edition where available © ProQuest LLC. No reproduction or digitization of the Microform Edition is authorized without permission of ProQuest LLC.

ProQuest LLC
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 - 1346 USA